STRUCTURAL FIRETRAINER® T-500

MINOT AFB, ND

OPERATION & MAINTENANCE MANUAL

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1.0 INTRODUCTION

1.1 GENERAL

This manual provides all of the necessary information to operate the:

Structural FireTrainer® T-500

LOCATED AT:

Minot AFB, ND

MANUFACTURED BY:

Kidde Fire Trainers Inc.

If you have any questions regarding the operation or maintenance of this trainer that are not answered in this manual contact the Kidde Fire Trainers Customer Support Group at:

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Utilization of the **Structural FireTrainer® T-500** for live-fire training exercises brings a level of versatility and realism that cannot be duplicated using conventional fire fighting training methods. The operator of the trainer will be able to vary the difficulty, complexity or scale of the fires to challenge the skills of the trainees.

To maximize training effectiveness and use the full potential of the system, a thorough understanding of the trainer system and controls is necessary. This section describes the layout of the manual and the Safety Guidelines for the Trainer. Read and understand the Safety Guidelines and system controls before operating the Trainer.

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1.2 How To Use This Manual

The Operation Manual for the **Structural FireTrainer® T-500** is presented in the following order:

Section 1	**Introduction**	This section provides a brief functional description how to use the manual and relevant safety guidelines and instructions.
Section 2	**System Description**	This section provides an equipment overview and a functional description of the major equipment components that are part of the site.
Section 3	**Operation**	Detailed operating instructions including Theory of Operation are given in this section.
Section 4	**Maintenance & Servicing**	This section provides information on maintaining the various systems, assemblies and components of the trainer.
Section 5	**Parts Lists**	This section provides figures and parts list tables for each of the **Structural FireTrainer® T-500** major assemblies.
Section 6	**Drawings & Reference Material**	This section contains interconnection drawings and reference material for the trainer.

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1.3 SAFETY

Normal operation of the training system presents potentially hazardous situations to student firefighting personnel. It is of the utmost importance that operators read, understand, and follow the safety precautions and guidelines detailed in the following chapters. Failure to follow these safety guidelines could place training area personnel at undue risk of personal injury or death.

1.3.1 SAFETY SYMBOLS

Various symbols are used throughout the manual to alert the reader to a potentially dangerous situation.

WARNING

THE WARNING SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

CAUTION

THE CAUTION SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT MAY CAUSE INJURY OR DAMAGE TO ASSOCIATED EQUIPMENT.

1.3.2 TRAINER PERSONNEL SAFETY RESPONSIBILITIES

Trainer Personnel (instructors, safety officers, system operators and maintenance personnel) are ultimately responsible for the safety of all personnel entering the training area and compliance to all applicable NFPA, OSHA, national, state and local codes and standards. The safety responsibilities of Trainer Personnel are summarized below:

The safety responsibilities of Trainer Personnel are summarized below:

1. Read and fully understand all provided trainer manuals and this safety handbook.
2. Ensure that the trainer is operated and maintained in accordance with all trainer manuals.
3. Discontinue trainer use if potentially hazardous conditions or anomalies occur.
4. Be familiar with each fireplace's behavior in all modes of operation under all conditions, and insure the safety of personnel with respect to potentially hazardous behaviors.
5. Ensure that personnel are properly trained in the use of self contained breathing apparatus (SCBA).
6. All personnel entering the training area shall wear full firefighting gear. This gear shall include: full-face mask, self-contained breathing apparatus (SCBA) and appropriate protective clothing. As in any fire situation, the exposure of personnel to smoke (real or synthetic), fumes, or other combustion products should be avoided.
7. Ensure that personnel are in a safe position prior to, and during, flame initiation.

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8. Ensure that the operator and instructors have a full knowledge and understanding of the training scenario prior to initiation. This should include all fireplace parameter settings as well as reflash capabilities or simulated flame effects.
9. At no time during live-fire operation should the trainer be left unmanned.
10. Ensure that only water is used as the extinguishing agent. Use of extinguishing agents other than water can result in diminished trainer life and performance and adversely impact safe trainer operation.
11. Ensure that at the completion of a training exercise, personnel without adequate protective clothing and SCBA do not enter a training structure until the ventilation system has been operated at maximum capacity for a minimum of 3 minutes to remove combustion by-products and smoke.
12. Ensure that proper communication is established and in operation between instructors and safety officers before and throughout each training exercise.

1.3.3 Student Safety Guidelines

Prior to conducting training, Trainer Personnel must inform students of the dangers involved in live fire training. The following is a list of safety guidelines that must be presented to instructors and students before use of the trainer.

Normal operation of the training system presents potentially hazardous situations to student firefighting personnel. It is of the utmost importance that operators read, understand, and follow the safety precautions and guidelines detailed on this page. Failure to follow these safety guidelines could place training area personnel at undue risk of personal injury or death.

1. Do not enter the training area until authorized to do so by Trainer Personnel.
2. Do not enter the training area without proper protective clothing and self-contained breathing apparatus (SCBA). Keep this equipment on during the entire live fire training exercise.
3. While within the training area, be alert at all times. Treat the live fire-training situation the same as an actual fire emergency. Be prepared for any aspect of flame behavior that is possible with an uncontrolled, fully involved fire.
4. Avoid contact with hot surfaces within the training area before, during, and after the live fire training exercise.
5. Avoid breathing smoke and products of combustion while outside the training area.
6. Minimize or avoid contact with clothing and equipment exposed to smoke, soot, and any other fire training residue. If contact cannot be avoided, clean exposed skin surfaces with soap and water as soon as practical.
7. In addition to the guidelines listed above, listen to and carefully follow all rules and directions provided by Trainer Personnel.

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1.3.4 SYSTEM SAFETY WARNING

HIGH VOLTAGES capable of causing serious injury or death are used in the training system equipment. Unless required for voltage measurement, all power should be turned off when servicing is being performed. When power must be on, use extreme care when working near exposed conductors.

FUEL GAS used in the trainer is capable of forming explosive mixtures with air that can be triggered by open flames or sparks. Even residual gas from piping sections can create a hazardous condition. Work only in well ventilated areas, with no exposed flames or possible ignition source.

OPEN FLAMES generated in the fireplace area during operation of this equipment can cause serious injury or death. Use extreme care when working in this area whenever training fires are in operation.

EXPOSURE to extremes in the environmental conditions such as high or low temperatures, high winds, etc. may occur during a training scenario. Extreme care should be used when training under adverse environmental conditions. Failure to exercise extreme care could result in serious injury or death.

WARNING

SAFE OPERATION OF THE TRAINING SYSTEM MANDATES THAT THE FOLLOWING SAFETY PRECAUTIONS BE TAKEN PRIOR TO ANY OPERATION OR ACTIVITY. FAILURE TO FOLLOW THESE PROCEDURES MAY RESULT IN SERIOUS INJURY OR DEATH.

1. All personnel entering the training area shall wear full firefighting gear. This gear shall include: full-face mask, self-contained breathing apparatus (SCBA) and appropriate protective clothing. As in any fire situation, the exposure of personnel to smoke (real or synthetic), fumes or other combustion products should be avoided.
2. All live fire training evolutions shall be conducted in accordance with all applicable NFPA Codes (such as 1402 1403, 1404, and 1500); OSHA; national; state; and local codes and standards.
3. If fuel leaks are detected, they must be repaired. Under no circumstance is trainer operation to be performed when a leak is present.
4. Keep flames and ignition sources away from fuel handling components. When ignited, the fuel may cause serious injury or death.
5. Ensure the upstream fuel supply valve is fully closed prior to the purging of any lines to atmosphere. Tag all valves closed to alert other personnel that service is being performed. Failure to comply may result in serious injury or death.
6. Only smoke generators approved by Kidde Fire Trainers for use in flame areas shall be used.
7. Only smoke fluid recommended by Kidde Fire Trainers shall be used in the smoke generator(s).

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1.3.5 Cold Weather Operation Guidelines

Kidde Fire Trainers strongly recommends that all **F**IRE**T**RAINER® systems not be operated for training purposes when the ambient temperature is below 32°F. Operation below this temperature may impact the effectiveness of student training.

As water is used as the primary extinguishing agent in most training scenarios, it should be apparent that freezing water and ice may become hazardous to trainees and other site personnel. It is with this safety consideration in mind that Kidde Fire Trainers issues this cold weather operation recommendation.

NOTE

THIS RECOMMENDATION DOES NOT AND SHOULD NOT BE CONSTRUED TO IMPLY THAT THE TRAINER WILL NOT OPERATE BELOW THE KIDDE FIRE TRAINERS RECOMMENDED OPERATING TEMPERATURE.

CAUTIONS

Each fireplace has a water bath fabricated from steel. The burner elements are located at the bottom of the bath. For this reason refer to the CAUTION located below.

CAUTION

WHEN NOT IN USE ALL FIREPLACE WATER BATHS SHOULD BE DRAINED OF WATER DURING COLD WEATHER PERIODS, WHEN THE ANTICIPATED AMBIENT TEMPERATURE IS GOING TO BE BELOW 32°F

WHEN TRAINING DURING THE ABOVE MENTIONED CONDITION, FILL WATER BATHS JUST PRIOR TO OPERATION AND DRAIN IMMEDIATELY UPON THE COMPLETION OF TRAINING.

FAILURE TO FOLLOW THESE CAUTIONS MAY RESULT IN SERIOUS DAMAGE TO THE ASSOCIATED EQUIPMENT.

1.3.4 Peripheral Equipment

- Integrated Sound System
- Smoke Generator
- Hartzell Ventilation Fan

NOTE

FOR THE ABOVE EQUIPMENT REFER TO SECTION 6 FOR SAFETY INFORMATION.

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2.0 SYSTEM DESCRIPTION

2.1 INTRODUCTION

The **Structural FireTrainer® T-500** is a live-fire-training system used to conduct training of firefighters and emergency rescue personnel in live-fire situations. The trainer is a system that generates propane fueled fires under the control of a PLC and an instructor. The trainer simulates Class A and B type fires. Class A fires involve ordinary dry combustible materials. Class B fires involve flammable liquids.

The **T-500 Live FIRE TRAINER** consists of one live fire training burn room. The Burn Room contains a fireplace platform with a water pan fabricated from steel sheet. At the bottom of the pan, a sparging system is installed through which the flammable gas is liberated. Gas/ oxygen mixing occurs above the water surface where a standing pilot initiates combustion. The liberation of gas through water greatly enhances the depth of flame and gives the impression of a substantial fire.

Burn room 1 is located on the first floor and is equipped with a fireplace platform for multiple mockup use. Depending upon the type of scenario a mockup is assembled on top of the fireplace platform and fires are created that engulf the mockup. This burn room is also equipped with a flashover and rollover effect. See Figure 2-1 & Table 2-1 for location. When the flashover effect is activated it produces a flame that extends across the ceiling. When the rollover effect is activated it produces a flame that extends up the wall to the ceiling. A Second Stage Gas Regulation Assembly is located on the outside wall of the first floor (Figure 2-1 & Table 2-1). The Second Stage Gas Regulation Assembly filters and regulates the pressure of the supplied gas.

Fire extinguishment is accomplished by applying extinguishing agent to the fire while the operator simulates the appropriate response of the fire.

Since the gas-fueled fires do not create smoke as a normal part of combustion, the burn room is equipped with a fixed smoke generator. See Figure 2-1 & Table 2-1 for locations. Smoke is generated using non-toxic mineral fluid and supplied to the training compartment to create obscuration normally associated with fire scenarios.

A T-500 Class A ventilation system is installed in the second floor Class A burn room. The system operates automatically and controls the environment to ensure student safety during live fire training drills See Figure 2-2.

The trainer is also equipped with an Integrated Sound Effects System that enhances the realism of the training system. The Integrated Sound Effects System produces audio effects that firefighters and emergency rescue personnel would experience during actual live fire situations. For further information refer to the Integrated Sound Effects Operation and Maintenance manual, located in Section 6 of this manual.

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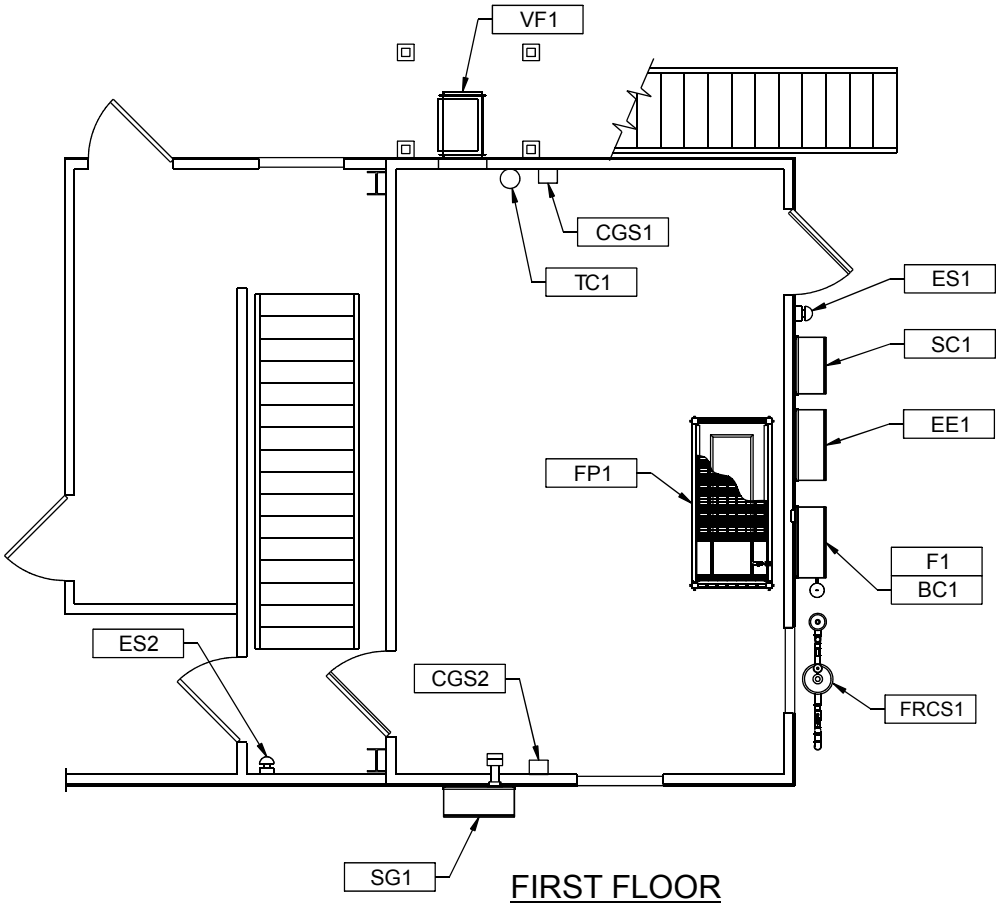


Figure 2-1. Training Building - First Floor

Table 2-1. Legend (refer to Figure 2-1)

Tag	Description	Tag	Description
FP1	Fireplace Platform	**FRCS 1**	Fuel Regulation & Control Station
BC1	Burner Control Assembly (for FP1)	**SG1**	Smoke Generator
EE1	Electrical Control Cabinet W/ Control Pendant	**TC1**	Temp Sensor (for FP1)
ES1 & 2	Emergency Stop (for FP1)	**VF1**	Vent Fan (for FP1)
F1	Flashover Effect Cabinet	**SC1**	Sound System Enclosure
GDS1 & 2	Combustible Gas Sensors (for FP1)		

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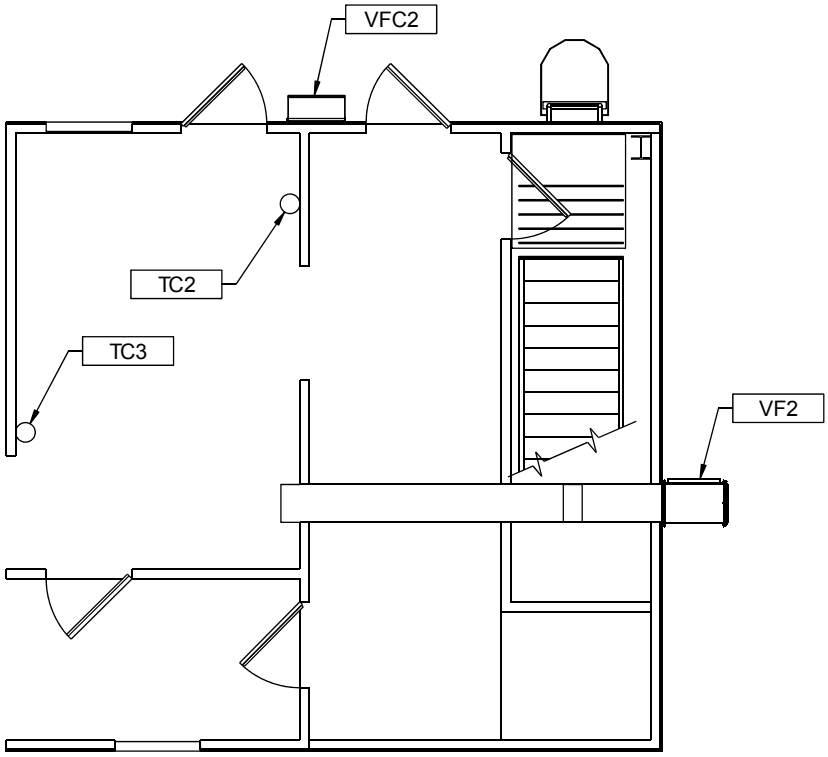


Figure 2-2. Training Building - Second Floor

Table 2-2. Legend (refer to Figure 2-2)

Tag	Description	Tag	Description
VF2	Vent Fan (for 2ND Floor)	**TC2 & 3**	Temp Sensors (for 2ND Floor)
VFC2	Vent Fan Electrical Control Enclosure		

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2.2 FIRE TRAINER COMPONENTS

2.2.1 VENTILATION/EXHAUST SYSTEM

The trainer contains four ventilation/exhaust fans. The main function of the ventilation/exhaust fan is to purge the building of all air impurities and to pull clean air into the burn area. The ventilation / exhaust fan accelerates and decelerates based on operation, warning, and alarm conditions.

When the Control Panel **STBY/ON/RESET** key is placed in the **ON** position the ventilation / exhaust fan turns on and will run an initial 30 second purge cycle at high speed. The Control Panel **PURGE** light flashes during the purge cycle.

⚠ CAUTION

NOTE! DURING THE PURGE CYCLE DO NOT TOUCH ANY BUTTONS OR SWITCHES. THIS MANY CAUSE THE PURGE CYCLE TO START OVER AGAIN!

At the end of the purge cycle, the fans go off. The system is now ready for operation.

In addition to the purge cycle the fan will also run at high speed if any of the following conditions occur:

- An emergency button is pressed
- An unsafe concentration of gas is detected
- The Pendant **SMOKE** switch is turned to **VENT.**
- A temperature of 500°F is exceeded
- A temperature of 700°F is exceeded

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2.2.1.1 EXHAUST FAN OPERATION-EMERGENCY

When an Emergency Stop button is activated, the exhaust fan in the structural trainer accelerates to 'high' speed and will continue to run at high speed until the emergency switch system is reset.

2.2.1.2 EXHAUST FAN OPERATION 10% LEL OF PROPANE GAS

When the gas detection system reads over 10% LEL the ventilation/exhaust fan will accelerate from low speed and the system will continue to run and attempt to remove the unburned gas buildup.

2.2.1.3 EXHAUST FAN OPERATION 25% LEL OF PROPANE GAS

When a reading above 25% LEL of propane gas is detected an emergency shutdown is initiated. All main flames and pilot flames are removed and the ventilation fan ramps to high speed. When the gas level drops below a reading of 10% LEL a system purge sequence is initiated. At the end of the 30 second system purge sequence, the trainer is ready for operation.

2.2.1.4 EXHAUST FAN OPERATION-SMOKE CLEAR

The Control Pendant has a three-position switch for the operation of the smoke system.

The positions of this switch are:

- SMK = Smoke On
- Center Pos. = Smoke/Vent Off
- Vent = Exhaust fan runs at high speed (Control Panel **PURGE** light goes on)

2.2.1.5 EXHAUST FAN OPERATION-500°F IS EXCEEDED

When the temperature in a fire room is detected as exceeding 500°F, the exhaust fan of the module will accelerate and continue to run at this speed until the temperature is detected as being less than 500°F. When the temperature drops below 500°F the fan will revert to low speed. A Message will appear at the Control Panel screen.

2.2.1.6 FAN OPERATION-700°F IS EXCEEDED

When the temperature in a burn room is detected as exceeding 700°F the exhaust fan will accelerate to high speed from the warning operating speed. When the temperature is detected as exceeding 700°F the burner(s) in the burn room turns off.

After a room temperature is detected as being greater than 700°F the fan runs at high speed and will continue to do so until the temperature drops below 700°, **STBY-ON-RESET** key switch is placed in the **RESET** position and released and all burner and pilot switches are in the **OFF** position.

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2.2.2 SMOKE GENERATION

To increase the realism of the fire training, a non-toxic smoke can be generated from a smoke generator. There is one smoke generator associated with each burn room (See Figure 2-1 & Table 2-1 for locations. The smoke generator is equipped with a nitrogen regulator and a nitrogen/CO_2 Bottle (CO_2 bottle by others). The smoke fluid is fed to the smoke generator from a tank located inside the smoke generator. An element heats the smoke fluid inside the smoke generator, causing it to vaporize creating a dense smoke effect. Smoke travels from the nozzle through a smoke duct in the wall and into the burn room. The smoke generation system can use nitrogen or CO_2 propellant to atomize the smoke fluid.

The smoke system must complete a pre-heat cycle before it is operational. The system must reach a temperature of 340-380°C and be pressurized by CO_2 or nitrogen to 95 psi before smoke can be generated. When the correct temperature has been reached, the Pendant **SMOKE** switch flashes.

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2.2.3 Manual Emergency E-Stop Button

When an E-Stop is pressed it shuts down all operating main and pilot flames. The exhaust fan will ramp up to full speed and remain at that speed until the activated push button is reset and the **STBY-ON-RESET** key switch is rotated to the **RESET** and released. The message **"System Emergency Stop Pressed"** will be displayed on the Control Panel screen.

2.2.4 Combustible Gas Detection

The **Structural FireTrainer® T-500** training system continuously monitors the gas levels inside the burn room. The burn room is equipped with two combustible gas sensors (See Figure 2-1 for locations and Figure 2-3 for detail). Each sensor is connected to a Transmitter located in the Electrical Control Cabinet. See Figure 2-1 for the Electrical Control Cabinet locations and Figure 2-4 for Transmitter locations. This system is hard-wired and fail-safe.

The gas sensor transmitter has two alarm levels; the first is a warning. The warning will trigger, when the presence of a combustible gas concentration exceeding 10% of the Lower Explosive Limit (LEL), for gas is detected in the burn area. This will accelerate the ventilation/exhaust system fan. The second alarm is triggered when the presence of a combustible gas concentration exceeds 25% of the LEL. When a concentration exceeding this amount is detected the system will automatically fault into an emergency stop condition.

The gas detection transmitter display indicates the percentage of LEL for the corresponding sensor and also displays alarm status.

The combustible gas sensor(s) in the burn area are fitted with protective splashguards that will prevent fire-extinguishing agent from impinging the sensing module readily.

⚠ CAUTION

THE GAS DETECTION HEADS ARE SENSING DEVICES. HARSH SOLVENTS AND SILICON BASED PRODUCTS CAN/WILL POISON THE GAS DETECTION HEADS. IF THE NEED ARISES TO USE THESE PRODUCTS WITHIN THE AREA OF THE GAS DETECTION HEADS MAKE SURE THAT THE HEADS ARE COVERED WELL, AND THAT THE AREA IS WELL VENTILATED.

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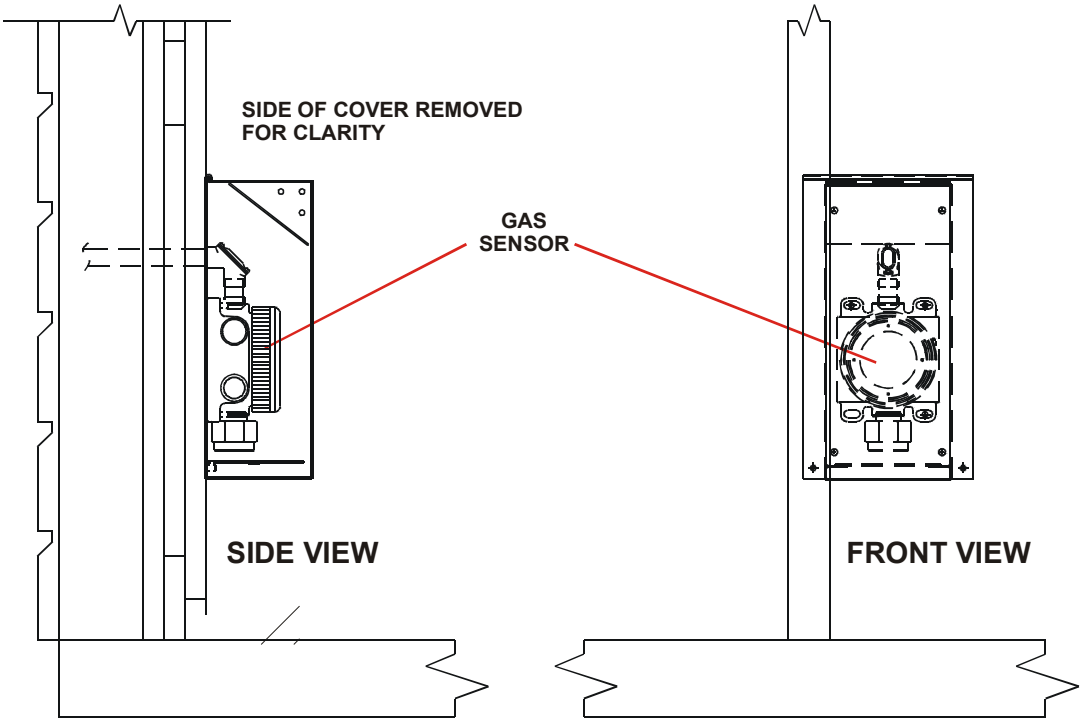
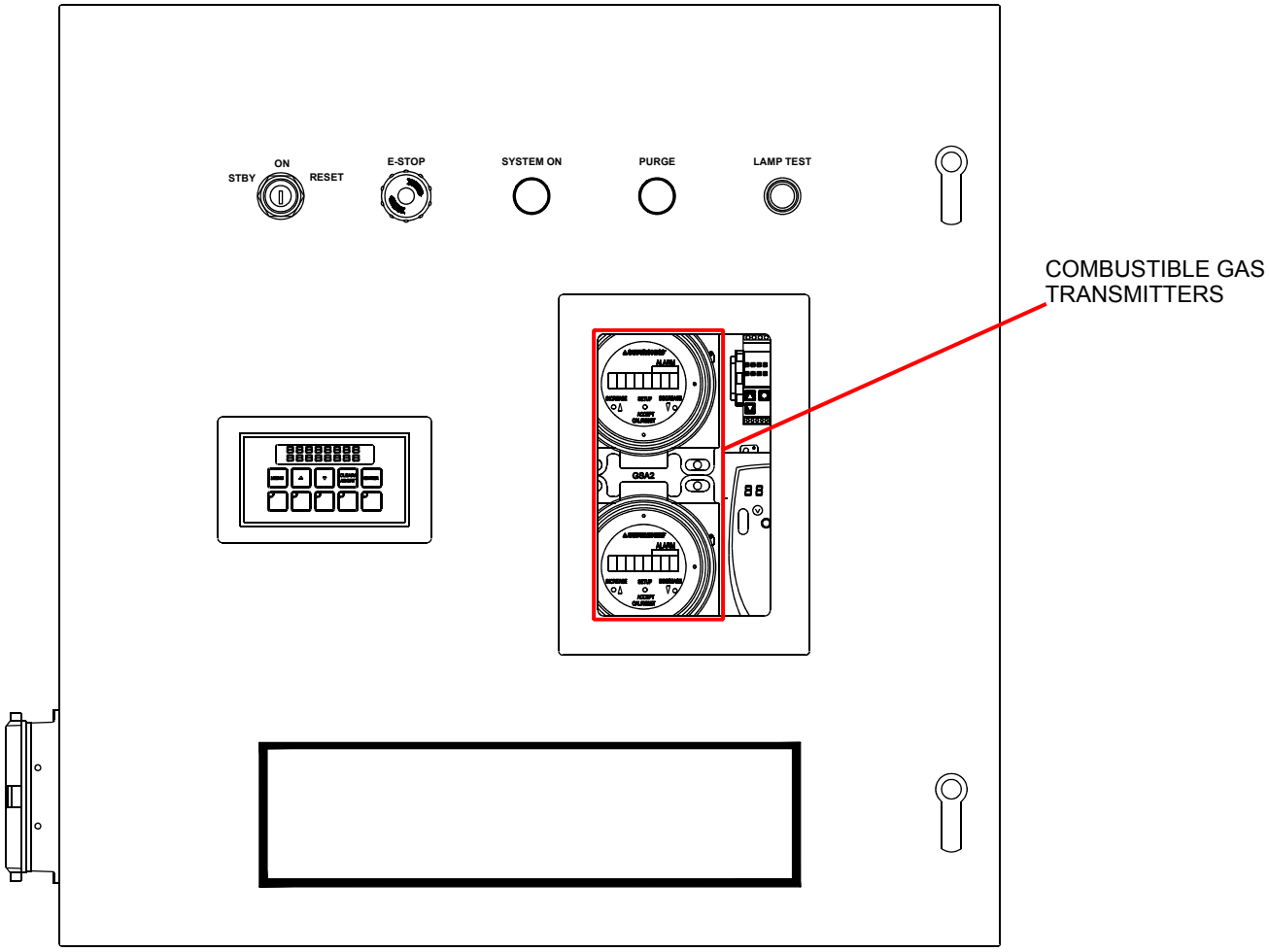


Figure 2-3. – Combustible Gas Sensor

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ELECTRICAL CONTROL CABINET

Figure 2-4. Gas Detection Transmitter

2.2.5 TEMPERATURE MONITORING

The temperature in the fire training burn room is monitored at all time when the system is in operation. The temperature-sensing device is a thermocouple. The thermocouple is mounted approximately five feet from the floor (See Figure 2-1 & Table 2-1 for locations). This thermocouple is hard wired to two alarms in the corresponding Electrical Control Panel. There is a 500 ° Fahrenheit and 700 °Fahrenheit alarm. With the thermocouple reading of 500°, a warning condition occurs, and the ventilation/exhaust fan accelerates. With the thermocouple reading at or above 700°F an alarm condition occurs, the system will automatically fault into an emergency stop condition. There is a 500 °Fahrenheit and 700 °Fahrenheit alarm. With the thermocouple reading of 500 a warning condition occurs, and the ventilation/exhaust fan accelerates. With the thermocouple reading at or above 700°F an alarm condition occurs, and the system will automatically fault into an emergency stop condition.

The thermocouple will be used to activate two levels. They are:

Warning Level

Alarm taken to PLC and causes associated ventilation/exhaust to ramp up, which will exhaust the excess heat from the burn area. As well a 500° F amber lamp will illuminate until the system drops below 500° F.

Alarm level

Hard wired into emergency stop circuit and monitored by PLC. At PLC the alarm causes the fan to run at high speed until alarm is cleared, shuts down all fireplaces and illuminates the red 700 level alarm light until emergency stop is reset. Set at 700° F.

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2.2.6 CONTROL PANEL

The Control Panel is located on the front of each Electrical control Cabinet. This panel is used to control and monitor all functions associated with each burn room. See Table 2-3 for Control/Indicator Descriptions and see Figure 2-5 for the Control Panel layout.

Table 2-3. Control Panel-Controls and Indicators Descriptions

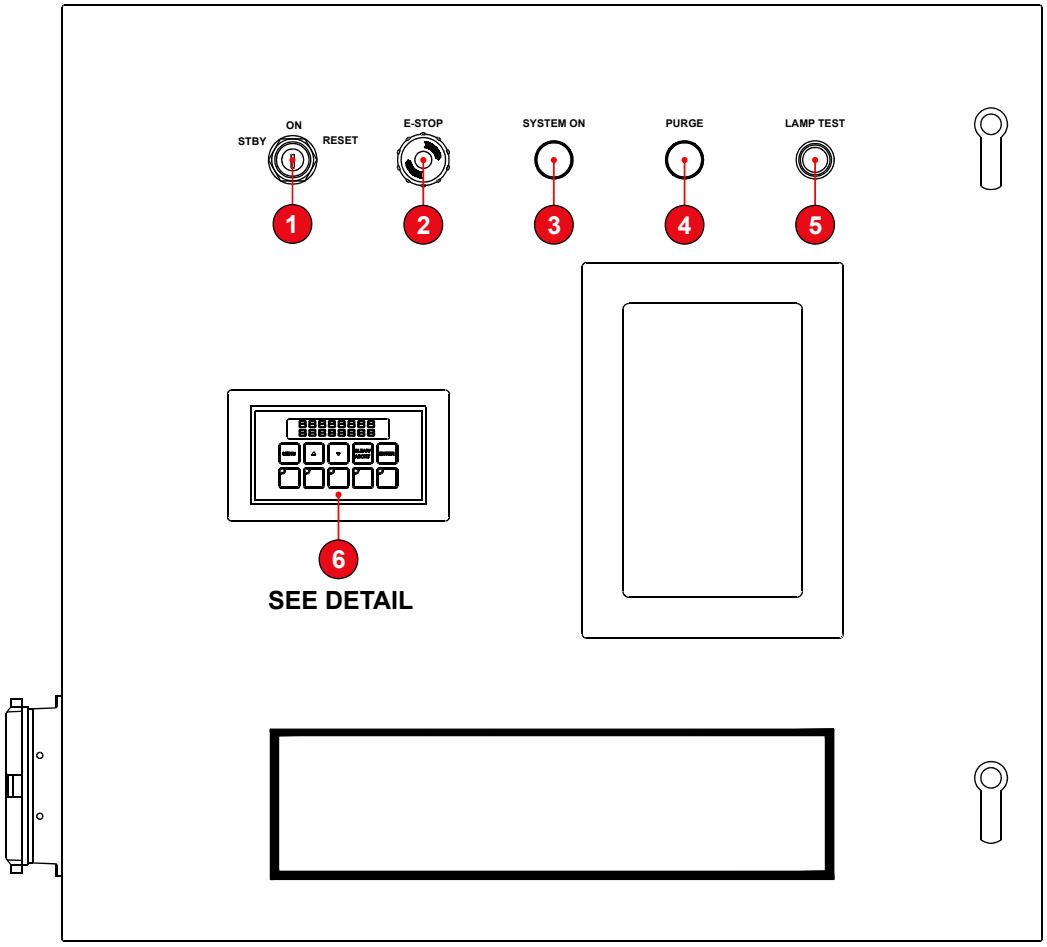
No.	Control	Indicator	Description
1	**X**		**STBY-ON-RESET** – This key switch has three positions: **STBY-** In this position power is applied to the combustible gas detection system. **ON-** When in this position, power is supplied to all trainer hardware. This is the normal switch position for training. **RESET-** This key switch must be rotated to this position and released after any type of emergency stop has occurred on the trainer. The condition that caused the emergency stop must be cleared before the reset is effective. The possible emergency stop conditions are: • Emergency Stop Pushbutton pressed. • Fire Room temperature exceeded 700°F. • Gas detection level exceeded 25% LEL.
2	**X**	**X**	**E-STOP** - Pressing this button will immediately shutdown all fires in the trainer. Operation can only be resumed by completing the following: • Clear the condition that caused the emergency stop. • Reset the E-Stop button (Rotate and release) • Turn the **STBY-ON-RESET** Key switch to the **RESET** position and release (Item 1).
3	**-**	**X**	**SYSTEM ON** - This lamp will turn on when the **STBY-ON-RESET** key switch has been turned to the **ON** position
4	**-**	**X**	**PURGE** - This lamp flashes during the 30 second purge cycle and will turn off when purge cycle is complete. This lamp will light solid when the Pendant **SMOKE** switch is placed in the **VENT** position.
5	**X**	**X**	**LAMP TEST** - Pressing this button will test all lamps on the Control Panel and Pendant.
6	**-**	**X**	**Operator Panel** – Used to display messages.

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Table 2-3. Control Panel-Controls and Indicators Descriptions

No.	Control	Indicator	Description
6A		**X**	Message Screen
6B	**X**	**-**	Press to clear messages
6C-D	**X**	**-**	Scroll Up & down through messages

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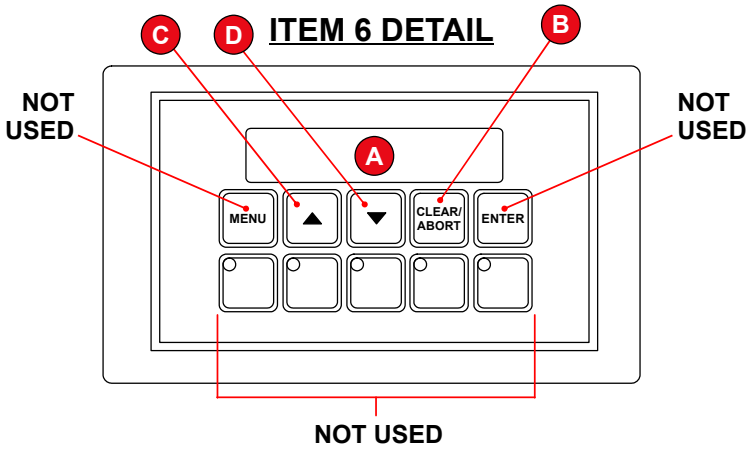


Figure 2-5. Control Panel

2.2.7 CONTROL PENDANT

The pendant is used to give the instructor full control of the training scenario. See Table 2-4 for Control/Indicator Descriptions and see Figure 2-6 for the Control Pendant layout.

Table 2-4. Control Pendant-Controls and Indicators Descriptions

No.	Control	Indicator	Description
1	X	X	**PILOT –** Turn switch to **ON** position and release to initiate the pilot flame. Flashing switch indicates pilot flame sequence is in progress. When lit bright and steady, indicates fireplace pilot flame sequence is complete and pilot flames are confirmed. Turn switch to **OFF** position and release, to turn pilot flame off.
2	X	X	**FLAME –** This three-position switch is used to control the fireplace burner main flame i.e. OFF, LOW or HIGH. The Green **ON/OFF** button must be held down to operate main flame. Turn switch to **OFF** position to turn main flame off.
3	X	X	**EFFECT –** This two-position switch is used to control the flashover effect (**FLSH**) and the rollover effect (**ROLL**). **FLSH –** When switch is held in this position, the flashover effect will be activated. The flashover effect will be active for about 5 sec. or until the switch is released. **ROLL –** When switch is held in this position, the rollover effect will be active until the switch is released.
4	X	X	**SMOKE –**This two-position switch is used to control the smoke generator (**SMK**) and the vent/exhaust fan (**VENT**). To generate smoke the Control Panel **STBY-ON-RESET** key switch must be in the **ON** position. **SMK –** Once indicator is flashing turn switch to this position to produce smoke. Switch indicator lights steady. Turn switch to center position to turn smoke off. Switch indicator light flashes. **VENT -** When the switch is turned to this position the vent/exhaust fan ramps up to high speed to clear the Burn Room. Turn switch to center position to return vent/exhaust fan to normal operation.

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Table 2-4. Control Pendant-Controls and Indicators Descriptions

No.	Control	Indicator	Description
5	**X**	**X**	**ON/OFF –** With pilot confirmed, Press and hold **ON/OFF** button to enable use of the **FLAME** selection switch. When **ON/OFF** button pressed light comes on bright and steady. When **ON/OFF** button released, the main flames extinguish and button flashes. Press and hold **ON/OFF** button again and main flames return to current **FLAME** switch setting.

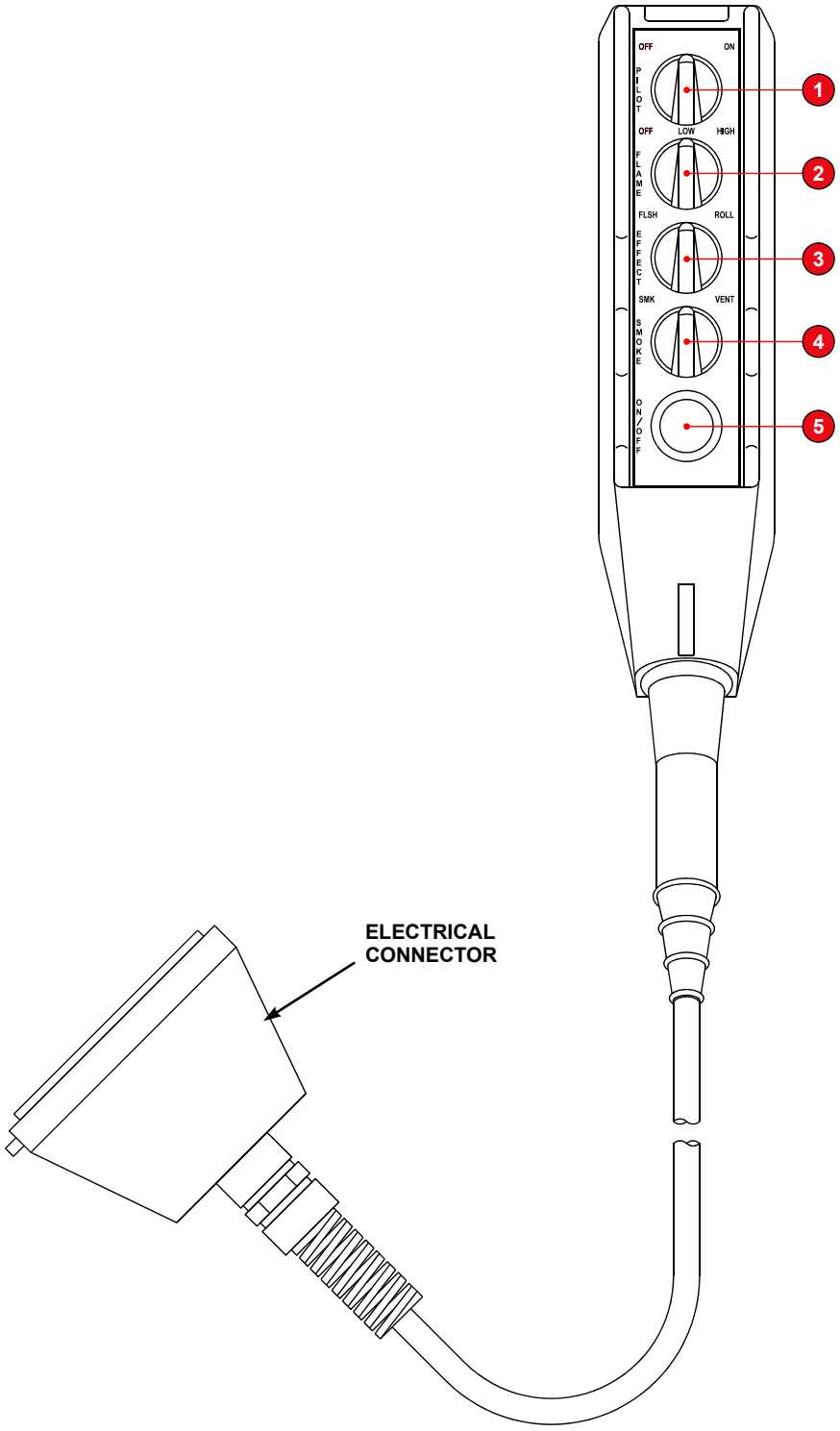


Figure 2-6. Control Pendant

2.2.8 WIRELESS CONTROL SYSTEM

The Wireless Control System assembly allows the instructor control of the burn room fireplace operation with the inherent mobility available without a cable connection. The Wireless Control System assembly consists of a hand held wireless transmitter and a wall mounted wireless receiver that connects to the connection on the Electrical Control Panel. Refer to Figure 2-7 for layout.

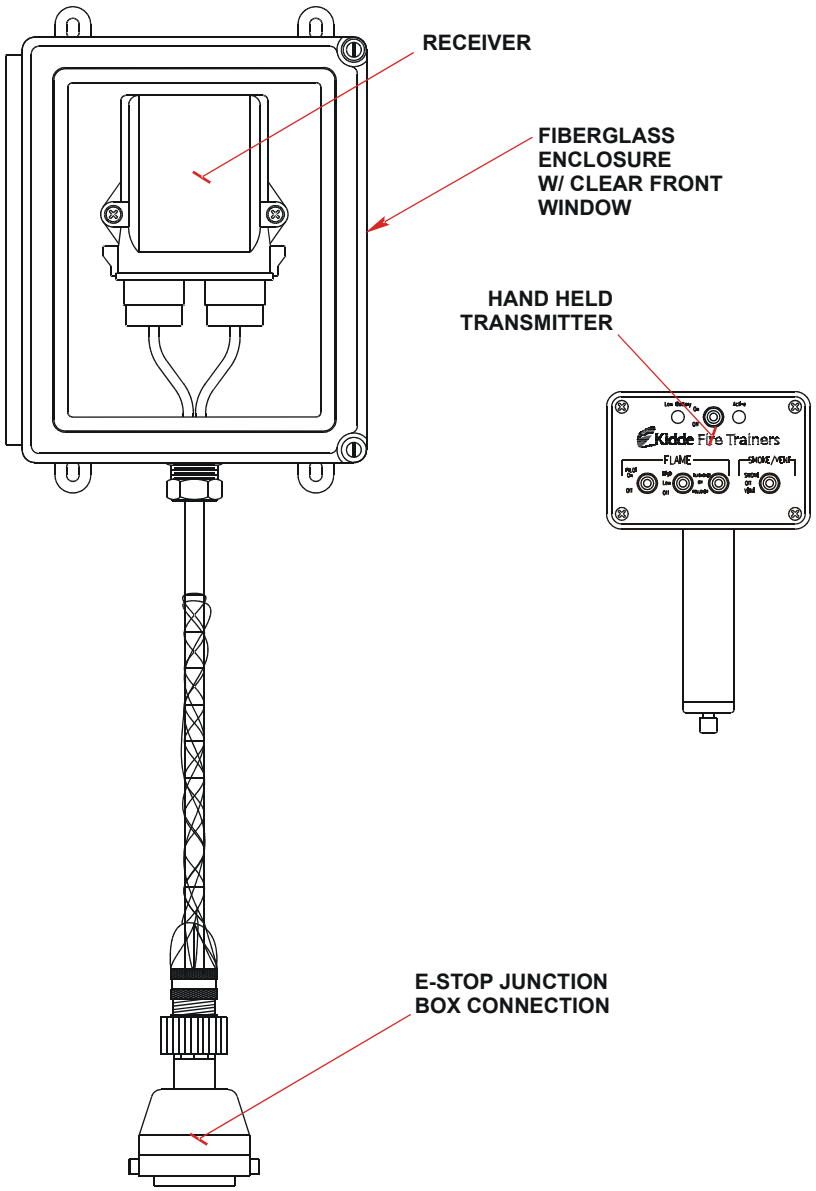
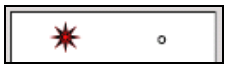


Figure 2-7. Wireless Control System

2.2.8.1 WIRELESS TRANSMITTER

The switches on the transmitter are used for Burn Room #3 operation. A trigger switch on the handle functions as a "Deadman" switch for the main flame. The trigger switch must be pulled in and held for main flame operation. See Table 2-5 for Control/Indicator Descriptions and see Figure 2-8 for the Wireless Transmitter layout.

Table 2-5. Wireless Transmitter-Controls and Indicators Descriptions

No.	Control	Indicator	Description
1,2		X	 **1 Slow Flashing Red, 2 Fast Flashing Yellow –** Low Battery. Unit will run approximately 10 hours after indication.
1,2		X	 **1 Slow Flashing Red, 2 Slow Flashing Yellow –** Stuck Transmitter Switch Detected. Ensure all switches are in .
1		X	 **Rapid Flashing Red for 10 Seconds -** Transmitter Failure
2		X	 **Solid Yellow –** A transmitter function is ON.
2		X	 **Fast Flashing Yellow –** Normal Operation. This light will flash several times per second, indicating that the transmitter is sending signals to the receiver. The light will remain on momentarily whenever a function changes.
3	X		**ON/OFF -** This two-position switch is used to enable operation of the wireless transmitter. The transmitter yellow indicator lamp will automatically stop flashing after sixty minutes of transmitter inactivity.

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Table 2-5. Wireless Transmitter-Controls and Indicators Descriptions			
No.	**Control**	**Indicator**	**Description**
4	**X**		**FLAME, PILOT/ON/OFF -** This two-position switch is used to control the pilot flame. Toggle the switch to the **ON** position to turn on the pilot flame. Toggle the switch to the **OFF** position to turn off the pilot flame.
5	**X**		**FLAME, HIGH/LOW/OFF -** This three-position switch is used to control the fireplace main flame i.e. **LOW** or **HIGH**. The pilot flame must be on and the **Deadman Trigger (6)** pulled in and held to operate the main flames. Releasing the **Deadman Trigger (6)** will turn the main flame off (pause). Pull the **Deadman Trigger (6)** in and hold again to turn main flame back on. Toggle the **FLAME, HIGH/LOW/OFF** switch to the **OFF** position to turn off main flame operation.
6	**X**		**Deadman Trigger** - This trigger must be pulled in and held to enable the main, flashover and rollover flames to operate. Releasing the trigger will turn the main flame off (pause). Pull trigger in and hold again to turn main flame back on.
7	**X**		**FLAME, FLASHOVER/OFF/ROLLOVER –** This three-position switch is used to control the **FLASHOVER** and **ROLLOVER** effects. **FLSHOVER –** When switch is held in this position, the flashover effect will be activated. The flashover effect will be active for about 5 sec. or until the switch is released. **ROLL –** When switch is held in this position, the rollover effect will be active until the switch is released. **!**
8	**X**		**SMOKE/VENT, SMOKE/OFF/VENT -** This three-position switch is used to control the smoke generator and vent fan in the associated burn room. To generate smoke the Control Panel **STBY-ON-RESET** key switch must be in the **ON** position. **SMOKE –** Toggle switch to this position to signal the smoke generator to emit smoke. Toggle switch to **OFF** position to stop emitting smoke. **VENT -** Toggle switch to this position to turn on the exhaust fan. Exhaust fan will ramp up to high speed to clear the Burn Room. When switch is toggled to **OFF** position the exhaust fan returns to current operating mode.

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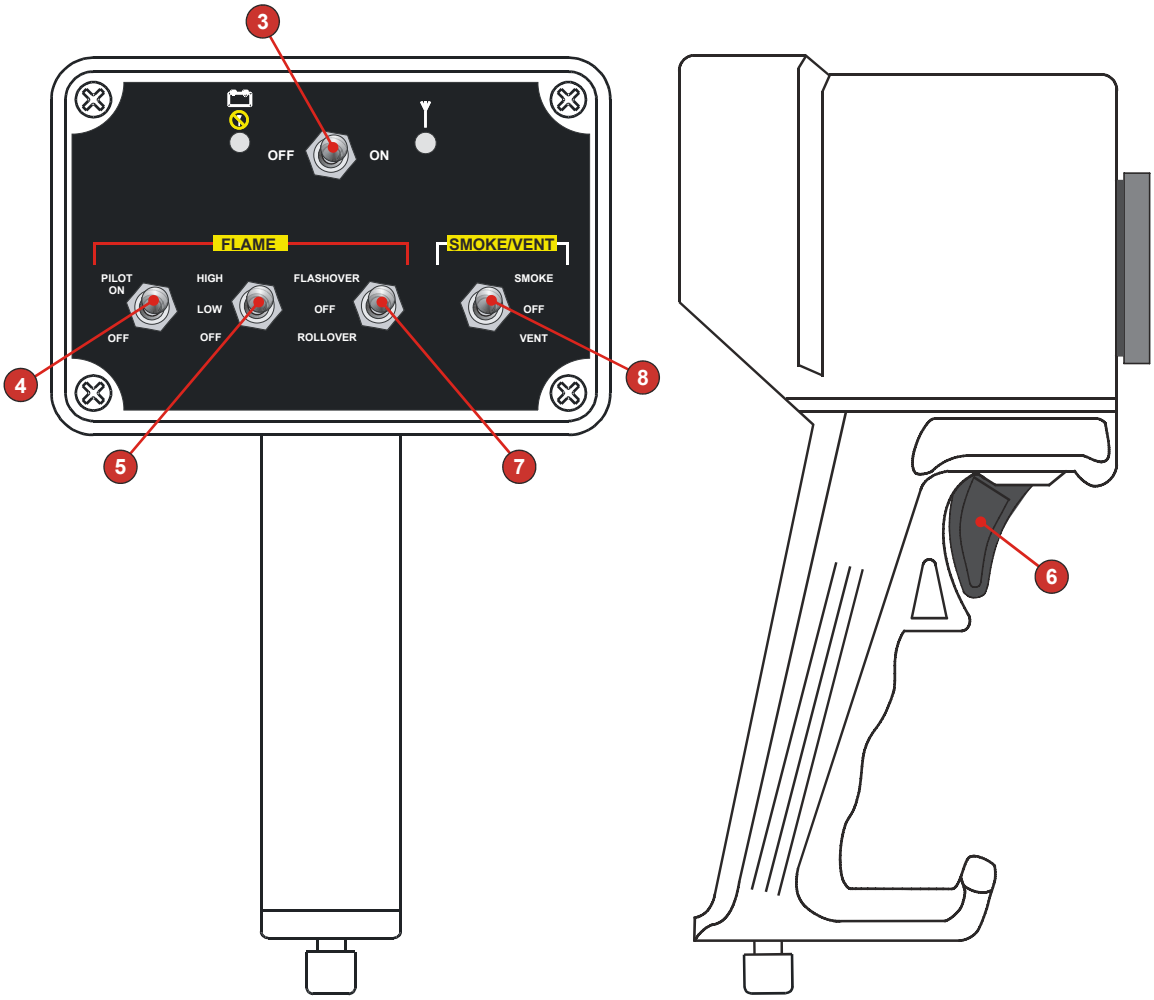


Figure 2-8. Wireless Transmitter

2.2.9 COMBUSTIBLE GAS SENSOR TRANSMITTER

There are two Combustible Gas Sensor Transmitters, located in each Electrical Control Cabinet, one for each Burn Room detection sensor and one in each of the Burn Room #2 & #3 equipment areas. When the **STBY-ON-REST** switch is in the **STBY** or **ON** position, the Transmitters continuously monitor the gas levels received from the detection sensors (See figures 2-2 & 2-3 for component locations). See Table 2-6 for Transmitter Indicators Descriptions and see Figure 2-9 for the Transmitter layout.

Table 2-6. Combustible Gas Detection Transmitter Indicators Descriptions

No.	Control	Indicator	Description
1	-	X	**SENSOR INPUT READING** – When **STBY-ON-REST** switch is in the **STBY** or **ON** position, this display is on and functions as a power indicator. This display provides a continuous reading of the gas detection sensor input in both the normal and calibration modes.
2	-	X	**ALARM HISTORY INDICATOR** – Asterisk indicates an alarm has activated since last reset. Blank indicates no alarm has energized since last reset.
3	-	X	**AUXILIARY ALARM STATUS INDICATOR** – Solid black bar indicates that the 25% LEL threshold has been exceeded. Blank indicates no alarm.
4	-	X	**LOW ALARM STATUS INDICATOR** – Solid black bar indicates that the 10% LEL threshold has been exceeded. Blank indicates no alarm.

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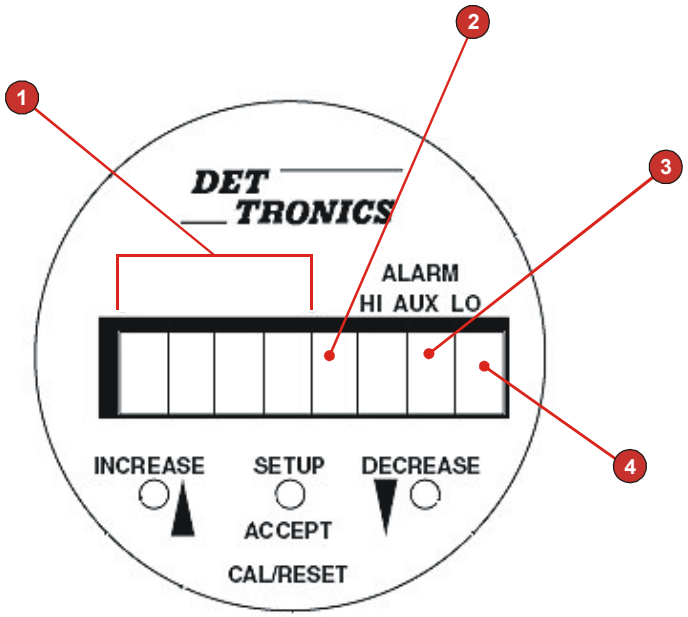


Figure 2-9. Combustible Gas Detection Transmitter

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2.2.10 Electrical Control Cabinet Assy. (Vent Fan)

A T-500 Class A ventilation system is installed in the second floor Class A burn room. The system operates automatically and controls the environment to ensure student safety during live fire training drills. See Table 2-7for Control/Indicator Descriptions and see Figure 2-10 for Control Cabinet layout.

Table 2-7. Control Cabinet - Controls and Indicators Descriptions

No.	Control	Indicator	Description
1	**X**		**HAND/OFF/AUTO** - This three-position switch is used to control the exhaust fan. HAND – This setting is the manual mode. Turning the switch to this position will run the exhaust fan at full speed. OFF - Turning the switch to this position will do the following: • Turn the exhaust fan off if it is running. • Not allow it to automatically operate. AUTO – When in this position the exhaust fan will operate based on the burn room temperature (+500°F & +750°F). See item numbers 2 and 3.
2		**X**	**TS1.** – This temperature controller will activate the exhaust fan when the temperature in the burn room rises above 500°F. The exhaust fan will run at medium speed from a temp of +500°F to 750°F (See item 3). The Exhaust fan will stop when the temperature lowers to below 500°F.
3		**X**	**TS2** - This temperature controller will increase the exhaust fan speed to full when the temperature in the burn room rises above 750°F. The Exhaust fan will reduce to medium speed when the temperature lowers below 750°F.
4		**X**	**ALARM** - This alarm will activate when the temperature in the burn room rises above 700°F. The alarm will go off when the temperature lowers to below 700°F.

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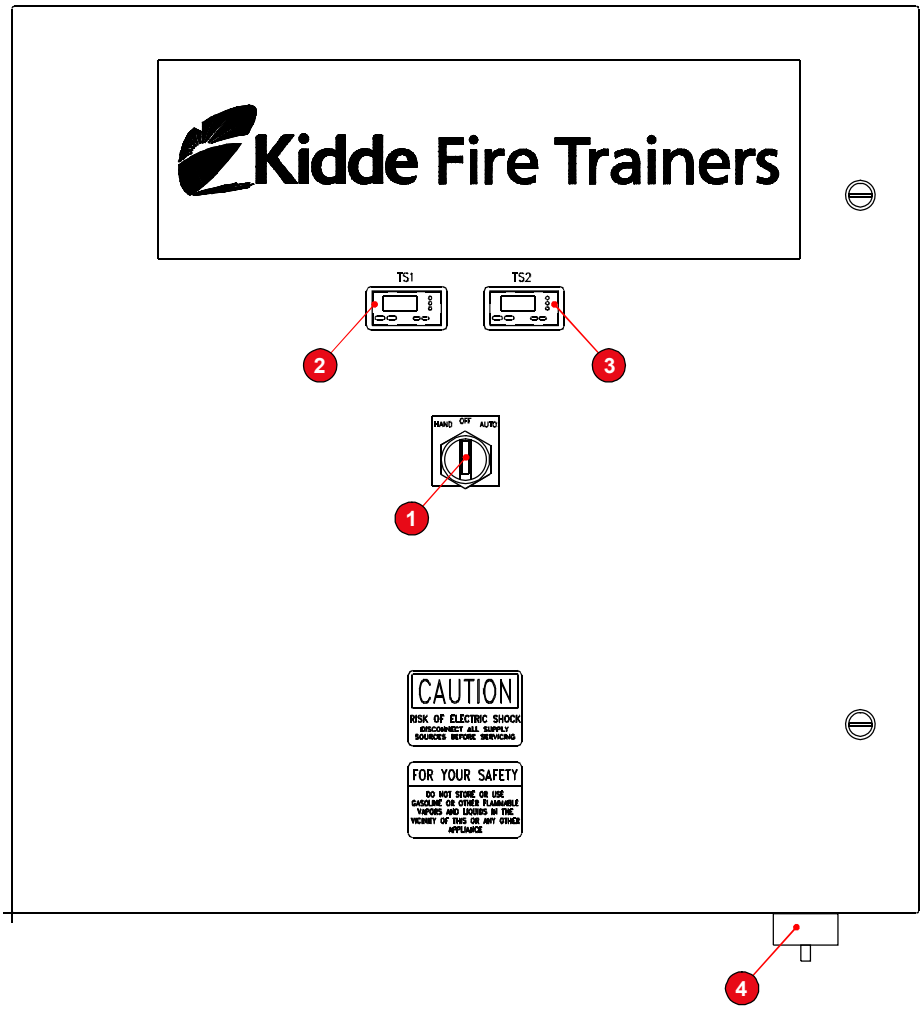


Figure 2-10 Electrical Control Cabinet Assy. (VF2)

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3.0 OPERATION

3.1 INTRODUCTION

This section describes the necessary procedures to operate the trainer.

3.2 PRE-OPERATIONAL CHECKOUTS

1. Do a visual inspection of burner control cabinets, electrical cabinet and exhaust fan.
2. Visually check main gas regulator.
3. Open tank valve(s) slowly.
4. Check gauge at regulator, it should read 5 psi.
5. Ensure 1/4 turn valve is open at each burner control cabinet.
6. Verify that fire pans are filled with water.

⚠ CAUTION

MAKE SURE THAT THE FIRE PANS ARE COMPLETELY FULL! DAMAGE MAY OCCUR IF MAIN BURNER IS LIT WITHOUT WATER IN THE PAN.

⚠ CAUTION

WHEN NOT IN USE THE FIREPLACE WATER BATH PANS SHOULD BE DRAINED OF WATER DURING COLD WEATHER PERIODS, WHEN THE ANTICIPATED AMBIENT TEMPERATURE IS GOING TO BE BELOW 32°F

WHEN TRAINING DURING THE ABOVE MENTIONED CONDITION, FILL WATER BATH PANS JUST PRIOR TO OPERATION AND DRAIN IMMEDIATELY UPON THE COMPLETION OF TRAINING.

FAILURE TO FOLLOW THESE CAUTIONS MAY RESULT IN SERIOUS DAMAGE TO THE ASSOCIATED EQUIPMENT.

7. Check the general area for debris and confirm that there are no obstacles that would prevent training.
8. Assemble desired Kidde Fire Trainers Prop on the fireplace pan assembly. Refer to paragraph 3.6 for instructions.
9. Verify that the emergency stop push buttons in the fire training compartments and Control Panels are pulled out.
10. Ensure that the smoke machines are full of fluid and that there is adequate nitrogen at a pressure of 70 psi.

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3.3 SYSTEM OPERATION

3.3.1 START UP

Refer to Section 2, Figures 2-5, 2-6, 2-7, 2-8 and Tables 2-3, 2-4 and 2-5 for descriptions and layout of the Control Panel, Wireless Transmitter and Pendant.

1. For wired control pendant operation, connect it to the electrical control panel. Ensure control pendant switches are all in the OFF position.
2. For wireless operation, at the associated wireless receiver ensure all switches are in the OFF position. Connect the receiver assembly to the fireplace electrical control panel.
3. Power the system on by turning on the main breaker, and turning the main disconnect to the ON position.
4. Verify that all the Emergency Stop switches are pulled out.
5. Turn the **STBY-ON-RESET** key switch to the **RESET** position, then release. When the key switch is released it will spring to the **ON** position. Verify that the **SYSTEM ON** lamp turns on.
6. Verify that the **PURGE** lamp flashes, and the exhaust fan turns on and runs for 30 second purge cycle. At the end of the purge cycle the exhaust fan will slow and the **PURGE** lamp will turn off.
7. Press the **LAMP TEST** button to verify that all Control Panel lamps and Pendant lamps are operational. Pendant lamps will go bright when **LAMP TEST** button is pressed.

The fire training compartment fires are now ready for operation.

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3.3.2 FIREPLACE OPERATION (WIRED CONTROL PENDANT)

Refer to Section 2, Figures 2-5, 2-6 and Tables 2-3 and 2-4 for descriptions and layout of the Control Panel and Pendant.

NOTE

IF AN E-STOP SWITCH IS ACTIVATED, THE CAUSE MUST BE IDENTIFIED AND CORRECTED, THE E-STOP RESET AND THE STBY-ON-RESET key switch rotated to the **RESET** position, then released**.**

1. At the Pendant rotate the **PILOT** switch to the **ON** position, then release. The **PILOT** switch indicator light will flash and after a short period of time the pilot flame will ignite and the **PILOT** switch indicator light will transition from flashing to bright solid and the green **ON/OFF** button will begin to flash.

NOTE

IF PILOT FLAME DOES NOT IGNITE, PILOT BUTTON WILL FLASH QUICKLY. AT THIS POINT REFER TO THE MESSAGE SCREEN ON THE CONTROL PANEL.

2. At the Pendant, press and hold down the green **ON/OFF** button.
3. Rotate the **FLAME** switch to the **LOW** or **HIGH** position as required. When the main flame ignites, the **FLAME** switch indicator light will transition from low to bright.
4. When the main flame is no longer required, shut it down by rotating the Pendant **FLAME** switch to the **OFF** position. Release the green **ON/OFF** button.
5. To shut the pilot flame(s) off, at the Pendant, rotate the **PILOT** switch to the **OFF** position, then release.

NOTE

THE MAIN FLAME WILL REMAIN OPERATING AS LONG AS THE GREEN ON/OFF BUTTON IS PRESSED. ONCE RELEASED THE MAIN FLAME WILL BE EXTINGUISHED, BUT THE PILOT BURNER WILL REMAIN OPERATING.

WHEN THE GREEN ON/OFF BUTTON IS PRESSED AND HELD AGAIN, THE MAIN FLAME WILL IGNITE AND RETURN TO THE HEIGHT THAT THE FLAME SWITCH IS CURRENTLY SET AT.

3.3.2.1 FLASHOVER FIRE

1. To activate the flashover fire, at the Pendant rotate the **EFFECT** switch to the **FLSH** position and hold. The flashover effect flame will be activated for about 5 seconds or until the **EFFECT** switch is release.

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NOTE

THE FLASHOVER FLAME CAN ONLY BE OPERATED WHEN THE GREEN ON/OFF BUTTON IS PRESSED.

3.3.2.2 ROLLOVER FIRE

1. To activate the rollover fire, at the Pendant rotate the **EFFECT** switch to the **ROLL** position and hold. The rollover effect flame will be activated as long as the **EFFECT** switch is held in the **ROLL** position.

NOTE

THE ROLLOVER FLAME CAN ONLY BE OPERATED WHEN THE GREEN ON/OFF BUTTON IS PRESSED.

3.3.2.3 SMOKE OPERATION

1. Once the Pendant **SMOKE** switch is flashing, smoke is generated by rotating the **SMOKE** switch to the **SMK** position.

3.3.2.4 EXHAUST FAN OPERATION

1. Turn on the exhaust fan by rotating the **SMOKE** switch to the **VENT** position.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

3.3.3 FIREPLACE OPERATION (WIRELESS CONTROL SYSTEM)

Refer to Section 2, Figure 2-5, 2-8 and Tables 2-3 & 2-5 for descriptions and layout of the Electrical Control Panel and Wireless Transmitter.

NOTE

IF AN E-STOP SWITCH IS ACTIVATED, THE CAUSE MUST BE IDENTIFIED AND CORRECTED, THE E-STOP RESET AND THE STBY-ON-RESET key switch rotated to the **RESET** position, then released**.**

WARNING

AT THE WIRELESS TRANSMITTER BE SURE FLAME AND SMOKE/VENT SWITCHES ARE IN THE OFF POSITION.

BE SURE THE CORRECT RECEIVER ASSEMBLY IS CONNECTED TO THE CORRESPONDING FIREPLACE ELECTRICAL CONTROL PANEL. EACH RECEIVER AND TRANSMITTER IS PAIRED TOGETHER VIA A UNIQUE ID CODE. EACH RECEIVER AND TRANSMITTER PAIR IS MARKED WITH THE FIREPLACE ELECTRICAL CONTROL PANEL LOCATION THE PAIR IS TO BE USED WITH.

1. At the Wireless Transmitter toggle **ON/OFF** switch to the **ON** position to enable the transmitter operation. The yellow indicator will begin to flash.
2. At the Wireless Transmitter toggle **PILOT, ON/OFF** switch to the **ON** position. After a short period of time the pilot flame will ignite
3. Once the pilot has ignited, at the Wireless Transmitter pull in and hold the **Deadman Trigger.** Toggle the **FLAME** switch to the **LOW** or **HIGH** position as required. When the **Deadman Trigger** is released, it will stop (pause) the main flame. Pull in and hold the **Deadman Trigger** again to resume main flame operation.
4. When the main flame is no longer required, at the Wireless Transmitter, shut it down by toggling the **FLAME** switch to the **OFF** position. Release the **Deadman Trigger**.

NOTE

THE MAIN FLAME WILL REMAIN OPERATING AS LONG AS THE DEADMAN TRIGGER IS PULLED IN AND HELD. ONCE RELEASED THE MAIN FLAME WILL BE EXTINGUISHED, BUT THE PILOT BURNER WILL REMAIN OPERATING.

WHEN THE DEADMAN TRIGGER IS PULLED IN AND HELD AGAIN, THE MAIN FLAME WILL IGNITE AND RETURN TO THE HEIGHT THAT THE FLAME SWITCH IS CURRENTLY SET AT.

5. To shut the pilot flame off, at the Wireless Transmitter, toggle the **PILOT** switch to the **OFF** position.

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3.3.3.1 FLASHOVER FIRE

1. To activate the flashover fire, at the Wireless Transmitter toggle the switch to the **FLASHOVER** position and hold. The flashover effect flame will be activated for about 5 seconds, unless the **FLASHOVER** switch is released before timeout is reached.

NOTE

THE FLASHOVER FLAME CAN ONLY BE OPERATED WHEN DEADMAN TRIGGER IS PULLED IN AND HELD.

3.3.3.2 ROLLOVER FIRE

2. To activate the rollover fire, at the Wireless Transmitter toggle the switch to the **ROLLOVER** position and hold. The rollover effect flame will be activated as long as the switch is held in the **ROLLOVER** position.

NOTE

THE ROLLOVER FLAME CAN ONLY BE OPERATED WHEN DEADMAN TRIGGER IS PULLED IN AND HELD.

3.3.3.3 SMOKE OPERATION

1. To emit smoke, at the Wireless Transmitter toggle the **SMOKE/VENT** switch to the **SMOKE** position. Toggle switch to **OFF** position to stop emitting smoke.

3.3.3.4 EXHAUST FAN OPERATION

1. Activate the exhaust fan by toggle the **SMOKE/VENT** switch to the **VENT** position. Exhaust fan will ramp up to high speed to clear the compartment.
2. Toggle switch to the **OFF** position to return exhaust fan to the current operating mode.

3.3.4 SOUND SYSTEM OPERATION

NOTE

FOR OPERATION REFER TO THE INTEGRATED SOUND EFFECTS O & M MANUAL LOCATED IN SECTION 6.

3.3.5 VENTILATION CONTROL CABINET OPERATION (2ND FLOOR)

See Section 2 Figures 2-10 and Table 2-7 for Ventilation Control Cabinet controls descriptions and layout.

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3.4 OPERATOR PANEL MESSAGES

Table 3-1. Panel Messages

Message	Reason	Remedy
Compartment Warning Temperature	Compartment Thermocouple reading 500°F, a warning condition occurs, and the ventilation/exhaust fan accelerates	Once Temperature drops below 500° ventilation/exhaust fan speed reduces.
	Compartment Thermocouple reading 700°F, alarm condition occurs, and the system will automatically fault into an emergency stop condition.	STBY-ON-RESET key must be placed in the RESET position and released to reset the system. If there are no other system errors the message will read “Kidde Fire Trainers Fire Training System”
Combustion Airflow Failure	The signal that indicates that the combustion air blower for the named fireplace is not present.	Check to see if the applicable combustion air blower is running. If it is, contact Customer Service at Kidde Fire Trainers. If the blower is not running, restart the system. After the system has restarted if the message is still present, contact Customer Service at Kidde Fire Trainers.
Smoke Generator Temp Out Of Range	The smoke generator temp. Sensor is indicating that the temperature is not within preset limits after the warm up period.	Turn off the smoke generator and turn on again. If the message is still displayed, contact Customer Service at Kidde Fire Trainers.
Pilot Time Out	Indicates the fireplace has reached the maximum time allowable in the pilot mode and the fireplace automatically shut down.	Restart

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Table 3-1. Panel Messages		
Message	**Reason**	**Remedy**
Mockup Pilot Ignition Failure	The signal that confirms that there is a pilot flame at the burner head is not present. The system will attempt to ignite the pilot three times before this message will appear.	If this message appears, restart system and attempt to start the pilot again. If the fault appears again, contact Customer Service at Kidde Fire Trainers.
Mockup Pilot Failed to Turn Off	The pilot flame is monitored for a short time period after the fireplace is turned off. If there is no pilot flame, but there is still a signal from the burner control unit indicating that the pilot flame is on, this message will appear.	Restart the system. After the system has restarted if the message is still present, contact Customer Service at Kidde Fire Trainers.
Mockup Pilot Request While Flame SW On	Pendant FLAME Switch is in the LOW or HIGH position.	Pilot will not ignite unless FLAME switch is in the OFF position.
Mockup Gas Pressure Out of Range	The main gas pressure has dropped below the preset low limit for correct system operation. The main gas supply has exceeded the preset high-pressure limit.	Verify that the main gas valve is open. If the problem persists, contact Kidde Fire Trainers Customer Service. Contact Kidde Fire Trainers Customer Service.
System Emergency Stop Pressed	The Emergency Stop button has been pushed in.	The push button must be pulled out and the STBY-ON-RESET key must be placed in the RESET position and released to reset the system. If system is ok the message will read “Kidde Fire Trainers Fire Training System”

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Table 3-1. Panel Messages		
Message	**Reason**	**Remedy**
Emergency Shutdown Circuit Failure	Emergency Circuit Relay has not Reset.	Turn the STBY-ON-RESET key to the RESET position and release to reset the system. If the message persists, Shut system down, and then restart. If the message persists, contact Kidde Fire Trainers Customer Service.
Gas Sensor 1 Warning Level	Gas sensor 1 detected a LEL gas level over 10% LEL).	Once the LEL drops below 10% the ventilation fan will reduce from high speed and message will read “Kidde Fire Trainers Fire Training System”. If problem persist refer to Section 4, paragraph 4.2.2 If the problem persists, contact Kidde Fire Trainers Customer Service.
Gas Sensor 2 Warning Level	Gas sensor 2 detected a LEL gas level over 10% LEL).	Once the LEL drops below 10% the ventilation fan will reduce from high speed and message will read “Kidde Fire Trainers Fire Training System”. If problem persist refer to Section 4, paragraph 4.2.2 If the message persists, contact Kidde Fire Trainers Customer Service.
Gas Sensor 3 Warning Level	Gas sensor 3 detected a LEL gas level over 10% LEL).	Once the LEL drops below 10% the ventilation fan will reduce from high speed and message will read “Kidde Fire Trainers Fire Training System”. If problem persist refer to Section 4, paragraph 4.2.2 If the message persists, contact Kidde Fire Trainers Customer Service.

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Table 3-1. Panel Messages		
Message	**Reason**	**Remedy**
Gas Sensor 1 High Level Shutdown	Gas sensor 1 detected a LEL gas level of 25% or higher. A system shutdown is initiated. All ventilation fans will remain on at high speed until the LEL level has dropped below 25% and then remain on for a further 30 seconds before automatically turning off.	Wait until the ventilation fans have turned off. And at the control panel reset system. If problem persist refer to Section 4, paragraph 4.2.2. If the problem persists, contact Kidde Fire Trainers Customer Service.
Gas Sensor 2 High Level Shutdown	Gas sensor 2 detected a LEL gas level of 25% or higher. A system shutdown is initiated. All ventilation fans will remain on at high speed until the LEL level has dropped below 25% and then remain on for a further 30 seconds before automatically turning off.	Wait until the ventilation fans have turned off. And restart system. If problem persist refer to Section 4, paragraph 4.2.2. If the problem persists, contact Kidde Fire Trainers Customer Service.
Gas Sensor 3 High Level Shutdown	Gas sensor 3 detected a LEL gas level of 25% or higher. A system shutdown is initiated. All ventilation fans will remain on at high speed until the LEL level has dropped below 25% and then remain on for a further 30 seconds before automatically turning off.	Wait until the ventilation fans have turned off. And restart system. If problem persist refer to Section 4, paragraph 4.2.2. If the problem persists, contact Kidde Fire Trainers Customer Service.

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3.3 EMERGENCY STOP LAMP INDICATIONS

The lamp on the Emergency Stop Push Button located on the Control Console will change status in response to several malfunctions that may occur. Table 3-1 lists these malfunctions with their reason and other possible indications.

Table 3-1-. Emergency Stop Lamp Indications

Lamp Indication	Possible Reason	Other Indications
Fast Flash	Vent Fan Purge Failure	On system startup when the key switch is initially switched on, the ventilation fan should come on for 30 seconds. The ventilation fan can be heard clearly when it ramps up to high speed.
Fast Flash	PLC Battery Low	Lamp on PLC processor located in the Electronic Enclosure will indicate low battery.
Fast Flash	E-Stop Circuit Failure	A hard wire or component failure has occurred in the E-Stop circuitry.
Fast Flash	High Temperature Shutdown	The thermocouple reading at or above 700°F an alarm condition occurs, and the system will automatically fault into an emergency stop condition The ventilation fan will ramp up to high speed. The ventilation fan lamp will fast flash.
Fast Flash	Smoke generator temperature out of range	The pendant SMK lamp will not go from slow flash to fast flash. The digital readout on the smoke generator will read a temperature other than 370-390°C.
Solid	Emergency Stop Push button activation	One of the E-Stop push buttons has been pushed in. Buttons are located: Refer to Section 2, for E-Stop Locations. All pilot and main flames that are on will be shut down.
Solid	High Gas Detection shutdown	The presence of a combustible gas concentration exceeds 25% of the LEL. When a concentration exceeding this amount is detected the system will automatically fault into an emergency stop condition. The ventilation fan will ramp up to high speed.

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3.5 System Shutdown

Once training has been completed for the day the following steps should be taken:

At Pendant:

1. Turn off **FLAME** switch.
2. Turn off **PILOT** switch.
3. Disconnect Pendant from Electrical Control Panel and safely store Pendant.

At Wireless Transmitter:

1. Toggle the **FLAME** switch to the **OFF** position. Release the **Deadman Trigger**.
2. Toggle the **PILOT** switch to the **OFF** position.
3. Toggle top switch to the **OFF** position.
4. Disconnect Wireless Receiver assembly from Electrical Control Panel and safely store Transmitter.

WARNING

To prevent unintentional operation, when the Wireless Control System is not being used, be sure to disconnect the Receiver assembly from the Electrical Control Panel and securely store the Transmitter.

At Control Panel:

1. Turn **STBY-ON-RESET** key to the **STBY** position.
2. Shut quarter turn gas valve at valve cabinet.
3. Turn off gas valve at tank.
4. Unplug instructor pendant and store in safe place.
5. Lock all cabinets
6. Inspect fire training compartments and props for sign of damage.
7. Verify that all equipment is stored safely, and that all cabinets are locked. Inspections of all equipment should be preformed on a daily schedule. These visual inspections are best done at the beginning and at the end of each training day.

CAUTION

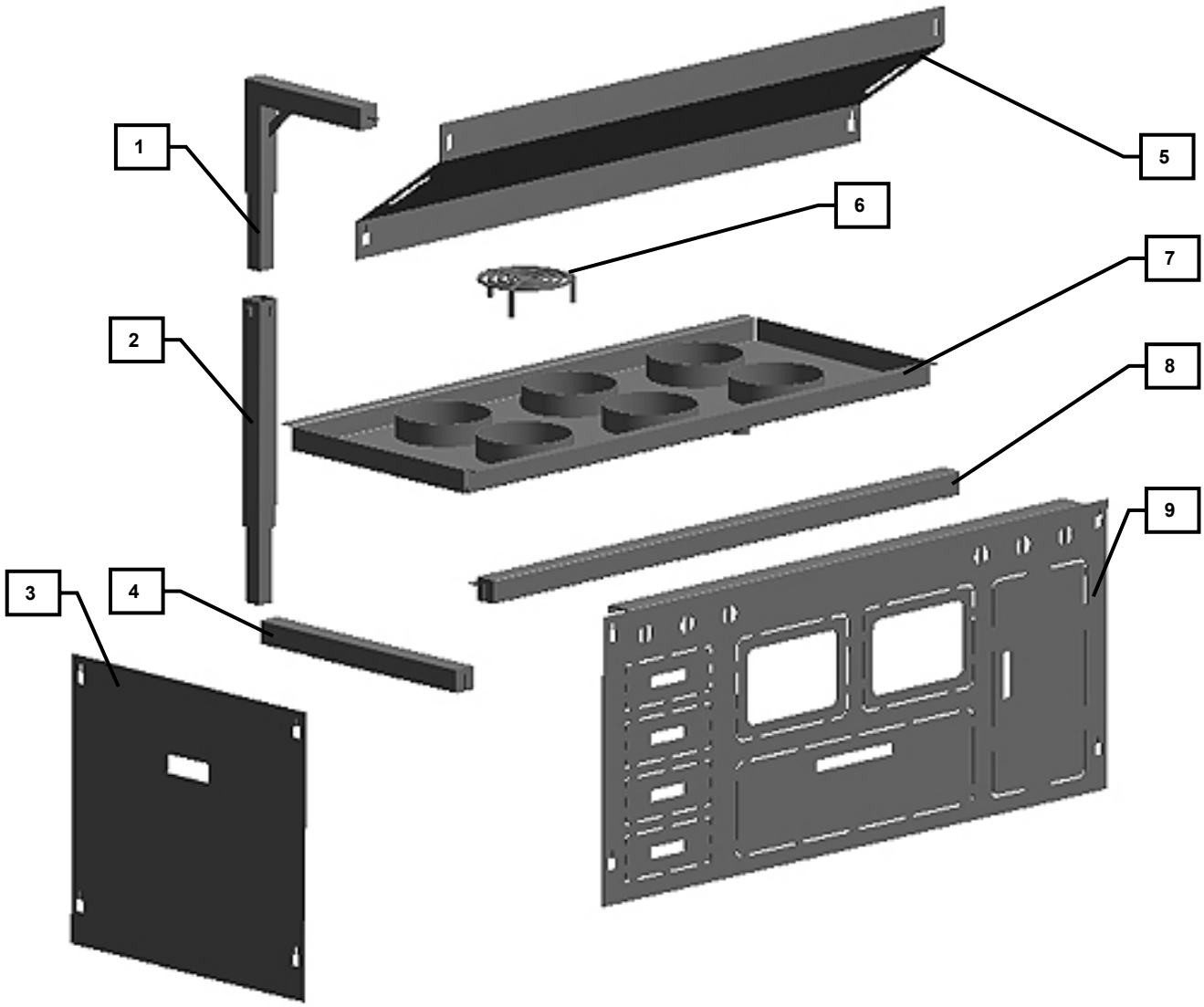
When not in use the fireplace water bath pans should be drained of water during cold weather periods, when the anticipated ambient temperature is going to be below 32°F

Failure to follow this caution may result in serious damage to the associated equipment.

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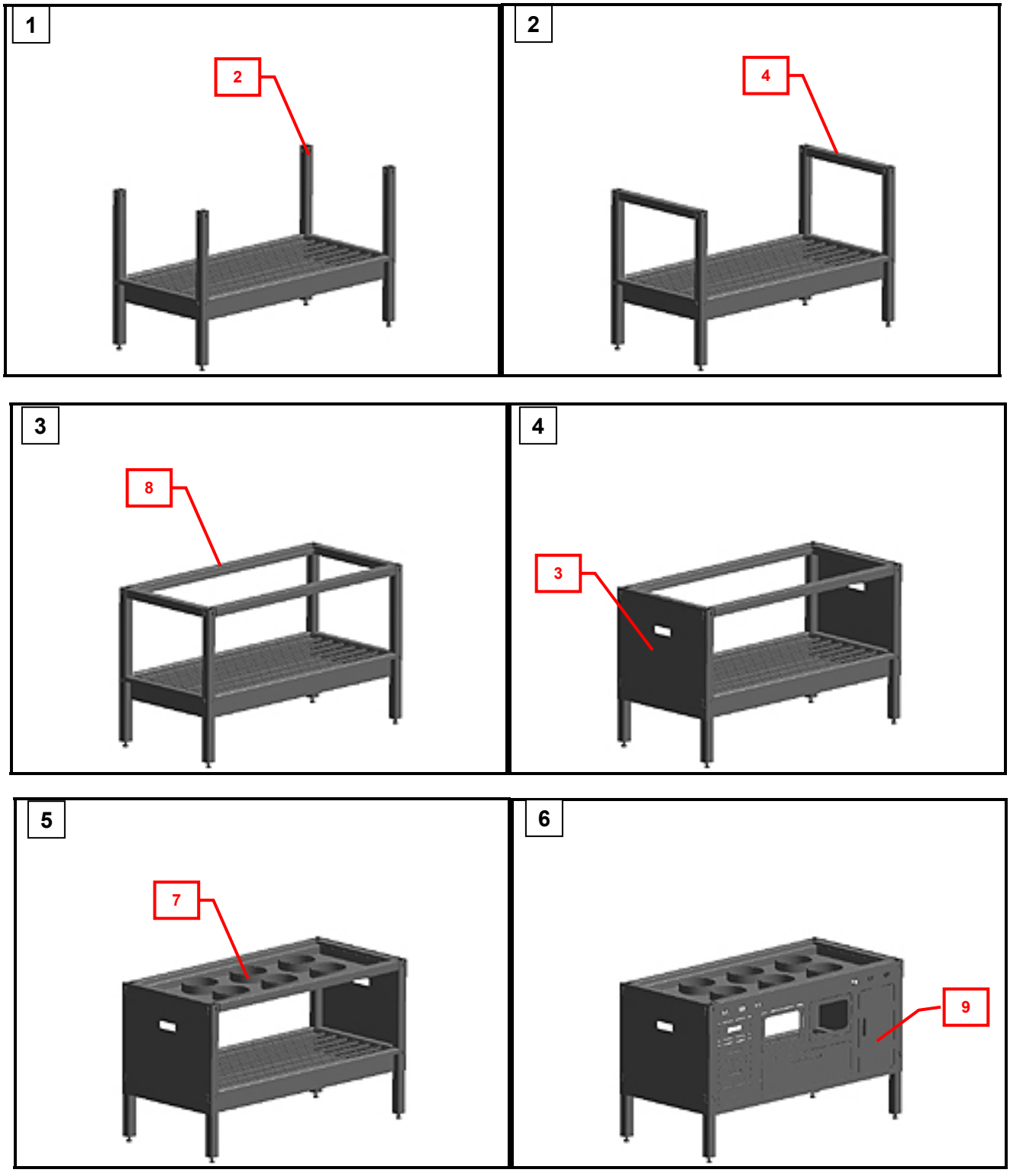
3.6 MOCK-UP ASSEMBLY INSTRUCTIONS

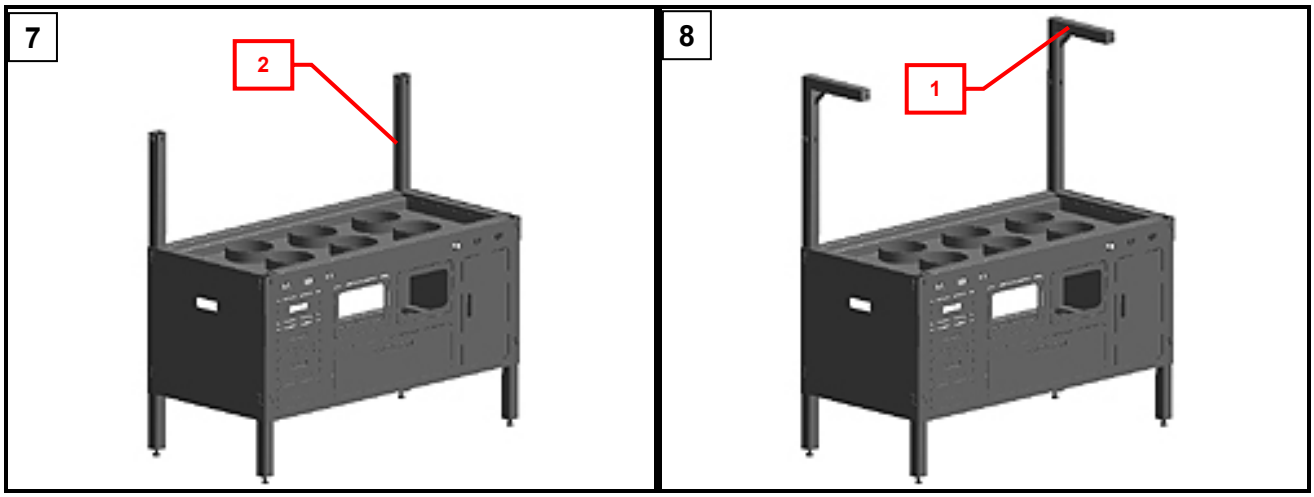
3.6.1 STOVE FPLC MOCK-UP

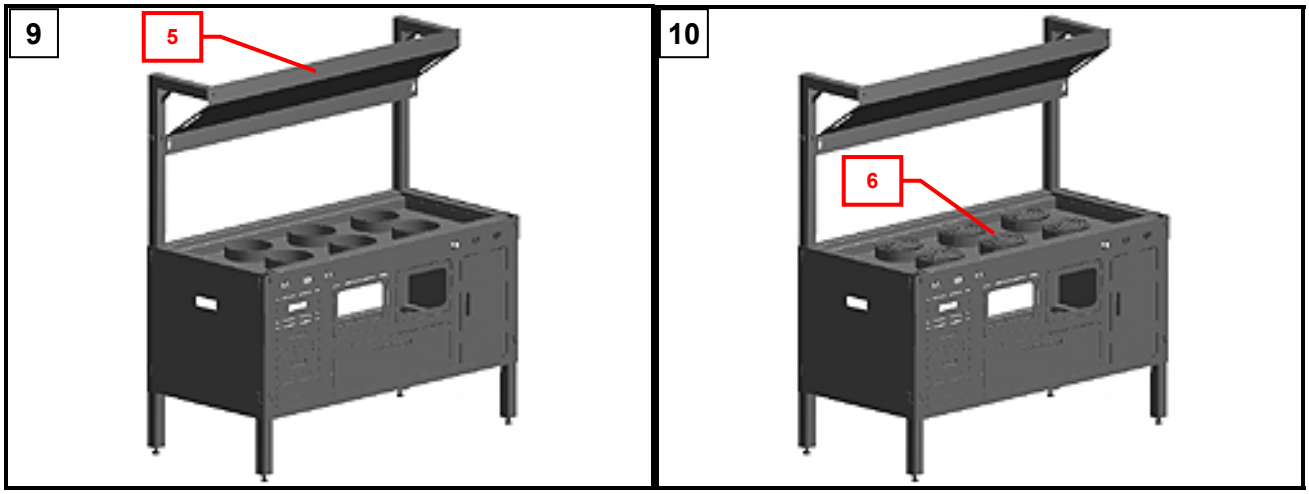


ITEM	DESCRIPTION	PART #	QTY
1	HOOD BRACKET	4D1-006-352	2
2	SIDE POST	4D1-006-349	6
3	STOVE SIDE PLATE	4D1-006-355	2
4	SHORT CROSS MEMBER	4D1-006-350-002	2
5	HOOD PLATE	4D1-006-353	1
6	LARGE STOVE COIL	4D1-006-358-002	6
7	KITCHEN STOVE TOP	4D1-006-356	1
8	LONG CROSS MEMBER	4D1-006-350-001	2
9	STOVE FRONT PLATE	4D1-006-354	1

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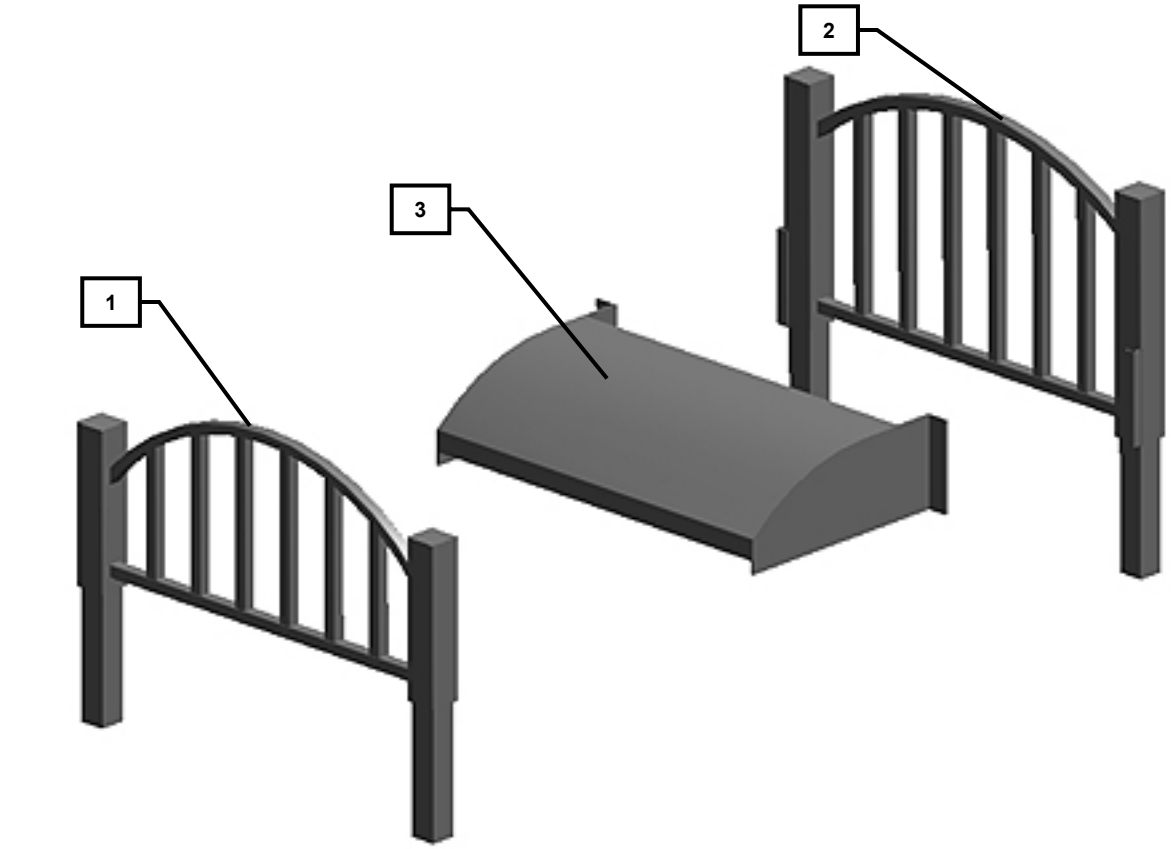


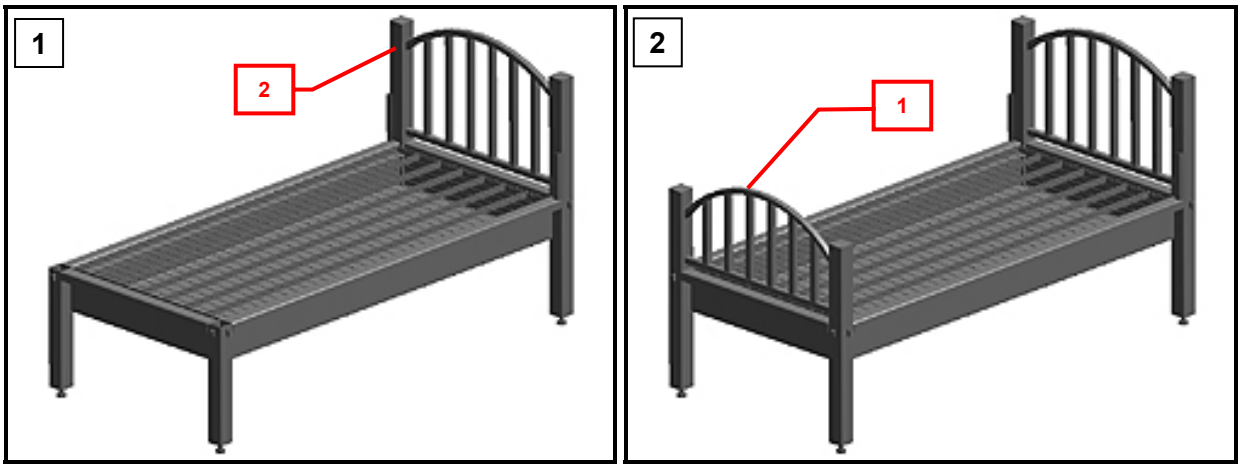


ITEM	DESCRIPTION	PART #	QTY
1	HOOD BRACKET	4D1-006-352	2
2	SIDE POST	4D1-006-349	6
3	STOVE SIDE PLATE	4D1-006-355	2
4	SHORT CROSS MEMBER	4D1-006-350-002	2
5	HOOD PLATE	4D1-006-353	1
6	LARGE STOVE COIL	4D1-006-358-002	6
7	KITCHEN STOVE TOP	4D1-006-356	1
8	LONG CROSS MEMBER	4D1-006-350-001	2
9	STOVE FRONT PLATE	4D1-006-354	1

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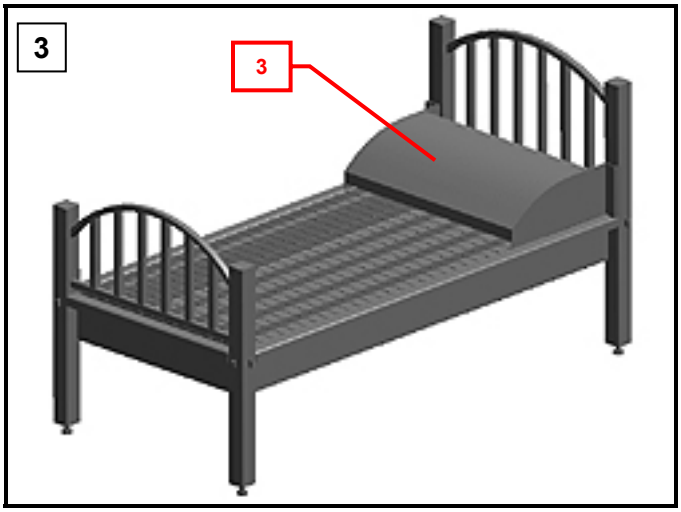
3.6.2 BED FPLC MOCK-UP





ITEM	DESCRIPTION	PART #	QTY
1	BED FOOTBOARD	4A1-006-478-D02	1
2	BED HEADBOARD	4A1-006-478-D01	1
3	PILLOW	4A1-006-478-D03	1

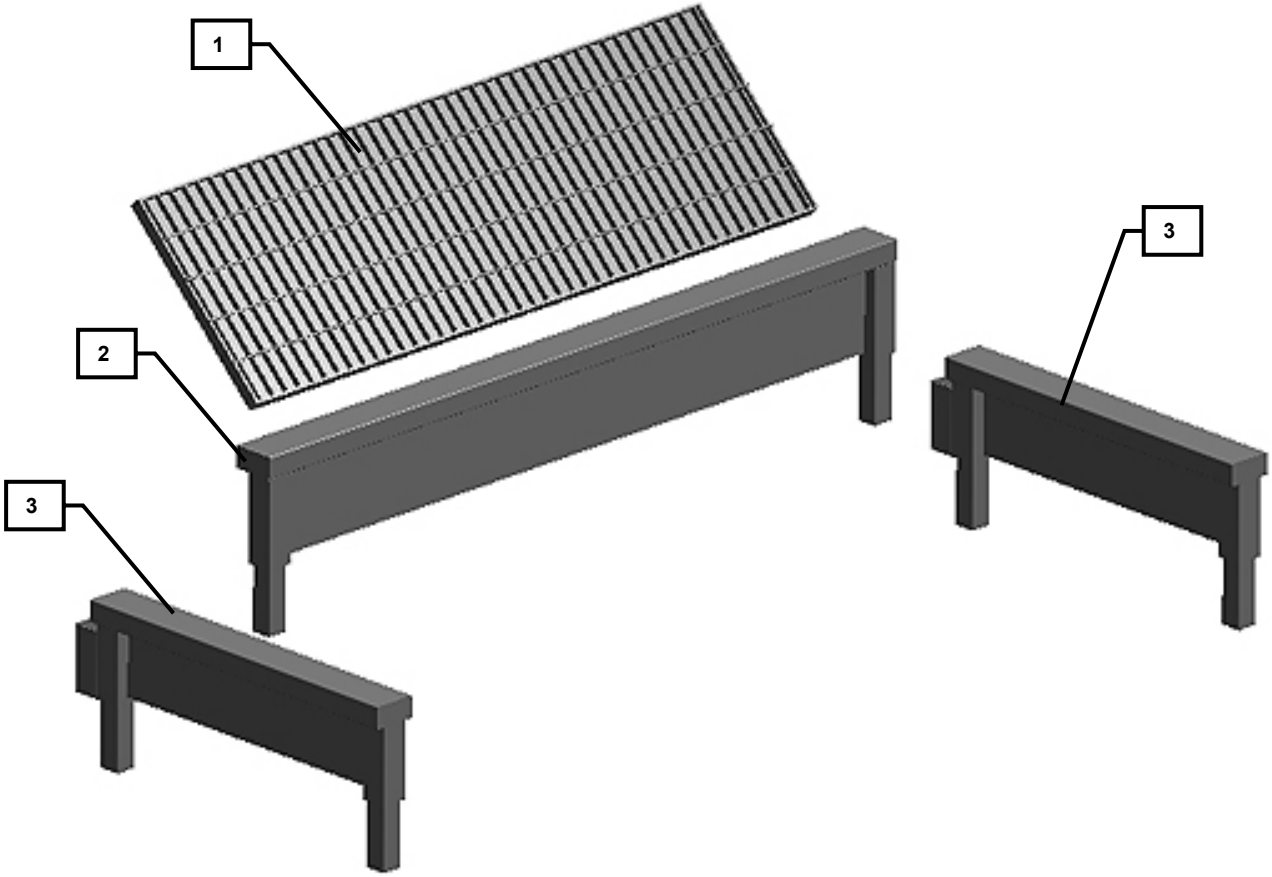
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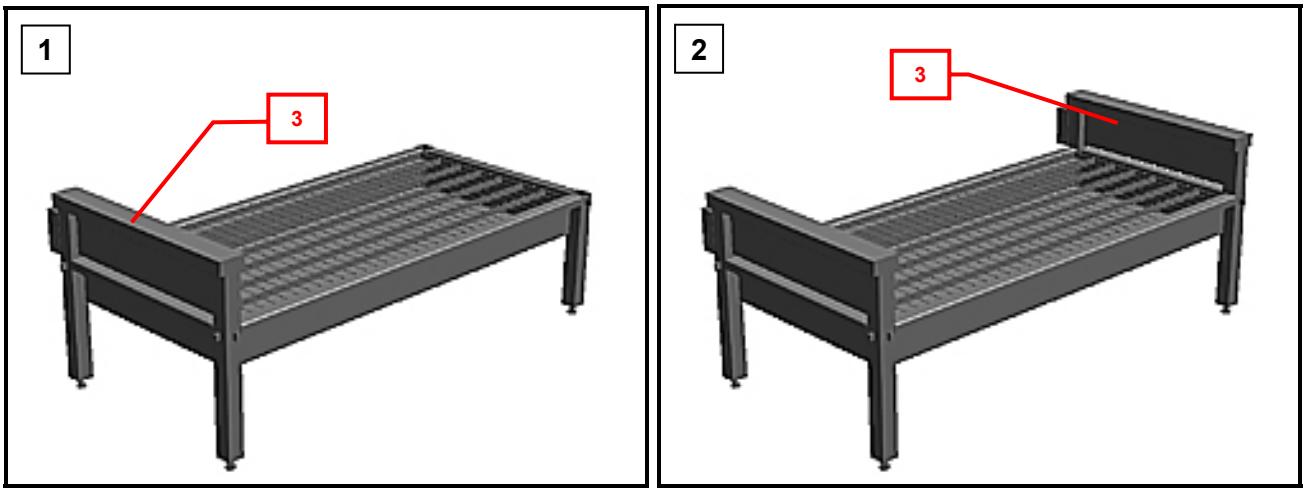


ITEM	DESCRIPTION	PART #	QTY
1	BED FOOTBOARD	4A1-006-478-D02	1
2	BED HEADBOARD	4A1-006-478-D01	1
3	PILLOW	4A1-006-478-D03	1

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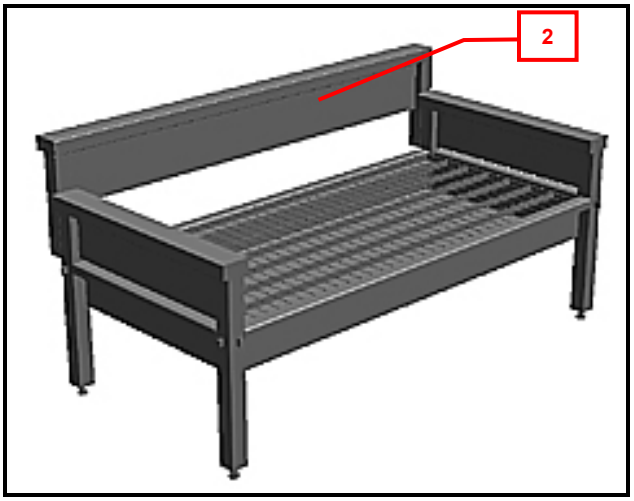
3.6.3 SOFA FPLC MOCK-UP

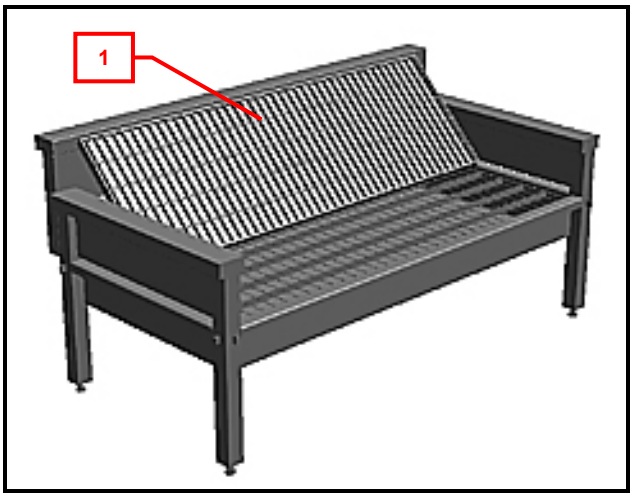




ITEM	DESCRIPTION	PART #	QTY
1	GRATING ASSEMBLY	4C1-006-335	1
2	BACKREST	4A1-006-480-D01	1
3	LEFT AND RIGHT ARMREST	4A1-006-480-D02	2

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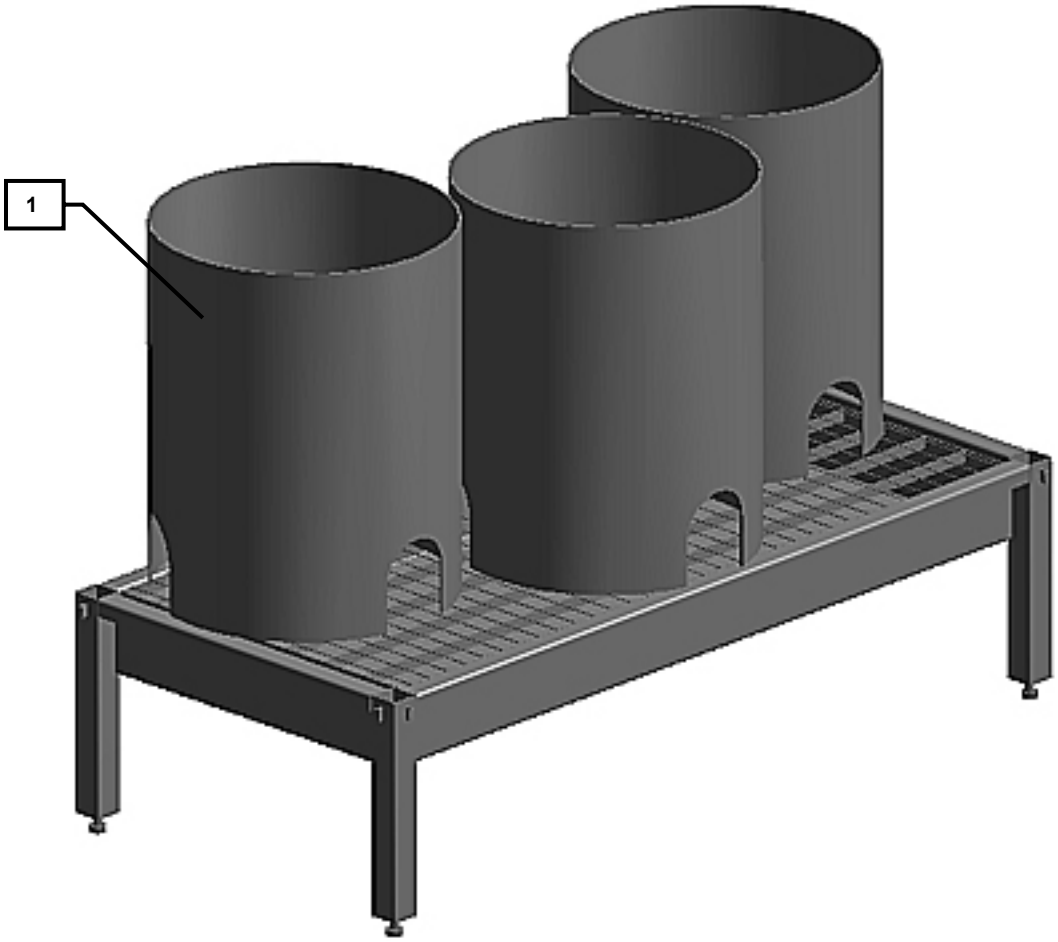




ITEM	DESCRIPTION	PART #	QTY
1	GRATING ASSEMBLY	4C1-006-335	1
2	BACKREST	4A1-006-480-D01	1
3	LEFT AND RIGHT ARMREST	4A1-006-480-D02	2

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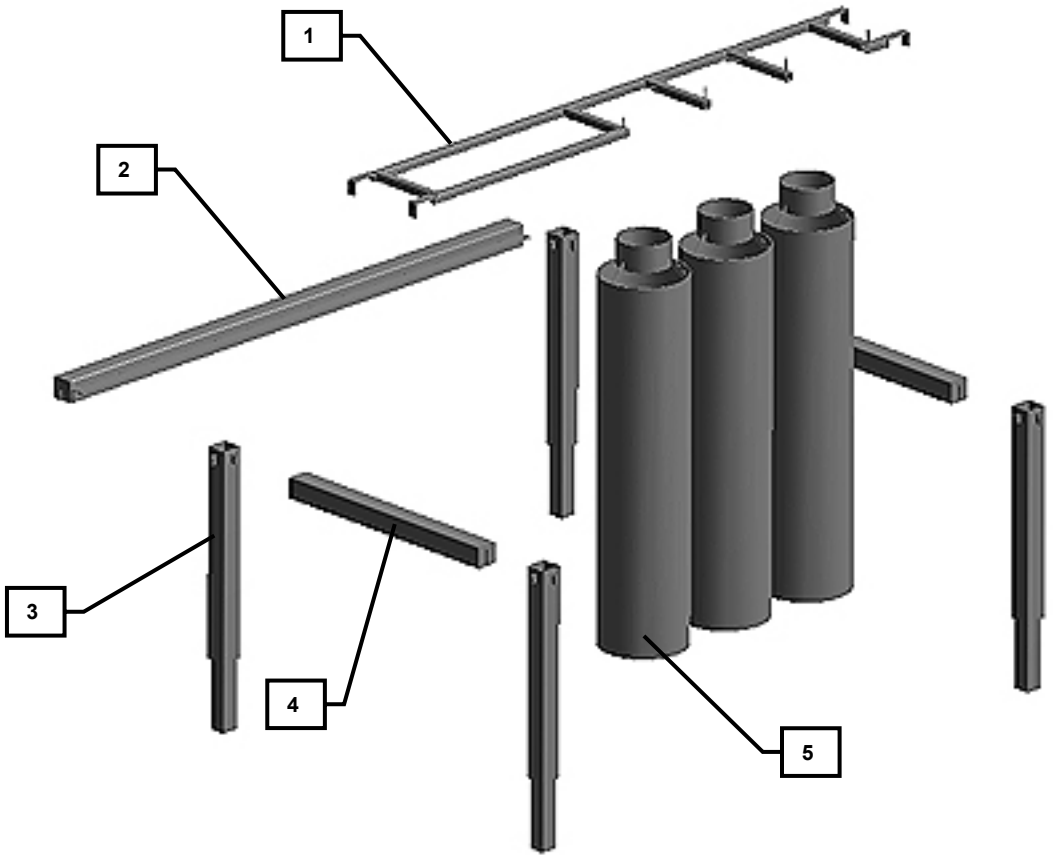
3.6.4 HAZMAT RACK FPLC MOCK-UP

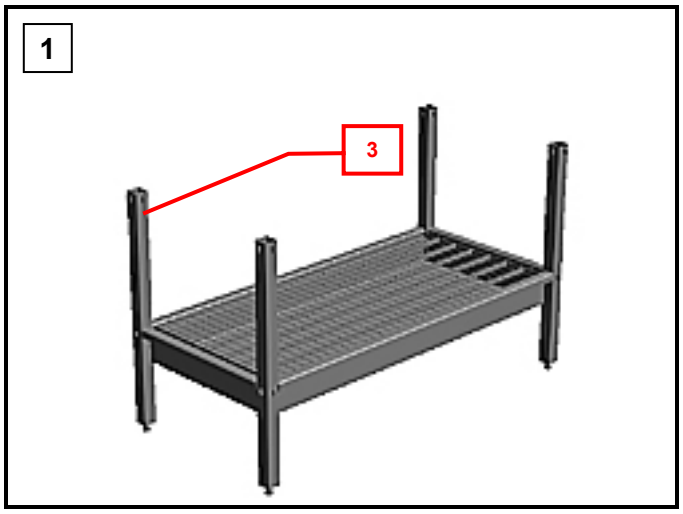


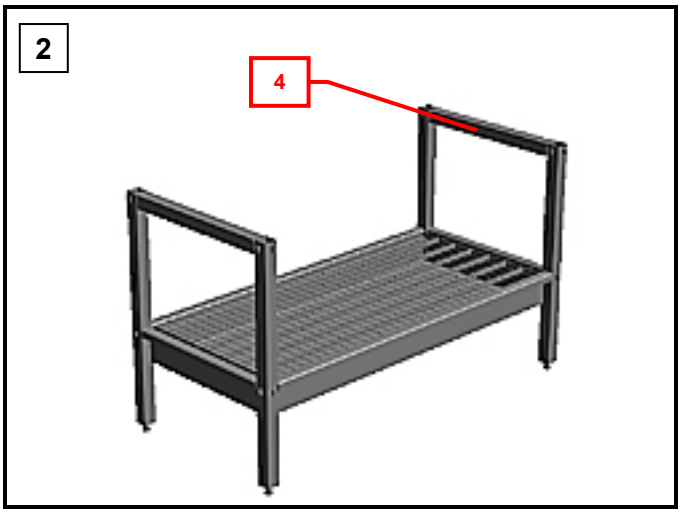
ITEM	DESCRIPTION	PART #	QTY
1	24” DIA. CYLINDER	ICS-336	3

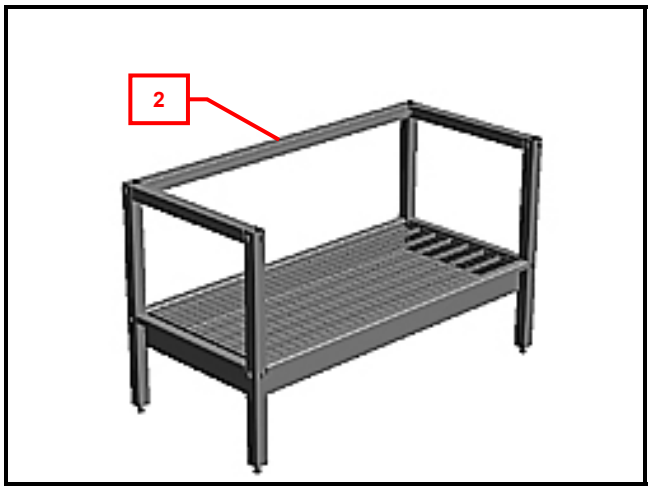
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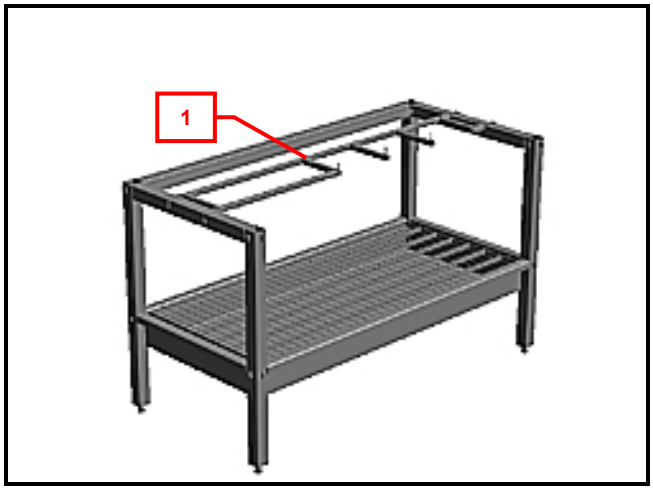
3.6.5 BOTTLE RACK FPLC MOCK-UP

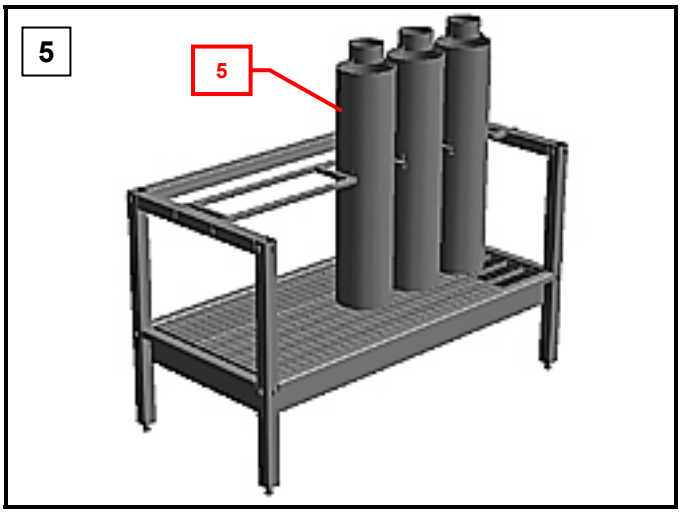








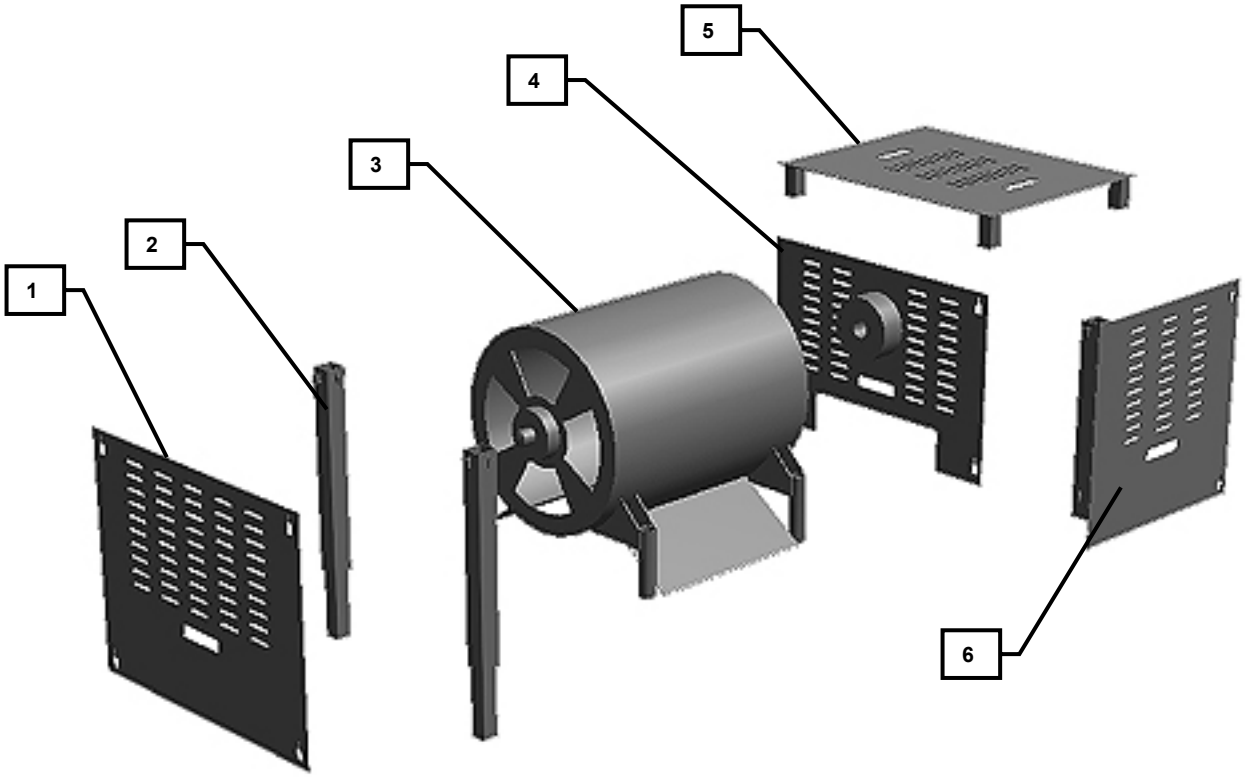


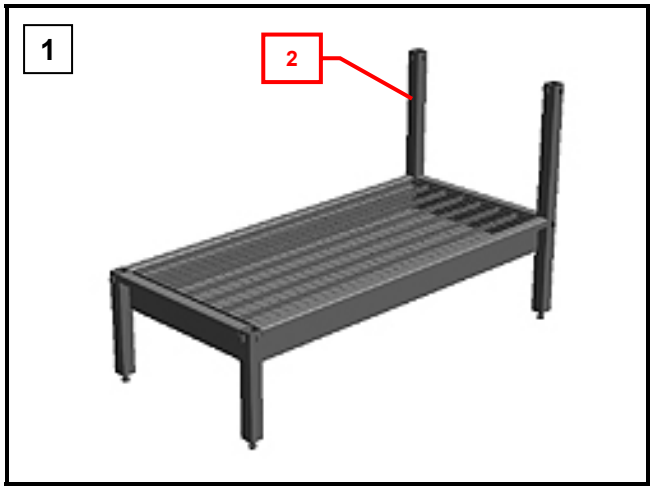


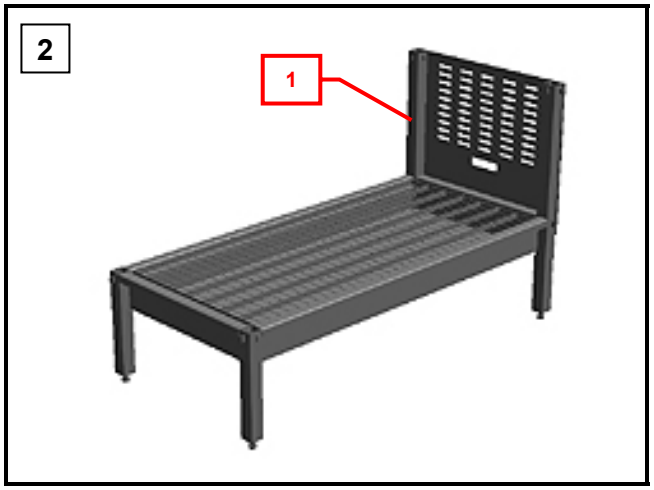
ITEM	DESCRIPTION	PART #	QTY
1	BOTTLE SUPPORT	4D1-006-347	1
2	LONG CROSS MEMBER	4D1-006-350-001	1
3	SIDE POST	4D1-006-349	4
4	SHORT CROSS MEMBER	4D1-006-350-002	2
5	BOTTLE	4D1-006-346	3
6	CHAIN	2C1-003-304	1

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3.6.6 MOTOR GENERATOR FPLC MOCK-UP

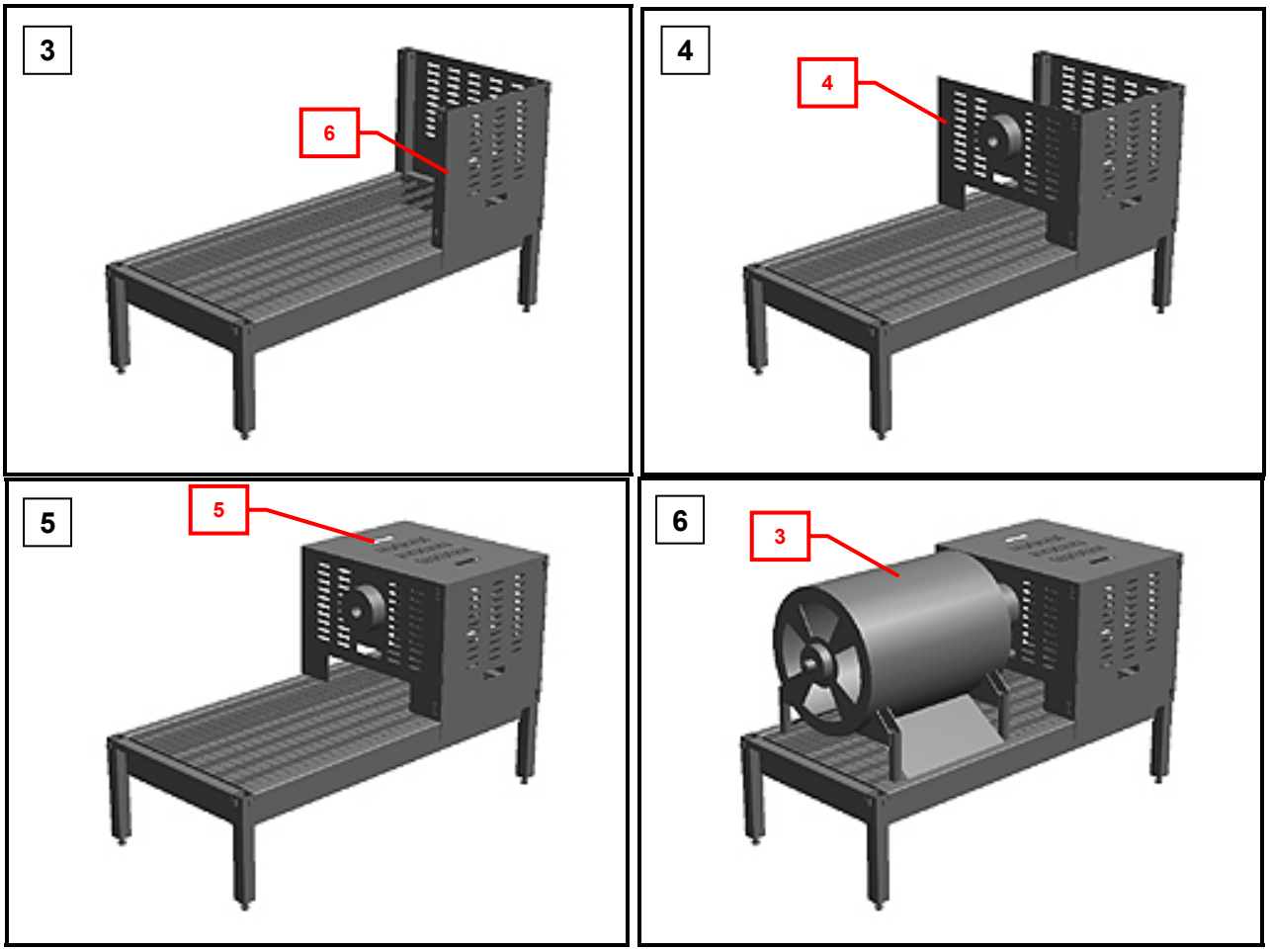






ITEM	DESCRIPTION	PART #	QTY
1	GENERATOR BACK SIDE PLATE	4D1-006-340	1
2	SIDE POST	4D1-006-349	2
3	ELECTRIC MOTOR GENERATOR	4D1-006-339	1
4	GENERATOR FRONT SIDE PLATE	4D1-006-343	1
5	GENERATOR TOP SIDE PLATE	4D1-006-342	1
	GENERATOR SIDE PLATE	4D1-006-341	1

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.



ITEM	DESCRIPTION	PART #	QTY
1	GENERATOR BACK SIDE PLATE	4D1-006-340	1
2	SIDE POST	4D1-006-349	2
3	ELECTRIC MOTOR GENERATOR	4D1-006-339	1
4	GENERATOR FRONT SIDE PLATE	4D1-006-343	1
5	GENERATOR TOP SIDE PLATE	4D1-006-342	1
	GENERATOR SIDE PLATE	4D1-006-341	1

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4.0 MAINTENANCE & SERVICING

4.1 MAINTENANCE

4.1.1 INTRODUCTION

A Live Fire Trainer requires maintenance to ensure that the mechanical and electrical components of the system are reliable and function properly. Kidde Fire Trainers, Inc. has a maintenance schedule that will limit the down time to a Live Fire Training System as long as the maintenance schedule is followed and proper care is taken. There are 4 levels of maintenance personnel. The levels are as follows:

LEVEL 1

This maintenance person can perform basic tasks, under the supervision of level 2 or level 3 Maintenance Personnel. This person shall have knowledge of the system, and have knowledge of the dangers that are involved in the system.

LEVEL 2

This maintenance person can perform intermediate tasks, under the supervision of level 3 Maintenance Personnel. This person shall have great knowledge of the system, and have knowledge of the dangers that are involved in the system. A level 2 Maintainer shall know how the system functions, and shall have basic trouble shooting skills.

LEVEL 3

This maintenance person can perform advanced tasks without any supervision. This person shall comply with the authority having jurisdiction, and shall be trained by Kidde Fire Trainers, Inc. Service Specialist. A level 3 Maintainer shall perform all trouble shooting, and shall take the responsibility for all maintenance aspects of the system

LEVEL 4

This is a Kidde Fire Trainers Service Specialist. This person shall comply with the authority having jurisdiction. A Kidde Fire Trainers service specialist shall inspect, calibrate, adjust, and verify that the safety system is in complete working operation. A Kidde Fire Trainers service specialist will also verify that the system functions properly and reliably.

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4.1.2 MAINTENANCE SCHEDULE

DAILY CHECK DESCRIPTION	LEVEL
Smoke Generator(s) (See Smoke Gen Manual in Sec. 6)	
Check Oil (Add if required) Kidde Type II Smoke Fluid.	**1**
Check Nitrogen/CO^2 Level. (Replace bottle if required)	**1**
Ventilation Fan (Refer to Section 6 for Info)	
Listen for excessive or unusual vibration or noise, report problems.	**1**
Emergency Stops	
Before Training begins randomly press an emergency stop to verify correct operation of safety system.	**1**
Fire Place Burn Pans	
Verify that water is above burner sparge in each of the pans. (Fill as required)	**1**
In seasons when overnight freezing may occur drain pans.	**1**

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WEEKLY CHECK DESCRIPTION	LEVEL
Smoke Generator(s) (See Smoke Gen Manual in Sec. 6)	
Check Oil (Add if required) Kidde Type II Smoke Fluid.	**1**
Check Nitrogen/CO^2 Level (Replace bottle if required).	**1**
Gas Detection System	
Perform a Combustible Gas Detection System test (Refer to 4.2.3)	**3**
Ventilation Fan (Refer to Section 6 for Info)	
Listen for excessive or unusual vibration or noise, report problems.	**2**
Visually inspect fan belt for signs of ware.	**2**
Visually inspect airflow switch for signs of damage.	**2**
Emergency Stops	
Before Training begins randomly press an emergency stop to verify correct operation of safety system.	**2**
Fire Place Burn Pans	
Verify that water is above burner sparge in each of the pans. (Fill as required)	**1**
In seasons when overnight freezing may occur drain pans.	**1**
Before training begins for the day inspect fireplace burn pans for unusual or abnormal conditions. (Report any potential problems)	**1**
Electrical Control Cabinet	
Verify that the BAT LED on the front of each PLC is not on (red). Replace if necessary. See 4.2.4 for procedure.	**3**

NOTE

REFER TO SECTION 6 FOR VENTILATION FAN LUBRICATION INFORMATION.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

MONTHLY CHECK DESCRIPTION	LEVEL
Smoke Generator(s) (See Smoke Gen Manual in Sec. 6)	
Remove carbon deposit from nozzle.	**3**
Ventilation Fan (Refer to Section 6 for Info)	
Listen for excessive or unusual vibration or noise, report problems.	**2**
Visually inspect fan belt for signs of ware.	**2**
Visually inspect airflow switch for signs of damage.	**3**
Emergency Stops	
Before Training begins randomly press an emergency stop to verify correct operation of safety system.	**3**
Fire Place Burn Pan	
Verify that water is above burner sparge in each pan. (Fill as required)	**1**
In seasons when overnight freezing may occur drain pans.	**1**
Before training begins for the day inspect fireplace burn pans for unusual or abnormal conditions (Report any potential problems).	**1**
Electrical Cabinet	
Visually inspect the cabinet internally and externally for signs of damage and corrosion.	**3**
Exposed Pipe Work	
Visually inspect pipe for signs of damage and corrosion.	**3**
Valve Cabinets	
Visually inspect the cabinet internally and externally for signs of damage and corrosion.	**3**

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

REFER TO SECTION 6 FOR VENTILATION FAN LUBRICATION INFORMATION.

3 MONTHLY CHECK DESCRIPTION	LEVEL
Combustible Gas Detection System	
Perform the Combustible Gas Detection System Adjustment (Refer to 4.2.2).	**3**

NOTE

REFER TO SECTION 6 FOR VENTILATION FAN LUBRICATION INFORMATION.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

ANNUAL CHECK DESCRIPTION	LEVEL
Smoke Generator(s) (See Smoke Gen Manual in Sec. 6)	
Check internal components for corrosion.	**4**
Fill Oil tank to proper level with Kidde Type II Smoke Fluid.	**1**
Clean nozzle of carbon deposit.	**4**
Open generator and inspect for oil and heat damage.	**4**
Temperature Read-Out	
Verify with a thermocouple simulator that operational set points trigger correctly.	**3**
Ventilation Fan (Refer to Section 6 for Info)	
Remove soot from exhaust duct.	**1**
Visually inspect fan belt for signs of ware.	**2**
Visually inspect airflow switch for signs of damage.	**3**
Visually inspect fan blade for signs of ware.	**3**
Verify that correct operation of warnings and alarms.	**4**
Emergency Stops	
Test each emergency stop under operational conditions.	**4**
Fire Place Burn Pans	
Visually Inspect fireplace.	**4**
Visually Inspect Burner Sprague.	**4**
Visually Inspect Burner Sprague connection.	**4**

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ANNUAL CHECK DESCRIPTION	LEVEL
Electrical Cabinet	
Visually inspect the cabinet internally and externally for signs of damage and corrosion.	**4**
Torque all terminal connections.	**4**
Measure & Record readings on all motor lines.	**4**
Measure & Record Amperage reading on main electrical feed.	**4**
Exposed Pipe Work	
Visually inspect pipe for signs of damage and corrosion.	**4**
Valve Cabinets	
Visually inspect the cabinet internally and externally for signs of damage and corrosion.	**4**
Replace Air Filters.	**4**
Measure & Record Pilot air pressure.	**4**
Measure & Record Combustion air pressure.	**4**
Verify Combustion air pressure switch is operating properly.	**4**
Pressure test gas piping.	**4**

NOTE

REFER TO SECTION 6 FOR VENTILATION FAN LUBRICATION INFORMATION.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

4.2 SERVICING

4.2.1 CONSUMABLE MATERIALS

Table 4-1 shows a list of the consumable materials along with recommended places to obtain items.

Table 4-1. Consumable Materials

Nomenclature	Part No./Specification	Distributor
Smoke Fluid Kidde Type II Smoke Fluid (Do Not Substitute)	1A1-002-017- 001 (six ½-gallon bottles) 002 (5-gallon bottle) 003 (55-gallon drum)	Kidde Fire Trainers inc. 17 Philips Parkway Montvale, NJ 07645 (201) 300-8100
Span Gas Cylinder, with span gas: 1.1% (±2%) (50% LEL) propane	226166-004	Detector Electronics Corp. 6901West 110th St. Minneapolis, MN55438 (952) 829-8750
Nitrogen	N/A	Local Supplier

4.2.2 COMBUSTIBLE GAS DETECTION SYSTEM ADJUSTMENT

GENERAL

The combustible gas sensors must be adjusted at periodic intervals (3 months) to ensure accuracy in their measurement of unburned gas levels. Unsafe levels could occur due to a leak in a supply line or when fires are being fought.

The combustible gas system consists of two sensors in each Burn Room, two transmitters in each Electronics Enclosure and a sensor/transmitter in Burn Rooms 2 & 3 equipment areas (see Figures 2-2 & 2-3).

A gas standard of a precise mixture of gas to air is injected into each sensor and the system automatically performs an adjustment when it is initiated using a magnet that is held underneath the transmitter unit for an approximate time period.

NOTE

ONLY USE THE TEST GAS PROVIDED

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

ADJUSTMENT GAS KIT

A kit is provided with all the components necessary to test the gas sensors for accuracy. The kit consists of:

- Test gas cylinders (2)
- Gas pressure gauge (1) and Sensor cap (1) connected by 3 ft. of plastic hose

Figure 4-1 shows the kit components ready for use. The gas pressure gauge and sensor cap assembly is screwed on to the top of one of the cylinders until it is tight. The knurled knob to the right of the gauge allows gas to flow to the sensor cap when it is opened.

ADJUSTMENT PROCEDURE

NOTE

PERFORM THE ADJUSTMENT PROCEDURE IN THE MORNING BEFORE COMMENCING TRAINING TO ENSURE THAT THE AIR AROUND THE SENOR IS RELATIVELY FREE OF CONTAMINANTS.

Refer to Figure 4-1 for details on Test Gas Kit.

1. Locate the sensors in the Burn Room and remove the protective cover by undoing the two nuts on the bottom of the cover. Rotate the cover outwards and lift up to remove.
2. Clip the gas sensor cap onto the bottom of the sensor.
3. Initiate adjustment by holding the Cal magnet under the correct transmitter for 7 seconds (See figure 4-2). The display will sequence through the Setup settings. When the transmitter enters the Calibration mode, the display will show the detected gas concentration and an alternating message “ZERO” “CAL”.
4. When the zero adjustment is complete the display message will now show the detected gas concentration and an alternating message: “APLY” “GAS”. Apply the calibration gas to the sensor by placing the calibration cup over the sensor. The display will show a rising gas concentration and an alternating message “GAS” “ON”
5. Continue applying calibration gas until the display shows the alternating message: “CAL “OK” then “RMV” “GAS”. At this point remove the calibration gas and if the calibration was successful the message displayed will be “XXXX SPAN” where XXXX will display a number that indicates sensor life. Any number over 100 indicates that the sensor is good. When the gas level falls below the lowest alarm set point, and if no faults are present, the transmitter automatically exits the Calibration mode.
6. If a fault is present the display will show the fault description then the alternating message RMV” “GAS. The fault must be cleared before a successful calibration can be completed. Refer to the Fault messages table 2-2.
7. Repeat steps 1-6 for the remaining burn room sensor.
8. Repeat steps 2-6 for burn room 2 equipment area sensor.

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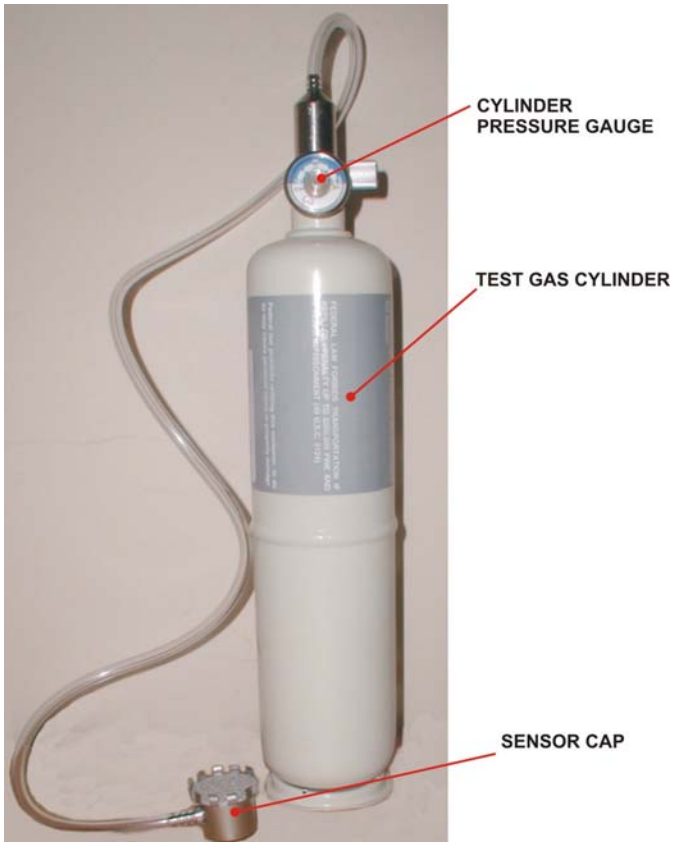


Figure 4-1. Test Gas Kit

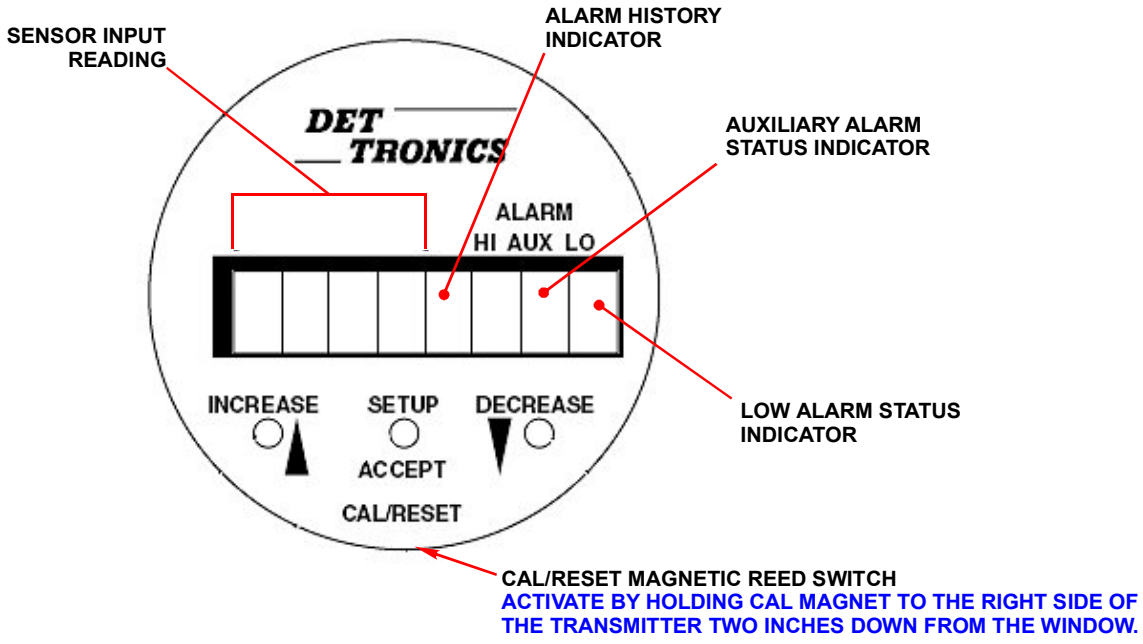


Figure 4-2. Gas Sensor Transmitter

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Table 4-2. Gas Sensor Fault Messages	
Fault Message Display	**Explanation and Corrective Action**
Blank Display	EEPROM sumcheck failure. Cycle power to clear fault. If fault persists, contact Kidde Fire Trainers, Inc. Service Department.
"COMPUTER" "FAILURE"	RAM or processor failure. Cycle power to clear fault. If fault persists, contact Kidde Fire Trainers, Inc. Service Department.
"WDT" "FAILURE"	Watchdog timer failure. Cycle power to clear fault. If fault persists, contact Kidde Fire Trainers, Inc. Service Department.
"EEPROM" "FAILURE"	Activate the Cal/Reset switch using the Cal Magnet, then perform Setup and Calibration procedures. If fault persists, contact Kidde Fire Trainers, Inc. Service Department.
"EXT RSET" "PROBLEM"	External reset button has been activated for 15 seconds or longer. Self-clearing when button is released.
"24V P.S." "FAILURE"	External 24 volt power supply is not in the 15 to 32 volt range. Check and correct input voltage. During normal operating mode, this fault is self clearing when the fault is corrected. If fault occurs during warm-up or calibration modes, activate the Cal/Reset switch using the Cal Magnet.
"5V P.S" "FAILURE"	Internal 5 volt analog power supply is not in the 4.75 to 5.35 volt range. During normal operating mode, this fault is self clearing when the fault is corrected. If fault occurs during warm-up or calibration modes, activate the Cal/Reset switch using the Cal Magnet. If fault persists, contact Kidde Fire Trainers, Inc. Service Department.
"SENSOR" "PROBLEM"	Sensor input fault. With 4 to 20 mA output sensors, the current output is below 2 mA or above 35 mA. In normal operating mode, the unit automatically goes through Warm-up when this fault clears. If this fault occurs at the end of the warm-up period or calibration procedure, recalibrate the sensor. If fault persists, contact Kidde Fire Trainers, Inc. Service Department.
"REPLACE" "SENSOR"	(In calibration mode) sensor is defective. Replace sensor and perform calibration procedure.
"CAL" "ABORTED"	(Cal Message) Time ran out while waiting for the gas reading to stabilize. Activate the Cal/Reset switch using the Cal Magnet.
"SENSOR" "E.O.L."	(Cal Message) Sensor reaching End Of Life. Consider replacement of the sensor in the next 1 or 2 calibrations.
"ZERO" "DRIFT"	Negative zero drift. Sensor input is –9% full scale or lower. Perform sensor calibration.

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4.2.3 COMBUSTIBLE GAS DETECTION SYSTEM TEST

In order to verify that the sensors are functioning correctly, a weekly test of the sensors is required. Refer to Figure 4-1 and Section 4.2.1 for additional information about the gas detection system.

1. Turn the pilot flame on for the Burn Room. The PILOT lamp for the selected fire will flash while the pilot is attempting to ignite and then transition to steady on when ignited.
2. Locate the sensors in the Burn Room and remove the protective covers by undoing the two nuts on the bottom of each cover. Rotate the cover outwards and lift up to remove.

NOTE

ONLY USE THE TEST GAS PROVIDED

3. Clip the gas sensor cap onto the bottom of one of the sensors (See figure 4-1).
4. Inject gas into the sensor by opening the valve on the cylinder.

NOTE

SLOWLY OPEN THE GAS VALVE SO THAT THE LEVEL DOES NOT RISE TOO QUICKLY.

5. While viewing the associated gas sensor transmitter, located in the Electrical Control Cabinet, the gas level display should rise. When the gas level reading reaches above 10% LEL the compartment ventilation fan will go to full speed. When the gas level reaches above 25% LEL an emergency shutdown is initiated. All main flames and pilot flames are removed and the ventilation fan remains at high speed.
6. Turn off the gas and remove the gas sensor cap from the gas sensor. When the gas level drops below a reading of 10% LEL a system purge sequence is initiated. At the end of the 30 seconds system purge sequence, the trainer is ready for operation.
7. Locate the remaining gas burn room gas sensor and repeat steps 1-6.
8. Repeat steps 2-6 for the burn room 2 equipment area sensor.

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4.2.4 PLC Battery Removal and Replacement

WARNING

DO NOT SUBSTITUTE ANY OTHER BATTERY FOR THE 1769-BA. SUBSTITUTING ANY OTHER BATTERY THAN THAT ALLOWED COULD RESULT IN AN UNSAFE CONDITION.

DO NOT INCINERATE OR EXPOSE BATTERY TO HIGH TEMPERATURES.

DO NOT SOLDER BATTERY OR LEADS; BATTERY COULD EXPLODE.

DO NOT OPEN, PUNCTURE, OR CRUSH BATTERY. BATTERY COULD EXPLODE AND TOXIC, CORROSIVE, AND FLAMMABLE CHEMICALS COULD BE EXPOSED.

DO NOT CHARGE BATTERY. EXPLOSION MAY RESULT OR CELL MAY OVERHEAT AND CAUSE BURNS.

DO NOT SHORT POSITIVE OR NEGATIVE TERMINALS TOGETHER BATTERY WILL HEAT.

CAUTION

TO PREVENT PROGRAM LOSS, ENSURE THE PROCESSOR IS POWERED UP DURING BATTERY REPLACEMENT.

The battery (P/N 1769-BA) is required for data backup in the event of a power failure will require replacement when the **BAT** Led on the front of the Controller is red. Follow the procedure below to replace the battery.

1. Verify that there are no gas detection LEL warning or shutdown messages for the applicable burn room.
2. To access the battery, remove the Controller door by sliding it forward.
3. Gently pull the battery out of the retaining clips.
4. Squeeze the retaining clips located on the end of the battery connector and remove connector from socket.
5. Insert the new battery connector into the connector socket until it clicks into place.

NOTE

THE CONNECTOR IS POLARIZED AND MAY ONLY BE INSERTED ONE WAY.

6. Insert the new battery into the battery retaining clips and close door.
7. Verify that the **BAT** LED is off and slide the door back into place on the controller. If the **BAT** LED it is not off, replace battery again. If problem persists contact Kidde Fire Trainers Customer Service.

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4.2.5 SMOKE FLUID RESERVOIR FILLING

NOTE

For Smoke Fluid Reservoir Filling Instructions, See Smoke Gen Manual in Sec. 6.

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5.0 PARTS LISTS

5.1 GENERAL

This section provides a figure and parts list table for each of the **Mobile FireTrainer® T-500** Trainer major assemblies.

5.2 BURNER CONTROL ASSEMBLY

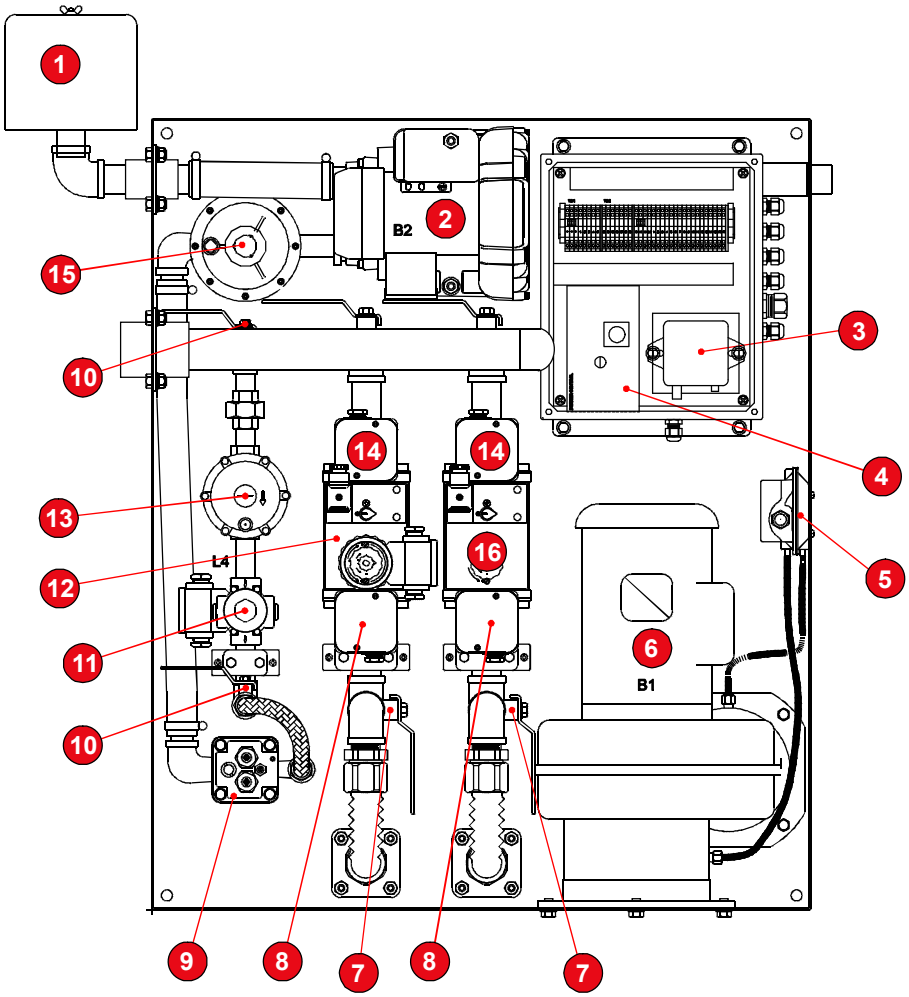


Figure 5-1. Burner Control Assembly (RF)

5.3 FLASHOVER VALVE CABINET ASSEMBLY RF

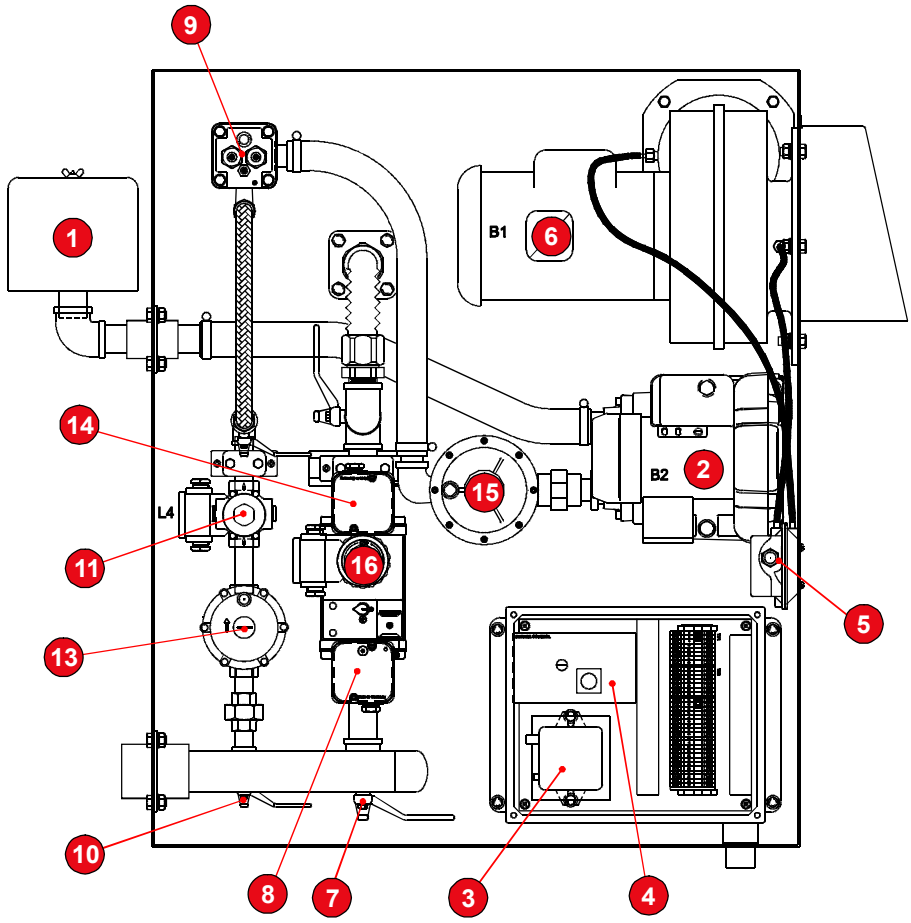


Figure 5-2. Flashover Valve Cabinet Assembly (RF)

Table 5-1. Burner Control Assembly W/ Flashover Valve Cabinet Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-1	**Burner Control Assembly W/ Rollover**	**4A1-006-364-001 & 002 Rev. C**	**06003**
Figure 5-2	**Flashover Valve Cabinet**	**4A1-006-312 Rev. C**	**06003**
1	Filter Element	515132	82877
2	Blower, 115VAC/60HZ, 1/8HP, 27 CFM MAX	R1102	24123
3	Transformer, IGTN, TZ17.5-12/100R, 115V	84331384	N/A
4	Controller, Burner, IFS 110IMT-10/2/2N,120V	84367720	N/A
5	Pressure Switch, Diff Air/.08-0.6WC	46010-1	N/A
6	Blower, Direct Drive, 1.5HP, 208VAC	SC-600	N/A
7	Valve, Ball, 1”NPT	73A-105-01	72219
8	Pressure Switch, 40-200 IWC SET	D217089	01443
9	Pilot Assembly, RF, 500mm	4A1-006-315-006	06003
10	Valve, Ball, 1/2”NPT	73A-103-01	72219
11	Valve, Solenoid, ½”NPT, 120VAC	46040-2	N/A
12	Valve, Solenoid, dmv-zrdle 702/6	D232214	N/A
13	Regulator, Gas Pressure, ½”NPT	2C1-003-737-003	06003
14	Pressure Switch, 2-20 IWC SET	D217087	N/A
15	Regulator, Air Pressure, 3/4”NPT	D229608	N/A
16	Valve, Dual Safety Shutoff, 1”NPT	D224846	N/A

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5.4 ELECTRICAL CONTROL CABINET (PANEL)

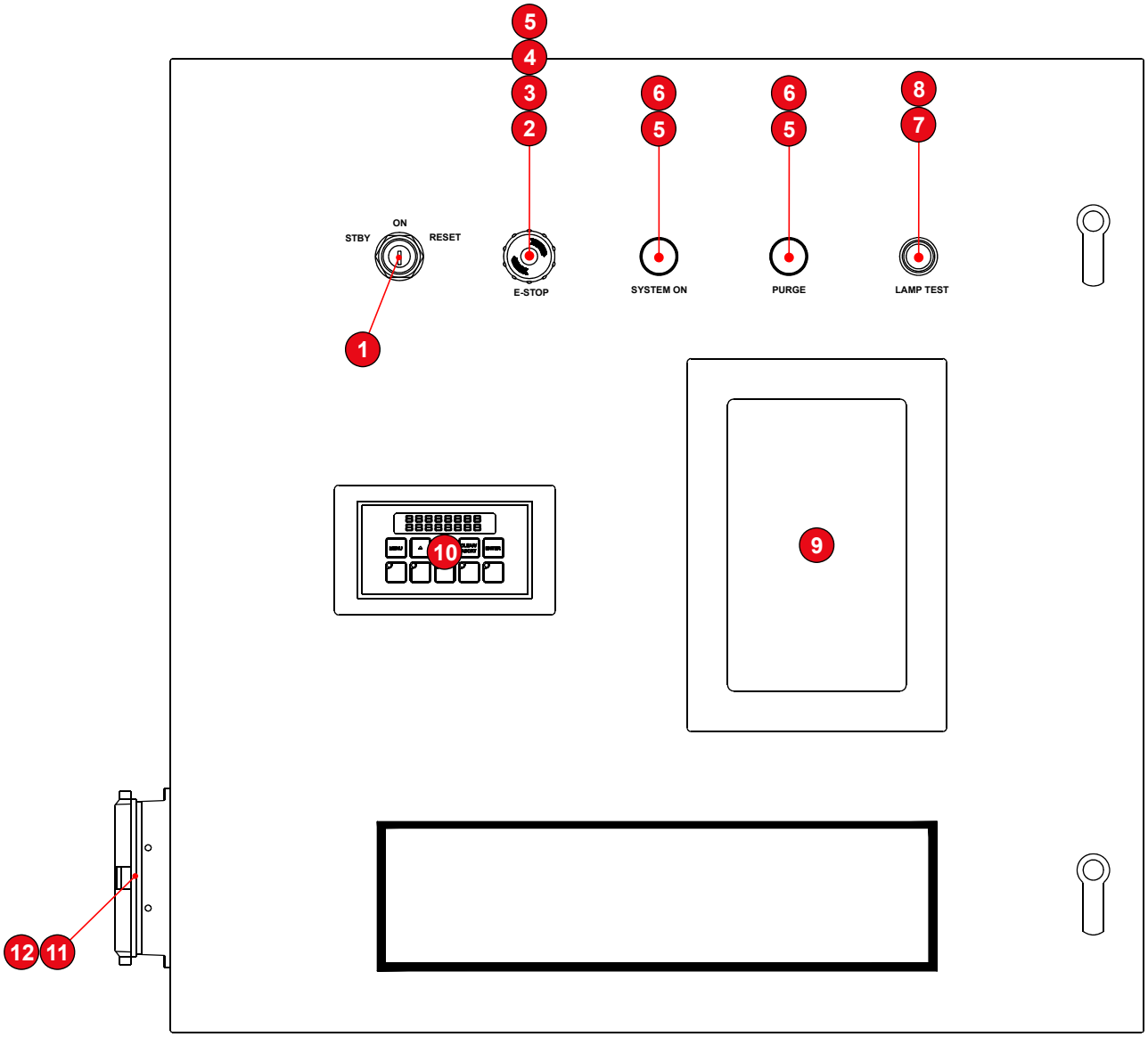


Figure 5-3. Electrical Control Cabinet (Panel)

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Table 5-2. Electrical Control Cabinet (Panel) Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-2	**Electrical Control Cabinet (Panel)**	**4A1-007-134 Rev B**	**06003**
1	Switch, Key, 3 Position	800T-J69-KE7B	01121
2	Operator, Mush, PSH-PUL, Illum, Red	800FP-LMP44	01121
3	Latch, Plastic	800F-ALP	01121
4	Block, Contact, 1 N.C.	800F-X01	01121
5	LED, INTGRTD, LTCH-MNT, 24V, GREEN	800F-N3G	01121
6	Light, Pilot, PLSTC, Green	800FP-P3	01121
7	PB, PLSTC, NON-ILUM, BLK	800FP-F2	01121
8	Block, Contact, 1 N.O.	800F-X10	01121
9	Kit, Window, SS, 11.50”x8.00”	APWK95NFSS	1WSN4
10	Panel, Operator, 2-Linex20-Char,LCD	OP-620	04YE1
11	Housing,BLKHD,MTG,SZ24B,LC, w/PL CVR	0900340302	0G8R9
12	Insert, Fem, Term, Screw, 1-24 Cont	09330242701	0G8R9

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5.5 ELECTRICAL CONTROL CABINET (ENCLOSURE)

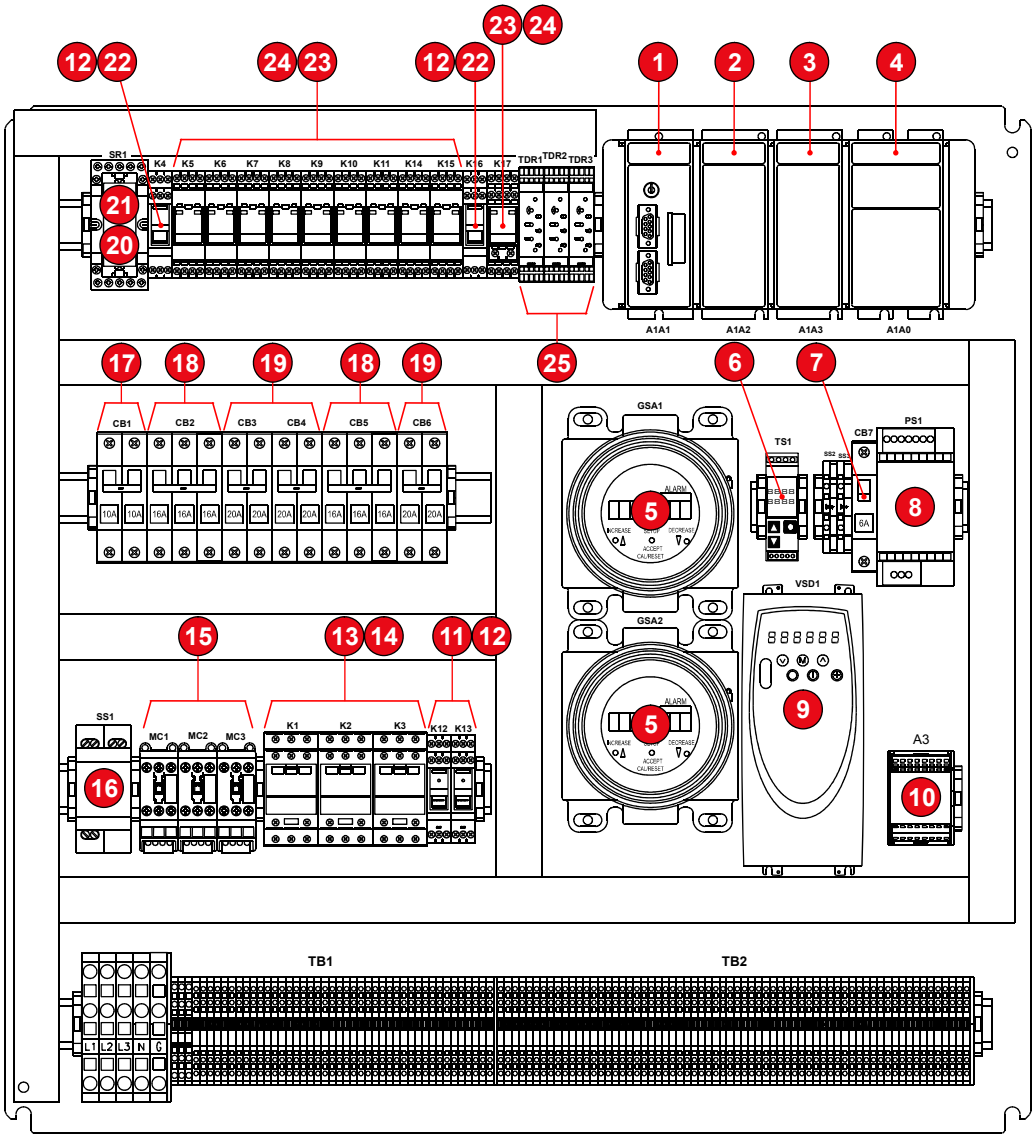


Figure 5-4. Electrical Control Cabinet Enclosure

Table 5-3. Electrical Control Cabinet (Enclosure) Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-4	**Electrical Control Cabinet (Enclosure)**	**4A1-007-134 Rev. B**	**06003**
1	PLC Processor, 512K	1769-L31	01121
-	Battery (PLC)	1769-BA	01121

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Table 5-3. Electrical Control Cabinet (Enclosure) Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
2	Module, 32-Input, 24 VDC	1769-IQ32	01121
3	24 VDC Output Module, 32 Point	1769-OB32	01121
4	Power Supply, 24VDC IN @ 5VDC OUT, 2A	1769-PB2	01121
5	Transmitter, Elect, w/ Relays, F/U9500A	006265-007	66008
6	Relay, TC,Alarm,IND,Controller,24V	DCL-33A-R/M1	N/A
7	Circuit Breaker, 1-Pole, 6 AMP	1GU6	0DU12
8	Power Supply, SW,24VDC,7.5A,180W	PSM24-180S	04YE1
9	AC Drive, SK,VR FRQ,240VAC,1PH,2HP	SKBD200150	3A0T3
10	Resistor Gate W/8 2.2K Res	289-113	65996
11	Relay, Plug-In, 2PT, 120VAC, W/LED	C12-A21X/120VAC	61526
12	Socket, Relay, 8-PIN	S12	61526
13	Relay, Plug-In, PWR, 3PDT,24VDC,W/LED	C5-A30FX/024VDC	61526
14	Socket, Relay, 11-PIN	S5-S	61526
15	Magnetic Contactor, 12A,24VDC	E111A12X3N	15605
16	Surge Supressor, 1PH, 2wire+G	16DT230	0TPE6
17	Circuit Breaker, 2-Pole, 10 AMP	2DU10	0DU12
18	Circuit Breaker, 3-Pole, 16 AMP	3DU16	0DU12
19	Circuit Breaker, 2-Pole, 20 AMP	2DU20	0DU12
20	Relay, Safety, 24VDC Coil	G7S-4A2B-DC24	61964
21	Socket, Safety Relay, Din Rail Mount	P7S-14F	61964
22	Relay, Plug-In, 2PDT,24VDC,W/LED	C12-A21FX/024VDC	61526
23	Relay, 2PTD, 24VDC, LED/Neon Ind	C7-A20FX/024VDC	61526
24	Socket, Relay, 8-PIN	S7-M	61526
25	Relay, Time Delay, DPDT	H3DS-MLC	61964

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5.6 CONTROL PENDANT ASSEMBLY

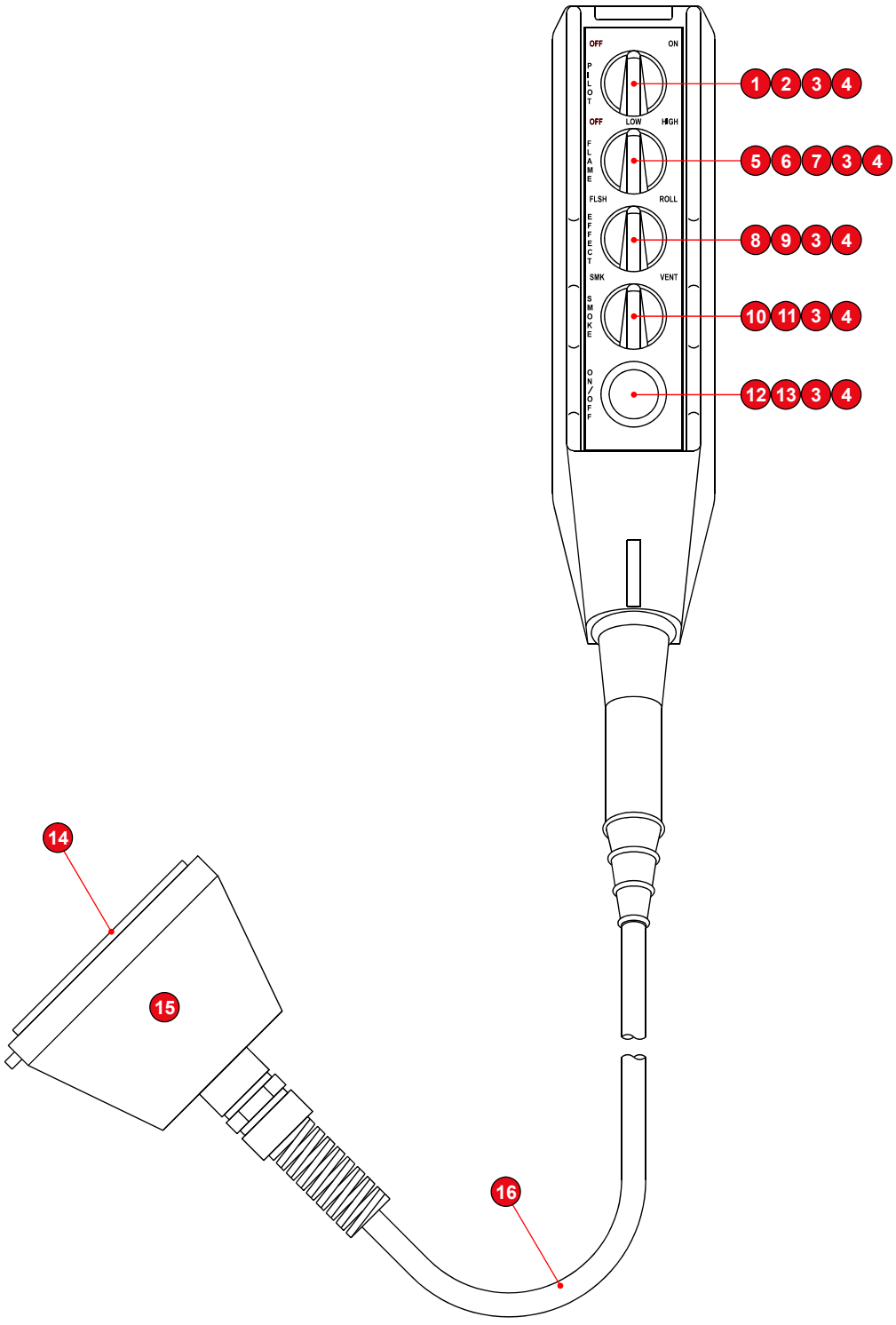


Figure 5-5. Control Pendant Assembly

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Table 5-4. Control Pendant Assembly Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-5	**Control Pendant Assembly**	**4A1-007-136**	**06003**
1	Switch, Selector, 3-POS, 90,SP-RN,ILLUM, BLU	800FP-LSB36N	01121
2	LED, INTGRTD,LTCH-MNT,24V,BLU	800F-N3B	01121
3	Block,Contact, 1 N.O.	800F-X10	01121
4	Latch, Plastic	800F-ALP	01121
5	Switch, Selector, 3-POS, 90D,MNTD,ILLUM, YEL	800FP-LSM35N	01121
6	LED, INTGRTD,LTCH-MNT,24V,YEL	800F-N3Y	01121
7	Block,Contact, 1 N.C.	800F-X01	01121
8	Switch, Selector, 3-POS, SP-RN,ILLUM, RED	800FP-LSB34N	01121
9	LED, INTGRTD,LTCH-MNT,24V,RED	800F-N3R	01121
10	Switch, Selector, 3-POS, 90D,MNTD,ILLUM, CLR	800FP-LSM37N	01121
11	LED, INTGRTD,LTCH-MNT,24V,WHT	800F-N3W	01121
12	PB,MOM,EXTD,ILLUM,GREEN	800FP-LE3	01121
13	LED, INTGRTD,LTCH-MNT,24V,GRN	800F-N3G	01121
14	Insert, Male, TERM,SCREW,1-24 CONT	09330242601	0G8R9
15	Hood, Side Entry, Size 24	09300242601	0G8R9
16	Cable, Tray, UNSHLD,27COND,20AW	NL-2720J	04569

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5.7 WIRELESS CONTROL SYSTEM

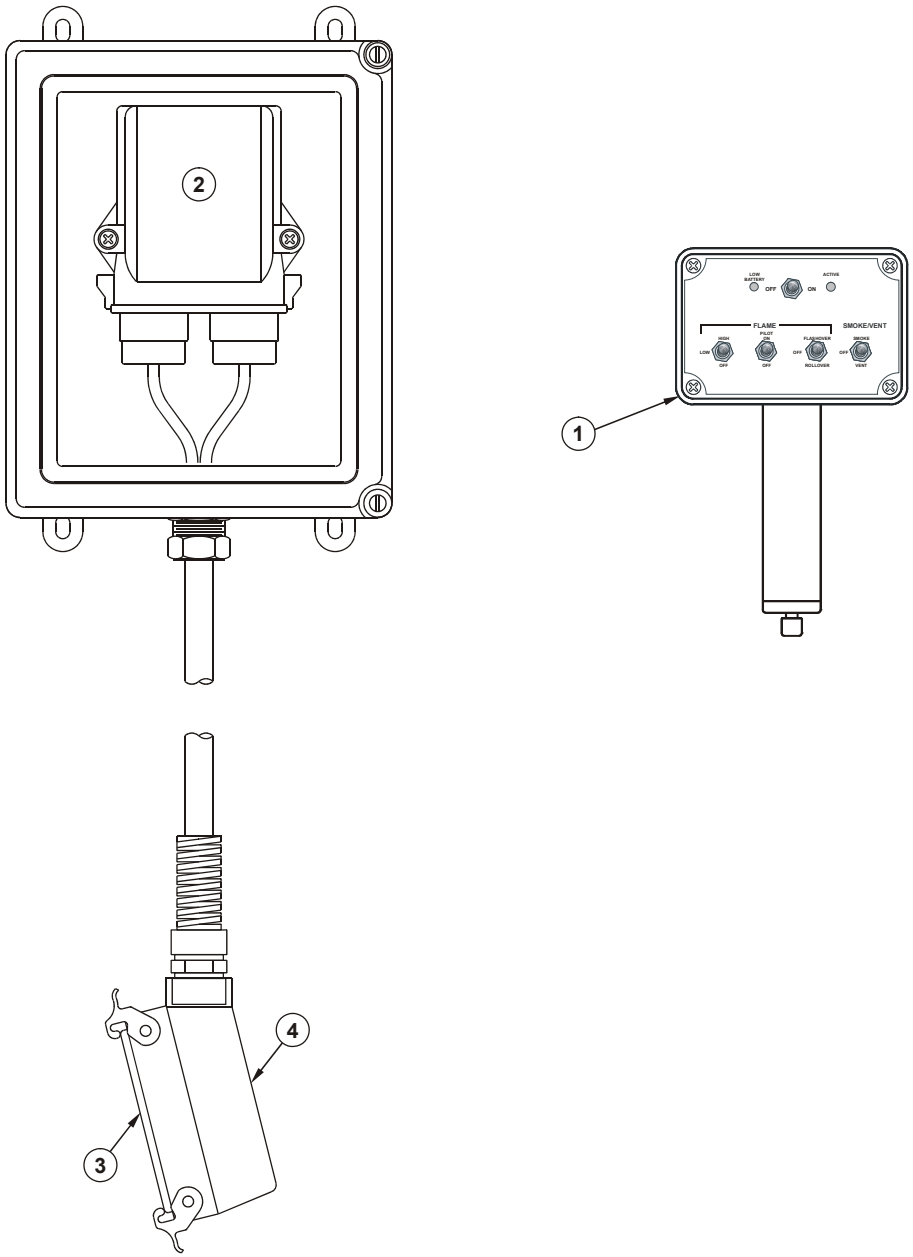


Figure 5-6. Receiver/Transmitter Assembly

Table 5-5. Receiver/Transmitter Assembly Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-6	**Receiver/Transmitter Assembly**	**4A1-008-035 Rev. -**	**06003**
1	Transmitter, Model T150, Kidde T-500	DFPE01202009A Ser No. is required along w/ part number	1SGP7
2	Receiver, Model R160, Kidde T-500	DFPE01202009B Ser No. is required along w/ part number	1SGP7
3	Insert, Male, 1-24 CT	09330242616	0G8R9
4	Hood Side Entry	09300241530	0G8R9

WARNING

EACH RECEIVER AND TRANSMITTER IS PAIRED TOGETHER VIA A UNIQUE ID CODE. WHEN REORDERING A REPLACEMENT TRANSMITTER OR RECEIVER YOU MUST INCLUDE THE SERIAL NUMBER ALONG WITH THE PART NUMBER FOUND IN TABLE 5-5.

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5.8 E-STOP ASSEMBLY

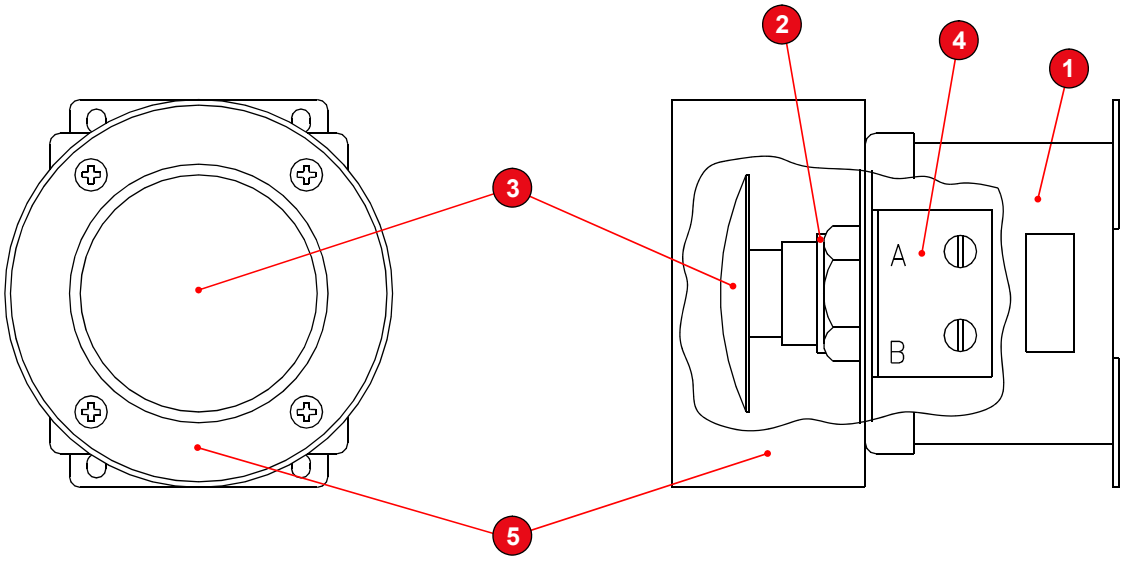


Figure 5-7. E-Stop Assembly

Table 5-6. E-Stop Assembly Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-7	**E-Stop Assembly**	**2A1-002-293 Rev. C**	**06003**
1	Enclosure, Switch	E1PBSS	1WSN4
2	Operator,Maint,Push-Pull	10250T5	15605
3	Button,Aluminum	10250TJ63	15605
4	Contact Block	10250T3	15605
5	Guard,Switch	3D1-002-183	06003

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5.9 GAS DETECTION SYSTEM ASSY. (BURN ROOMS)

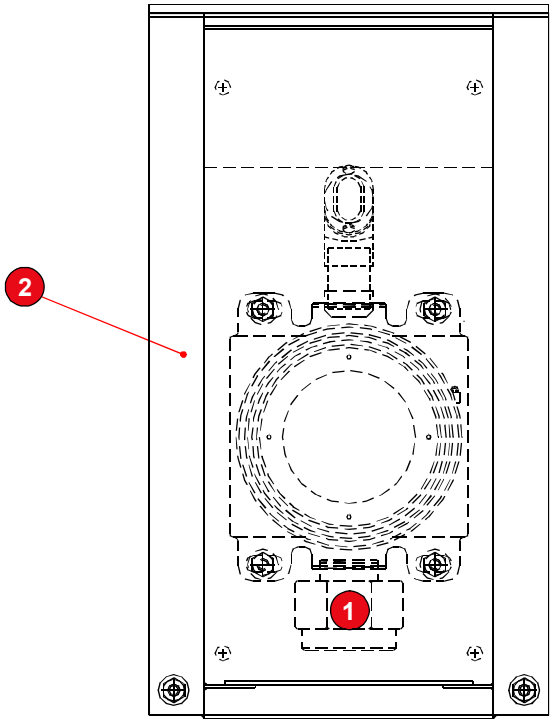


Figure 5-8. Gas Detection System Assembly

Table 5-7. Gas Detection System Assembly Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-8	**Gas detection System Assembly**	**4A1-006-903**	**06003**
1	Sensor, Gas Combustible	006824-001	66008
2	Cover, Head, Propane detection	4D1-006-367	06003

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5.10 TEMPERATURE SENSOR ASSEMBLY

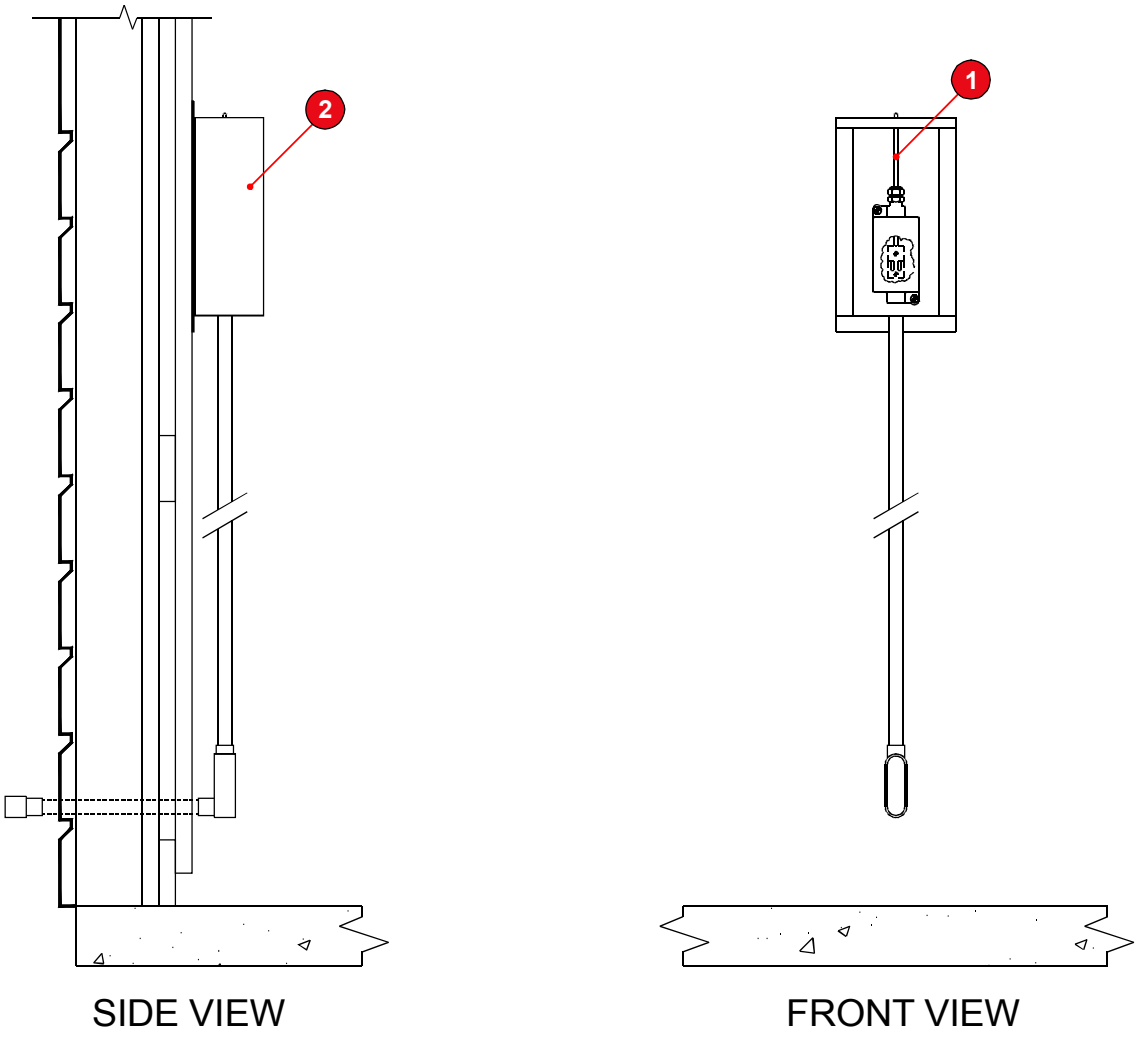
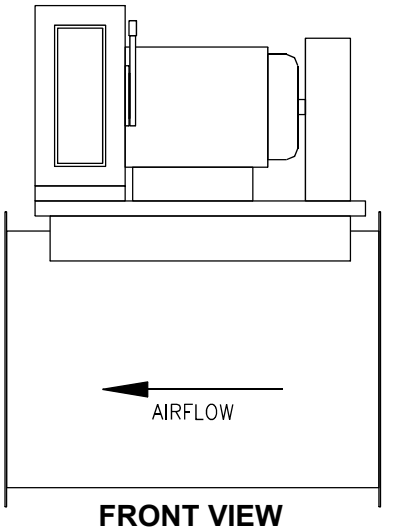


Figure 5-9. Temperature Sensor assembly

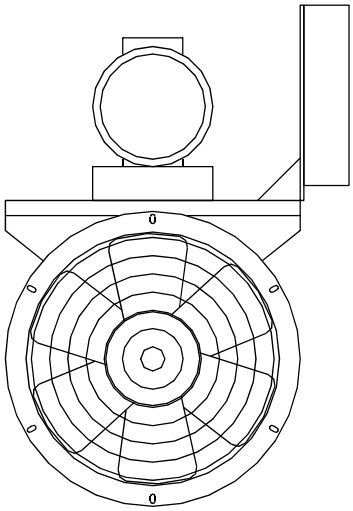
Table 5-8. Temperature Sensor Assembly Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-9	**Temperature Sensor Assembly**	**3A1-002-944 Rev. E**	
1	Thermocouple, 6' LG	2C1-003-822	06003
2	Heat Shield	3D1-002-945	06003

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5.11 HVC EXHAUST FAN ASSEMBLY



FRONT VIEW



SIDE VIEW

Figure 5-10. High Temp Exhaust Fan Assembly (Wall Mount)

Table 5-9. High Temp Exhaust Fan Assembly (Wall Mount) Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-10	**HVC Exhaust Fan Assembly**	**3C1-003-371-B08 Rev. C**	**06003**

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.12 ELECTRICAL CONTROL CABINET (2ND FLOOR)



Figure 5-11. Electrical Control Cabinet Panel

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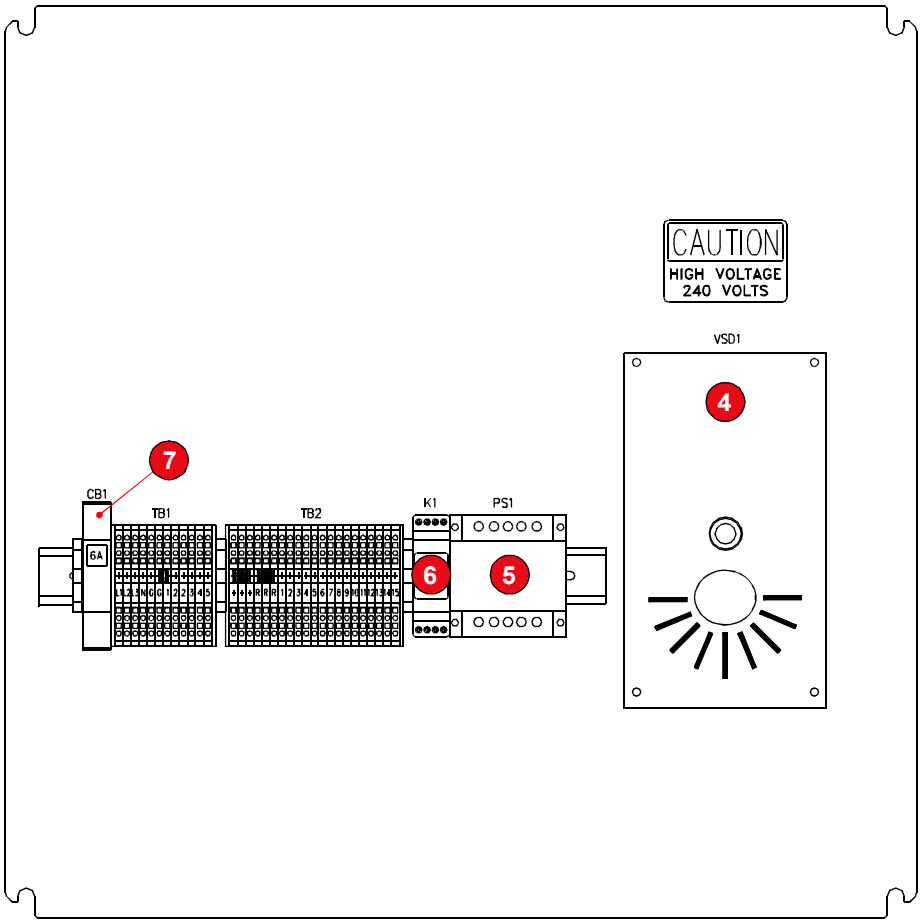


Figure 5-12. Electrical Control Cabinet Enclosure

Table 5-10. Electrical Control Cabinet Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-11	**Panel**	**4A1-006-525**	**06003**
1	Switch, Selector, 3-POS, Non-Illum, WHT	800T-J2A	01121
2	Temperature Controller, 24 VDC	SD31-LJJA-AA0R	79221
3	Electric Buzzer, Piezo	623-2029	3SUZ0
Figure 5-12	**Enclosure**	**4A1-006-525**	**06003**
4	Variable Frequency AC Drive, 230V, 1-3 PH	GS2-23P0	04YE1
5	Power supply, 100W, 24DC, 4.2AMP	SDP4-24-100RT	55814
6	Relay, Plug-In, 2PDT, 24VDC, w/ LED	C7-A20FX/024VDC	1663B
-	Socket, Relay, 8 Pin	S7-M	1663B
7	Circuit Breaker, One-Pole, 6AMP	1BU6	0DU12

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.13 BED FPLC MOCK-UP

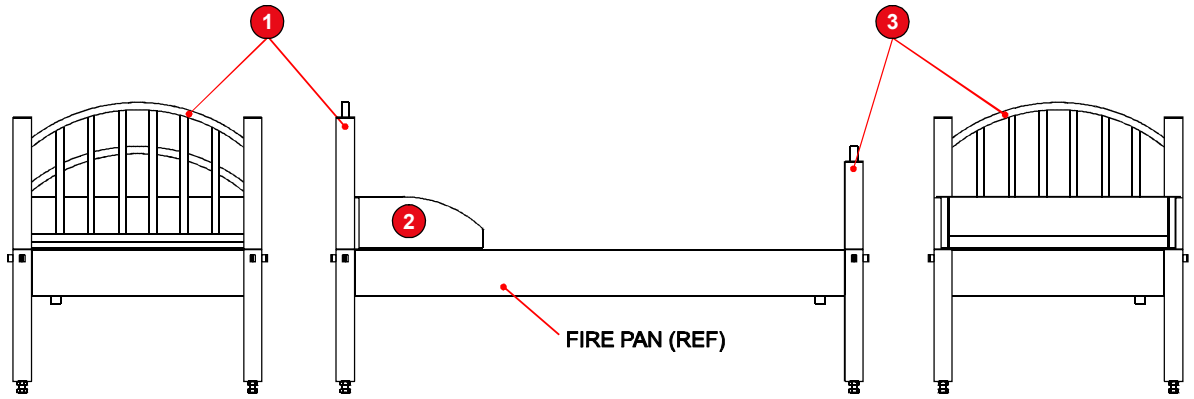


Figure 5-13. BED FPLC Mock-UP

Table 5-11. Bed FPLC Mock-UP Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-13	**Bed FPLC Rack Mock-UP**	**4A1-006-337**	**06003**
1	Headboard	4A1-006-337-D01	06003
2	Pillow	4A1-006-337-D03	06003
3	Footboard	4A1-006-337-D02	06003

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.14 STOVE FPLC MOCK-UP

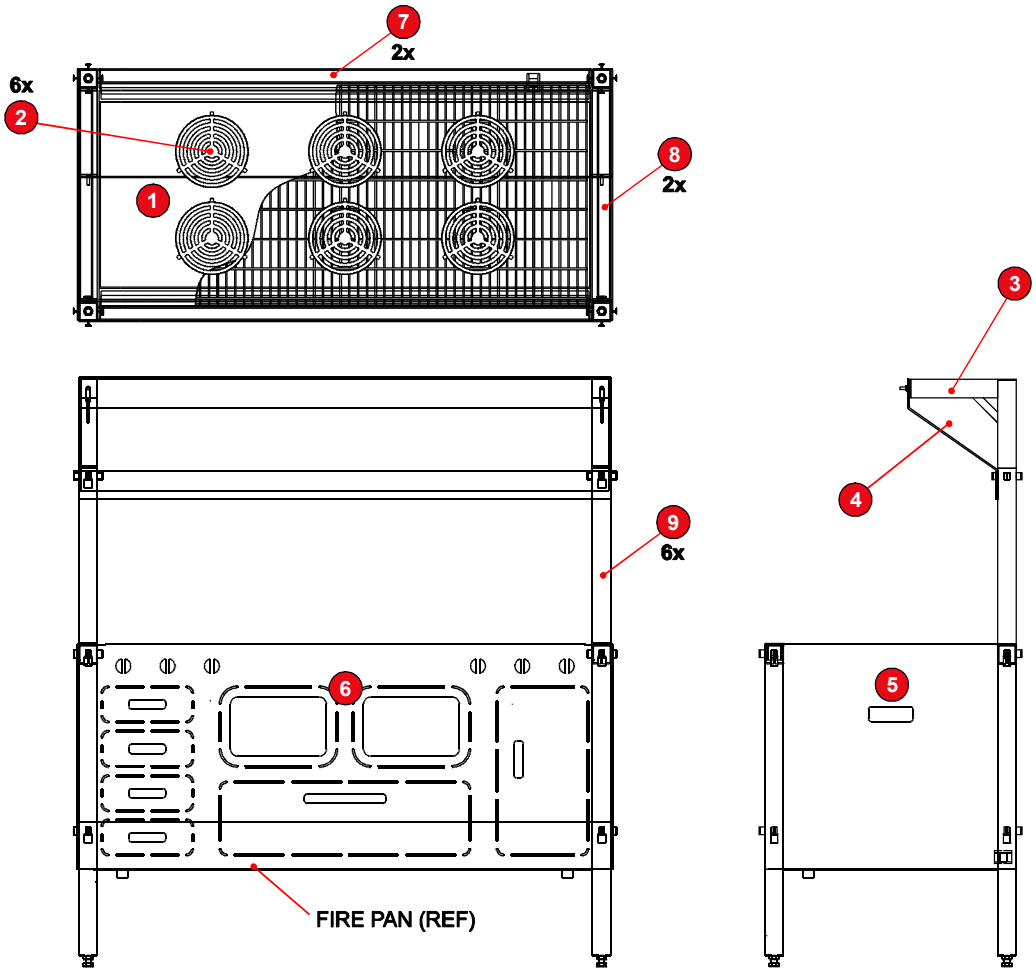


Figure 5-14. Stove FPLC Mock-UP

Table 5-12. Stove FPLC Mock-UP Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-14	**Stove FPLC Mock-UP**	**4A1-006-351**	**06003**
1	Top, Stove	4D1-006-356	06003
2	Stove Coil, Large	4D1-006-358-002	06003
3	Hood Bracket	4D1-006-352	06003
4	Hood Plate	4D1-006-353	06003
5	Side Plate, Stove	4D1-006-355	06003
6	Front Plate, Stove	4D1-006-354	06003
7	Cross Member, Long	4D1-006-350-001	06003
8	Cross Member, Short	4D1-006-350-002	06003
9	Side Post	4D1-006-349	06003

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.15 ELECTRIC MOTOR GENERATOR

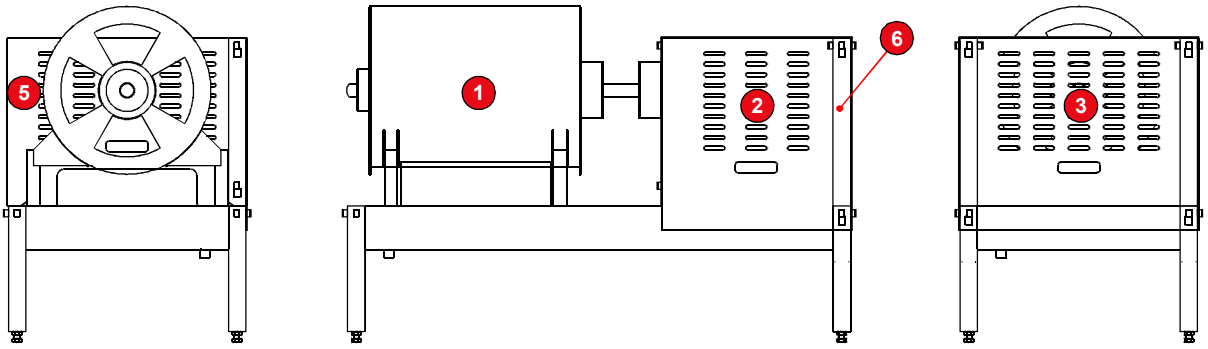


Figure 5-15. Electric Motor Generator FPLC Mock-UP

Table 5-13. Electric Motor Generator FPLC Mock-UP Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-15	**Motor Generator FPLC Mock-UP**	**4A1-006-483**	**06003**
1	Motor Generator Assembly	4A1-006-339	06003
2	Side Plate, Motor Generator	4A1-006-341	06003
3	Rear Plate, Motor Generator	4A1-006-340	06003
4	Front Plate, Motor Generator	4A1-006-343	06003
5	Top Plate, Motor Generator	4A1-006-342	06003
6	Side Post	4D1-006-349	06003

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.16 GAS BOTTLES

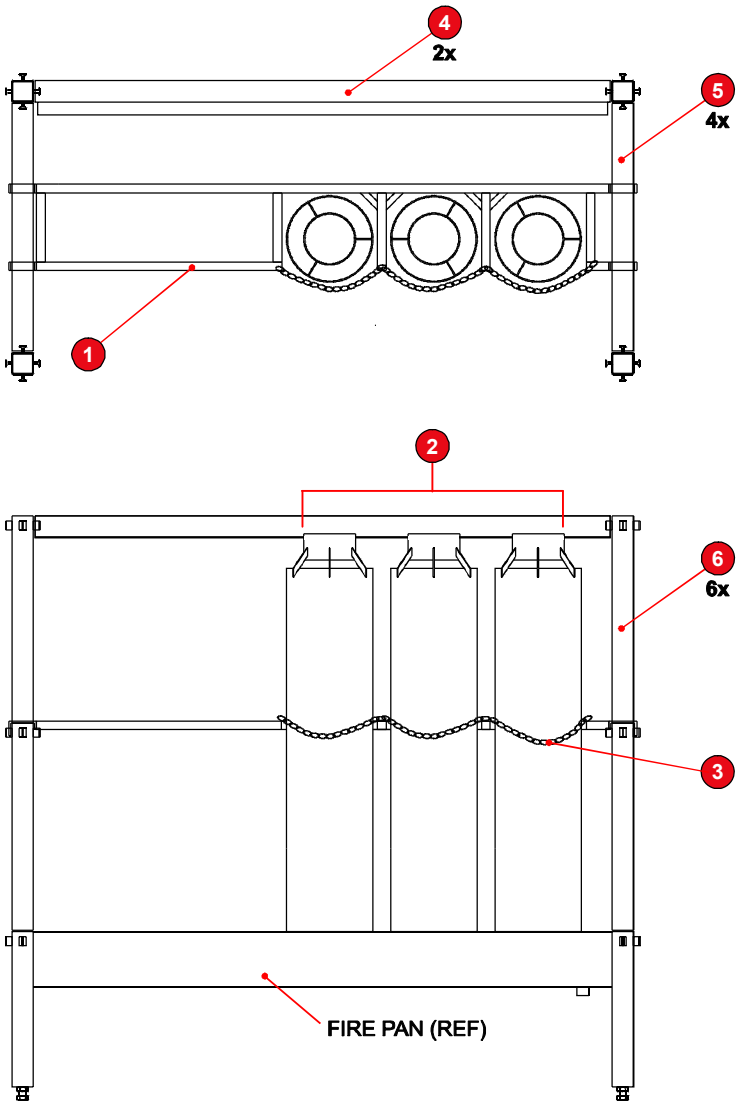


Figure 5-16. Gas Bottle FPLC Mock-UP

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Table 5-14. Bottle Rack FPLC Mock-UP Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-16	**Bottle Rack FPLC Mock-UP**	**4A1-006-479**	**06003**
1	Bottle Support	4D1-006-347	06003
2	Gas Bottles	4D1-006-346	06003
3	Chain, Link	2C1-003-304	36003
4	Cross Member, Long	4D1-006-350-001	06003
5	Cross Member, Short	4D1-006-350-002	06003
6	Side Post	4D1-006-349	06003

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.17 HAZMAT

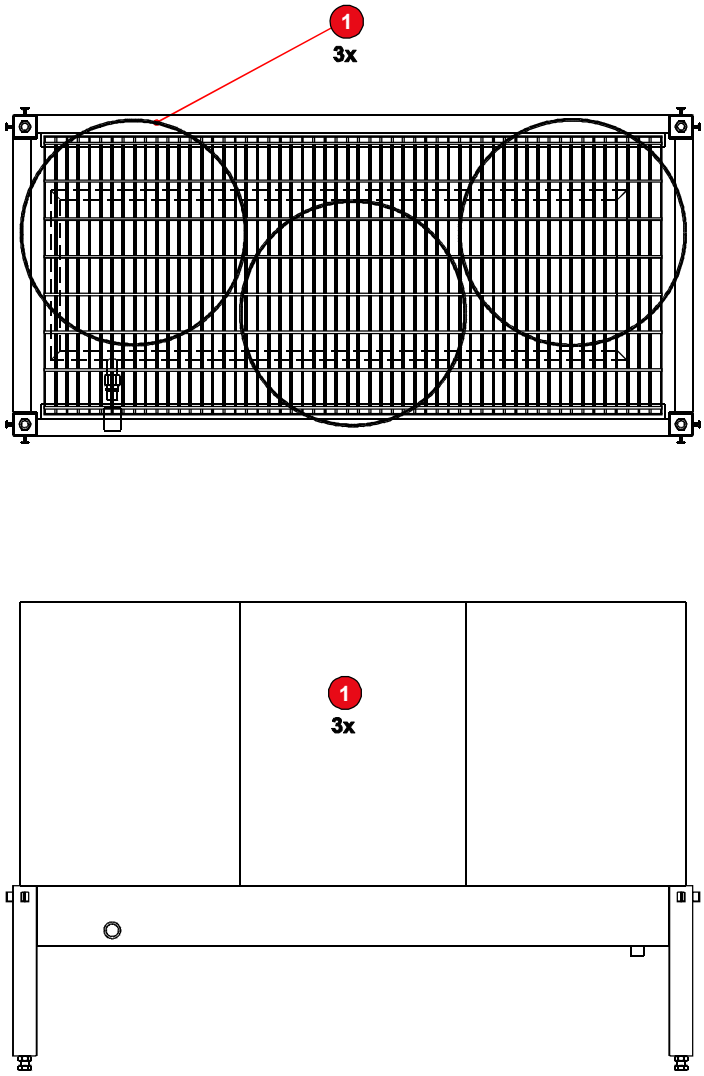


Figure 5-17. Hazmat FPLC Mock-UP

Table 5-15. Hazmat FPLC Part List			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-17	**Cylinders, 24” DIA.**	**ICS-336**	**06003**

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.18 SOFA

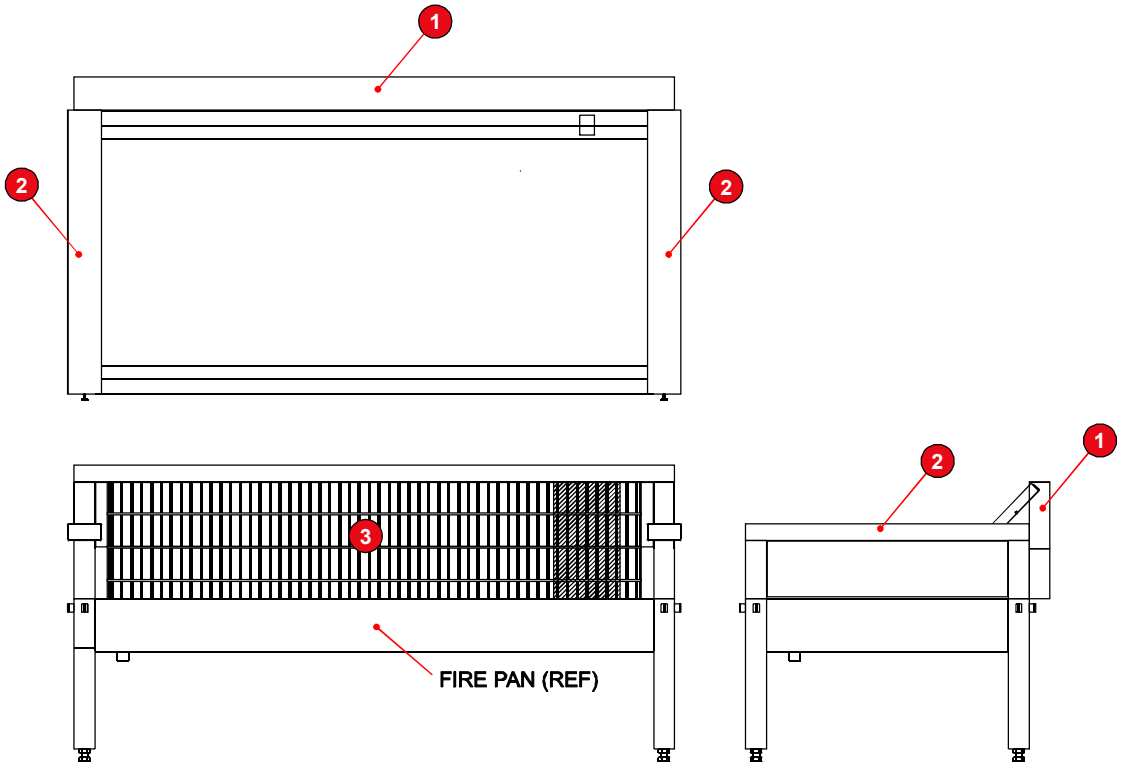


Figure 5-18. Sofa FPLC Mock-UP

Table 5-16. Sofa FPLC Mock-UP Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-18	**Sofa FPLC Mock-UP**	**4A1-006-480**	**06003**
1	Back Rest	4A1-006-480-D01	06003
2	Arm Rest, Left/Right	4A1-006-480-D02	06003
3	Grate, Backrest	4C1-006-335	06003

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.19 PAN ASSEMBLY

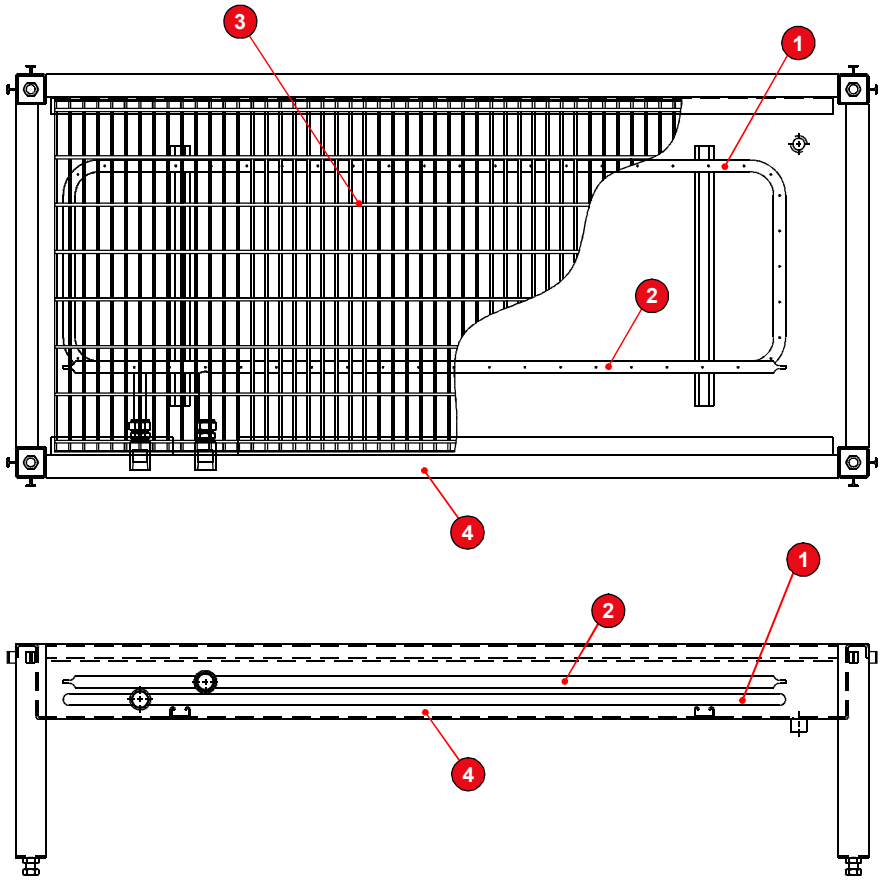


Figure 5-19. Pan Assembly

Table 5-17. Pan Assembly Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-19	**Pan Assembly**	**-**	**06003**
1	Burner Element Assembly	4A1-006-476	06003
2	Rollover Burner Element	3A1-003-289	06003
3	Grate	4C1-007-928	06003
4	Pan Assembly	4A1-006-365	06003

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5.20 SECOND STAGE REGULATING STATION

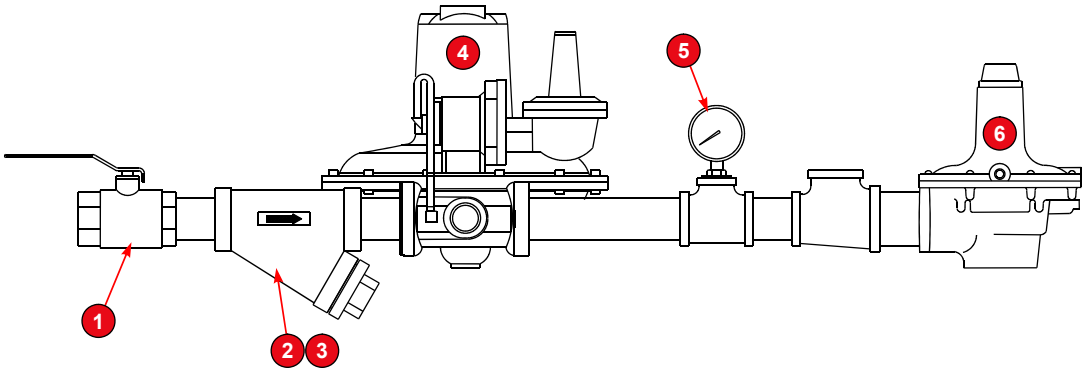


Figure 5-20. Second Stage Regulating Station Assembly

Table 5-18. Second Stage Regulating Station Assembly Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-20	**Second Stage Regulating Station Assy.**	**4A1-006-344**	**06003**
1	Ball Valve, BRZ, 1-1/2”NPT	80-107-01	72219
2	Y-Strainer Filter for a 2C1-004-059-004 Y-Strainer	Order by Description	06003
3	Gasket, Y-Strainer, 600#, 1-1/2”NPT	2C1-004-069-004	06003
4	Reg, Gas, 1-1/2”NPT, 10PSI IN/5PSI OUT	CL34-2	N/A
5	Gauge, Liq Fil, 0-15PSI, ¼” NPT Bot	4088K28	39428
6	Valve, Relief, 2”NPT, 4-10PSI SPRG	289H-4	22508

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5.21 SOUND SYSTEM

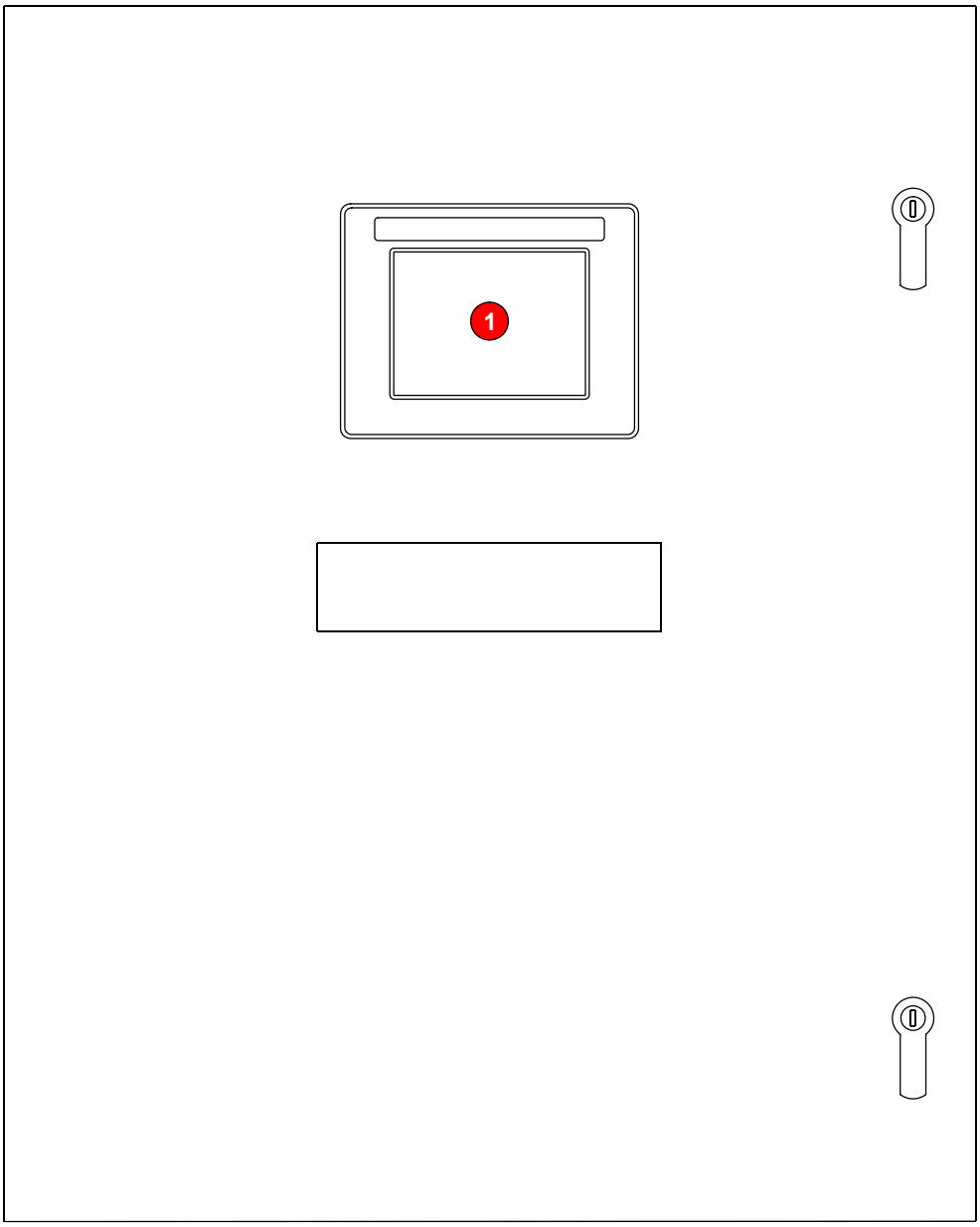


Figure 5-22. Sound System Control Console (Front Panel)

Table 5-19. Control Console (Front Panel) Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 5-22	**Control Console (Front Panel)**	**4A1-007-416**	**06003**
1	Graphic, HMI, Mono, W/Touch Screen, 5.6”	eTOP05-0045	1SP80

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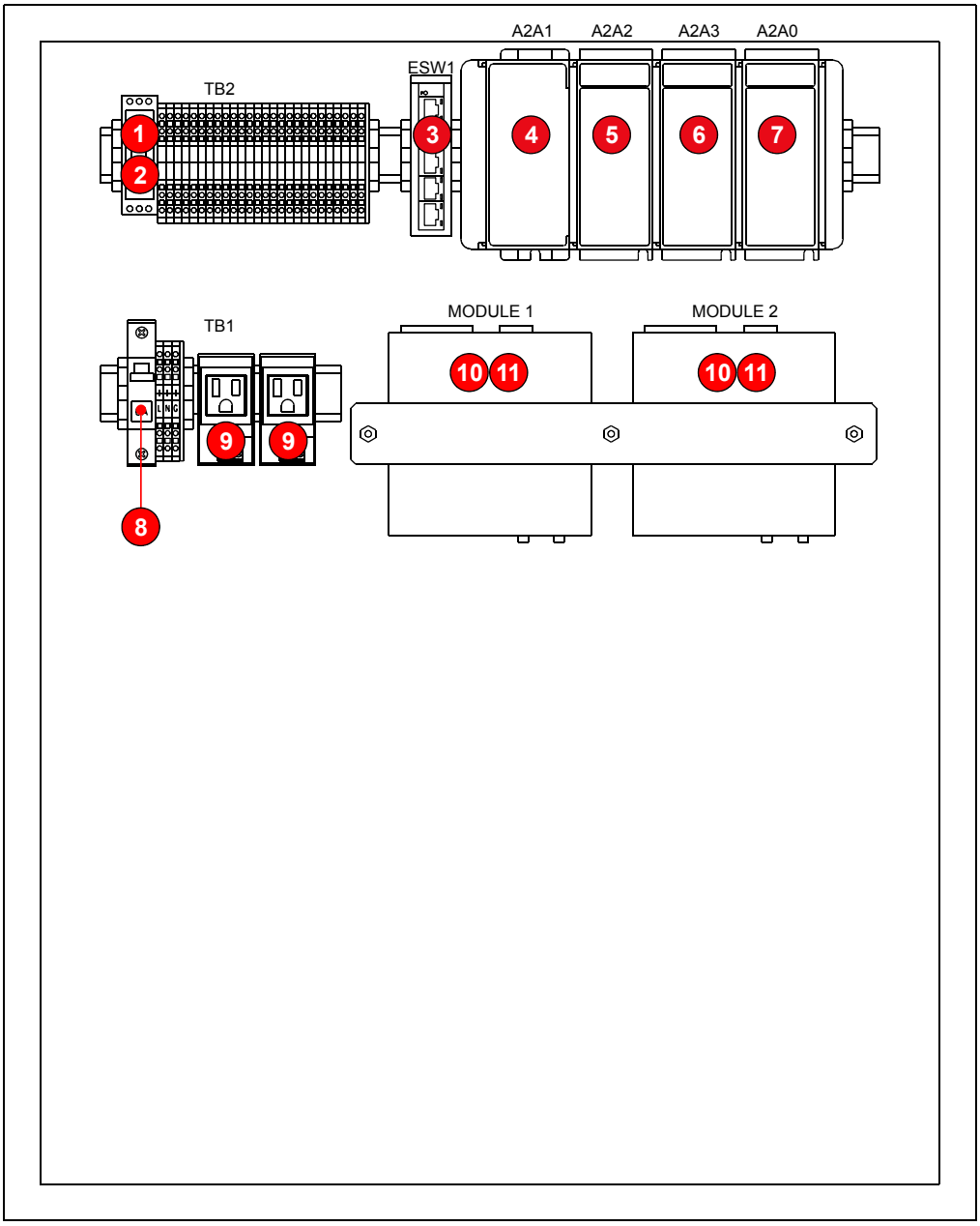

A2A1
A2A2
A2A3
A2A0
ESW1
TB2
1
2
3
4
5
6
7
MODULE 1
MODULE 2
TB1
10
11
10
11
9
9
8

Table 5-20. Control Console (Internal Components) Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 4-23	**Control Console (Observation Room)**	**4A1-007-366**	**06003**
1	Relay, 2PDT, 24VDC,LED/NEON IND	C7-A20DX/024VDC	61526
2	Socket, Relay, 8-Pin	S7-M	61526
3	Switch, Ethernet, Spider 5TX	943824-001	3V1N9
4	PLC Processor, 750K, Ethernet, CF Card	1769-L32E	01121
5	DC Input Module	1769-IQ16	01121
6	AC/DC Output Module	1769-OW16	01121
7	Power Supply, 24VDC in, @ 5VDC Out, 2A	1769-PB2	01121
8	Circuit Breaker, One-Pole, 6AMP	1492-CB1H060	01121
9	Receptacle, Din Rail, Mounted	991548	63279
10	Sound Module W/ Amplifier	QV531QV5M14F8	N/A
11	Flash Card, Compact, 64MB (T2 Sound)	CF/64S	N/A

Table 5-21. Speakers Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
-	Speakers, 2-Way Wthrprf, 6.5”, 5-150W	C1-AW650BLK	01BK1
-	Speakers, 80HM, 30W	DMS-3P	06003

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Section 6 Drawings & Reference Material

6.1 INTRODUCTION

This section contains drawings and reference material that support the **Structural FireTrainer® T-500**.

Refer to Table 6-1 for a list of drawings that have been supplied in this section.

Refer to Table 6-2 for a list of reference materials that have been supplied in this section.

Table 6-1. Drawings

Drawing Number	Title
IW4B1-007-653	MINOT T-500 Interconnect Wiring Diagram
WD4A1-007-134 (Rev E)	I/O Rack Electrical Control Cabinet, Interconnect Wiring Diagram
WD4A1-006-364 (Rev B)	Burner Control Assembly Wiring Diagram
WD4A1-006-312 (Rev B)	Flashover Burner Control Assembly Wiring Diagram
WD4A1-006-525	Electrical Control Cabinet Wiring Diagram (2ND Floor)

Table 6-2. Reference Materials

Item	Title
Integrated Sound Effects System	Operation and Maintenance Manual
Smoke Generator	Operation and Maintenance Manual
Hartzell Ventilation Fan	Installation, Operation & Maintenance Manual Recommended Safety Practices

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DWG NO IW4B1-007-653 SH 1

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD

Legend of Common Reference Designators

AD	Agent Detection
B	Blower Motor
CAB	Combustion Air Blower
CB	Circuit Breaker
CGDS	Combustible Gas Drawn Sample
CGS	Combustible Gas Sensor
D	Diode
DS	Door Switch
ES	E-Stop
F	Flashover Effect
FHM	Fireplace Hardware Module
FP	Fireplace Mockups
FRCS	Fuel Regulator & Control Station
FS	Flame Safeguard
HOA	Hand-Off-Automatic Switch
I	Indicator
IG	Ignitor
JB	Junction Box
K	Relay
KS	Keyswitch
L	Solenoid Valve
LCP	Local Control Panel
LT	Lamp/Light
MC	Magnetic Contactor Enclosure
ME	Master I/O Rack Enclosure
MV	Motorized Valve
MVA	Motorized Valve Assy
OS	Operator Station
P	Plug
PND	Pendant, Control
POC	Proof of Closure
PRS	Pressure Switch
PS	Power Supply
PT	Pressure Transducer
R	Receptacle
RBC	Remote Burner Control
S	Switch
SCP	Scenario Control Panel
SD	Smoke Damper
SG	Smoke Generator
SP	Service Panel
SPK	Speaker
SSOV	Safety Shutoff Valve
T	Transformer
TB	Terminal Block
TC	Thermocouple
TDR	Time Delay Relay
TS	Temperature Switch
VF	Ventilation Fan
VFC	Ventilation Fan Control
W	Cable
Z	Zener Diode

Installation: MINOT AFB T-500 INSTALLATION

Page	Page Title	Revision
1	STATUS REVISION OF PAGES	–
2	MULTI-PROP, ELECT ENCL TO BURNER CONTROL	–
3	MULTI-PROP, PENDANT, WIRELESS AND E-STOP CONTROL	–
4	MULTI-PROP, VENT FAN CONTROL	–
5	SOUND SYSTEM INTERFACE	–
6	CLASS A VENT FAN SYSTEM	–

CABLE NUMBER (W'S)	*DESCRIPTION*	*PART NUMBER*
W3, W7, W10, W20-W22	7 COND, 18 AWG	221807
W2, W6, W9	7 COND, 14 AWG	221407
W1, W8, W11	18 COND, 18 AWG	221818
TC1, TC2, TC3	THERMOCOUPLE	TT-K-20
W4, W5	4C SHLD, 18 AWG	2890
W12, W13, W14, W15	3C, 18 AWG	221803

UNLESS OTHERWISE SPECIFIED, ALL WIRE IS TYPE THHN OF SPECIFIED COLOR AND GAUGE.

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PROJECT	T5_MINOT_AFB	INSTALLATION	FIELD_WIRE	Kidde Fire Trainers	
		LOCATION/S		TITLE	MINOT AFB T-500 INSTALLATION INTERCONNECT DIAGRAMS REVISION STATUS OF PAGES
4B1-007-653	SO 3925	DWN KR	04/30/09	SIZE D	CAGE CODE 06003 / DWG NO IW4B1-007-653
NEXT ASSY	USED ON	CHKD EC	05/04/09		
APPLICATION		APVD AC	05/04/09	SCALE NA	CAD FILE: IW4B7653- / SHEET 1 OF 6

MULTI-PROP FIREPLACE BURN ROOM

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD

NOTE:FIELD MODIFICATION TO BC1
ADD JUMPER TB2:L1 TO TB2:1
RELOCATE W6:1 TO TB1:1
W12:1 TO TB1:2
W13:1 TO TB1:4

=FIELD_WIRE+SP1

CB1 (30 AMP)

1	2001 /10/BLK	TB1 L1	LINE 1
2	2002 /10/RED	L2	LINE 2
N	2004 /10/WHT	N	NEUTRAL
G	2005 /10/GRN	G	GROUND

=FIELD_WIRE+EC1
ELECTRONIC I/O RACK ENCL

NEUTRAL	N	W1:2 /18/BLK	TB2 N	NEUTRAL
GROUND	G	W1:G /18/GRN	G	GROUND
PILOT START	18	W1:3 /18/BLK	1	PILOT START
PILOT RESET	19	W1:4 /18/BLK	2	PILOT RESET
BURNER PILOT PRVD	20	W1:5 /18/BLK	3	BURNER PILOT PRVD
			8	BURNER MAIN VALVE
BURNER LOW GAS VALVE	22	W1:7 /18/BLK	9	BURNER LOW GAS VALVE
BURNER HI GAS VALVE	23	W1:8 /18/BLK	10	BURNER HI GAS VALVE
ROLLOVER VALVE OPEN	24	W1:9 /18/BLK	11	ROLLOVER VALVE ENABLE
			12	ROLLOVER VALVE ON
BURNER COMBUSTION AIR BLOWER	TB1 5	W2:1 /14/BLK	TB2 13	BURNER COMBUSTION AIR BLOWER
	6	W2:2 /14/BLK	14	
	7	W2:3 /14/BLK	15	
BURNER PILOT AIR BLOWER	8	W2:4 /14/BLK	16	BURNER PILOT AIR BLOWER
	9	W2:5 /14/BLK	17	
GROUND	G	W2:G /14/GRN	G	GROUND
+24 VDC	TB2 +	W3:1 /18/BLK	TB1 +	+24 VDC
TIE POINT	49	W3:2 /18/BLK	2	
TIE POINT	50	W3:3 /18/BLK	3	

=FIELD_WIRE+BC1
BURNER CONTROL

BURNER CONTROL ASSY
P/N: 4A1-006-364
W/D: WD4A1-006-364

=FIELD_WIRE+FLASH
FLASHOVER

NEUTRAL	N	W8:2 /18/BLK	TB2 N	NEUTRAL
GROUND	G	W8:G /18/GRN	G	GROUND
PILOT START	21	W8:3 /18/BLK	1	PILOT START
PILOT RESET	29	W8:4 /18/BLK	2	PILOT RESET
FLASHOVER PILOT PRVD	25	W8:5 /18/BLK	3	FLASHOVER PILOT PRVD
FLASHOVER MAIN VALVE	26	W8:6 /18/BLK	8	FLASHOVER MAIN VALVE
FLASHOVER ON/OFF VALVE	27	W8:7 /18/BLK	9	FLASHOVER ON/OFF VALVE
FLASHOVER COMBUSTION AIR BLOWER	TB1 12	W9:1 /14/BLK	TB2 13	FLASHOVER COMBUSTION AIR BLOWER
	13	W9:2 /14/BLK	14	
	14	W9:3 /14/BLK	15	
FLASHOVER PILOT AIR BLOWER	8	W9:4 /14/BLK	16	FLASHOVER PILOT AIR BLOWER
	9	W9:5 /14/BLK	17	
GROUND	G	W9:G /14/GRN	G	GROUND
TIE POINT	TB2 49	W10:1 /18/BLK	TB1 A	+24 VDC
TIE POINT	50	W10:2 /18/BLK	B	
SYSTEM GAS PRESSURE	15	W10:3 /18/BLK	2	
SYSTEM CAB	16	W10:4 /18/BLK	3	

I/O RACK ELECT ENCL ASSY
P/N: 4A1-007-134
W/D: WD4A1-007-134 REV. D

FLASHOVER VALVE CAB ASSY
P/N: 4A1-006-312
W/D: WD4A1-006-312

NOTE:FIELD MODIFICATION TO FLASH1
ADD JUMPER TB2:L1 TO TB2:1
REMOVE ALL JUMPERS FROM TB1:+
RELABEL AS TB1:A W5:1
AND TB1:B W7:1, AND TB1:C OPEN.
RELOCATE W6:1 TO TB1:1

=FIELD_WIRE+EC1

I/O RACK ELECT ENCL ASSY
P/N: 4A1-007-134
W/D: WD4A1-007-134 REV. D

=FIELD_WIRE+BURN_ROOM

COMBUSTIBLE GAS SENSOR #1 INPUT	TB2 38	W4:1 /18/BLK	CGS1 B	COMBUSTIBLE GAS SENSOR #1
	39	W4:2 /18/WHT	W	
	40	W4:3 /18/RED	R	
	41	W4 SH	CONNECT CABLE SHIELD AT TERMINAL TB2:41 ONLY	
COMBUSTIBLE GAS SENSOR #2 INPUT	TB2 42	W5:1 /18/BLK	CGS2 B	COMBUSTIBLE GAS SENSOR #2
	43	W5:2 /18/WHT	W	
	44	W5:3 /18/RED	R	
	45	W5 SH	CONNECT CABLE SHIELD AT TERMINAL TB2:45 ONLY	
COMPARTMENT T/C INPUT	TB2 9	TC1:+ /20/YEL	TC1 +	COMPARTMENT TEMPERATURE
	10	TC1:- /20/RED	-	

=FIELD_WIRE+SG

AC LINE POWER	TB1 10	W6:1 /14/BLK	TB2 L1	AC LINE POWER
AC LINE POWER	11	W6:2 /14/BLK	L2	AC LINE POWER
GROUND	G	W6:G /14/GRN	G	GROUND
SMOKE HEAT ENABLE	TB2 1	W7:1 /18/BLK	TB1 1	SMOKE HEAT ENABLE
24V RTN	R	W7:2 /18/BLK	2	24V RTN
+24 VDC	1	W7:3 /18/BLK	3	+24 VDC
SMOKE TEMP READY	21	W7:4 /18/BLK	4	SMOKE TEMP READY
SMOKE EMIT	22	W7:5 /18/BLK	5	SMOKE EMIT
24V RTN	R	W7:6 /18/BLK	6	24V RTN

SMOKE CABINET ASSY
P/N: 4C1-007-552
WD: WD4A1-006-310

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PROJECT	T5_MINOT_AFB	INSTALLATION	FIELD_WIRE	Kidde Fire Trainers	
		LOCATION/S		TITLE	MINOT AFB T-500 INSTALLATION INTERCONNECT DIAGRAMS MULTI-PROP ELECT ENCL TO BURNER CONTROL
4B1-007-653	SO 3925	DWN	KR	04/30/09	SIZE D / CAGE CODE 06003 / DWG NO IW4B1-007-653
NEXT ASSY	USED ON	CHKD	EC	05/04/09	
APPLICATION		APVD	AC	05/04/09	SCALE NA / CAD FILE: IW4B7653- / SHEET 2 OF 6

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD

MULTI-PROP FIREPLACE BURN ROOM

=FIELD_WIRE+EC1

PILOT 0 START SWITCH	A2A2	16	23	J1	3
PILOT 0 STOP SWITCH	A2A2	20	24		4
PILOT 0 INDICATOR	A2A3	24	25		5
+24V DC	TB2 +	A3 8	2.2K A3 1		
VALVE 0 LOW SWITCH	A2A2	24	26		6
VALVE 0 HIGH SWITCH	A2A2	28	27		7
VALVE 0 INDICATOR	A2A3	28	28		8
	A3 8	2.2K	A3 3		
VALVE 2 LOW SWITCH ROLL	A2A2	26	29		9
VALVE 3 LOW SWITCH FLASH	A2A2	25	30		10
VALVE 2 INDICATOR	A2A3	30	31		11
VALVE 3 INDICATOR	A2A3	29	32		
	A3 8	2.2K	A3 5		
SG START / STOP SWITCH	A2A2	2	33		13
VENT START / STOP SWITCH	A2A2	1	34		14
SMOKE INDICATOR	A2A3	4	35		15
	A3 8	2.2K	A3 7		
OPERATOR PRESENCE SWITCH	A2A2	7	36		16
OPERATOR PRESENT	A2A3	2	37		17
	A3 8	2.2K	A3 9		
REMOTE CONTROL ID		32			12
+24V DC		+			1
24V RET		-			2

P1

CONTROL PENDANT

PILOT: Off / On — B

MAIN: Off / Low / High — A

EFFECT: Flash / Roll — R

Smoke / Vent — W

DEADMAN — G

CONTROL PENDANT ASSY.
P/N: 4A1-007-136

CONTROL PENDANT CONNECTION
(SHOWN FOR REFFERENCE ONLY)

=FIELD_WIRE+BURN_ROOM1

E-STOP CHAIN	TB2 6	W21:1 /18/BLK	E1 A1
	46	W21:2 /18/BLK	B1
	47		B2
E-STOP CHAIN	7	W21:4 /18/BLK	A2

COMPARTMENT EMERGENCY STOP #1
P/N: 2A1-002-293

		W22:1 /18/BLK	E2 A1
	47	W22:2 /18/BLK	B1
	48		B2
E-STOP CHAIN	8	W22:4 /18/BLK	A2

COMPARTMENT EMERGENCY STOP #2

I/O RACK ELECT ENCL ASSY
PN: 4A1-007-134
WD: WD4A1-007-134 REV. D

CONTROL PENDANT

PILOT START / STOP, BLUE ILLMINATED SELECTOR, 3-POS, SPRING RETURN

FLAME CONTROL, AMBER ILLUMINATED SELECTOR, 3-POS MAINTAINED

FLASHOVER / ROLLOVER EFFECT, RED ILLUMINATED SELECTOR, 3-POS SPRING RETURN

SMOKE / VENT, WHITE ILLUMINATED SELECTOR, 3-POS MAINTAINED

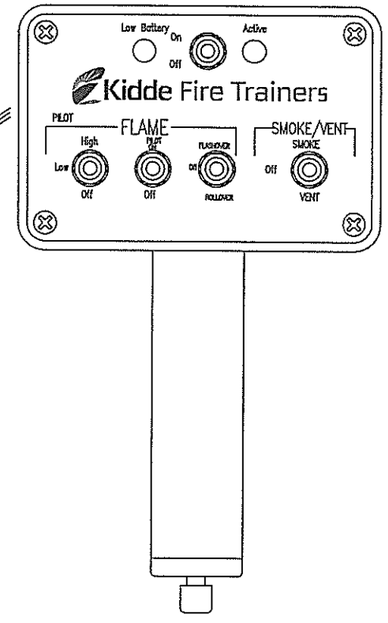
OPERATOR PRESENT, GREEN ILLUMINATED BUTTON

WIRELESS PENDANT R160 RECEIVER

P1

6	BLUE/RED	A1	LOW FIRE
7	WHT/RED	A9	HIGH FIRE
10	ORG/BLK	A10	FLASHOVER
9	ORG/RED	A12	ROLLOVER
3	ORG/GRN	A7	PILOT START
4	RED/GRN	A8	PILOT STOP
13	WHT/RED/BLK	B1	SMOKE
14	RED/BLK/WHT	B2	VENT
16	BLK/RED	B12	DEADMAN
12	RED		
1	RED	A6	+24 VDC
2	BLACK	A3	24V RTN

T-500 WIRELESS PENDANT ASSY
PN:4A1-007-063



PROPRIETARY INFORMATION OF Kidde Fire Trainers
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PROJECT	T5_MINOT_AFB	INSTALLATION	FIELD_WIRE	Kidde Fire Trainers	
		LOCATION/S		TITLE	MINOT AFB T-500 INSTALLATION INTERCONNECT DIAGRAMS MULTI-PROP PENDANT,WIRELESS AND E-STOP CONTROL
4B1-007-653	SO 3925	DWN	KR 04/30/04	SIZE D	CAGE CODE 06003 DWG NO IW4B1-007-653
NEXT ASSY	USED ON	CHKD	EC 05/04/09		
APPLICATION		APVD	AC 05/04/09	SCALE NA	CAD FILE: IW4B7653- SHEET 3 OF 6

KITCHEN FIREPLACE

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD

=FIELD_WIRE+EC1

MAIN FLAME ENABLE A2A3 12 → LOW SP FAN K17 13/14 → TB2 R

SYSTEM VENT ENABLE A2A3 16 → HIGH SP FAN K16 4/5 → TB2 R

VENT FAN PROVED A2A2 30 → LOW SP FAN K17 1/9

TB1 L1, L2, G → CB6 20 AMP → LINE IN L1, L3, G

FAULT T5/T6

DIGITAL OUTPUT B3

+24V POWER B2

ENABLE/RESET B4 — LOW SP FAN K17 2/1 — HIGH SP FAN K16 5/9 — +24V DC (KEY SW) TB2 1

RUN/FORWARD B5 — LOW SP FAN K17 6/10 — HIGH SP FAN K16 7/8

PRESET SPEED #2 B7

PRESET SPEED #3 T4

T1 → TB2 R

LINE OUT U, V, W → TB1 15, 16, 17

G

6001 /14/BLK

6002 /14/RED

6003 /14/BLU

6004 /14/GRN

I/O RACK ELECT ENCL ASSY
P/N: 4A1-007-134
W/D: WD4A1-007-134 REV. D

=FIELD_WIRE+VENT_FAN1

DS1: L1/T1, L2/T2, L3/T3

6005 /14/BLK

6006 /14/RED

6007 /14/BLU

6004 /14/GRN

VF1: T1, T2, T3 — M

EXHAUST FAN
2 HP
3450 RPM
6.8 AMP

EXHAUST FAN ASSY,208VAC,3P,2HP,RF
P/N: 4A1-006-370-018

VSD PARAMETERS SETTING (Commander SK)

Par	Description	Setting	
	Level 1 parameters		
1	Minimum set speed (Hz)	0.0	FIELD SETTING
2	Maximum set speed (Hz)	60.0	
3	Acceleration rate (s/100Hz)	10.0	
4	Deceleration rate (s/100Hz)	10.0	
5	Drive configuration	Pr	
6	Motor rated current (A)	5.5	
7	Motor rated speed (rpm)	0	
8	Motor rated voltage (V)	230	
9	Motor power factor	.85	
10	Parameter access	L3	
	Level 2 parameters		
11	Start/Stop logic select	4	
12	Brake controller enable	dIO	
13	Not used		
14	Not used		
15	Jog reference (Hz)	1.5	FIELD SETTING
16	Analog input 1 mode (mA)	VoLt	
17	Enable negative preset speeds	OFF	
18	Preset speed 1 (Hz)	0	
19	Preset speed 2 (Hz)	25	
20	Preset speed 3 (Hz)	60	
21	Preset speed 4 (Hz)	45	
22	Load display units	A	
23	Speed display units	Fr	
24	Customer defined scaling	1.000	
25	User security code	0	
26	Not used		
27	Power up keypad reference	0	
28	Parameter cloning	no	
29	Load defaults	no	
30	Ramp mode select	1	
31	Stop mode select	1	
32	Dynamic V to f select	OFF	
33	Catch a spinning motor select	0	
34	Terminal B7 mode select	dig	
35	Digital output control (terminal B3)	AT.LD	
36	Analog output control (terminal B1)	Fr	
37	Maximum switching frequency (kHz)	3	
38	Autotune	0	
39	Motor rated frequency (Hz)	60.0	
40	Number of motor poles	Auto	
41	Voltage mode select	Ur I	
42	Low frequency voltage boost (%)	3.0	
43	Serial communications baud rate	19.2	
44	Serial comms address	1	

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PROJECT	T5_MINOT_AFB	INSTALLATION	FIELD_WIRE	
		LOCATION/S		
4B1-007-653	SO 3925	DWN	KR	05/01/09
NEXT ASSY	USED ON	CHKD	EC	05/04/09
APPLICATION		APVD	AC	05/04/09

Kidde Fire Trainers

TITLE: MINOT AFB T-500 INSTALLATION
INTERCONNECT DIAGRAMS
KITCHEN, VENT FAN CONTROL

SIZE	CAGE CODE	DWG NO
D	06003	IW4B1-007-653

SCALE NA | CAD FILE: IW4B7653- | SHEET 4 OF 6

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD

=FIELD_WIRE+EC1

TB1
1 — 501 /18/BLK
N — 502 /18/WHT
G — 503 /18/GRN

NOTE:K8 USED FOR ROLLOVER INTERLOCK SINCE K7 IS USED FOR SOUND SYSTEM INTERFACE 4A1-007-134 REV E

TB2 + — 505 /18/BLU — K7 (MAIN VALVE BURNER LOW ON) 6 4 — 506 /18/BLU — TB2 51 — W11:1 /18/BLK

NOTE:WIRES TO K7,K11,&SR1 ARE NOT PART OF T500 I/O ASSY AND ARE FOR THE SOUND ASSY.

K11 (FLASHOVER ENABLE) 6 4 — 507 /18/BLU — TB2 52 — W11:2 /18/BLK

SR1 (EMERGENCY STOP) 43 44 — 508 /18/BLU — TB2 53 — W11:3 /18/BLK

NOTE: SOUND SYSTEM POWERED OFF IN STANDBY TO PREVENT E.STOP ALARM

TB2 1 — W11:4 /18/BLK
R — W11:5 /18/BLK
G — W11:G /18/GRN

I/O RACK ELECT ENCL ASSY
PN: 4A1-007-134
WD: WD4A1-007-134 REV. D

=CONTROLASSY+SOUND_SYSTEM

TB1
L — LINE POWER
N — NEUTRAL
G — GROUND

TB2 1 — W11:1 /18/BLK
- 0 A2A2 SOUND INPUT 0
- MAIN FLAME ENAB K1 (7 8) — TB2 R — 24V RET

TB2 2 — W11:2 /18/BLK
- MAIN FLAME ENAB K1 (5 1) — 2 A2A2 SOUND INPUT 2
- MAIN FLAME ENAB K1 (5 3) — 3 A2A2 SOUND INPUT 3

TB2 4 — W11:3 /18/BLK
- 1 A2A2 SOUND INPUT 1
- 11 A2A2 E-STOP INPUT

TB2
1 — W11:4 /18/BLK
R — W11:5 /18/BLK
G — W11:G /18/GRN

SOUND MODULE 1 (FIRE SOUNDS)
R: + TB2 10 — W12:1 /18/BLK — SPEAKERS WEATHERPROOF SPK1 FOWARD BURN
R: − TB2 11 — W12:2 /18/BLK
L: + TB2 12 — W13:1 /18/BLK — SPEAKERS WEATHERPROOF SPK2 REAR BURN
L: − TB2 13 — W13:2 /18/BLK

MODE SWICTH

SOUND MODULE 2 (BACKGROUND SOUNDS)
R: + TB2 14 — W14:1 /18/BLK — SPEAKERS WEATHERPROOF SPK3 FOWARD BURN
R: − TB2 15 — W14:2 /18/BLK
L: + TB2 16 — W15:1 /18/BLK — SPEAKERS WEATHERPROOF SPK4 REAR BURN
L: − TB2 17 — W15:2 /18/BLK

DWG: WD4A1-007-416
BOM : 4A1-007-416-003

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PROJECT	T5_MINOT_AFB	INSTALLATION	FIELD_WIRE	Kidde Fire Trainers
		LOCATION/S		TITLE: MINOT AFB T-500 INSTALLATION INTERCONNECT DIAGRAMS SOUND SYSTEM INTERFACE
4B1-007-653	SO 3925	DWN KR	04/30/09	SIZE D · CAGE CODE 06003 · DWG NO IW4B1-007-653
NEXT ASSY	USED ON	CHKD EC	05/04/09	
APPLICATION		APVD AC	05/04/09	SCALE NA · CAD FILE: IW4B7653- · SHEET 5 OF 6

DWG NO	IW4B1-007-653	SH	6

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD

=FIELD_WIRE+VENT_FAN_CONT

=FIELD_WIRE+SP1

CB1 L1 T1 / L2 T2 / L3 15 AMP T3 / N / G

TB1 L1 L2 L3 N G

D1
POWER SUPPLY
24VDC 4.2A 100W
L LINE +24 VDC +
N NEUTRAL
G GROUND 24V RET -

CB1 1 6 AMP 2

TB1 L1 N G — TB1 2

VSD2
FORWARD DI1
INPUT
L1 L2 L3 G
VARIABLE SPEED AC MOTOR DRIVE
COMMON DCM
(HAND) FULL SPEED (SW3) DI6
(+500F) MEDIUM SPEED (SW1) DI4
(+750F) HIGH SPEED (SW2) DI5
T1 T2 T3 G
OUTPUT TO VENT FAN

TS1 AND TS2
PRESS SET-UP ARROW TO 750
PRESS UP/DOWN FOR 3 SEC TO ENTER SETUP PAGE

SET	v
L in	H
SP.Lo	-328
SP.hi	2500
Ot 1	Pr.AL
Ot 2	Pr.AL
dSP1	OFF
dSP2	OFF
LOC	0

PRESS INFINITY FOR 3 SEC TO ENTER OPERATIONS

SET	v
A1.hi	500
A1.Lo	-328
A2.hi	750
A2.Lo	-328

ALL OTHER SETTINGS REMAIN AT DEFAULT

TB2 + (3X) — R (3X)

TB2 5 (2X) — D1:A — D2 A C D1:C

TB2 1 COMMON

TB2 2 (HAND) FULL SPEED (SW3)

3 (+500F) MEDIUM SPEED (SW1)

4 (+750F) HIGH SPEED (SW2)

TB2 + +24 VDC

TB2 R 24V RET

HOA SWITCH
PB1 Hand Off Auto
3 4 1 2

S1

FUNCTION	1-2	3-4
HAND	CLSD	OPEN
OFF	OPEN	OPEN
AUTO	OPEN	CLSD

TS1
1 + TC+ 10
2 - TC- 11
5 +500 F R1 6
3 +750 F R2 4

TS2
1 + TC+ 10
2 - TC- 11
5 +500 F R1 6
3 +750 F R2 4

BURN COMPARTMENT TEMPERATURE #1
TB2 8 TB2 9 — T/C:2 + -

BURN COMPARTMENT TEMPERATURE #2
TB2 10 TB2 11 — T/C:3 + -

K1 8 7 ALARM (+750 F)

K1 (contacts)

CP	5 3	6 4
	627	629
CP	5 1	6 2

ALARM (+750 F) K1 5 3 — TB2 5 FORWARD (TIE POINT)

K1 4 6 618 — TB2 13

AUDIBLE ALARM AH1 1 2

VSD1 PARAMETERS SETTING

P 0.00	Voltage	208
P 0.01	Amps	4
P 0.02	Frequency	60
P 0.03	Motor RPM	1750
P 0.04	Max RPM	1750
P 1.00	Stop Metho	00
P 1.01	Accel Time	10
P 1.02	Decel Time	10
P 2.00	V/Hz Set	02
P 3.00	Oper Comm	02
P 3.01	Input (DI1)	01
P 3.02	Input (DI3)	00
P 3.03	Input (DI4) (Low)	03
P 3.04	Input (DI5) (Med)	04
P 3.05	Input (DI6) (Hi)	05
P 3.11	Output 1	00
P 3.12	Output 2	06
P 4.00	Com Source	01
P 5.01	Preset 1 (Med)	45
P 5.02	Preset 2 (Hi)	45
P 5.03	Preset 3 (Hi)	60
P 5.04	Preset 4 (Lo)	25
P 5.05	Preset 5 (Hi)	60
P 5.06	Preset 6 (Hi)	60
P 5.07	Preset 7 (Hi)	60
P 6.00	Overload	00
P 6.01	Au. Restart	05
P 6.02	Pwr Loss	01
P 8.00	Display	01

NOTE:
SET VSD1 BASE FREQUENCY TO 25HZ IN AUTO MODE
WHEN BURN ROOM TEMPS ARE BELOW FIRST SETPOINT 500 DEG.

PN: 4A1-006-525
WD:4A1-006-525

DS1 G — L1 T1 T1 / L2 T2 T2 / L3 T3 T3 — VFC2 — M1 — 1 HP 1750 RPM 4.8 AMP

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PROJECT	T5_MINOT_AFB	INSTALLATION	FIELD_WIRE	Kidde Fire Trainers
		LOCATION/S	ENCLOSURE	TITLE: MINOT AFB T-500 INSTALLATION INTERCONNECT DIAGRAMS WIRING DIAGRAM
4B1-007-653	SO 3925	DWN KR	09/16/09	SIZE D — CAGE CODE 06003 — DWG NO IW4B1-007-653
NEXT ASSY	USED ON	CHKD EC	09/16/09	
APPLICATION		APVD AC	09/16/09	SCALE NA — CAD FILE: IW457653- — SHEET 6 OF 6

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ADD AC DRIVE, RELAYS FOR BURNER.	01/19/07	LMI/SH
B	AS BUILT UPDATED	02/08/08	EC
C	AS BUILT UPDATED	03/28/08	EC
D	UPDATED VENT FAN CONTROL WIRING	07/17/08	LMI/EC
E	GAS VALVE INTERLOCK ADDED	03/08/09	KR/LI

Legend of Common Reference Designators

AD	Agent Detection
B	Blower Motor
CAB	Combustion Air Blower
CB	Circuit Breaker
CGDS	Combustible Gas Drawn Sample
CGS	Combustible Gas Sensor
D	Diode
DS	Door Switch
ES	E-Stop
F	Flashover Effect
FHM	Fireplace Hardware Module
FP	Fireplace Mockups
FRCS	Fuel Regulator & Control Station
FS	Flame Safeguard
HOA	Hand-Off-Automatic Switch
I	Indicator
IG	Ignitor
JB	Junction Box
K	Relay
KS	Keyswitch
L	Solenoid Valve
LCP	Local Control Panel
LT	Lamp/Light
MC	Magnetic Contactor Enclosure
ME	Master I/O Rack Enclosure
MV	Motorized Valve
MVA	Motorized Valve Assy
OS	Operator Station
P	Plug
PND	Pendant, Control
POC	Proof of Closure
PRS	Pressure Switch
PS	Power Supply
PT	Pressure Transducer
R	Receptical
RBC	Remote Burner Control
S	Switch
SCP	Scenario Control Panel
SD	Smoke Damper
SG	Smoke Generator
SP	Service Panel
SPK	Speaker
SSOV	Safety Shutoff Valve
T	Transformer
TB	Terminal Block
TC	Thermocouple
TDR	Time Delay Relay
TS	Temperature Switch
VF	Ventilation Fan
VFC	Ventilation Fan Control
W	Cable
Z	Zener Diode

Installation: IO RACK ELECTRONIC ENCL ASSY

Page	Page Title	Revision
1	REVISION STATUS OF PAGES	E
2	PLC POWER, KEY SWITCH, E. STOP	D
3	SYSTEM SWITCHES AND INDICATORS	E
4	COMBUSTIBLE GAS DETECTION, HIGH TEMP SHUTDOWN	B

VSD PARAMETERS SETTING (Commander SK)

Par	Description	Setting	
	Level 1 parameters		
1	Minimum set speed (Hz)	0.0	
2	Maximum set speed (Hz)	60.0	
3	Acceleration rate (s/100Hz)	10.0	
4	Deceleration rate (s/100Hz)	10.0	
5	Drive configuration	Pr	
6	Motor rated current (A)	5.50	FIELD SETTING
7	Motor rated speed (rpm)	3450	FIELD SETTING
8	Motor rated voltage (V)	230	FIELD SETTING
9	Motor power factor	.85	FIELD SETTING
10	Parameter access	L3	
	Level 2 parameters		
11	Start/Stop logic select	4	
12	Brake controller enable	diS	
13	Not used		
14	Not used		
15	Jog reference (Hz)	1.5	
16	Analog input 1 mode (mA)	VoLt	
17	Enable negative preset speeds	OFF	
18	Preset speed 1 (Hz)	0	
19	Preset speed 2 (Hz)	25	FIELD SETTING
20	Preset speed 3 (Hz)	60	FIELD SETTING
21	Preset speed 4 (Hz)	60	FIELD SETTING
22	Load display units	A	
23	Speed display units	Fr	
24	Customer defined scaling	1.000	
25	User security code	0	
26	Not used		
27	Power up keypad reference	0	
28	Parameter cloning	no	
29	Load defaults	no	
30	Ramp mode select	1	
31	Stop mode select	1	
32	Dynamic V to f select	OFF	
33	Catch a spinning motor select	0	
34	Terminal B7 mode select	dig	
35	Digital output control (terminal B3)	At:Ld	
36	Analog output control (terminal B1)	Fr	
37	Maximum switching frequency (kHz)	3	
38	Autotune	0	
39	Motor rated frequency (Hz)	60.0	
40	Number of motor poles	Auto	
41	Voltage mode select	Ur I	
42	Low frequency voltage boost (%)	3.0	
43	Serial communications baud rate	19.2	
44	Serial comms address	1	

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL WIRE IS TYPE THHN OF SPECIFIED COLOR AND GAUGE.

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PROJECT	T500_IO	INSTALLATION	WIRING_DIAG	Kidde Fire Trainers	
		LOCATION/S		TITLE	IO RACK ELECTRONIC ENCL ASSY WIRING DIAGRAMS REVISION STATUS OF PAGES
4A1-007-134	T-500	DWN	B AQUINO	05/18/06	SIZE D | CAGE CODE 06003 | DWG NO WD4A1-007-134
NEXT ASSY	USED ON	CHKD	B AQUINO	05/18/06	
APPLICATION		APVD	LMI	05/18/06	SCALE NA | CAD FILE: WD4A7134E | SHEET 1 OF 4

DWG NO WD4A1-007-134 | SH 2

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ADD AC DRIVE, RELAYS FOR BURNER.	01/19/07	LMI/SH
B	AS BUILT UPDATED	02/08/08	ECLMI
C	AS BUILT UPDATED	03/28/08	EC/LMI
D	UPDATED VENT FAN CONTROL WIRING	07/17/08	LMI/EC

STANDBY POWER

208-240VAC 1 Ø 40A OR 3 Ø 30A

GROUND

NEUTRAL (X2)

CB1 10 AMP

CB2 16 AMP

CB3 20 AMP

CB4 20 AMP

CB5 16 AMP

CB6 20 AMP

KEY SW PWR K1 252

OPERATE PWR K3 269

SMK GEN PWR K2 260

MC2 257

MC1 253

MC3 258

FIREPLACE POWER (E-STOP)

BURNER BLOWER

PILOT BLOWER

SMOKE GEN POWER

FLASHOVER BLOWER

EXHAUST VENT

KEY SWITCH

FUNCTION	B1–B2	A3–A4
OFF	OPEN	OPEN
ON	CLSD	OPEN
RESET	CLSD	CLSD

VSD1 — LINE IN / LINE OUT

LOW SP VENT K17 276

HIGH SP VENT K16 273

FAULT

DIGITAL OUTPUT PRESET SP #2 ENABLE/RESET

DIGITAL OUTPUT PRESET SP #3 RUN/FORWARD

0V DC

VARIABLE SPEED DRIVE 1/3 PHASE 230V 2HP 7A P/N: SKD0200150

LOW SP VENT K17 273

HIGH SP VENT K16 276

LOW SP VENT K17 276

A2A2 VENT PROVED

+24V DC

** SEE FRONT PAGE FOR VSD1 PARAMETERS SETTING.

SS1 240 VAC SURGE SUPRESSOR

DOOR GROUND E2

ENCLOSURE GROUND E3

PS1 POWER SUPPLY 24VDC, 7.5A PSM24-180S

CB7 6 AMP

+24V DC

24VDC SURGE SUPRESSOR (VARISTOR) SS2

24VDC SURGE SUPRESSOR (AVALANCHE DIODE) SS3

24R

A1 IO RACK ASSSY

DS1 DISPLAY PANEL PN OP-620

OP-4CBL-2

A1A1 PLC PROCESSOR 1769-L31 RS232C

A1A2 DC INPUT 32 PT MODULE 1769-IQ32

A1A3 DC OUTPUT 32 PT MODULE 1769-OB32

A1A0 POWER SUPPLY 1769-PB2

NOTE:ADD 1769-IQ32 TO RACK FOR -002 OPTION

SYSTEM POWER

LT1

KS1 KB1A

EStop Reset

MANUAL RESET K4

LAMP TEST PB1

LAMP TEST A2A2 6

SYSTEM KEY ON A2A2 5

CABINET E.STOP ES1

E.STOP CHAIN

GSA1 K14 460

GSA2 K15 470

ES1_LT

ESTOP INDICATOR A2A3 1

E.STOP ID 0 A2A2 0

Estop Control SR1

24V RET

MANUAL RESET K4 253

WATCHDOG TDR1 451

E.STOP STATUS A2A2 4

OPERATE POWER A2A3 0

VENT INDICATOR A2A3 3

PURGE LAMP LT2

HIGH SP VENT ENAB A2A3 16

MED SP VENT ENAB A2A3 9

PILOT ENABLE A2A3 12

KEY SW PWR K1

PILOT BLOWER MC1

TDR2

TDR3

BURNER BLOWER MC2

FLASH BLOWER MC3

SMK GEN PWR K2

HIGH SP VENT K16

LOW SP VENT K17

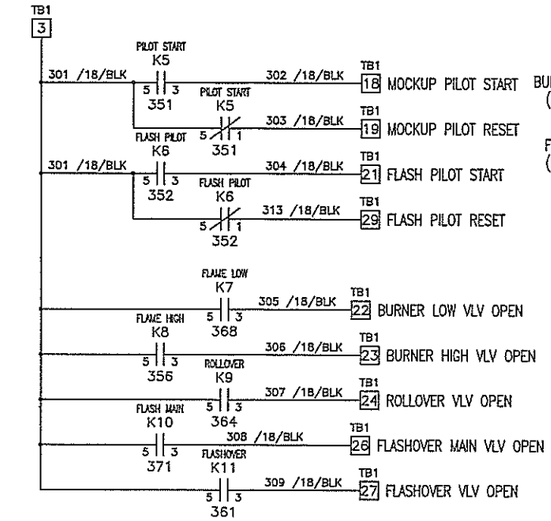
TDR2 254

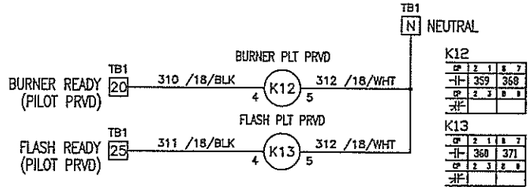
TDR3 255

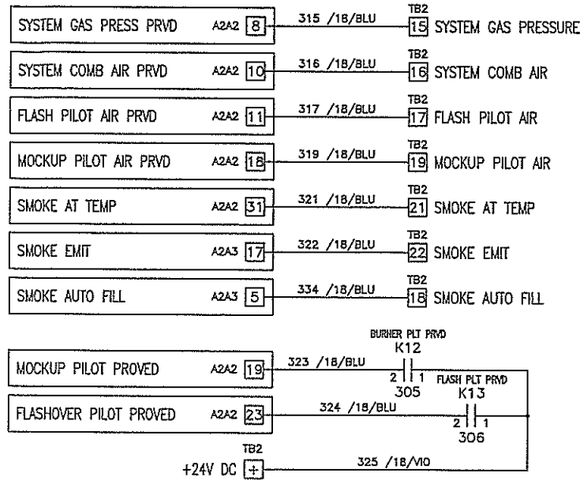
PROJECT	T500_IO	INSTALLATION	WIRING_DIAG	Kidde Fire Trainers	
		LOCATION/S	ELEC_ENCL	TITLE	IO RACK ELECTRONIC ENCL ASSY WIRING DIAGRAMS PLC POWER, KEY SWITCH, E.STOP
4A1-007-134	T-500	DWN B AQUINO 11/11/05		SIZE D	CAGE CODE 06003
NEXT ASSY	USED ON	CHKD B AQUINO 11/11/05		DWG NO WD4A1-007-134	
APPLICATION		APVD LMI 11/11/05		SCALE NA	CAD FILE: WD4A7134E
				SHEET 2 OF	

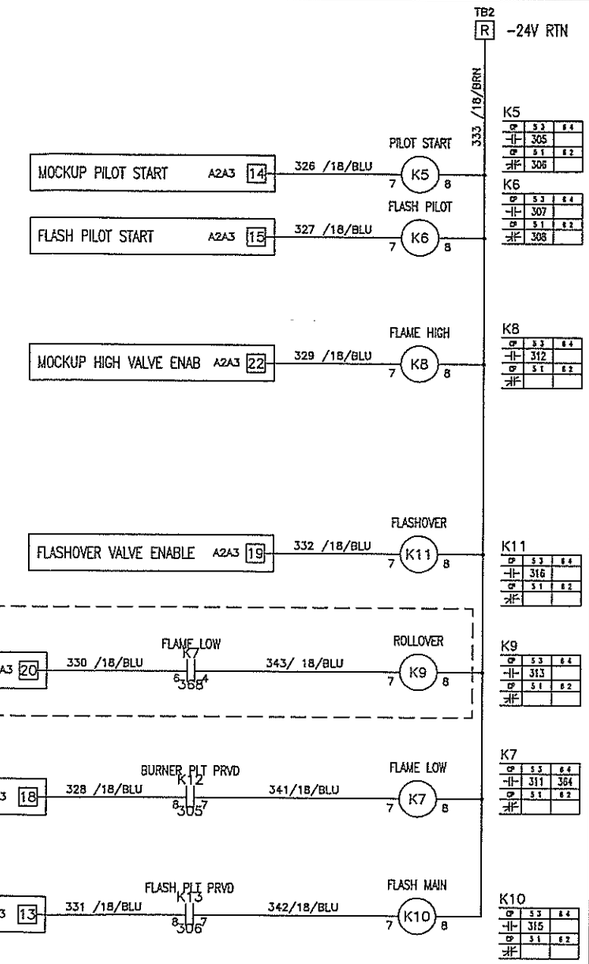
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REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ADD AC DRIVE, RELAYS FOR BURNER.	01/19/07	LMI/SH
B	AS BUILT UPDATED	02/08/08	EC
C	AS BUILT UPDATED	03/28/08	EC
E	K12 AND K13 INTERLOCKED TO GAS VLAVES	03/08/09	KR/LI







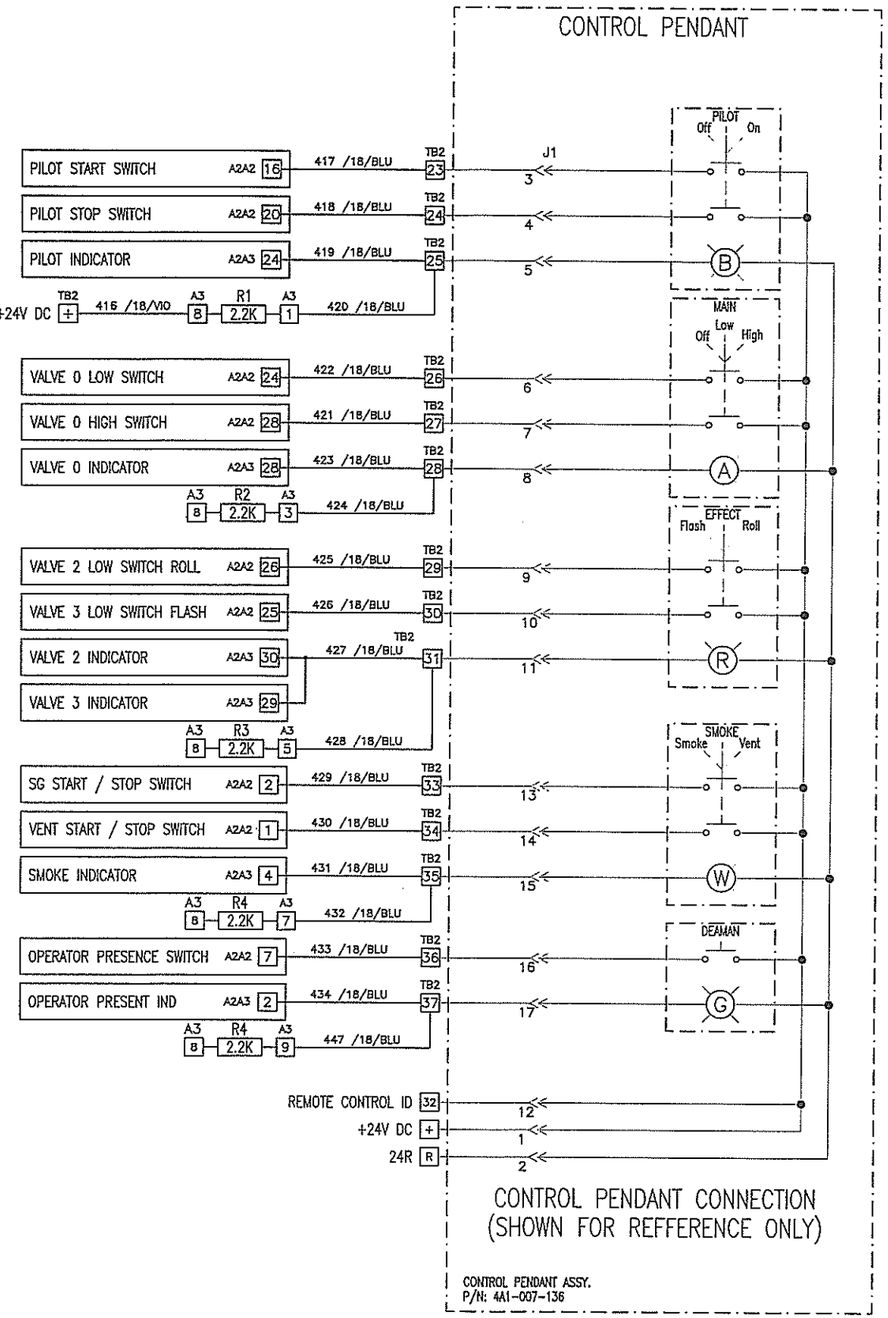


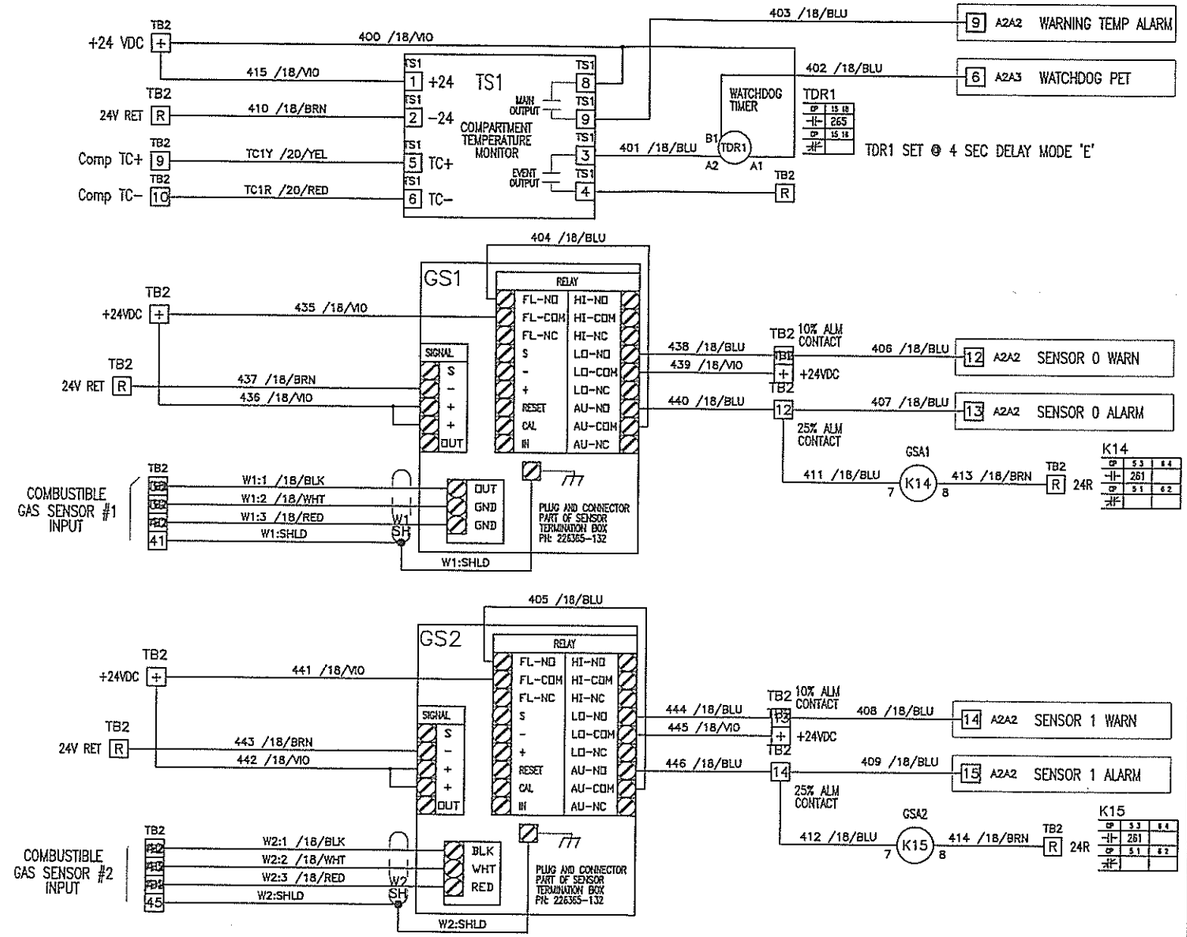
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PROJECT	T500_IO	INSTALLATION	WIRING_DIAG	Kidde Fire Trainers	
		LOCATION/S	ELEC_ENCL	TITLE	IO RACK ELECTRONIC ENCL ASSY
					WIRING DIAGRAMS
					SYSTEM SWITCHES AND INDICATORS
4A1-007-134	T-500	DWN	B AQUINO	11/11/05	SIZE D / CAGE CODE 06003 / DWG NO WD4A1-007-134
NEXT ASSY	USED ON	CHKD	B AQUINO	11/11/05	
APPLICATION		APVD	LMI	11/11/05	SCALE NA / CAD FILE: WD4A7134E / SHEET 3 OF 4

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ADD AC DRIVE, RELAYS FOR BURNER.	01/19/07	LMI/SH
B	AS BUILT UPDATED	02/08/08	EC





NOTES:

REFER TO THE "SETUP SECTION OF 95-8444 U9500 USER MANUAL.
1 - ALARM RELAYS FOR THIS ASSEMBLY ARE NON-LATCHING AND NORMALLY ENERGIZED.
2 - ALARM RELAYS SETPOINTS FOR THIS ASSEMBLY ARE AS FOLLOWS:

LOW RELAY 10% LEL
AUX RELAY 25%LEL
HIGH RELAY 60%LEL

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PROJECT	T500_IO
INSTALLATION	WIRING_DIAG
LOCATION/S	ELEC_ENCL
4A1-007-134	T-500
NEXT ASSY	USED ON
APPLICATION	

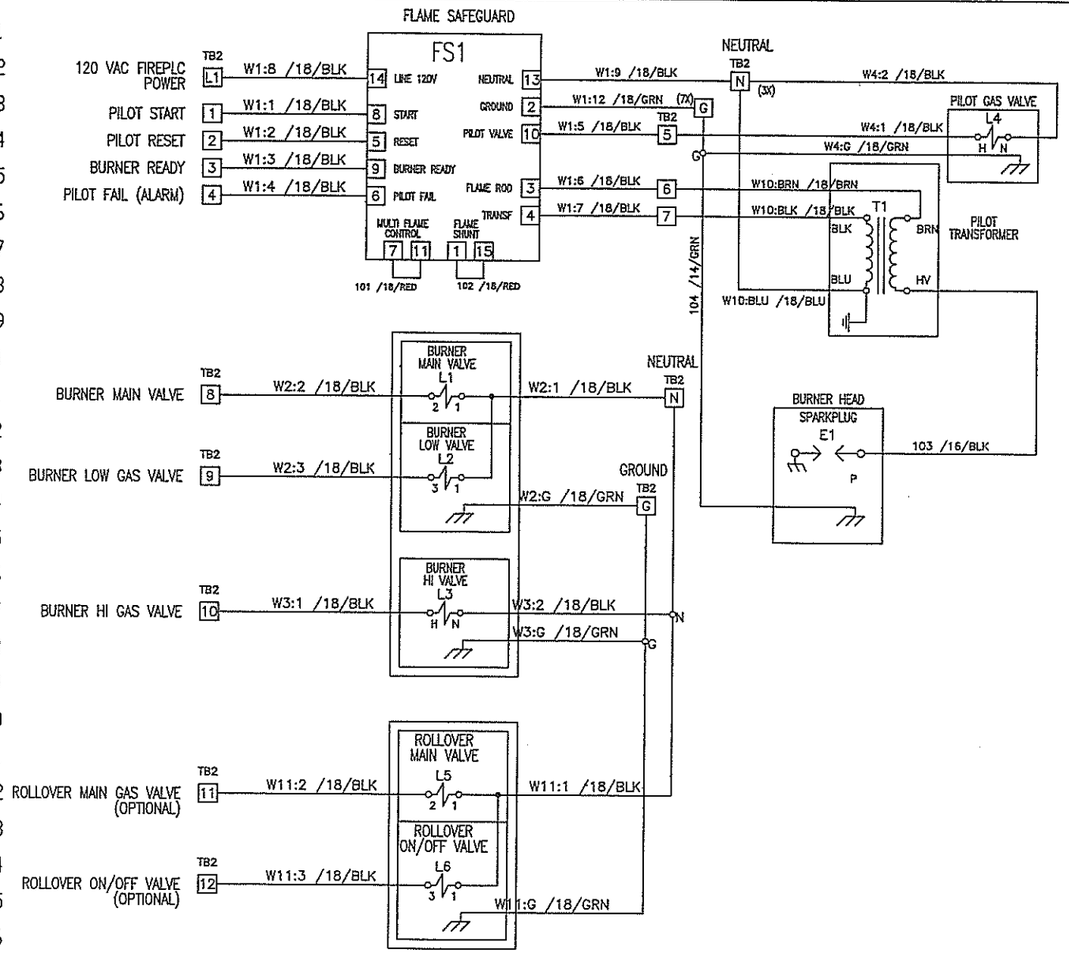
DWN	B AQUINO	05/08/06
CHKD	D MUSTO	05/08/06
APVD	LMI	05/08/06

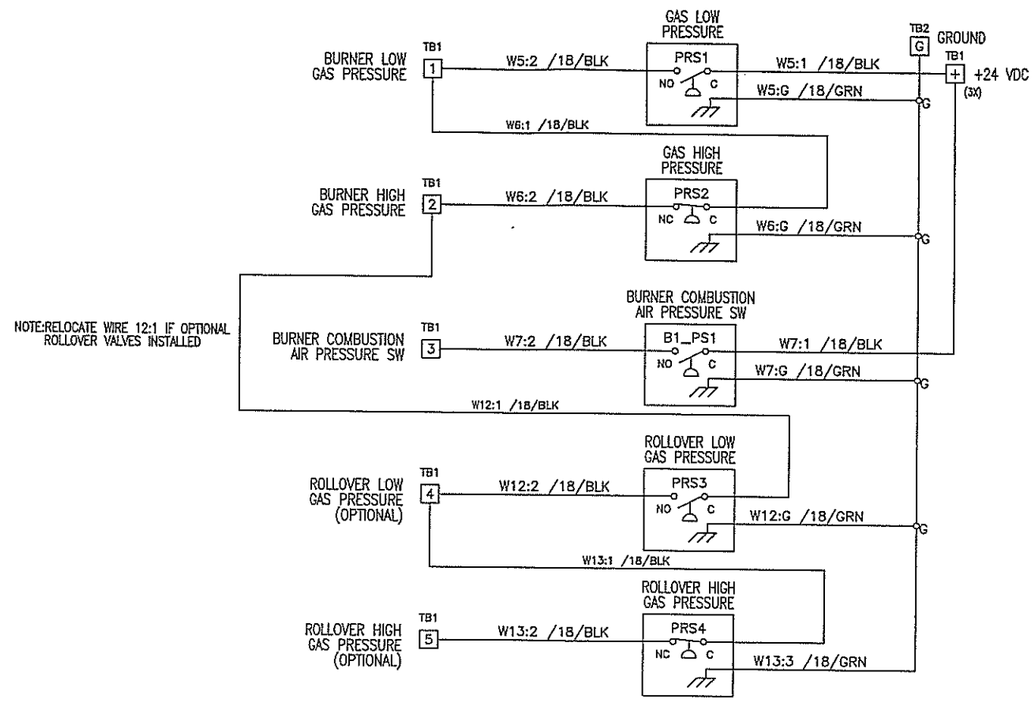
Kidde Fire Trainers

TITLE: IO RACK ELECTRONIC ENCL ASSY
WIRING DIAGRAMS
COMB GAS DETECTION

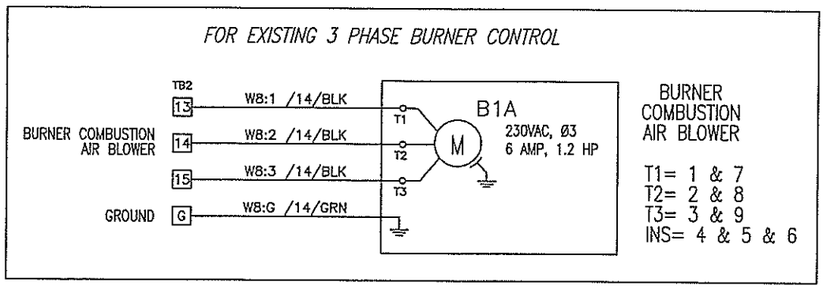
SIZE	CAGE CODE	DWG NO
D	06003	WD4A1-007-134

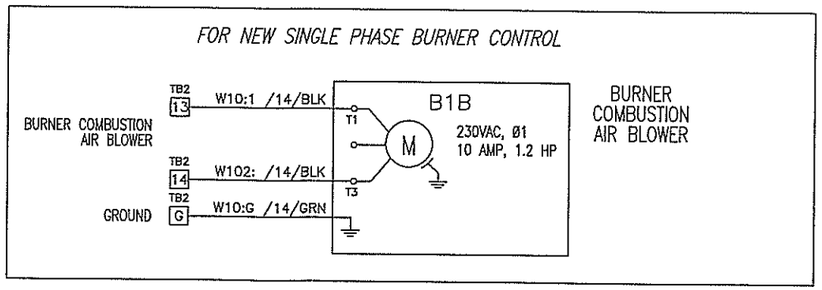
SCALE NA | CAD FILE: WD4A7134E | SHEET 4 OF 4

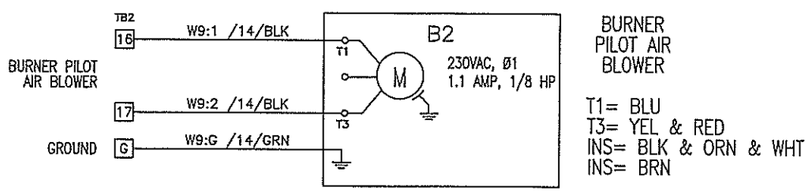




NOTE:SET GAS LOW PRESSURE SWITCHES TO 20"
SET HIGH PRESSURE SWITCHES TO 140"







REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ADDING COMBUSTION BLOWER REF	01/16/08	SW/SH
B	RE-WIRE PRESSURE SWITCHES IN SERIES	11/13/08	KR/EC

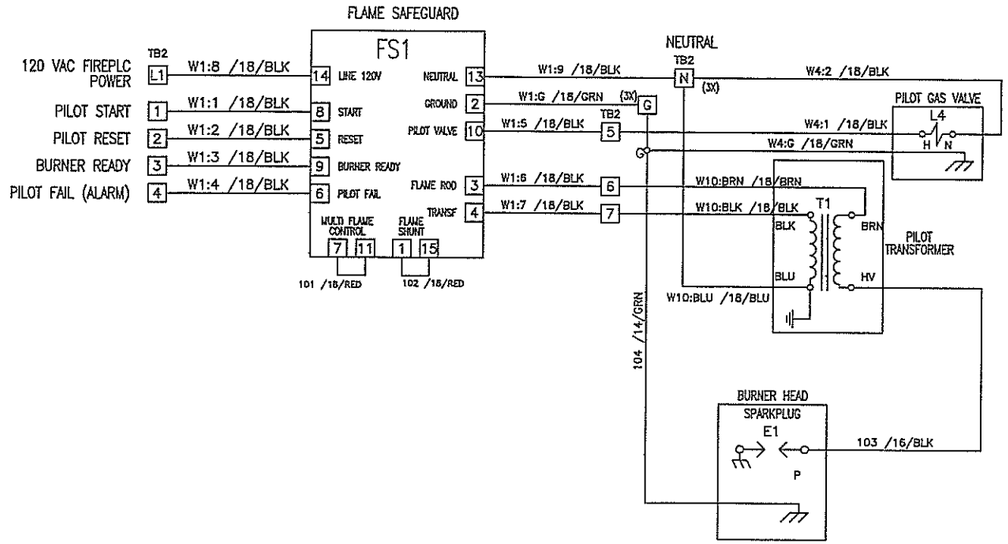
DWG NO WD4A1-006-364 SH 1

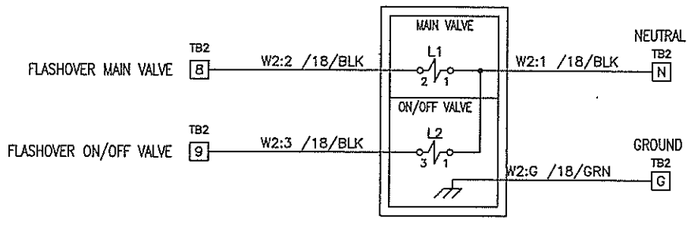
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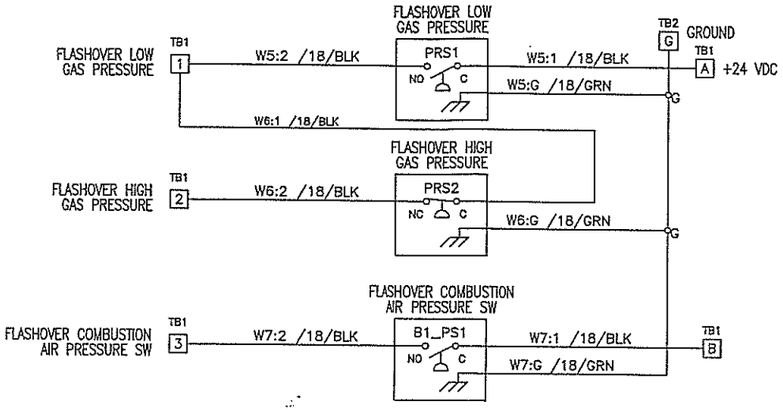
PROJECT	ICS_BURNER		INSTALLATION	WIRING_DIAG	Kidde Fire Trainers		
			LOCATION/S	ENCLOSURE	TITLE	BURNER CONTROL ASSY	
						WIRING DIAGRAM	
4A1-006-364	ICS-RF & ML	DWN	S HUYNH	05/10/04	SIZE D	CAGE CODE 06003	DWG NO WD4A1-006-364
NEXT ASSY	USED ON	CHKD	S HUYNH	06/16/04			
APPLICATION		APVD	D BRADY	06/16/04	SCALE NA	CAD FILE: WD4A6364B	SHEET 1 OF 1

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ADDING COMBUSTION BLOWER REF	01/06/08	SW/SH
B	RE-WIRE PRESSURE SW IN SERIES	11/13/08	KR/EC

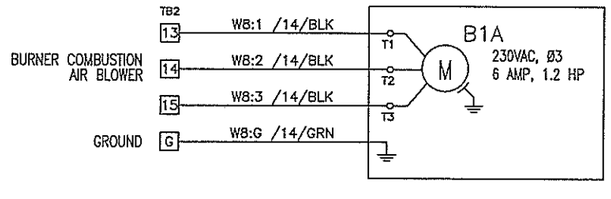






NOTE: SET LOW PRESURE SWITCH TO 20"
SET HIGH PRESSURE SWITCH 10 140"

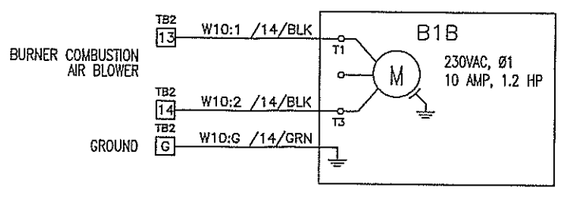
FOR EXISTING 3 PHASE BURNER CONTROL



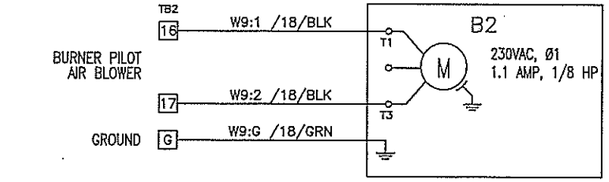
BURNER COMBUSTION AIR BLOWER

T1= 1 & 7
T2= 2 & 8
T3= 3 & 9
INS= 4 & 5 & 6

FOR NEW SINGLE PHASE BURNER CONTROL



BURNER COMBUSTION AIR BLOWER



BURNER PILOT AIR BLOWER

T1= BLU
T3= YEL & RED
INS= BLK & ORN & WHT
INS= BRN

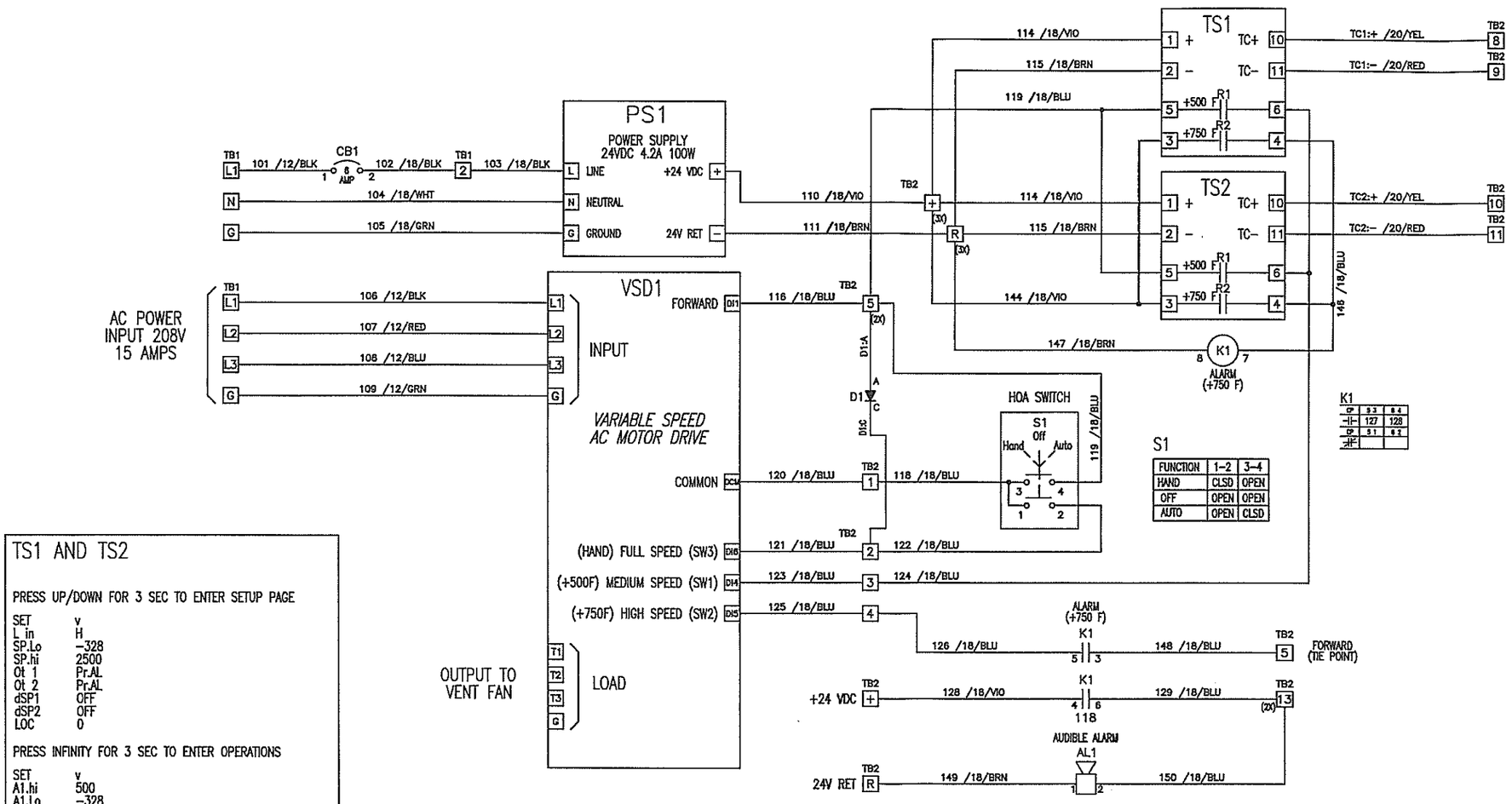
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PROJECT	ICS_FLASHOVER	INSTALLATION	WIRING_DIAG	Kidde Fire Trainers	
		LOCATION/S	ENCLOSURE	TITLE	FLASHOVER VALVE CAB ASSY
4A1-006-312	ICS-RF & ML	DWN S HUYNH	05/10/04		WIRING DIAGRAM
NEXT ASSY	USED ON	CHKD S HUYNH	06/16/04	SIZE D / CAGE CODE 06003 / DWG NO WD4A1-006-312	
APPLICATION		APVD D BRADY	06/16/04	SCALE NA / CAD FILE: WD4A5312B	SHEET 1 OF 1

DWG NO WD4A1-006-525 SH 1

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	EC030415	03/07/05	DB/BA



TS1 AND TS2

PRESS UP/DOWN FOR 3 SEC TO ENTER SETUP PAGE

SET	v
L in	H
SP.Lo	-328
SP.hi	2500
Ot 1	Pr.AL
Ot 2	Pr.AL
dSP1	OFF
dSP2	OFF
LOC	0

PRESS INFINITY FOR 3 SEC TO ENTER OPERATIONS

SET	v
A1.hi	500
A1.Lo	-328
A2.hi	750
A2.Lo	-328

ALL OTHER SETTINGS REMAIN AT DEFAULT

K1

	3	4
	127	128
	5	6

VSD1 PARAMETERS SETTING

P 0.00	Voltage	208
P 0.01	Amps	4
P 0.02	Frequency	60
P 0.03	Motor RPM	1750
P 0.04	Max RPM	1750
P 1.00	Stop Metho	00
P 1.01	Accel Time	10
P 1.02	Decel Time	20
P 3.00	Oper Comm	02
P 3.01	Input (DI1)	00
P 3.02	Input (DI3)	00
P 3.03	Input (DI4) (Low)	03
P 3.04	Input (DI5) (Med)	04
P 3.05	Input (DI6) (Hi)	05
P 3.11	Output 1	00
P 3.12	Output 2	06
P 4.00	Com Source	01
P 5.01	Preset 1 (Med)	45
P 5.02	Preset 2 (Hi)	60
P 5.03	Preset 3 (Hi)	60
P 5.04	Preset 4 (Hi)	60
P 5.05	Preset 5 (Hi)	60
P 5.06	Preset 6 (Hi)	60
P 5.07	Preset 7 (Hi)	60
P 6.00	Overload	00
P 6.01	Au. Restart	05
P 6.02	Pwr Loss	01
P 8.00	Display	00

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PROJECT	RF_VENTCONTL
INSTALLATION	WIRING_DIAG
LOCATION/S	ENCLOSURE
4A1-006-525	VARIOUS
NEXT ASSY	USED ON
APPLICATION	

DWN	S HUYNH	12/08/04
CHKD	B AQUINO	12/09/04
APVD	D BRADY	12/09/04

Kidde Fire Trainers

TITLE: ELECTRICAL CONTROL CABINET ASSY. RF-VF WIRING DIAGRAM

SIZE	CAGE CODE	DWG NO
D	06003	WD4A1-006-525

SCALE NA | CAD FILE: WD4A525A | SHEET 1 OF 1

INTEGRATED SOUND EFFECTS SYSTEM

OPERATION & MAINTENANCE MANUAL

KFT Fire Trainer, LLC
(Formerly known as Kidde Fire Trainers, Inc.)
5 Pearl Court, Unit D
Allendale, NJ 07401 USA
Tel: 1-800-899-1701 EXT. 502

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DOCUMENT NO.: 3925
DATE: 05-13-10 / 05-24-2016 / 05-29-2026
Revision: New existing manual, Revision B cover page update

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Sound System Wiring Diagram

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1.0 INTRODUCTION

1.1 GENERAL

This manual provides all of the necessary information to operate the:

INTEGRATED SOUND EFFECTS SYSTEM

MANUFACTURED BY:

Kidde Fire Trainers Inc.

If you have any questions regarding the operation or maintenance of this system that are not answered in this manual contact the Kidde Fire Trainers Customer Support Group at:

KIDDE FIRE TRAINERS INC.

17 PHILIPS PARKWAY
MONTVALE, NJ 07645
USA
TEL: (201)300-8100
WEB SITE: WWW.KIDDEFT.COM
E-MAIL: INFO@KIDDE.COM

To maximize the effectiveness and use the full potential of this system, a thorough understanding of the system and controls is necessary. This section describes the layout of the manual and the Safety Guidelines for the system. Read and understand the Safety Guidelines and system controls before operating the Integrated Sound Effects System.

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1.2 How To Use This Manual

The Operation Manual for the **Integrated Sound Effects System** is presented in the following order:

Section 1	**Introduction**	This section provides a brief functional description how to use the manual and relevant safety guidelines and instructions.
Section 2	**System Description**	This section provides an equipment overview and a functional description of the major equipment components that are part of the system.
Section 3	**Operation**	Operating instructions are given in this section.
Section 4	**Parts Lists**	This section provides figures and parts list tables for each of the major assemblies.

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1.3 SAFETY

It is of the utmost importance that operators read, understand, and follow the safety precautions and guidelines detailed in the following chapters. Failure to follow these safety guidelines could place training area personnel at undue risk of personal injury or death.

1.3.1 SAFETY SYMBOLS

Various symbols are used throughout the manual to alert the reader to a potentially dangerous situation.

WARNING

THE WARNING SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

CAUTION

THE CAUTION SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT MAY CAUSE INJURY OR DAMAGE TO ASSOCIATED EQUIPMENT.

1.3.2 TRAINER PERSONNEL SAFETY RESPONSIBILITIES

Trainer Personnel (instructors, safety officers, system operators and maintenance personnel) are ultimately responsible for the safety of all personnel entering the training area and compliance to all applicable NFPA, OSHA, national, state and local codes and standards. The safety responsibilities of Trainer Personnel are summarized below:

The safety responsibilities of Trainer Personnel are summarized below:

1. Read and fully understand this manual prior to operation.
2. Ensure that the system is operated and maintained in accordance with the manual.
3. Discontinue system use if potentially hazardous conditions or anomalies occur.

1.3.4 SYSTEM SAFETY WARNING

HIGH VOLTAGES capable of causing serious injury or death are used in the system. Unless required for voltage measurement, all power should be turned off when servicing is being performed. When power must be on, use extreme care when working near exposed conductors.

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2.0 SYSTEM DESCRIPTION

2.1 INTRODUCTION

This section provides a description of the Integrated Sound Effects System and its major components along with a description of the operator controls and indicators for each major component.

2.2 SYSTEM DESCRIPTION

The Integrated Sound Effects System enhances the realism of the structural **FIRE TRAINER® T-500** training system. The Integrated Sound Effects System produces audio effects that firefighters and emergency rescue personnel would experience during actual live fire situations.

There is a Sound System enclosure and Operator Interface Panel enclosure used for each **FIRE TRAINER® T-500** burn room. Two speakers are located in each fire training compartment, one for ambient sounds and one for effects sounds.

The fireplace I/O enclosure (PLC) interfaces with the corresponding Sound System enclosure.

2.3 EQUIPMENT DESCRIPTION

2.3.1 SOUND SYSTEM ENCLOSURE

The Sound System enclosure houses the receivers and amplifier used to deliver the sound effects to the burn room speakers.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

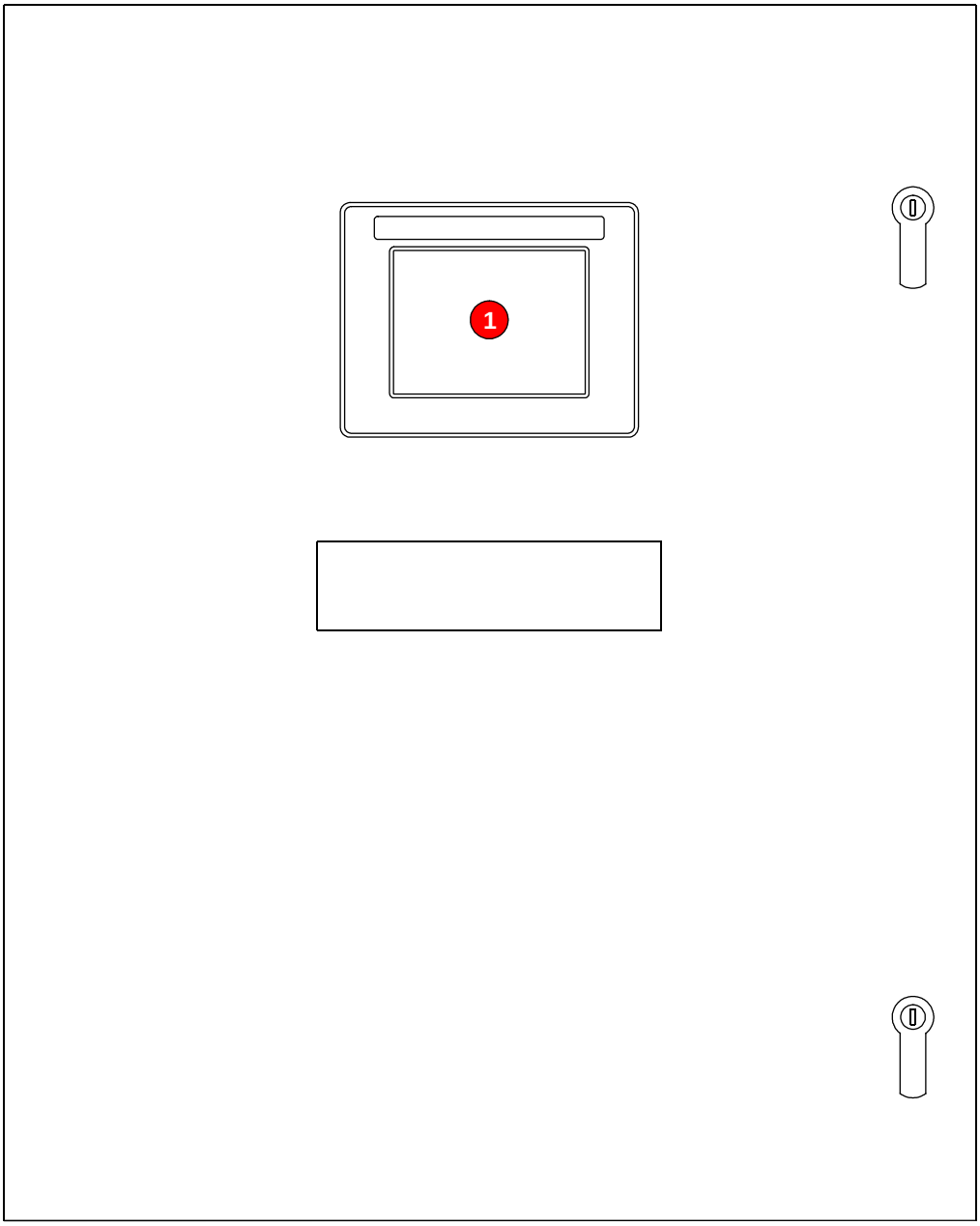


Figure 2-1. Sound Effects Control Enclosure

Table 2-1. Sound Effects Control Enclosure -Controls and Indicator Descriptions

FIG. 2-1 ITEM NO.	CONTROL	INDICATOR	NOMENCLATURE	DESCRIPTION
1	X	X	Operator Interface Panel	Provides the operator a touch screen graphical interface to control and monitor the Integrated Sound Effect System.

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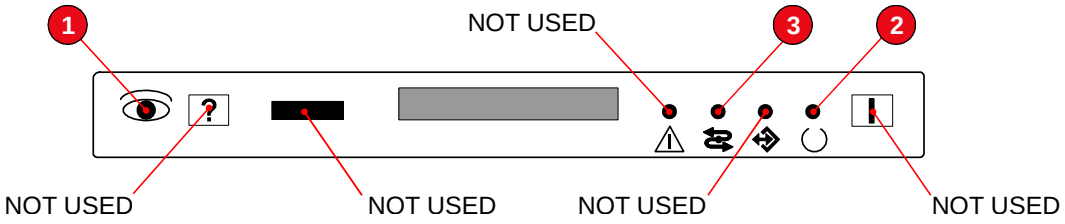
2-2 System Description

CAUTION

BE CAREFUL NOT TO DAMAGE THE TOUCH SCREEN PANEL. NEVER PUSH DOWN ON THE TOUCH SCREEN WITH FORCE, AND NEVER PRESS IT USING A HARD OR POINTED OBJECT.

2.3.1.1 TOUCH SCREEN DISPLAYS

INITIALIZING SCREEN



LED	Color	Status	Meaning
	red	OFF	No Touch Screen Hardware Problems Detected
		BLINK	Touch Screen Battery Low
		ON	Touch Screen Hardware Fault
	green	OFF	No Touch Screen Activation
		ON	During any Touch Screen Activation
	green	OFF	Touch Screen Hardware Fault (Power Cable)
		ON	Touch Screen in Operation
	green	BLINK	Touch Screen Communication Error (Communication Cable)
		ON	Touch Screen Communication OK

INITIALIZING SCREEN

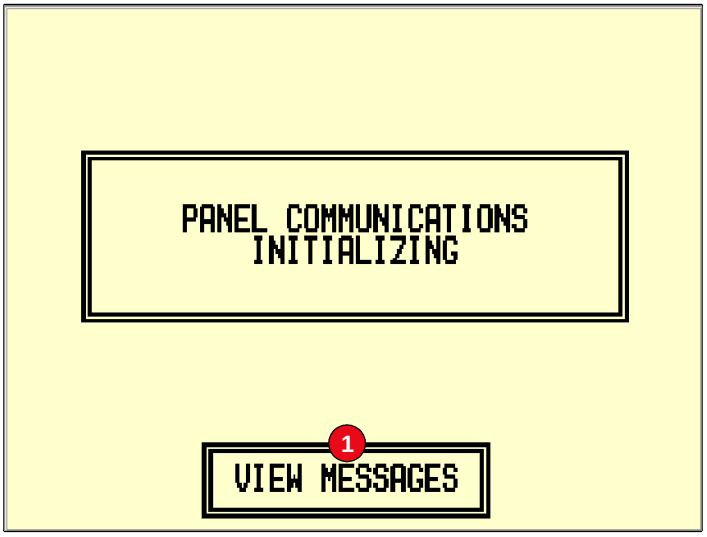


Figure 2-3. Initializing Screen

Table 2-2. Logon Screen - Controls and Indicators Descriptions

FIG. 2-3 ITEM NO.	CONTROL	INDICATOR	NOMENCLATURE	DESCRIPTION
-		X	**PANEL COMMUNICATIONS INITIALIZING**	This screen appears when power is applied to the **T-4000 M PLUS**. Once Panel Communications Initialization is completed (about 2-3 seconds), the Logon Screen should appear (Figure 2-4) If the Logon Screen does not appear touch the center of the Panel Communications Initialization Screen. If panel communications has failed, the Panel Communications Failure screen will appear. **During Operation -** If the Panel Communications Initialization Screen appears during operation touch the center of the Screen. If panel communications has failed, the Panel Communications Failure screen will appear.
1	X	-	**VIEW MESSAGES**	When touched, displays message screen. Refer to Message Screen - Controls and Indicators Descriptions.

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LOGON SCREEN

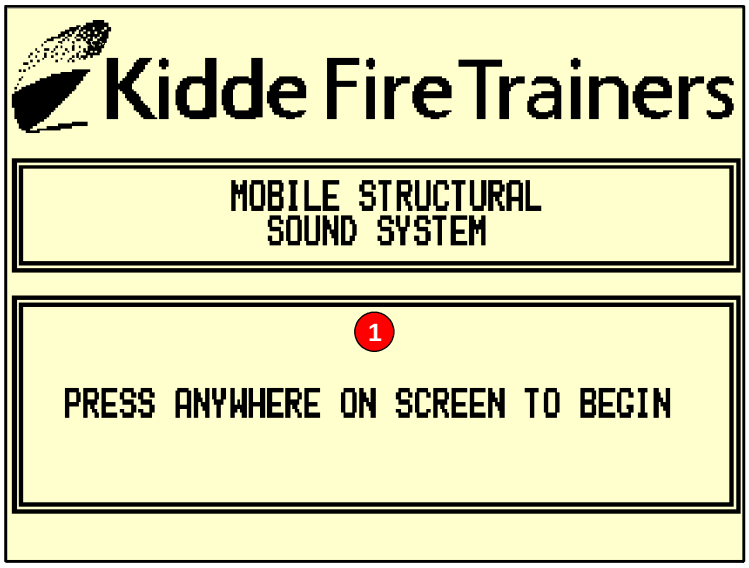


Figure 2-4. Logon Screen

Table 2-3. Logon Screen - Controls and Indicators Descriptions

FIG. 2-4 ITEM NO.	CONTROL	INDICATOR	NOMENCLATURE	DESCRIPTION
1	X	-	**Press Anywhere on Screen to Begin**	This screen appears when power is applied to the **MOBILE FIRETRAINER® T-4000 M PLUS**. Touch to access Sound Effects Screen.

SOUND EFFECT SCREEN

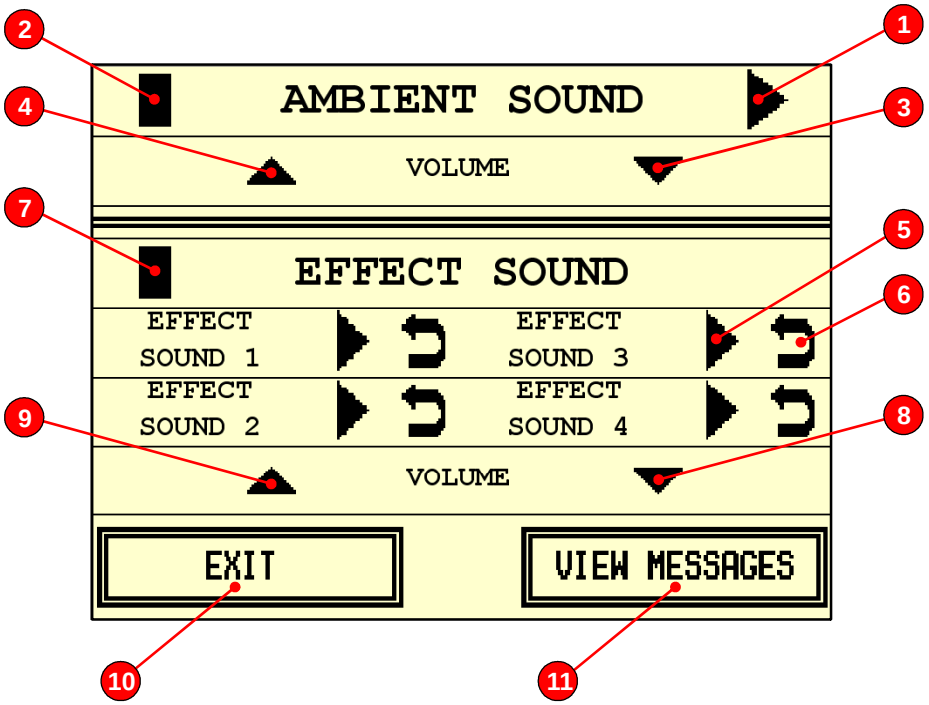


Figure 2-5. Sound Effects Screen (Typical)

Table 2-4. Sound Effects Screen - Controls and Indicators Descriptions

FIG. 2-5 ITEM NO.	CONTROL	INDICATOR	NOMENCLATURE	DESCRIPTION
1	X	X	**Ambient Sound Play**	Touch to set ambient sound to on. Ambient sound will play automatically during fireplace/flashover operation. Button indicates it has been selected.
2	X	-	**Ambient Sound Stop**	Touch to turn off ambient sound.
3	X	X	**Lower Ambient Sound Volume**	Touch to lower the ambient sound volume. Button indicates it has been selected.
4	X	X	**Raise Ambient Sound Volume**	Touch to raise the ambient sound volume. Button indicates it has been selected.
5	X	X	**Effect Sound Play**	Touch the play button of the desired effect sound. When selected effect sound plays once and then stops. Button indicates it has been selected.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Table 2-4. Sound Effects Screen - Controls and Indicators Descriptions

FIG. 2-5 ITEM NO.	CONTROL	INDICATOR	NOMENCLATURE	DESCRIPTION
6	**X**	**X**	**Effect Sound Loop**	Touch the loop button of the desired effect sound. When selected effect sound plays continuously. When selected button indicates it has been selected.
7	**X**	**-**	**Effect Sound Stop**	Touch to turn off effect sound.
8	**X**	**X**	**Lower Effect Sound Volume**	Touch to lower the effect sound volume. When selected button indicates it has been selected.
9	**X**	**X**	**Raise Effect Sound Volume**	Touch to raise the effect sound volume. When selected button indicates it has been selected.
10	**X**	**-**	**Exit**	Touch to return to Logon Screen.
11	**X**	**-**	**View Messages**	When touched, displays message screen. Refer to Message Screen - Controls and Indicators Descriptions.

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System Message Screen

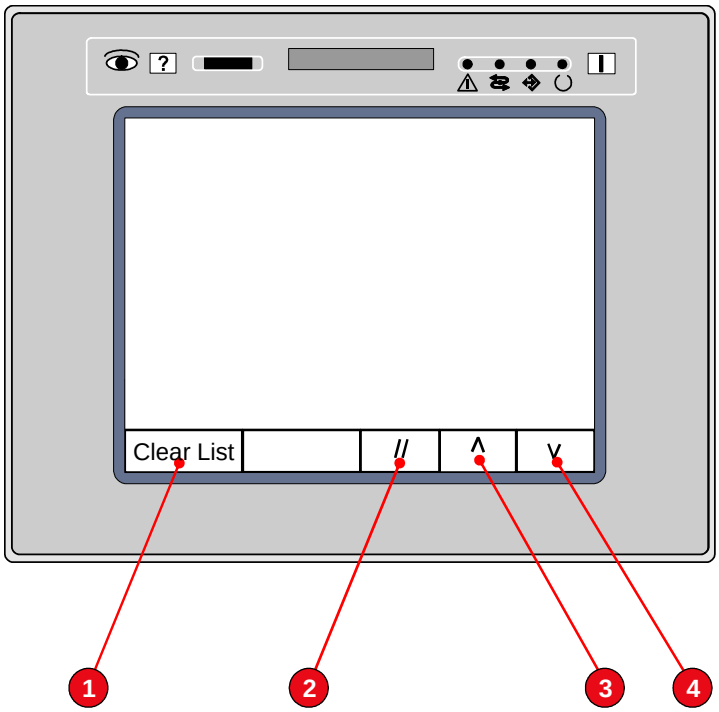


Figure 2-6. Message Screen

Table 2-5. Message Screen - Controls and Indicators Descriptions

Fig. 2-6 Item No.	Control	Indicator	Nomenclature	Description
1	X	-	Clear List	When touched clears all messages from screen.
2	X	-	//	When touched, returns to previous screen.
3	X	-	^	Used to scroll up.
4	X	-	v	Used to scroll down.

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EMERGENCY SHUTDOWN SCREEN

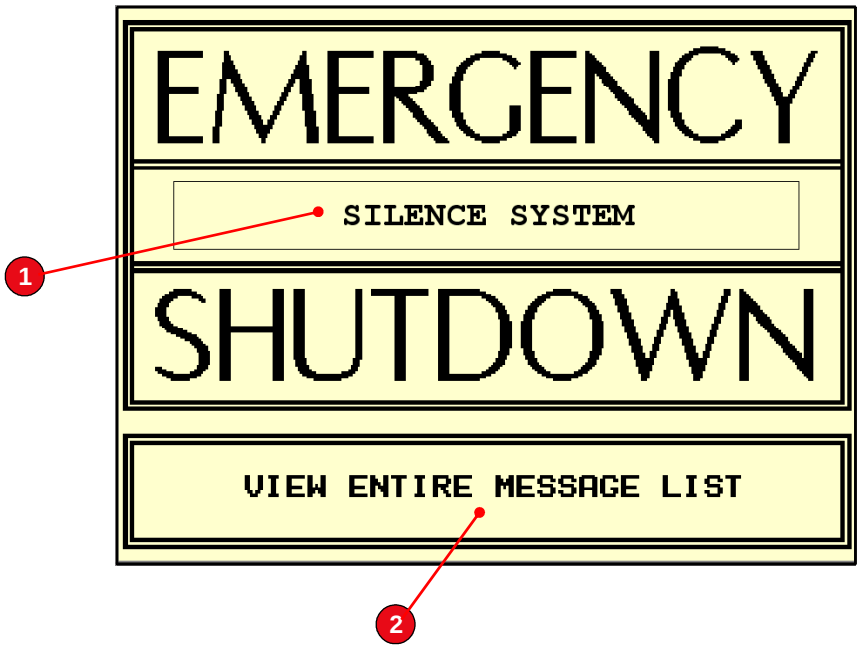


Figure 2-7. Emergency Shutdown Screen

Table 2-6. Emergency Shutdown Screen - Controls and Indicators Descriptions

FIG. 2-7 ITEM NO.	CONTROL	INDICATOR	NOMENCLATURE	DESCRIPTION
1	X	-	**Silence System**	When a **MOBILE FIRETRAINER® T-4000 M PLUS**. emergency shutdown has been initiated, an audible alarm is played over the Integrated Sound Effect System speakers. Touch to turn off the alarm.
2	X	-	**View Entire Messages List**	When touched, displays message screen. Refer to Message Screen - Controls and Indicators Descriptions.

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3.0 OPERATION

3.1 INTRODUCTION

This section provides startup, shutdown and operating procedures for the proper and safe use of the Integrated Sound Effects System.

3.2 STARTUP

The Integrated Sound Effects System is enabled when power is applied to the structural **FIRE TRAINER® T-500**. Once power is applied to the structural **FIRE TRAINER® T-500** follow the steps below to operate the Integrated Sound Effects System.

1. Verify that the Sound Effects touch screen illuminates and when Panel Communications Initializing is complete the Logon Screen comes up (Refer to Figure 3-1

⚠ CAUTION

BE CAREFUL NOT TO DAMAGE THE TOUCH SCREEN PANEL. NEVER PUSH DOWN ON THE TOUCH SCREEN WITH FORCE, AND NEVER PRESS IT USING A HARD OR POINTED OBJECT.

2. At Logon screen display, touch “PRESS ANYWHERE ON SCREEN TO BEGIN” button. A Sound Effects screen is displayed as shown in Figure 3-2.

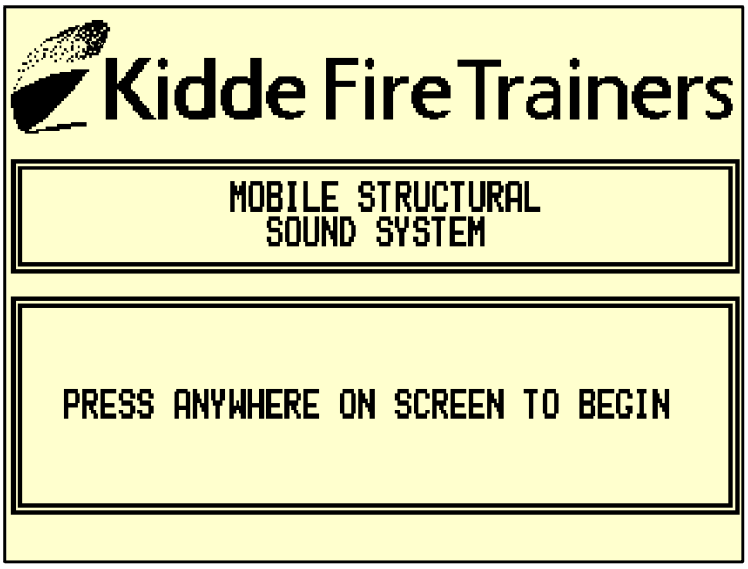


Figure 3-1. Logon Screen

3.3 OPERATION

3.3.1 SOUND EFFECT SCREEN

AMBIENT SOUND

Refer to Figure 3-2

1. Touch the ambient sound play button (1) to set the ambient sound to on. The ambient sound will play during fireplace and flashover operation.
2. Touch the volume up (4) or down (3) button to adjust ambient sound volume.
3. Touch the ambient sound stop button (2) to disable sound during fireplace operation.

EFFECT SOUND

1. Use the Effect Sound speaker selection switch and select the front speaker only, front & rear speakers or the rear speaker only to play the Effect Sounds. Refer to Section 2, Figure 2-2, Table 2-1 for location and description.

Refer to Figure 3-2

2. Choose the desired effect sound by touching the corresponding play button (5). The effect sound that was selected will play once and stop. For continuous play touch the corresponding effects sound loop button (6).
3. Touch the volume up (9) or down (8) button to adjust effects sound volume.
4. Touch the effects sound stop button (7) to turn off the effects sound.

GENERAL

Refer to Figure 3-2

1. Touch the EXIT button (10) to return to the Logon screen.
2. Touch the VIEW MESSAGES button (11) to access the message screen.

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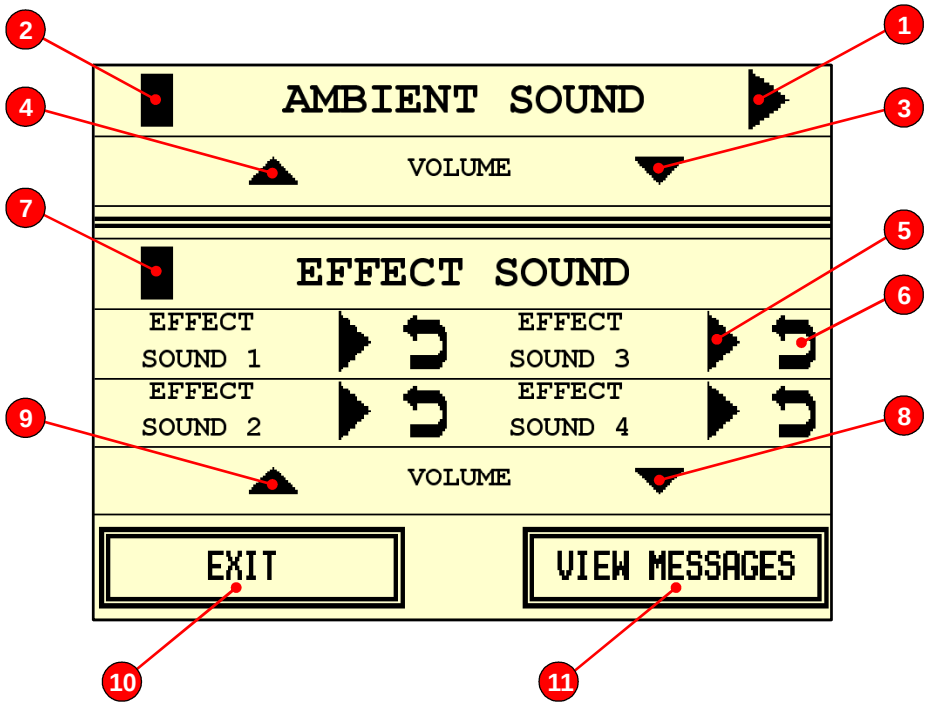


Figure 3-2. Sound Effects Screen (Typical)

3.3.2 EMERGENCY SHUTDOWN SCREEN

When a **FIRE TRAINER® T-500**. emergency shutdown has been initiated, The Integrated Sound Effects System generates an audible alarm sound that is played over all training area Integrated Sound Effects System speakers. The alarm sound can be turned off by touching the “Silence System” button (1) on the Integrated Sound Effects System Emergency Shutdown screen (Figure 3-3). The alarm sound ceases when the emergency stop situation is cleared.

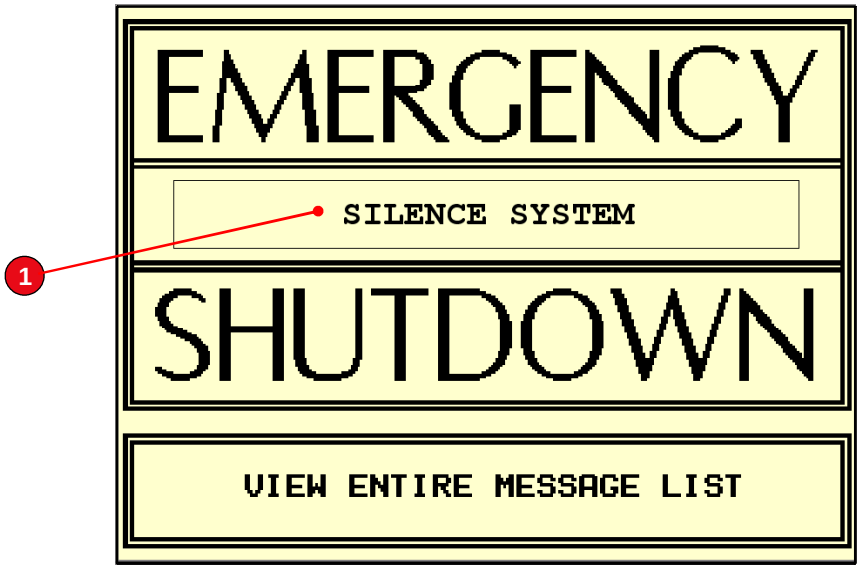


Figure 3-3. Emergency Shutdown Screen

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4.0 PARTS LISTS

4.1 INTRODUCTION

This section provides a figure and parts list table for the Integrated Sound Effects system.

4.2 SOUND SYSTEM ENCLOSURE

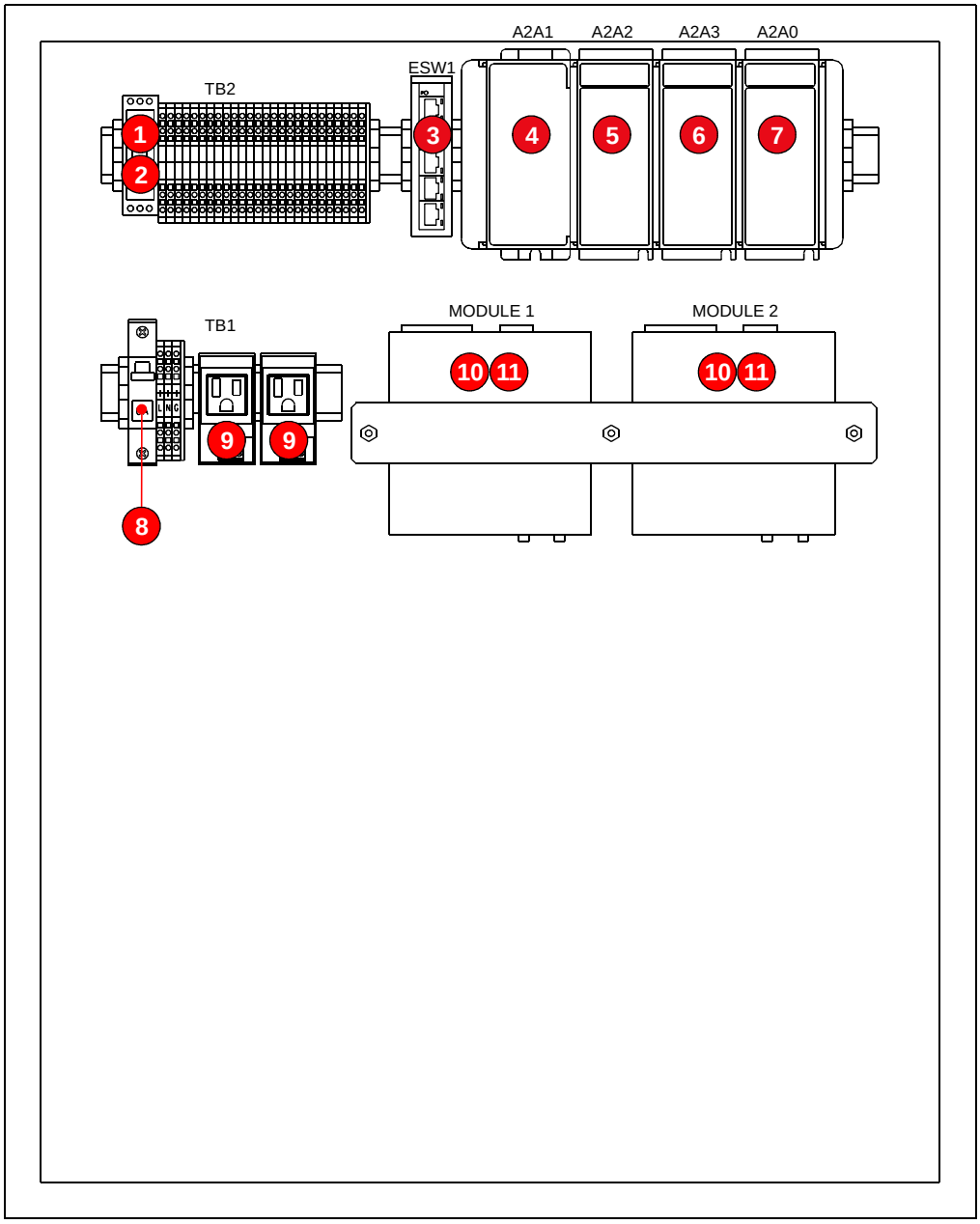


Figure 4-1. Sound System Enclosure

Table 4-1. Sound System Enclosure Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 4-1	**Sound System Assy.**	**4A1-007-416 –003 Rev. B**	**06003**
1	Relay, Plug-In, 2PDT, 24VDC, W/LED	C7-A20FX/024VDC	61526
2	Socket, Relay, 8-Pin, 2PDT, 10A/380V	S7-M	61526
3	Switch, Ethernet, Rail, Spider 5TX	943824-001	3V1N9
4	PLC Processor, 750K, Ethernet, CF Card	1769-L32E	01121
5	Module, 16-Input, 24VDC	1769-IQ16	01121
6	Module, Relay OTPT, 16PT Compact I/O	1769-OW16	01121
7	Power Supply, 24VDC IN @ 5VDC OUT, 2A	1769-PB2	01121
8	Circuit Breaker, 1-pole, 6A	1GU6	0DU12
9	Module, Outlet, Single, 120VAC, 15A	9915480000	61526
10	Sound Module W/Amplifier	QV531QV5M14F8	N/A
11	Flash Card, Compact, 64MB (T2 Sound)	CF/64F	N/A

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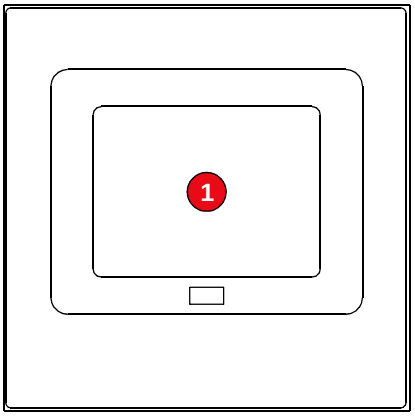


Figure 4-2. Operator Interface Panel Enclosure

Table 4-2. Operator Interface Panel Enclosure Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
Figure 4-2	**Operator Interface Panel Enclosure**	**4A1-007-416 –003 Rev. B**	**06003**
1	Graphics, HMI, MONO, W/TCHSCRN, 5.6”	Etop05-0045	1sp80

4.3 SPEAKERS

Table 4-3. Speakers Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
-	Speakers, 2-Way Wthrprf, 6.5”, 5-150W	C1-AW650BLK	01BK1

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DWG NO WD4A1-007-416 | SH 1

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	UPDATED AS BUILT	01/28/08	DM/SH

120V AC
10 AMP
60Hz

TB1 L 139 /18/BLK → CB1 (6 AMP, 1, 2) → 142 /18/BLK → R1 L, N, G
TB1 N 140 /18/WHT → R1 N
TB1 G 141 /18/GRN → R1 G → R2 G

R1 L N G
R2 L N G

**NOTE:
RECEPTACLES R1 AND R2 FOR
SOUND MODULE POWER SUPPLIES

NOTE:INPUTS ARE AT 5 VDC LEVELS
SOME FLASH MEMORY CARDS
MAY BE INCOMPATABLE WITH THESE UNITS

P1 *SOUND MODULE 1*

MODULE 0 SOUND 0 OUTPUT	A2A3 0	113 /24/RED	1	MODULE 0 SOUND 0 INPUT
MODULE 0 SOUND 1 OUTPUT	A2A3 1	114 /24/GRY	2	MODULE 0 SOUND 1 INPUT
MODULE 0 SOUND 2 OUTPUT	A2A3 2	115 /24/GRY	3	MODULE 0 SOUND 2 INPUT
MODULE 0 SOUND 3 OUTPUT	A2A3 3	116 /24/GRY	4	MODULE 0 SOUND 3 INPUT
MODULE 0 SOUND 4 OUTPUT	A2A3 4	117 /24/GRY	5	MODULE 0 SOUND 4 INPUT
MODULE 0 STOP SOUND	A2A3 5	118 /24/GRY	14	MODULE 0 STOP SOUND INPUT
MODULE 0 VOL UP OUTPUT	A2A3 6	119 /24/GRY	12	MODULE 0 VOL UP INPUT
MODULE 0 VOL DOWN OUTPUT	A2A3 7	120 /24/GRY	13	MODULE 0 VOL DOWN INPUT
VDC1 INPUT	A2A3 V1	121 /24/GRY	9	MODULE 0 0V SIGNAL

AMBIENT SOUND

RIBBON CABLE

P2 *SOUND MODULE 2*

MODULE 1 SOUND 0 OUTPUT	A2A3 8	122 /24/RED	1	MODULE 1 SOUND 0 INPUT
MODULE 0 SOUND 1 OUTPUT	A2A3 9	123 /24/GRY	2	MODULE 1 SOUND 1 INPUT
MODULE 1 SOUND 2 OUTPUT	A2A3 10	124 /24/GRY	3	MODULE 1 SOUND 2 INPUT
MODULE 1 SOUND 3 OUTPUT	A2A3 11	125 /24/GRY	4	MODULE 1 SOUND 3 INPUT
MODULE 1 SOUND 4 OUTPUT	A2A3 12	126 /24/GRY	5	MODULE 1 SOUND 4 INPUT
MODULE 1 STOP SOUND	A2A3 13	127 /24/GRY	14	MODULE 1 STOP SOUND INPUT
MODULE 1 VOL UP OUTPUT	A2A3 14	128 /24/GRY	12	MODULE 1 VOL UP INPUT
MODULE 1 VOL DOWN OUTPUT	A2A3 15	129 /24/GRY	13	MODULE 1 VOL DOWN INPUT
VDC2 INPUT	A2A3 V2	130 /24/GRY	9	MODULE 1 0V SIGNAL

EFFECTS SOUNDS

SPEAKER
R+ 131 /18/RED → TB2 10
R- 132 /18/BLK → TB2 11
MODULE 1 RIGH SPEAKER
L+ 133 /18/RED → TB2 12
L- 134 /18/BLK → TB2 13
MODULE 1 LEFT SPEAKER

MODE SWITCH
SS1
Both
Stair / Rear
1A 2A 1B 2B

NOTE:WHEN USED WITH A T500 SYSTEM(SEE 4A1-007-003)
REMOVE SS1 AND WIRE NUMBERS
143 AND 144
RUN WIRES 135 AND 137 DIRECTLY TO THE TB

SPEAKER
R+ 135 /18/RED → 1A → 143 /18/RED → TB2 14
R- 136 /18/BLK → 2A → TB2 15
MODULE 2 RIGH SPEAKER
L+ 137 /18/RED → 1B → 144 /18/RED → TB2 16
L- 138 /18/BLK → 2B → TB2 17
MODULE 2 LEFT SPEAKER

DS1
TOUCH SCREEN DISPLAY PANEL
P/N: 4C1-006-198
24 VDC @ 0.6 AMP
Ethernet
+24V 0V GND
24 0V G

ESW1
ETHERNET SWITCH
Port 1
Port 2
CAT5 Patch Cable
+24V 0V GND
24 0 G

A2 IO RACK ASSY (CompactLogix)

A2A1	A2A2	A2A3	A2A0
1769-L32E PLC PROCESSOR	1769-IQ16 16 POINT 24 VDC INPUT MODULE	1769-OW16 16 PT AC/DC RELAY OUTPUT	1769-PB2 POWER SUPPLY 24VDC In 2A 5 VDC Output
Ethernet	24V RET C1 / 24V RET C2		+24V DC +V C / 24V RET C / GND G

DASH -002 OPTIONAL
DS2
TOUCH SCREEN DISPLAY PANEL
P/N: 4C1-006-198
24 VDC @ 0.6 AMP
+24V 0V GND
24 0V G

+24V DC INPUT
TB2 + (x3) 101 /18/VIO
TB2 R (x3) 102 /18/BRN
TB2 G (x3) 103 /18/GRN
101 /18/VIO → ESW1 +24V
102 /18/BRN → ESW1 0V
103 /18/GRN → ESW1 GND
101 /18/VIO → A2A1 +24V DC
102 /18/BRN → A2A1 24V RET
103 /18/GRN → A2A1 GND
101 /18/VIO → DS2 +24V
102 /18/BRN → DS2 0V
103 /18/GRN → DS2 GND

SOUND TRIGGER INPUT 0	A2A2 0	104 /18/BLU	TB2 1 MAIN FLAME INPUT
SOUND TRIGGER INPUT 2	A2A2 2	107 /18/BLU → K1 (1,2,5) MAIN FLAME ENAB → 108 /18/BLU	TB2 2 FLASHOVER INPUT
SOUND TRIGGER INPUT 3	A2A2 3	109 /18/BLU → K1 (3,1,2,5) MAIN FLAME ENAB	
SOUND TRIGGER INPUT 4	A2A2 4	110 /18/BLU	TB2 3 SOUND TRIGGER INPUT 4
SOUND TRIGGER INPUT 1	A2A2 1	111 /18/BLU	TB2 4 E-STOP SOUND TRIGGER / E-STOP INPUT
E-STOP INPUT	A2A2 11	112 /18/BLU	

MAIN FLAME INPUT 105 /18/BLU → K1 (7, 8) MAIN FLAME ENAB → 106 /18/BRN → TB2 R 24V RET

CP	23	24	
	125		
CP	21	22	
	124		

SPARE TERMINALS
TB2 5 6 7 8 9

PROPRIETARY INFORMATION OF Kidde Fire Trainers
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PROJECT	ML_SOUND_SYS	INSTALLATION	WIRING_DIAG	Kidde Fire Trainers	
		LOCATION/S	ENCLOSURE	TITLE	CONTROL CONSOLE ASSY, ML SOUND&FIRE
					WIRING DIAGRAM
4A1-007-416	VARIOUS	DWN	S HUYNH 08/14/07	SIZE D	CAGE CODE 06003 / DWG NO WD4A1-007-416
NEXT ASSY	USED ON	CHKD	L INGLESE 08/14/07		
APPLICATION		APVD	D MUSTO 08/14/07	SCALE NA	CAD FILE: WD4A7416A / SHEET 1 OF 1

Kidde WM Smoke System

Overview

Your smoke system is based upon the best selling and proven ViCount unit. It is designed to produce a temperature resistant smoke to add realism to Hot Fire Trainers. The smoke is resistant to 356—392°F.

Principle of Operation

This system produces an smoke effect entrained in an inert carrier gas. Any inert gas can be used, but generally N_2 or CO_2 are recommended. Generally, for multiple systems we would recommend the use of N_2 only.

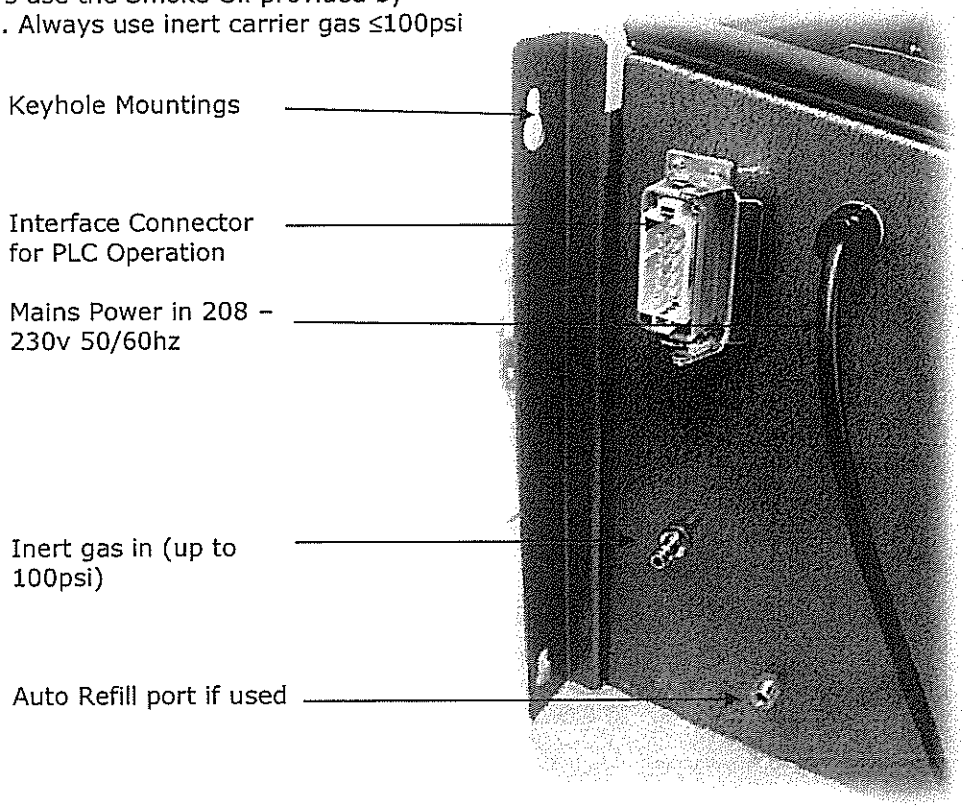
Inert gas is mixed with the Smoke Oil to form a fine mist, which is then heated within a precision, machined heat exchanger block. There the oil is vaporised, and when the oil exits the nozzle of the heat exchanger it expands, cools and condenses to form an artificial smoke.

Warning

In operation the exposed nozzle of the heat exchanger will get extremely hot and measures should be taken to guard it accordingly. It will remain hot for several hours after disconnection.

Never refill the Smoke System with oil when the system is operating. (a pressure relief slot is machined in the Filler Cap should that will release gas in the event of the tank being pressurised

Always use the Smoke Oil provided by Kidde. Always use inert carrier gas ≤100psi



Preparation

Mount and secure the unit on the wall using the keyhole mountings on the chassis. Note that the nozzle of the heat exchanger should line up with your entry point into the simulator. Ensure that the unit is mounted horizontally.

Remove the Brass Filler cap and fill the Smoke System with Smoke Oil—available through Kidde, paying attention to the sight glass on the side of the unit. Replace the Brass Filler cap, ensuring a seal is made.

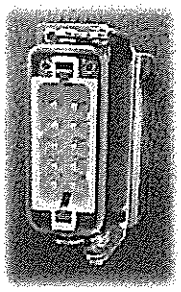


Your inert gas (regulated to give a pressure output of 0—100psi (0—6.9 bar) should now be connected to the Smoke System via the gas hose and quick connector supplied. Set the pressure at 100 psi.

Connect the Smoke System to a suitable 208 -230v, 50-60 hz supply

To produce smoke

Power up the Smoke System by applying a suitable control voltage (24vDC) from the PLC (see wiring schematic 16423 enclosed) across Pins 1 & 2 on the Interface connector



The Digital Display inside the unit will light. The Green display is the target temperature of the heat exchanger (in Celsuis), the Red display is the actual temperature of the heat exchanger.

After 3-4 minutes Relay 2 within the Smoke System will change state to indicate that "Minimum" working temperature has been achieved and pins 3 and 4 on the Interface connector will short out. It is recommended however that the PLC should be configured to ensure that a Smoke on signal is not sent to the unit for a further 1 minutes to allow the heat exchanger within the unit to reach a very stable condition.

To produce smoke, apply a suitable control voltage (24vDC) from the PLC across Pins 5 & 6 on the interface connector. The pressurising solenoid within the smoke system will open, allowing inert gas into the oil tank, and smoke will be produced, thickening after a few seconds .

To stop producing smoke:

Remove the control voltage across pins 5 & 6 on the Interface connector. The gas that is left in the oil tank will now purge through the heat exchanger so the oil tank is in an unpressurised state.

To adjust the volume of smoke

The volume of smoke that the unit will produce is determined by the pressure of the inert gas supply. To produce the maximum volume of smoke operate at 100 psi, providing that the Red Display temperature does not fall below 340C. If it falls below this temperature, reduce the operating pressure until continuous smoke is achieved.

Smaller quantities of smoke can be produced by reducing the pressure.

Auto Refill Function

If you have multiple Smoke Systems you can make use of the Auto Refill function.

If connected to a suitable pressurised oil source, your Smoke System can be configured to refill automatically at the end of the training day, or at times agreed with Kidde. An oil supply must be connect to Oil Port via 1/4" OD tube, and the PLC should be programmed only to allow the re-fill valve to operate at pre determined periods, when smoke is not being requested. A 24vDC input is required across Interface Pins 6 & 7 to energise the refilling solenoid. The solenoid will automatically close when maximum fluid level has been reached.

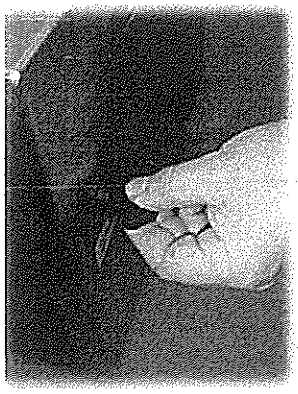
Cover Removal, Local Control and Testing

In normal circumstances access to the inside of the Smoke System is un-necessary, as all connections are made externally, and access to the oil tank for topping up is also external.

However, the unit has the ability to make smoke independently of the PLC if required, and to do this it is necessary to remove the cover. Ensure the unit is electrically isolated before removing the cover.

Removal of the Cover:

- Undo the Key locks by rotating the locks 90° in the direction indicated on the label
- Pull back on the cover (a)
- Ease the lower lip of the Cover away from the chassis (b)
- Remove the Cover from the chassis as indicated (c)





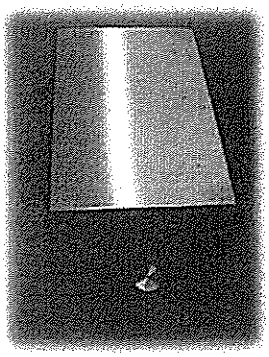


To Replace the Cover

- Position the cover as indicated
- Push down firmly on the front edge of the cover whilst pushing the cover forwards (the front edge of the cover should be in line with the angle supports (d)
- Whilst pushing forward (e) push down on the rear face (f) so that the lower lip of the cover is engaged below the chassis.
- Whilst pushing the cover forwards, push in and turn key locks to engage cover







Internal Controls

Red switch (power on)

Digital display will light provided there is power to the Smoke System when this switch is put in on (1) position. Take extreme care if operating the unit with the cover removed as electrical

Black Switch (smoke on 1, smoke off 0)

This will enable smoke to be produced providing:

- Inert gas is supplied to the Smoke System
- The Smoke System is filled with Smoke Oil



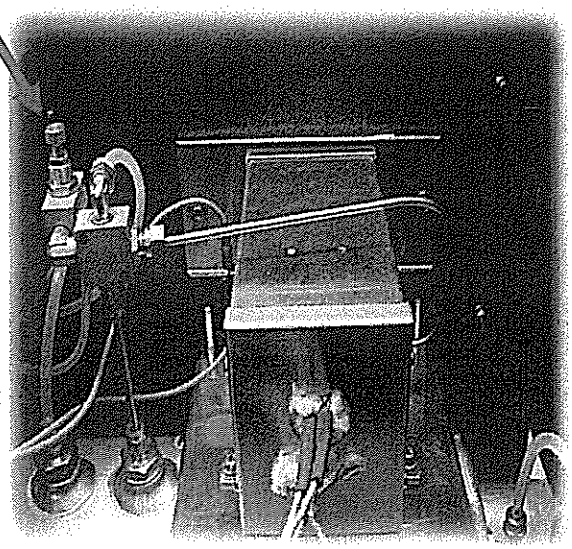
The temperature of the heat exchanger is > 340C. For optimum performance however, ensure that the unit has achieved full temperature (Green display≥ 390C)

Adjustment of Smoke Quality

The smoke quality can be adjusted via the Metering valve if required, but this should only be done be Kidde technicians.

To thicken the smoke:

- Slacken locknut
- Turn valve clockwise 180°
- Re-test, checking smoke quality from the nozzle of the heat exchanger (smoke should be dry to the touch 2' in front of the nozzle)
- If necessary continue adjustment. Tighten the locknut when smoke quality has been set as desired



To thin the smoke:

- Slacken locknut
- Turn valve anti-clockwise 180°
- Re-test, checking smoke quality from the nozzle of the heat exchanger (smoke should be dry to the touch 2' in front of the nozzle)
- If necessary continue adjustment. Tighten the locknut when smoke quality has been set as desired

To Drain away any condensation or to drain the Oil Tank

A removable plate is fitted that can be removed to allow easy cleaning of the inside of the smoke system, and access to the Drain Plug for the pressure tank. If draining the oil tank slacken off the Filler Cap to allow for faster drainage.

Mains Cable

Use 2.5mm sq 3-Core 3183Y on 110v/2.2kw Machines RS 397-233
Use 1.5mm sq 3-Core 3183Y on all other VICounts RS 379-205
Use Cable gland RS 389-606 for both size cables

Cable Sizes

1.5mm Tri-rated wire for all machines
1.5/2.5mm equipment wire for all machines

3
Float switch is CLOSED with all tank EMPTY and opens when FULL.

Use Pink High Temp. Cable for these links RS 359-179

Heat Exchanger

Mains Supply

Earth Post

TB5

Thermocouple

1100 W Heater

1100 W Heater

Valther Panel Plug 700110 704310 702810

Brown
Blue
Green
Yellow
Red
White
+24Vdc

Float Switch

Orange
Orange

Ready Relay R/L 2 (N/O)
Smoke Relay R/L 3 (N/O)
Power-ON Relay R/L 1 (N/O)

BLUE
BLUE
BLUE
BROWN
BROWN
BROWN

R/L4 (N/O) Refill Relay
Refill Solenoid

Unmetered Fan Outlet

Mains Switch

Neutral Tab 3
Neutral Tab 2
Line Tab 1

Cap 0.47uF

TB 4
(F3)
TB 5
TB 6
TB 7

1 Smoke Switch 2

(F2)

S.S.R. TD24/25 3-32Vdc

Solenoid
Earth

*TLK 39 110 - 240V range

MODULE TLK 39*

Fuse Ratings all VICount Machines

Voltage	Power	F1	F2	F3
110v	1.1Kw	12.5A	6.3A	2A
110v	2.2Kw	25A	6.3A	2A
230v	1.1Kw	10A	6.3A	2A
230v	2.2Kw	12.5A	6.3A	2A

Fuse	25A	RS247-3999
Fuse	12.5A	RS209-9282
Fuse	10A	RS413-248
Fuse	6.3A	RS247-4059
Fuse	2A	RS413-204

R/L 2 = Ready contacts (0 Volts) (FINDER 4031 8240 5000) Ac Coil
R/L 1 = Power ON contacts M15M A H 001. 8. 24Vdc Coil
R/L 3 = Make Smoke contacts M15M A H 001. 8. 24Vdc Coil
R/L 4 = Auto Refill contacts M15M A H 001. 8. 24Vdc Coil
Base ZD1 Pin 11 Comm Pin 14 N/O

Note :- Side Covers Must be Earthed

Note :- R/L 4 should not be energised when smoke is being produced

Issue No.	Alteration	By	Date	Issue No.	Alteration	By	Date

Customer KIDDE	Drawn by K. Clark	dimensions in mm unless otherwise stated	Projection	Dwg No. 16423
	Date 11.3.08	ORIGINAL SCALE N/A		
Job No.	Chk'd by T.A.D.	Title WALL MOUNT SMOKE SYSTEM KIDDE SPEC., WIRING DIAGRAM		
Project No.	Date 11.3.08			

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MATERIAL SAFETY DATA SHEET

PRODUCT NAME:
Kidde Fire Trainers Type II Smoke Fluid, Part Number 1C1-002-012

MSDS No: KFTI-003

EFFECTIVE DATE: 01/14/00; Revised 9/30/04; Revised 03/29/06

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1. CHEMICAL PRODUCT AND COMPANY INFORMATION

PRODUCT NAME	CHEMICAL NAME
Kidde Fire Trainers Type II Smoke Fluid	White Mineral Oil, N. F.

PURCHASED UNDER PART NUMBERS:

1A1-002-017-001 - Kidde FT Type II Smoke Fluid, (6) ½ gallon bottles
1A1-002-017-002 - Kidde FT Type II Smoke Fluid, 5-gallon container
1A1-002-017-003 - Kidde FT Type II Smoke Fluid, 55 gallon drum

SUPPLIERS NAME

Kidde Fire Trainers, Inc.

SUPPLIER ADDRESS

17 Philips Parkway
Montvale, New Jersey 07645
Telephone: 201-300-8100 Fax: 201-300-8101

Emergency: CHEM TREC 1-800-424-9300 (U.S. and Canada)

The Emergency Telephone number is to be used only in the event of a chemical emergency involving a spill, leak, fire, exposure, accident, or other incident with this product.

PRODUCT USE

Artificial smoke for use during live fire fighting.

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2. COMPOSITION/INFORMATION ON INGREDIENTS

SUBSTANCE DESCRIPTION	PERCENT	CAS#
White Mineral Oil, N.F.	100%	8042-47-5

3. HAZARD IDENTIFICATION

APPEARANCE AND ODOR

Raw Material is a clear transparent liquid, essentially odorless.

STATEMENT OF HAZARD

May cause mild skin irritation to sensitive individuals.

FIRE AND EXPLOSION HAZARDS FROM RAW MATERIALS

Keep sealed containers of raw material away from heat and flames.

PRIMARY ROUTES OF EXPOSURE

The primary routes of exposure to this product are dermal contact and inhalation of mists or fog.

SKIN CONTACT – ACUTE

Skin contact may cause mild irritation to sensitive individuals.

EYE CONTACT – ACUTE

Eye contact may cause mild irritation to sensitive individuals.

INGESTION - ACUTE

Excessive ingestion may cause irritation of the gastrointestinal system.

This product is a fully refined white mineral oil meeting the requirements of the National Formulary for Light Mineral Oil and the requirements of the Food and Drug Administration for White Mineral Oil according to 21 CFR 172.878 and 21CFR 178.3620(a).

4. FIRST AID MEASURES

INHALATION

Not Applicable

SKIN CONTACT FIRST AID

Wash all affected areas with soap and water.

EYE CONTACT FIRST AID

Immediately flush with running water. Hold the eyelids apart during the flushing to ensure rinsing the entire surface of the eye and lids with water. Get medical attention if irritation occurs and persists.

INGESTION FIRST AID

If excessive ingestion occurs, get medical attention by calling a physician or poison control center immediately. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, keep head below hips to reduce the risk of aspiration

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NOTE TO PHYSICIAN

This product is a food additive grade mineral oil

5. FIRE FIGHTING MEASURES FOR USE ON RAW MATERIALS

EXTINGUISHING MEDIA

Use water, fog, dry chemical, foam or carbon dioxide extinguishing agents.

FIRE FIGHTING PROCEDURES

As in any fire, prevent human exposure to fire, smoke, fumes or products of combustion. Evacuate nonessential personnel from the fire area. Fire fighters should wear full face, self-contained breathing apparatus and turn-out gear. If possible, move containers from the fire area. If not leaking, keep the fire exposed containers cool with water fog or spray to prevent rupture due to excessive heat. High pressure water may spread product from broken containers increasing contamination of fire hazard.

FIRE & EXPLOSION HAZARDS OF RAW MATERIALS

Keep sealed containers of raw materials away from heat and flames.

OTHER FIRE & EXPLOSION HAZARDS

No other fire or explosion hazards of this product are known.

MATERIAL SAFETY DATA SHEET

PRODUCT NAME:
Kidde Fire Trainers Type II Smoke Fluid, Part Number 1C1-002-012

MSDS No: KFTI-003

EFFECTIVE DATE: 01/14/00; Revised 9/30/04; Revised 03/29/06

HAZARDOUS PRODUCTS/COMBUSTION

Decomposition of this product under open fire conditions can produce carbon monoxide and carbon dioxide.

NFPA HEALTH RATING – 0 **NFPA FLAMMABILITY RATING – 1**

NFPA REACTIVITY RATING – 0 **NFPA OTHER – ND**

6. ACCIDENTAL RELEASE MEASURES FOR MATERIAL CLEANUP

Isolate spill area and restrict nonessential personnel. All personnel involved in spill cleanup should follow appropriate industrial water. hygiene practices (see section 8). Stop source of spill. Dike area to prevent spill from spreading. Soak up liquid with a suitable absorbent such as clay, saw dust, or kitty litter. Sweep up absorbed material and place in a chemical waste container for disposal. CAUTION! Spill area may be slippery. Cover spill area with a slurry of powdered household detergent and water. Use stiff brush to work slurry into cracks and crevices. Allow to stand for 2-3 minutes, then flush with water.

7. HANDLING AND STORAGE

HANDLING

Containers should be located in an area where they can be rotated regularly (first in, first out) and visually inspected for dents and bulging on a weekly basis. If bulged drums are found, they should be vented in an open area by removing the two-inch bung very slowly. The two-inch bung should not be removed completely until there is no sound of pressure being released. The bung can then be removed but should be done slowly with care . Emptied container may retain product residues. Follow warnings (if any) and precautions even after container is emptied.

STORAGE

Containers should be stored in a cool, dry well ventilated area away from flammable or oxidizing materials and sources of heat or flame. Exercise due caution to prevent damage to or leakage from the container.

Prolonged storage at elevated temperatures under wet alkaline or acid conditions should be avoided to assure product integrity. Care should be taken to prevent moisture condensation in the container.

MAXIMUM STORAGE TEMPERATURE

66°C (150°F)

GENERAL COMMENTS

This product is a fully refined white mineral oil meeting the requirements of the National Formulary XVII and requirements of the Food and Drug Administration as per 21 CFR 172.878.

8. PHYSICAL AND CHEMICAL PROPERTIES

VAPOR PRESSURE (mm Hg)

Less than 0.005 at 20°C

BOILING POINT

Greater than 230°C (450°F)

SPECIFIC GRAVITY (Water =1)

Less than 1 at 15°C

SOLUBILITY IN WATER (by weight %)

Negligible at 25°C

COEFFICIENT OF OIL/WATER

N/D

MELTING POINT

Not applicable

FLASH POINT (Method used – ASTM D-92)

Greater than 188°C (370°F)

LOWER EXPLOSION LIMIT

None Established

ODOR THRESHOLD (ppm)

None Established

SOLUBILITY IN OTHER

Not Determined

POUR POINT

None Established

Ph FACTOR

N/D

UPPER EXPLOSION LIMITS

N/D

AUTO IGNITION TEMPERATURE

N/D

9. STABILITY AND REACTIVITY

STABILITY

Product is stable under normal conditions

INCOMPATIBILITIES

This product is incompatible with strong oxidizers, strong acids and strong alkalis. It hydrolyses slowly at ambient temperatures in acidic or alkaline aqueous solutions.

POLYMERIZATION

Hazardous polymerization is not expected to occur.

10. TOXICOLOGICAL INFORMATION

CARCINOGENICITY

IARC.....NO	OSHA.....NO
NTP.....NO	ACGIH.....NO

TOXICOLOGICAL – CHRONIC

No data available

TOXICOLOGICAL – ACUTE

No data available

11. DISPOSAL CONSIDERIOSNS

WASTE DISPOSAL

Materials that cannot be used or chemically reprocessed should be disposed of in accordance with all applicable regulations. Product containers designed for single use should be thoroughly emptied before disposal. NOTE! State and local regulations may be more stringent than federal.

This product, if unused, does not meet the EPA's RCRA criteria as either a listed or a characteristic hazardous waste. Generators of waste are required to evaluate their materials for compliance with RCRA and local disposal procedures and regulations.

12. TRANSPORT INFORMATION

SHIPPING INFORMATION

D.O.T.: Not Regulated

Reportable Quantity: Not Applicable

Freight Classification: 65 Petroleum Oil N.O.I.B.N.

Special Transportation Notes: None

MATERIAL SAFETY DATA SHEET

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MSDS No: KFTI-003

EFFECTIVE DATE: 01/14/00; Revised 9/30/04; Revised 03/29/06

REQUIRED LABELS

LABEL INFORMATION

This product is made in a manner to insure quality, purity and stability. Handle carefully to preserve its original quality. May contain up to 10 ppm tocopherol as stabilizer. Hazard warning label not required under OSHA Hazard Communication Standard 29 CFR 1910.1200.

SPILLS OR LEAKS

Shut off leak and dike up spills. Absorb with an inert material such as sand, earth, or vermiculite, and dispose of in compliance with Federal, State, and local regulations

13. REGULATORY INFORMATION

White mineral oil is subject to the following:

ENVIRONMENTAL LIST

RTECS – 53012

MA.LIST

NJ Right-To-Know 8042-47-5

PA.LIST

TSCA – Not Listed

FOR REFERENCE PURPOSES ONLY!

MATERIAL SAFETY DATA SHEET

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MSDS No: KFTI-003

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NOTICE

The information herein is presented in good faith and believed to be accurate as of the effective date shown above. However, no warranty, expressed or implied, is given. Regulatory requirements are subject to change and may differ from one location to another; it is the buyer's responsibility to ensure that its activities comply with federal, state or provincial and local laws.

PREPARED BY:

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AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC.

Recommended SAFETY PRACTICES

For Users and Installers of Industrial and Commercial Fans

AMCA Publication 410-96

FOREWORD

i. This publication has been prepared by the Air Movement Division of the Air Movement and Control Association International, Inc. (AMCA). The information contained in this publication has been derived from many sources. The suggestions made necessarily should be general in their meaning and cannot be applied literally to all specific situations or conditions.

ii. ***The safe installation and operation of fans is the responsibility of the system designer, installer, maintainer, and user.*** From the initial system design through the life of the equipment, safety should be a foremost consideration. Some areas which require some special attention include system design, layout and construction, fan performance specifications, foundation and installation details, storage procedures, start-up and commissioning procedures, operation, maintenance, and repair. Specific safety requirements are mandated by federal, state, and local codes. *Recommended Safety Practices for Users and Installers of Industrial and Commercial Fans* is published by AMCA for assistance. System designers, installers, maintainers, and users should consult and properly comply with all applicable codes and guidelines.

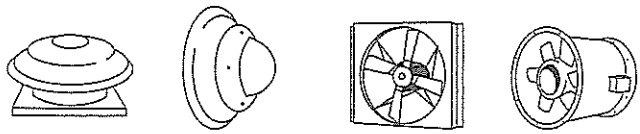
iii. The safety recommendations contained herein are intended to assist designers, installers, maintainers, or other users of air moving devices in the safe operation and use of the devices mentioned. These recommendations do not represent the only methods, procedures, or devices appropriate for the situations discussed. Caution should be used at all times when working in or around moving parts.

iv. AMCA disclaims any and all warranties, expressed or implied, regarding the products sold by the manufacturer with which this booklet has been provided. Further, AMCA recommends that competent personnel be consulted in deciding what is the preferred or recommended safety procedure in a particular instance where the guidelines contained in this booklet are unclear or in any way incomplete.

v. AMCA has offered the information within this booklet to assist in the safe operation, maintenance, and use of the products sold by members of AMCA. In so doing, AMCA does not assume any legal duties of the designer or manufacturer to instruct or warn about their product. AMCA expressly disclaims liability for any injury or damage arising out of the operation or use of the product or the guidelines contained herein.

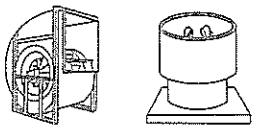
vi. These recommended safety practices were adopted by the AMCA membership on April 28, 1996.

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Power Roof Ventilator | *Wall Exhauster* | *Propeller Fan* | *Axial Fan*

1. INTRODUCTION



Centrifugal Fan | *Upblast Roof Exhauster*

1.1 Fans and other air moving devices are made in a wide variety of types, sizes, and arrangements. This publication addresses the proper use and installation of industrial and commercial fans. It is not intended to address residential and consumer fans.

1.2 Various "size" factors are important when assessing potential for injury; some factors are: diameter of impeller (wheel, rotor, propeller), rotational inertia, voltage, and current.

1.3 This guide is intended to assist in the safe installation of air moving equipment and to warn operating and maintenance personnel of the commonly recognized hazards associated with this equipment.

1.4 ***Handling and installation should always be performed only by experienced and trained personnel who are aware of the hazards associated with rotating equipment. Failure to comply with these practices may result in death or serious bodily injury.*** In addition to following the manufacturer's installation instructions, care should be taken to ensure compliance with specific safety requirements mandated by federal, state, and local codes. Industry safety standards and practices published by AMCA and by other recognized agencies and associations should be consulted and followed where applicable.

2. PERSONNEL SAFETY ACCESSORIES

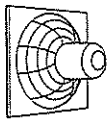
2.1 GENERAL

2.1.1 Protective devices are incorporated as standard construction on some types of fans but on many fans, these devices are offered as optional accessories. This is done because the need for the devices and the design required will frequently depend upon the type of system, fan location, and operating procedures being employed. Proper protective safety devices; company safety standards; specific safety requirements mandated by federal, state, and local codes; and industry safety standards and practices published by AMCA and by other recognized agencies and associations should be determined by the user, who should specify and obtain the appropriate devices from the fan manufacturer or others, and should not allow operation of the equipment without them. Examples of available devices include the following:

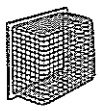
2.2 FAN GUARDS

2.2.1 All fans have moving parts which require guarding in the same way as other moving machinery. Fans located less than seven (7) feet above the floor require special consideration. Specific safety requirements should comply with mandated federal, state, and local codes; and industry safety standards and practices published by AMCA and by other recognized agencies and associations should be followed.

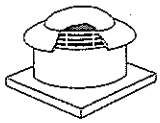
2.2.2 Roof-mounted fans and other fans which are not generally accessible may not require safety guards which might otherwise be appropriate. Where accessibility to these fans is occasional or infrequent, the expense of permanent guarding may be reduced through the use of lockout switches and suitable warnings. In such cases, maintenance personnel should engage the lockout switch before undertaking any maintenance or repairs. As is the case with other machinery involving moving parts, common sense and caution will preserve personal safety.



Industrial Type Guard For Propeller Fan



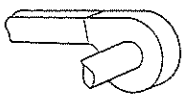
Maximum Safety Guard for Propeller Fan



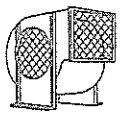
Screen on Roof Ventilator

2.3 INLET AND OUTLET GUARDS

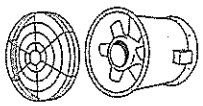
2.3.1 Axial and centrifugal fans are often connected directly to ductwork which will prevent contact with the internal moving parts; when an exposed inlet or outlet represents a hazard, a suitable guard should be installed.



Centrifugal Fan Protected by Ductwork



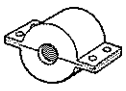
Inlet or Outlet Guard on Centrifugal Fan



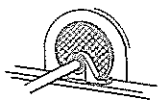
Guard for Axial Fan With Non-Ducted Inlet or Outlet

2.4 DRIVE GUARDS

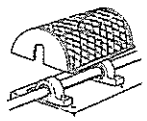
2.4.1 Fans may be driven directly from the motor shaft or through a belt drive. Where the bearing assembly, rotating shaft, sheaves, or belts are exposed, a suitable guard may need to be provided. Some example guards are shown below.



Drive Coupling Guard

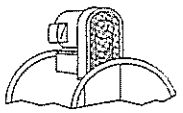


Heat Slinger Guard (Shaft and bearing guard omitted for clarity)



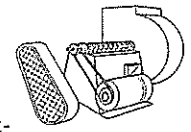
Shaft and Bearing Guard

2.4.2 Drive guards may be required for tubular centrifugal or axial fans to cover the exposed drive sheave and belts outside the fan housing.



Drive Guard - Axial Fan

2.4.3 A typical centrifugal fan drive guard may vary with the arrangement. Safety guards should be used when drive systems are accessible to personnel. In restricted areas, omission of the back cover may be acceptable.



Drive Guard - Centrifugal Fan

2.4.4 Dampers and their linkage may operate suddenly without warning at high speeds. Dampers and their linkage contain pinch points which should be identified and guarded.

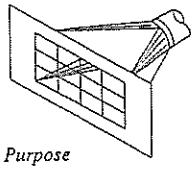
3. HIDDEN DANGERS

3.1 GENERAL

3.1.1 In addition to the obvious hazards associated with the moving parts of rotating machinery, fans present additional potential hazards that are not so obvious and should be considered by the system designer and user for safe operation.

3.2 SUCTION AND AIR PRESSURE

3.2.1 Fans operate by creating suction and air pressure which can be hazardous. Solid objects can be drawn into a fan's inlet and then become dangerous projectiles when they are exhausted through the fan's outlet. ***Solid objects can also cause fan failure or impeller failure due to imbalance or damage to the impeller blades.*** Personnel in close proximity to a fan inlet can be overcome by the suction, and drawn into the fan.



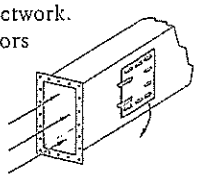
Special Purpose Intake Screen

3.2.2 Whenever there is a possibility that solid objects can be drawn into a remote intake, the intake should be guarded at all times. Before a guard is removed, the fan should be disconnected and the power supply locked out.

3.2.3 Where fans are installed over an occupied area, safety guards should be provided to prevent dropped objects from entering this area during installation and maintenance.

3.2.4 Access doors to a fan or duct system should never be opened while the fan is operating or coasting to a stop. On the downstream (or pressure) side of the system, releasing the door with the system in operation may result in an explosive opening. On the upstream (or suction) side, the inflow may be sufficient to draw in tools, clothing, and other materials. The power supply should always be locked out prior to accessing a fan or ductwork.

3.2.5 Fan design sometimes requires access doors to be supplied with internal components such as a plug to fill a hole in the fan casing.These doors can often be heavy and difficult to handle. Care should be exercised when opening, removing, and installing these components.



Bolted Access Door in Duct

3.3 WINDMILLING

3.3.1 Even when the power supply is locked out, fans may cause injury or damage if the impeller is subject to "windmilling" which is the turning of the impeller and drive components due to a draft in the system. To guard against this hazard, the impeller should be secured to physically restrict rotational movement.

3.4 TEMPERATURE

3.4.1 Many fans, fan motors, and fan components run at temperatures that could burn someone who comes in contact with the hot areas, including discharged or leaking gases. If this potential hazard is present, steps should be taken so that personnel working near the fan are aware of the danger and can exercise caution.

3.5 FAN NOISE AND ENVIRONMENT

3.5.1 Some fans can generate sound that could be hazardous to exposed personnel. Sound pressure can be measured in the field, but obtaining accurate data is difficult. The environment in which the fan operates can impact the ability to obtain accurate fan sound readings. Consult the manufacturer for fan sound data. It is the responsibility of the system designer, installer, user, and maintainer to comply with specific safety requirements mandated by federal, state, and local codes; and to follow industry safety standards and practices published by AMCA and by other recognized agencies and associations, regarding personnel safety from exposure to fan noise associated with use and exposure to the equipment.



Hearing Protection

3.6 STROBOSCOPIC EFFECT

3.6.1 The stroboscopic effect of certain lights in combination with certain fan speeds may cause a rotating assembly to appear stopped. In these cases, irregular markings can be placed on the moving parts to prevent this type of effect. Personnel should be warned that the fan may be in motion even if it appears not to be.

3.7 SPECIAL PURPOSE FANS AND SYSTEMS

3.7.1 The hidden dangers associated with Special Purpose Fans used in special systems are covered in Section 6.

4. POWER ISOLATION

4.1 Every fan should be installed with a suitable device allowing it to be completely disconnected or isolated from the power supply.

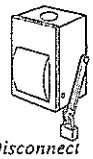
4.2 Many fans are started by remote switches or push-buttons,

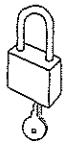
by interlocks with other equipment, or by automatic controls. Before performing any maintenance, inspection, or other activity which will require removal of guards, ductwork, access doors, etc., or exposure of moving parts, the fan power supply should be locked out and the fan tagged out of service.

4.3 In some installations other equipment, such as gas burners, may be interlocked with the fan so that disconnecting the fan will automatically shut off the burner or other device. Maintenance on systems of this type should be performed only under the supervision of competent engineering personnel and in accordance with applicable codes and standards.



Remote Switch


Disconnect Switch


Lock Carried by Maintenance Personnel

4.4 In cases where the fan is power driven by a source other than an electric motor, appropriate provisions should be made for the isolation or disengagement of the power supply.

5. START-UP CHECK LIST

5.1 GENERAL

5.1.1 Before putting any fan into initial operation, the manufacturer's instructions should be followed. Transportation, handling, and installation can cause fasteners to loosen, and cause misalignment of fan components. Carefully follow this check list when commissioning equipment.

5.1.2 Lock out the primary and all secondary power sources.

5.1.3 A complete inspection should be made of all of the ductwork and the interior of the fan. Make certain there is no foreign material which can be drawn into or blown through the fan or ductwork. Appropriate protective measures and safety practices should be observed when entering or working within these areas. These measures might include the use of goggles, respirators, or other personal protective devices.

5.1.4 Make sure the foundation or mounting arrangement and the duct connections are adequately designed and installed per drawings and in accordance with recognized acceptable engineering practices and with the fan manufacturer's recommendations.

5.1.5 Check and tighten all bolts, fasteners, and set screws as necessary.

5.1.6 Check the fan assembly and bearings for proper grounding to prevent static electricity discharge.

5.1.7 Ensure power and drive components such as motor starter, variable frequency drive, or hydraulic power unit are properly sized, matched, and connected to the fan.

5.1.8 Check bearings for recommended lubricant and lubrication amount.

5.1.9 Spin the rotating assembly to determine whether it rotates freely, without hitting anything, and is not grossly out of balance.

5.1.10 Inspect impeller for proper rotation for the fan design.

5.1.11 Check alignment of drives and all other components.

5.1.12 Check the belt drive for proper sheave selection and installation and make sure the sheaves are not reversed (excessive speeds could develop).

5.1.13 Check for recommended belt tension.

5.1.14 Properly secure all safety guards.

5.1.15 Assure that all appropriate warnings have been put in place.

5.1.16 Secure all access doors to the fan and ductwork.

5.1.17 Momentarily energize the fan to check the direction of rotation. Listen as the fan coasts to a stop for any unusual noise, identify the source, and take corrective action as necessary.

5.1.18 Switch on the electrical supply and allow the fan to reach full speed. Check carefully for:

(1) Excessive vibration
(2) Unusual noise
(3) Proper belt alignment
(4) Proper lubrication
(5) Proper amperage, voltage, or power values.
(6) If any problem is indicated, SWITCH OFF IMMEDIATELY.
(7) Lock out the power supply. Secure the fan impeller if there is a potential for windmilling. Check carefully for the cause of the trouble, correct as necessary, and repeat check list procedure.

5.2 Even if the fan appears to be operating satisfactorily, shut down after a brief period, lock out the power supply, and recheck items 5.1.5 through 5.1.17 as the initial start-up may have loosened the bolts, fasteners, and set screws.

5.3 The fan may now be put into operation, but during the first eight hours of running, it should be closely observed and checked for excessive vibration and noise. At this time checks should also be made of motor input current and motor and bearing temperatures to ensure that they do not exceed manufacturer's recommendations.

5.4 After eight hours of operation, the fan should be shut down and the power locked out. Check list items 5.1.5 through 5.1.17 should be inspected and adjusted, if necessary.

5.5 After twenty-four (24) hours of satisfactory operation, the fan should be shut down (locked out) and the drive belt tension should be readjusted to recommended tension.

5.6 After commissioning and start-up, the fan should be operated and maintained in accordance with the manufacturer's and component manufacturer's recommendations. Some basic guidelines for WARNING SIGNS and ROUTINE MAINTENANCE are included in Sections 7 and 8 of this publication. These sections are meant as a supplement to other publications and are not intended to replace the manufacturer's instructions.

6. SPECIAL PURPOSE FANS

6.1 Most fans are designed to handle clean air at standard temperatures between 32°F and 120°F. These fans should not be placed in systems or used for other than their design intended use. *Special Purpose Fans* are designed for use in systems that may include extreme temperatures, explosive, toxic, or special gases, material handling, corrosive environments, or other special hazards which should be carefully considered. Specific safety requirements should comply with mandated federal, state, and local codes; and industry safety standards and practices published by AMCA and by other recognized agencies and associations should be followed.

6.2 Where the system will handle explosive or flammable materials (i.e., dust, fumes, vapors or gases), fans of spark-resistant construction should be used.

6.3 Fans connected by ductwork or other piping may contain gases other than air which are hazardous. In these cases, procedures should be established to prevent exposure of personnel working on or near the fan, and by maintenance personnel who may need to enter the fan. Appropriate personal protective equipment as determined by the material safety data sheet, and system operators should be utilized. Appropriate environmental protective measures should also be taken.

6.4 Fan inlet boxes, housings, ductwork, and other system components which are large enough to permit entry should be considered confined spaces. System areas may also serve as low points where heavy gases, liquids, or other substances may accumulate and present explosive, fire, health, or suffocation hazards. Appropriate protective measures and safety practices should be observed when entering or working within these areas.

6.5 Material-handling fans are specially designed to allow the fan to handle a specific type of material without excessive accumulation of material on the fan impeller. Fans handling corrosive gases or erosive materials should be checked periodically. If loss of material is evident, the fan should be shut down, power supply locked out, and tagged out of service. The manufacturer or other qualified personnel should be consulted to determine if the fan is

within safety limits for operation. To ensure satisfactory operation it is essential to observe the manufacturer's limitations concerning the type of material to be handled by the fan.

6.6 Fan ratings and maximum speed limits are typically based on the use of air at 70°F. At temperatures above the normal range (specified by the manufacturer), a reduction should be made in the maximum speed limit. Information on this reduction and on other precautions to be taken for high temperature applications should be obtained from the fan manufacturer. Personnel working near high temperature fans should be aware that coming in contact with the fan's housing, ductwork, or handled gases could result in serious burns. Where the danger of burns is not apparent, appropriate warnings should be posted. Appropriate protective apparel should be worn whenever working in close contact with heated housings or ductwork.

6.7 Corrosive contaminants can be formed when moisture combines with an active airborne chemical. Fans subjected to corrosive contaminants will corrode; however, suitable protective coatings or material, if used in the fan construction, can delay corrosion. Protected fans should be regularly inspected to ensure that the protection remains effective. Personnel working in environments with airborne chemicals may require personal protective apparel equipment.

6.8 Where liquid can accumulate within the fan, provide for the installation of adequately sized drains.

6.9 In those applications where there is a potential for chemical build-up (such as grease, creosote, etc.), periodic cleaning and proper drainage are necessary to avoid a fire hazard.

7. WARNING SIGNS

7.1 GENERAL

7.1.1 A change in the operating characteristics of a fan may indicate the need for maintenance. Sudden changes may indicate severe problems or dangerous conditions developing. Investigate any changes in the operational characteristics or unusual symptoms of the fan. Refer to AMCA Publication 202, *Troubleshooting*, for a more detailed explanation of investigating procedures. Consult your manufacturer or other qualified consultant with questions concerning changes observed.

7.2 EXCESSIVE VIBRATION

7.2.1 Operational vibration levels are one of the best indicators of the condition of the blower. Careful observation and monitoring of vibration levels can detect a minor problem in the early stages of development when correction is less costly and easier. Recommended maximum vibration levels should be obtained from the equipment manufacturer.

7.2.2 If excessive vibration is observed, stop the fan and lock it out until the cause is corrected. Check for material build-up on the impeller. Generally this will show up as material flaking off the fan impeller and causing an imbalance which may lead to catastrophic failure of the fan or its components. Excessive vibration can also be caused by looseness in the drive train, loose fasteners, misalignment or impeller damage. Contact the fan manufacturer or other qualified consultant to determine the maximum vibration level if it is not included in maintenance instructions.

7.3 NOISE

7.3.1 Changes to the sound level may indicate maintenance is needed. Some unusual noises often heard include: bearing noise indicating the bearings need lubricant or replacement; scraping or ticking noise indicating the rotating parts are hitting the stationary parts; squealing indicating the belt drive needs tensioning; repeated changing pitch of the blower indicating operation of the blower at too low a flow. If any of these noises or any other unusual noises are detected, their cause should be determined and corrective action taken as necessary.

7.4 HIGH MOTOR TEMPERATURES

7.4.1 Check that cooling air to the motor has not been diverted or blocked by dirty guards or similar obstacles. Check the input amperage. An increase in amperage may indicate that some major change has occurred in the system.

7.5 HIGH BEARING TEMPERATURES

7.5.1 This condition is usually caused by improper lubrication; this can be either "over," "under," or "unsuitable" lubrication. In every case, if the cause of the trouble is not easily seen, experienced personnel should examine the equipment before it is put back in operation.

7.6 POOR PERFORMANCE

7.6.1 Too much flow or pressure or too little flow or pressure is often a symptom of a change in the operating system. A fan will typically operate at the same performance in a static system. Some typical causes include: operating of the fan backwards after maintenance procedures; filters dirty or not in place; change or blockage in the ductwork; change in speed of the fan (switching the sheaves); loss or failure of the impeller. All of these causes and many others will affect the flow and pressure produced by the fan.

8. ROUTINE MAINTENANCE

8.1 A preventive maintenance program is an important aspect of an effective safety program. Consult your manufacturer or other qualified consultant with questions concerning changes observed during periodic inspections and routine maintenance.

8.2 The fan manufacturer's operating and maintenance recommendations, as well as the components manufacturer's instructions (such as motor, bearing, drives, etc.) should be strictly followed.

8.3 *Maintenance should always be performed by experienced and trained personnel who are aware of the hazards associated with rotating equipment.* Do not attempt any maintenance on a fan unless the fan power supply has been locked out and tagged out and the impeller has been secured.

8.4 When performing maintenance functions which include disassembly of the fan, careful consideration should be given to the size, weight, center of gravity, and lifting means of the fan components. It should also be noted that the outboard bearing on some fans such as arrangements 1, 8, 9, and 10 is often cap-loaded. Removal of the securing means may result in a sudden change in impeller position.

8.5 Historical data is often the best indicator for determining the operational condition of the fan. Maintenance logs which include relubrication, vibration levels, temperature levels, power requirements, inspections, and other pertinent records should be maintained and consulted as necessary when assessing the condition of the fan.

8.6 Under normal circumstances, handling clean air, the system should require cleaning only once a year. However, the fan and system should be checked at regular intervals to detect any unusual accumulation.

8.7 The fan impeller should be specially checked for build-up of material or dirt which may cause an imbalance with resulting undue wear on bearings and belt drives. A regular maintenance program should be established as needed to prevent material build-up.

8.8 Periodic inspection of the rotating assembly should be made to detect any indication of weakening of the rotor because of corrosion, erosion, or metal fatigue. Where signs of deterioration are found, lock out and tag out the impeller until the unit has been inspected and approved by a qualified consultant.

INSTALLATION, OPERATION,

& MAINTENANCE

MANUAL

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HARTZELL FAN, INC., PIQUA, OH 45356
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HARTZELL WARRANTY

LIMITED WARRANTIES

Hartzell represents to Buyer that any goods to be delivered hereunder will be produced in compliance with the requirements of the Fair Labor Standards Act of 1938 as amended.

Hartzell also warrants to Buyer its goods to be free from defects in workmanship and material under normal use and service for one (1) year after tender of delivery by Hartzell. No warranty extends to future performance of goods and any claims for breach of warranty or otherwise accrues upon tender of delivery.

The foregoing constitute Hartzell's sole and exclusive warranties and are in lieu of all other warranties, whether written, oral, express, implied or statutory.

LIMITATION OF LIABILITY FOR BREACH OF WARRANTY

Hartzell's obligation for any breach of warranty is limited to repairing or replacing, at its option, without cost to Buyer at its factory any goods which shall, within such a warranty period, be returned to it with transportation charges prepaid, and which its examination shall disclose to its satisfaction to have been defective. Any request for repair or replacement should be directed to Hartzell Fan, Inc., P.O. Box 919, Piqua, Ohio 45356. Hartzell will not pay for any repairs made outside its factory without its prior written consent. This does not apply to any such Hartzell goods which have failed as a result of faulty installation or abuse, or incorrect electrical connections or alterations, made by others, or use under abnormal operating conditions or misapplication of the goods.

LIMITATION OF LIABILITY

To the extent the above limitation of liability for breach of warranty is not applicable, the liability of Hartzell on any claim of any kind, including negligence, for any loss or damage arising out of or connected with, or resulting from the sale and purchase of the goods or services covered by these Terms and Conditions of Sale or from the performance or breach of any contract pertaining to such sale or purchase or from the design manufacture, sale, delivery, resale, installation, technical direction installation, inspection repair, operation or use of any goods or services covered by these Terms and Conditions shall, in no case exceed the price allocable to the goods or services which gave rise to the claim and shall terminate one year after tender of delivery of said goods or services.

In no event whether as a result of breach of contract, or warranty or alleged negligence, defects, incorrect advice or other causes, shall Hartzell be liable for special or consequential damages, including, but not limited to, loss of profits or revenue, loss of use of the equipment or any associated equipment, cost of substitute equipment, facilities or services, down time costs, or claims of customers of the Buyer for such damages. Hartzell neither assumes nor authorizes any person to assume for it any other liability in connection with the sale of its goods or services.

NO IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS

HARTZELL DOES NOT WARRANT THAT SAID GOODS ARE OF MERCHANTABLE QUALITY OR THAT THEY ARE FIT FOR ANY PARTICULAR PURPOSE. THERE IS NO IMPLIED WARRANTY OF MERCHANTABILITY AND THERE IS NO IMPLIED WARRANTY OF FITNESS.

Safety Accessories, Application and Use Warning

The safe installation and operation of equipment supplied by Hartzell Fan, Inc. is the responsibility of the system designer, installer, maintainer, and user. Since the application and use of its equipment can vary greatly, Hartzell Fan, Inc. offers various product types, optional safety accessories, and sound performance data per laboratory tests. Hartzell Fan, Inc. sells its equipment with and without safety accessories, and accordingly, it can supply such safety accessories only upon receipt of an order. The need for safety accessories will frequently depend upon the type of system, fan location and operating procedures being employed. The proper protective safety accessories to meet company standards, local codes, and the requirements of the Occupational Safety and Health Act must be determined by the user since safety requirements may vary depending on the location and application of the equipment. If applicable local conditions, standard, codes or OSHA rules require the addition of the safety accessories, the user should specify and obtain the required safety accessories from Hartzell Fan, Inc. and should not allow the operation of the equipment without them.

Owners, employers, users, and installers should read "RECOMMENDED SAFETY PRACTICES FOR USERS AND INSTALLERS OF INDUSTRIAL AND COMMERCIAL FANS" published by the Air Movement Control Association, Inc., 30 West University Drive, Arlington Heights, Illinois 60004. A copy of this publication is enclosed with each fan shipped from Hartzell Fan, Inc., and is available upon request at Hartzell's office in Piqua, Ohio 45356 (937-773-7411).

Please contact Hartzell Fan, Inc. or your local Hartzell representative for more information on product types, safety accessories, and sound performance estimates.

Remember, the selection of safety accessories and the safe installation, application and operation of equipment supplied by Hartzell Fan, Inc. is <u>your</u> responsibility. This warning supersedes all previous editions.

INSTALLATION, OPERATION, & MAINTENANCE MANUAL

INTRODUCTION:

The purpose of this manual is to aid in the proper installation and operation of fans manufactured by **HARTZELL FAN, INC.** These instructions are intended to supplement good general practices and are not intended to cover detailed instruction procedures, because of the wide variety and types of fans manufactured by **HARTZELL FAN INC.**

The safe installation and operation of fans is the responsibility of the system designer, installer, maintainer, and user. From the initial system design through the life of the equipment, safety should be a foremost consideration. Some areas, which require some special attention, include system design, layout and construction, fan performance specifications, foundation and installation details, storage procedures, start-up and commissioning procedures, operation, maintenance, and repair.

Handling and installation should always be performed by experienced and trained personnel who are aware of the hazards associated with rotating equipment. Failure to comply with these practices may result in death or serious bodily injury.

Contact your local Hartzell representative for further assistance.

SHIPMENT AND RECEIVING:

All equipment shipped from Hartzell Fan, Inc. is prepared for shipment in accordance with the requirements of the commercial carrier and/or any special considerations required by the nature of the product.
The Bill of Lading or Express Receipt is an acknowledgement by the Transportation Company of the receipt in GOOD CONDITION, meeting the above requirements for the shipment covered by our invoice.

Our responsibility for this shipment has now ceased. We will not be responsible for loss or damage when you give the Transportation Company a clear receipt. (Standard conditions of sale are F.O.B. factory, unless other terms have been quoted and purchased.)

Thoroughly inspect all shipments as soon as received. Keep a record of all equipment received, including inspection details and date of receipt, because of the possibility of partial shipments.

If any of the items called for in this Bill of Lading or Express Receipt are short or damaged, do not accept them until the Freight or Express Agent makes a Damage or Short Shipment Notification on your freight bill or express receipt.

If any concealed loss or damage is discovered, **NOTIFY YOUR FREIGHT OR EXPRESS AGENT AT ONCE** and request an inspection. This is absolutely necessary. Unless you do this, the transportation companies will not entertain any claim for loss or damage. If the agent will not make an inspection, then you should make An Affidavit to the effect that you notified the agent on that particular date and the agent failed to show up. This, with your aforementioned documentation, will properly support your claim.

We will assist you in every possible manner in collecting claims for loss or damage, however this does not make us responsible for collection of claims or replacement of material.

HANDLING:

Handle your equipment with care. Some fans are provided with lifting lugs or holes for easy handling. Others must be handled using nylon straps or well-padded chains and cables, which protect the fan's coating and housing. Spreader bars should be used when lifting large parts.

Axial fans should be lifted by using straps around the fan housing only. **DO NOT LIFT AXIAL FANS BY THE MOTOR, MOTOR BASE, PROP OR FLANGES.**

Centrifugal fans are best lifted using one strap under the fan's scroll and another strap around the bearing base. **DO NOT LIFT CENTRIFUGAL FANS BY THE FAN SHAFT, WHEEL, FLANGES OR INLET SUPPORT.**

Roof ventilators should be lifted by using straps around the fan housing or base only. Spreader bars should also be used to avoid damage to stack caps or hoods. **DO NOT LIFT ROOF VENTILATORS BY THE STACK CAP OR HOOD.**

STORAGE:

If fans are stored for any length of time, they should be stored in a clean, dry location to prevent rust and corrosion. Outdoor storage is not recommended. When outdoor storage is necessary, they should be protected from the elements. Cover the fan inlet and outlet, grease the bearings, and keep motors dry and clean.

EXTENDED STORAGE:

Fans are to be stored in their original containers or equivalent protection and should be kept in a clean, dry, protected warehouse where exercised control over temperature, dust, dew point, shock, and vibration is reasonably maintained.

A) Temperatures: Between 50 degrees F and 120 degrees F.

B) Maximum relative humidity of 60%

C) Shock or vibration: 2 mils maximum to prevent bearings from brinelling. Exceeding this limit will require vibration-dampening material under the units.

Motor bearings (and fan bearings on belt drive units) are to be greased at the time of going into extended storage. Motor shaft (and fan shaft on belt drive units) are to be manually rotated every month and additional grease added, purging some of that in the bearing cavity every six (6) months. ***Grease in bearings is to be purged at time of removal from storage, making sure that an ample supply of fresh grease is in each grease cavity. Grease used must be compatible with that already in motor and fan bearings. (See Page 14)***
All motors with space heaters are to have the heaters connected if storage conditions exceed 60% relative humidity and/or if temperatures are below 50 degrees F.

Motor windings should be meggered at the time the equipment is put in storage. At the time of removal from storage, the resistance reading must not have dropped more than 50% from the initial reading. Contact Hartzell Fan, Inc., Warranty and Service Department, if the motor resistance is less than 50% of the initial reading. NOTE: ***Motors in storage may absorb moisture in their windings resulting in a significant loss of insulation resistance.*** **THE APPLICATION OF POWER TO A MOTOR WITH INSUFFICIENT INSULATION RESISTANCE MAY RESULT IN DAMAGE TO THE MOTOR OR OTHER EQUIPMENT.**

Storage records complying with the above requirements should be maintained.

If an extended warranty is desired, contact your local sales representative for charges and details.

INSTALLATION:

Centrifugal Fans should always be mounted to a flat, level, and rigid structure. The fan base should be shimmed and leveled. Gaps between the foundation and fan base should be grouted. This will ensure permanent alignment and a smooth-running, vibration-free fan, as well as minimize maintenance costs. Failure to properly install the fan base can contribute to excessive vibration.

Poured concrete foundations are recommended, wherever practical, for floor mounted fans. If vibration isolators are required, they should be installed between the fan and the foundation. Tighten all mounting bolts securely with lock washers and lock nuts.

Fans mounted off ground level should be rigidly mounted to a structural platform and be placed as near as possible to or over, a solid wall or column. Supports for suspended fans must be crossbraced for live load support to prevent side sway.

For roof mounted fans, place the fan curb panel on the roof curb, level, and then anchor the unit to the curb using lag screws, neoprene washers, and flat washers. *It is recommended that stacks be independently mounted to the roof and use guy wires to prevent side sway.* AVOID SUPPORTING A STACK DIRECTLY ON THE FAN FLANGE.

In both axial and centrifugal fans, inlet and outlet ducts should be independently supported, and never supported by the fan flanges. Flexible duct connections are recommended. These connections will also minimize noise.
The independent mounting of stacks and ducts to the fan flanges will assure that the fan will not be subjected to external forces which may twist or deform fan housing. This also will ensure that the impeller will not strike the housing or cause misalignment of the sheaves and bearings.

It is recommended that access doors be placed in ductwork just ahead of the fan inlet and just behind the fan outlet for ease of inspection and maintenance. Access doors in a duct system should never be opened with the fan running, to avoid possible equipment damage and personal injury.

No turns in duct should be closer than two and one half fan diameters away from the inlet or outlet of a fan. Walls or flat surfaces should also not be closer than one fan diameter from the inlet side.

Branch entries into the main duct should be spaced such that they do not enter directly opposite each other. A maximum of 45 degree angle between main branch and entering branch should be used as a guideline. *Avoid sudden changes in duct size.* Use a 14 degree included angle, or less, in reducing a duct to a fan, and not more than 30 degrees included angle to the discharge. Elbow turns should be kept to a centerline radius of at least one and one-half diameters of the duct.

Electrical connections for the fan motor must be connected by a qualified electrician, conforming to the National Electrical code and local codes and practices. When making electrical connections, the motor must be connected following electrical characteristics as indicated by the motor nameplate, and adhere to the wiring diagram on the motor nameplate or in the motor terminal box. *Motors may fail immediately if improperly connected.* It is also recommended that an overload device to protect the motor be installed between the current supply and the motor. Recommended tolerances for overload devices should be plus 10 % of the motor full load amperage rating, including the allowance for the motor service factor.

NOTE: *It is common for motors to draw several times full load amperage during startup, for approximately fifteen seconds. Larger sized fans may require twenty seconds or longer.*

STARTUP:

Lock out the primary and all secondary power sources.

A complete inspection should be made of all ductwork and the fan interior. Make certain there is no foreign material, which can be drawn into or blown through the fan or ductwork. Appropriate protective measures and safety practices should be observed when entering or working within these areas. These measures may include the use of goggles, respirators, or other personal protective devices.

Make sure the foundation or mounting arrangement and the duct connections are adequately designed and installed per drawings and in accordance with recognized acceptable engineering practices.

Check and tighten all bolts, fasteners, and set screws as necessary. NOTE: *Forces encountered during shipment, handling and rigging can disturb factory settings.*

Check the fan assembly and bearings for proper grounding to prevent static electrical discharge. NOTE: *Especially important when using FRP fans.*

Ensure power and drive components such as motor starter, variable frequency drive, or hydraulic power unit are properly sized, matched, and connected to the fan.

Check bearings for recommended lubricant and lubrication amount.

Clearance should be checked between the impeller and fan housing. Spin the impeller to determine whether it rotates freely, without hitting anything, and is not grossly out of balance. NOTE: *Prior to shipment, all fans have been thoroughly inspected and have passed stringent operation and balance test.*

Inspect impeller for proper rotation for fan design. Arrows to show direction of rotation and airflow are attached to the fan housings.

Check alignment of drives and all other components.

Check the belt drive for proper sheave selection and installation and make sure the sheaves are not reversed (excessive speeds could develop).

Check for recommended belt tension.

Properly secure all safety guards.

Assure that all appropriate warnings have been put in place.

Secure all access doors to the fan and ductwork.

Restore power and momentarily energize the fan to check the direction of rotation. Listen as the fan coasts to a stop for any unusual noise, identify the source, and take corrective action as necessary.

Switch on the electrical supply and allow the fan to reach full speed. Check carefully for: (1) Excessive vibration (2) Unusual noise (3) Proper belt alignment and tension (4) Proper lubrication (5) Proper amperage, voltage, or power values. If any problem is indicated, SWITCH OFF IMMEDIATELY.
Lock out power supply. Secure the fan impeller if there is a potential for windmilling. Check carefully for cause of the trouble, correct as necessary, and repeat check list procedure. (Ref. Page 15 Trouble shooting chart)

NOTE: *The fan should not need balancing, as it was balanced at the factory to be within stringent vibration levels before shipment. However, there are several things that may cause vibration, such as rough handling in shipment and erection, weak foundations, and alignments. It is recommended that the vibration levels be checked with a vibration analyzer to verify that the vibration is within levels recommended later in this manual. (See Page 13, Table 1.)*

Even if the fan appears to be operating satisfactorily, shut it down after a brief period, lock out the power supply, and recheck startup proceedures, as the initial start-up may have loosened the bolts, fasteners, and set screws.

STARTUP CONT:
The fan may now be put into operation, however during the first eight hours of operation, it should be closely observed and checked for excessive vibration and noise. At this time checks should be made of motor input current and motor and bearing temperatures to ensure that they do not exceed manufacture's recommendations.

After eight hours of operation, the fan should be shut down and the power locked out. Recheck startup proceedures and adjust, as necessary.

After twenty-four hours of satisfactory operation, the fan should be shut down (locked out) and the drive belt tension should be readjusted to recommended tension. (See Page 12) NOTE: (*It is normal for belt drive fans to squeal briefly during startup.)*

MAINTENANCE:

Periodic inspection of all the fan parts is the key to good maintenance and trouble-free fan operation. Frequency of inspection must be determined by the user and is dependent upon the severity of the application. Prepare a maintenance schedule and make sure it is strictly adhered to.

NEVER SERVICE OR ADJUST ROTATING EQUIPMENT WHILE IT IS IN OPERATION. LOCK OUT THE POWER SOURCE BEFORE PERFORMING MAINTENANCE.

Regular fan maintenance should include the following:

1) Check the fan impeller for any buildup of foreign material or wear from abrasion. Both can cause excessive vibration which will lead to damage of the impeller and other fan components. Replace the impeller if excessive wear is noticed. Carefully clean the impeller of any foreign material.

2) Check V-belt drives for proper alignment, tension, and excessive wear.

3) Lubricate the fan and motor bearings.

4) Consult the trouble shooting guide for excessive vibration or noise, insufficient performance, or when the fan does not operate. (See Page 15 Trouble shooting chart)

5) Tighten all bolts and setscrews.

BEARINGS AND LUBRICATION:

All Hartzell belt drive fan bearings are heavy duty; self-aligning ball or roller type, depending on the fan size, motor horsepower, and performance, and are relubricable for continuous service.

Selection of the correct bearing grease and greasing intervals depends on several things. Extreme high or low temperatures, dirty or damp surroundings, and excessive vibration exceeding the "fair" levels in Table 1 are all things that will require more frequent greasing or special greases. (Refer to Page 14 for recommended greases and greasing intervals.)

The motor bearings and the fan bearings on the belt drive fans should be greased at regular intervals. Motor manufacturers' greasing instructions and recommendations should be followed closely. Avoid the use of a pressure greasing system which tends to fill the bearing chamber completely. Do not overgrease. Use only 1 or 2 shots with a hand gun in most cases. Maximum hand gun rating 40 P.S.I. Rotate bearings during lubrication where good safety practice permits. NOTE: On motors with non-regreasable sealed bearings, no lubrication is required for the life of the bearings.

The most frequent cause of bearing failure is not greasing often enough, using incompatible greases or excessive belt tension. Excessive vibration, especially if the bearing is not rotating, will also cause bearings to fail. Bearings must also be protected from water and moisture to avoid internal corrosion.

NOTE: *It is typical for some anti friction bearings to exhibit a running surface temperature in excess of 140 deg. F. This surface will be too hot to touch, but is not cause for alarm.*

BEARING REPLACEMENT:

Fan bearings on belt drive fans should not need to be replaced for many years if the above recommendations are strictly adhered to. However, use the following procedure when bearing replacement is necessary.

1. Lock out power source.
2. Gain access to the fan bearings. On duct, ductaxial, and vaneaxial fans, the fan probably will have to be removed from the duct system. Remove the bearing cover, if any.
3. Loosen the belts by shifting the motor.
4. Remove the impeller and disconnect the lube tubes.
5. Remove the shaft and bearing assembly. Note the position of the bearings' shims.
6. Measure the location of the bearing to the impeller end of the shaft and the bearing spacing.
7. Loosen all bearing/shaft setscrews or other locking device.
8. Remove bearings (may have to be pressed off the shaft).
9. Polish the shaft with fine emery paper (240 Grit or finer) and file the setscrew dimples smooth.
10. Install new bearings on the shaft, making sure that the collars are together, (i.e., facing each other on the shaft) and set screws are in line with each other. Lightly seat one setscrew on each bearing to hold in the approximate position from #5) above.
11. Mount the shaft/bearing assembly in the fan, on shims, with bolts. Do not tighten yet. Just snug up. Loosen setscrews.
12. Center the shaft in the housing (both ends) as closely as possible and shim where necessary. The fan impeller may need to be temporarily installed to get its clearances equal.
13. Tighten the bearing mounting bolts.
14. With a soft-faced mallet, tap the shaft between the bearings while turning the shaft by hand to seat the bearing races. The shaft must turn freely. Tighten all bearing setscrews. Spin the shaft again. Is it free?
15. Reinstall the lube tubes.
16. While rotating shaft purge bearings with fresh grease (Ref. Page 14).
17. Install bearing cover, impeller, and belts, and adjust the motor to get proper belt tension. Also, make sure that the sheaves are properly aligned. (See V-belt drives , Page 10 & 11).
18. If a new shaft is supplied, then ignore items #6 through #8.

Carefully following this procedure will ensure trouble-free service.

V-BELTS DRIVES:

V-belts on Hartzell belt drive fans are oil, heat, and static resistant type, and oversized for continuous duty. With proper installation and maintenance, years of operating efficiency can be added to the life span of the V-belt drive.

V-belt drives should be completely guarded.

A noisy V-belt indicates the need for attention. V-belt noise can be caused by the slapping of the belts against the drive guard or other obstruction. Check for an improperly installed guard, loose belts, buildup of foreign material in the sheave grooves, or excessive vibration. The cause of excessive vibration should be determined and corrected. **NOTE:** ***It is normal for belts to squeal briefly at startup.***

Check belt tension often. Ideal tension is the tension at which the belt will not slip under peak load conditions. Over-tensioning shortens belt and bearing life. Keep belts free from foreign material which may cause slipping. The use of belt dressing is not recommended. (See Page 12 for recommended tensioning.)

Inspect sheaves often. Keep all sheaves' grooves smooth and uniform. Burrs and rough spots along the sheave rim can damage belts. Dust, oil, and other foreign matter can lead to pitting and rust, and should be avoided as much as possible. Badly worn grooves or a shiny groove bottom indicates that either the sheave, the belt, or both are badly worn. Replace either or both belts and worn sheaves.

Check sheave alignment. Sheaves that are not aligned properly cause excessive belt wear and sheave wear.

V-BELT DRIVE REPLACEMENT:

Whether you are just installing new belts or a completely new drive, worn bearings, bent shafts, or other components that might cause future problems should be replaced at this time. If installing belts only, check existing sheaves carefully for worn grooves or other damage.

Although alignment is not as critical in V-belt drives as in others, proper alignment is essential to long belt and sheave life. **NOTE:** ***Loose belts or misalignment can cause fan vibration.***

First, make sure that drive shafts are parallel. ***The most common causes of misalignment are nonparallel shafts and improperly located sheaves.*** Where shafts are not parallel, belts on one side are drawn tighter and pull more than their share of the load. As a result, these belts wear out faster, requiring the entire set to be replaced before it has given maximum service. If misalignment is in the sheave, belts will enter and leave the grooves at an angle, causing excessive belt and sheave wear. (See Page 11, Figure 1).

Shaft alignment can be checked by measuring the distance between the shafts at three or more locations. If the distances are equal, then the shafts will be parallel.

To check the location of the sheaves on the shaft, a straightedge or a piece of string can be used. If the sheaves are properly lined up, the string will touch them at the points indicated by the arrows (See Page 11, Figure 2). Rotating each sheave a half revolution will determine whether the sheave is wobbly or the shaft is bent. Correct any misalignment.

Always use matched belts and never mix new and used belts on a drive. Always replace belts with the right type of V-belt. Install belts correctly. When belts are forced into the sheave with a screwdriver or other wedge, the outer fabric is often ruptured and cords broken. It is well worth the time to move the driver unit closer so the V-belts can be slipped easily into the sheave groove without damage. Shorten the center distance between the driven and driver sheave so the belts can be put on without the use of force. While the belts are still loose on the drive, rotate the drive until all the slack is on one side. Then increase the center distance until the belts are snug. The drive is now ready for tensioning. Tension the belts as indicated on Page 12. **NOTE:** ***Never "roll" or "pry" the belts into the sheave grooves.*** This can damage the belt cords and lead to belt turnover, short life, or actual breakage. Moreover, it is both difficult and unsafe to install belts this way. Keep take-up rails, motor base, or other means of center distance adjustment free of dirt, rust, and grit. Lubricate adjusting screws and slide rails from time to time.

Tighten all sheave bolts and setscrews before reconnecting the power. **NOTE:** ***All sheave bolts, setscrews, and belts should be checked and tightened if necessary after two days of initial operation.***

IMPORTANT: DO NOT INCREASE THE FAN SPEED BEFORE FIRST CONTACTING THE HARTZELL FACTORY. ARBITRARILY SPEEDING UP THE FAN CAN CAUSE MOTOR OVERLOAD, FAILURE, AND POTENTIAL IMPELLER FAILURE.

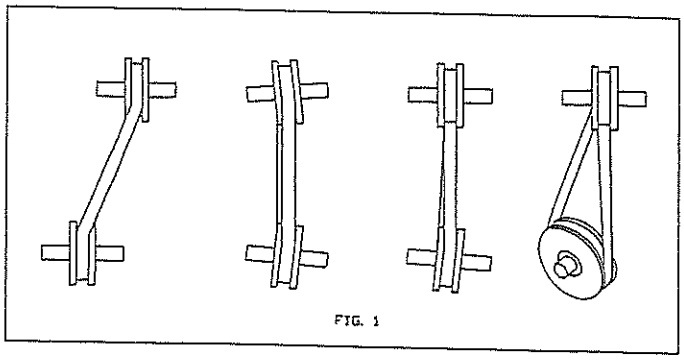


FIG. 1

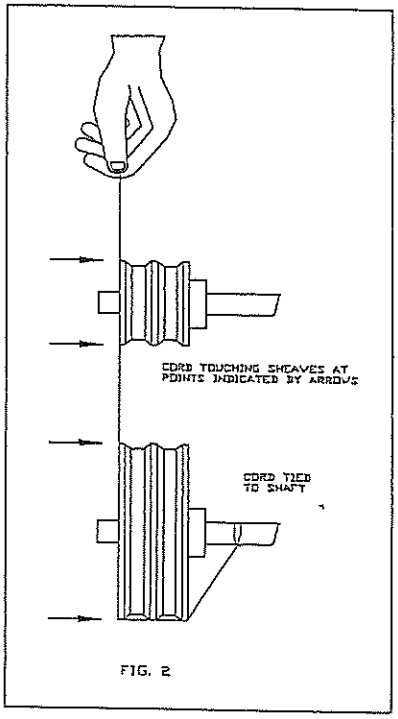


FIG. 2

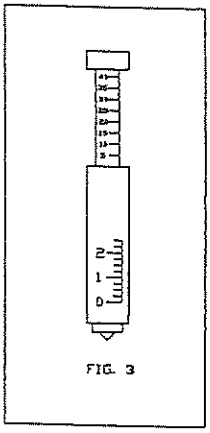


FIG. 3

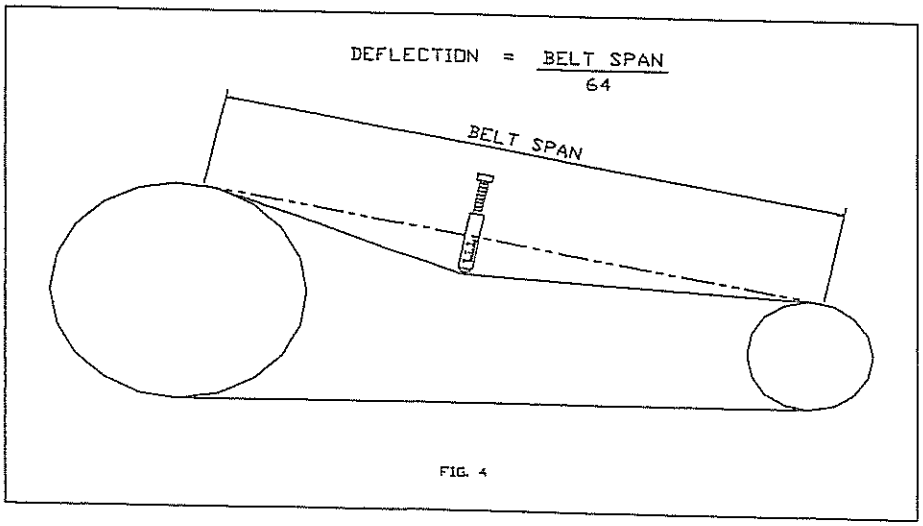


FIG. 4

INSTALLING, TENSIONING AND CHECKING V-DRIVES

GENERAL DRIVE TENSIONING GUIDELINES:

1. IDEAL TENSION IS THE TENSION AT WHICH THE BELT WILL NOT SLIP UNDER PEAK LOAD CONDITIONS.
2. OVER TENSIONING SHORTENS BELT AND BEARING LIFE.
3. KEEP BELTS FREE FROM FOREIGN MATERIAL WHICH MAY CAUSE SLIPPING.
4. MAKE PERIODIC V-DRIVE INSPECTION, TENSION WHEN SLIPPING. THE USE OF BELT DRESSING IS NOT RECOMMENDED.
5. BEFORE INSTALLING A NEW SET OF V-BELTS, CHECK THE CONDITION OF THE SHEAVES, DIRTY OR RUSTY SHEAVES IMPAIR THE DRIVES EFFICIENCY AND ABRADE THE BELTS, RESULTING IN PREMATURE FAILURE. ALSO, WORN SHEAVES CAN SHORTEN THE BELT LIFE BY AS MUCH AS 50%.
6. DO NOT USE A NEW OR USED BELT AS REPLACEMENT FOR A UNITOF A SET. IF A BELT BREAKS A NEW SET OF MATCHED BELTS IS NECESSARY. ALWAYS REPLACE BELTS WITH THE SAME KIND THAT WERE ON THE FAN BEFORE.
7. AFTER PROPERLY TENSIONING THE BELTS, DOUBLE-CHECK TO BE SURE THE SHEAVE GROOVES ARE CORRECTLY ALIGNED, AND THAT ALL SHAFTING IS PARALLEL.

INSTALLATION AND CHECKING METHODS:

I. VISUAL METHOD

1. WHEN INSTALLING BELTS, REDUCE THE CENTER DISTANCE SO THAT THE BELTS MAY BE PLACED IN THE SHEAVE GROOVES WITHOUT FORCING. ARRANGE THE BELTS SO THAT THE TOP AND BOTTOM SPANS HAVE ABOUT THE SAME AMOUNT OF SAG. APPLY TENSION TO THE BELTS BY INCREASING THE CENTER DISTANCE UNTIL BELTS ARE SNUG AND HAVE A LIVE SPRINGING ACTION WHEN STRUCK WITH THE HAND.
2. OPERATE THE DRIVE A FEW MINUTES TO SEAT THE BELTS IN THE SHEAVE GROOVES. OBSERVE THE OPERATION OF THE DRIVE UNDER ITS HIGHEST LOAD CONDITIONS (USUALLY STARTING). A SLIGHT BOWING OF THE SLACK SIDE OF THE DRIVE INDICATES ADEQUATE TENSION. EXCESSIVE BOWING OR SLIPPAGE INDICATES INSUFFICIENT TENSION. IF THE SLACK SIDE REMAINS TAUT DURING THE PEAK LOAD, THE DRIVE IS TOO TIGHT.
3. NEW DRIVE TENSION SHOULD BE CHECKED SEVERAL TIMES DURING THE FIRST 24 HOURS OF OPERATION, BY OBSERVING THE SLACK SIDE SPAN.

II. TENSIONING GAGE METHOD

WHEN A TENSION GAGE IS AVAILABLE AND THE CENTER OF THE BELT SPAN IS ACCESSIBLE, THE FOLLOWING METHOD MAY BE USED. TO DETERMINE THE POUNDS FORCE REQUIRED TO TENSION A DRIVE WITH A BELT TENSIONER, PROCEED AS FOLLOWS:

1. MEASURE THE BELT SPAN AS SHOWN AND CALCULATE THE DEFLECTION INCHES USING THE GIVEN EQUATION. SET LARGE O-RING FOR CALCULATED INCHES OF DEFLECTION.
2. SET SMALL O-RING AT O AND PRESS DOWN THE BELT TENSIONER AT CENTER OF BELT SPAN AS SHOWN.
   A. ON A SINGLE BELT DRIVE, DEPRESS BELT TENSIONER UNTIL THE LARGE O-RING IS EVEN WITH BOTTOM OF A STRAIGHT EDGE PLACED ACROSS THE OUTSIDE EDGE OF THE TWO SHEAVES.
   B. ON MULTIPLE BELT DRIVE, DEPRESS BELT TENSIONER UNTIL LARGE O-RING IS EVEN WITH THE TOP OF THE NEXT BELT. AVERAGE READINGS FROM ALL BELTS IS THE VALUE TO USE IN THE TABLES BELOW.
3. REMOVE TENSION GAGE AND OBSERVE THE NEW POSITION OF THE SMALL O-RING IS SET AT THE NUMBER OF DEFLECTION POUNDS FOR THE SET NUMBER OF INCHES.
4. COMPARE THIS READING, OR THE AVERAGE OF SEVERAL READINGS IN THE CASE OF MULTIPLE BELTS, TO THE NEW/USED VALUES IN THE TABLES BELOW FOR THE PROPER BELT CROSS SECTION. IF READINGS DO NOT FALL IN THIS RANGE, READJUST THE BELT TENSION AS DESCRIBED IN THE VISUAL METHOD AND REPEAT MEASUREMENT.

EXAMPLE:

1. BELT SPAN =1625.6mm AND SMALL SHEAVE IS 203.2mm P.D. WITH COG BELTS.
2. 1625.6mm/64=25.4mm REQUIRED DEFLECTION.
3. SET LARGE O-RING AT 25.4mm ON GAGE INCH SCALE.
4. SET SMALL O-RING AT ZERO ON PLUNGER.
5. PRESS DOWN ON BELTS WITH GAGE UNTIL LARGE O=RING IS EVEN WITH THE NEXT BELT OR A STRAIGHT EDGE, WHICHEVER THE CASE MAY BE. WITH MULTIPLE BELTS, SEVERAL READINGS ARE NEEDED TO GET AN AVERAGE.
6. USE THE POUND FORCE READING OR AVERAGE OF SEVERAL READINGS REQUIRED FOR 25.4mm DEFLECTION IN THE TABLES BELOW.
7. THE "B" BELT TABLE FOR 203.2mm P.D. SMALL SHEAVE SHOULD HAVE A DEFLECTION FORCE BETWEEN 3.311kg AND 4.672kg .
8. INCREASE OR DECREASE THE TENSION ON BELTS UNTIL THE DEFLECTION FORCE IS BETWEEN 3.311kg AND 4.672kg.

(REF. Page 11)
(Figures 3 & 4)

CROSS SECTION	SMALLEST SHEAVE DIA. mm	RPM RANGE	BELT DEFLECTION FORCE			
			STD. BELTS		COG BELTS	
			USED	NEW	USED	NEW
A, AX	76.2-96.5	1000-2500	1.678	2.495	1.860	2.767
		2501-4000	1.270	1.905	1.542	2.268
	96.5-121.9	1000-2500	2.041	3.084	2.268	3.357
		2501-4000	1.724	2.585	1.950	2.903
	127-177.8	1000-2500	2.449	3.629	2.585	3.810
		2501-4000	2.132	3.175	2.313	3.447
B, BX	86.4-106.7	860-2500	2.404	3.583	2.223	3.266
		2501-4000	2.087	3.084	1.905	2.812
	111.8-147.3	860-2500	2.404	3.583	3.221	4.763
		2501-4000	2.041	3.039	3.221	4.128
	147.3-223.5	860-2500	2.858	4.264	3.856	5.715
		2501-4000	2.722	4.037	3.311	4.672
C, CX	177.8-228.6	500-1740	5.216	7.711	6.668	9.888
		1741-3000	4.264	6.260	5.398	7.938
	241.3-406.4	500-1740	6.396	9.525	7.212	10.659
		1741-3000	5.670	8.391	6.713	9.798
D	304.8-406.4	200-850	11.29	16.78	11.34	16.78
		851-1500	9.62	14.20	9.53	14.06
	457.2-508	200-850	13.79	20.50	13.65	20.41
		851-1500	11.70	17.24	11.34	17.24

CROSS SECTION	SMALLEST SHEAVE DIA. mm	RPM RANGE	BELT DEFLECTION FORCE			
			STD. BELTS		COG BELTS	
			USED	NEW	USED	NEW
3V, 3VX	55.9-61.0	1000-2500	N/R	N/R	1.497	2.223
		2501-4000	N/R	N/R	1.270	1.950
	72.4-97.8	1000-2500	1.633	2.313	1.905	2.359
		2501-4000	1.361	1.996	1.724	2.540
	104.6-175.3	1000-2500	2.223	3.311	2.404	3.583
		2501-4000	1.996	2.994	2.223	3.311
5V, 5VX	111.8-170.2	500-1749	N/R	N/R	4.627	6.895
		1750-3000	N/R	N/R	3.992	5.987
		3001-4000	N/R	N/R	2.540	3.856
	180.3-276.9	500-1740	5.761	8.573	6.713	10.024
		1741-3000	5.080	7.575	6.214	9.117
	299.7-406.4	500-1740	7.031	10.61	7.756	11.57
		1741-3000	6.622	9.888	7.620	11.34
8V	317.5-431.8	200-850	14.97	22.36	N/A	N/A
		851-1500	12.16	18.05	N/A	N/A
	457.2-569	200-850	18.05	26.85	N/A	N/A
		851-1500	16.01	23.90	N/A	N/A

MOTORS:

The fundamental principle of electoral maintenance is KEEP THE MOTOR CLEAN AND DRY. This requires periodic inspection of the motor, the frequency of which depends upon the type of motor and the service.

Periodic checks of voltage, frequency, and current of a motor while in operation is recommended. Such checks assure the correctness of frequency and voltage applied to the motor, and yield an indication of the fan load. Comparison of this data with previous data will give an indication of the fan performance. Any serious deviations should be investigated and corrected.

Fractional motors usually have prelubricated sealed bearings with no grease fittings and are lubricated for life.

Lubricate integral horsepower motors per the motor manufacturer's recommendations. Lubrication frequency depends on the motor horsepower, speed, and service. Use compatible greases. (See Page 14). Do not overgrease.

If the motor is totally enclosed fan cooled (TEFC), nonventilated (TENV), or air over (TEAO), it is recommended that the condensation drain plugs be removed. This is not necessary with motors equipped with automatic drains, which should be left in place as received.

VIBRATION:

Excessive fan vibration can be caused by many things. **ALL POSSIBLE SOURCES OF THE EXCESSIVE VIBRATION MUST BE CHECKED OUT AND CORRECTIVE ACTION TAKEN IMMEDIATELY TO CORRECT THE PROBLEM.** See the fan trouble-shooting chart (Table 2) for possible causes of excessive fan vibration.

A vibration analyzer will be of great assistance in determining the amount of vibration. The following values give an indication of the fan vibration condition. Vibration readings should be taken on the fan bearings (if possible, on belt drive fans), or motor (on direct drive fans), or on the fan housing of duct, ductaxial, and vaneaxial fans. Horizontal, vertical, and axial readings should be taken.

TABLE 1

Vibration displacement in mils, "Double Amplitude"

FAN RPM	GOOD	FAIR	SLIGHTLY ROUGH	ROUGH
870	1.7	3.3	6.5	above 6.5
1160	1.2	2.5	5.0	above 5.0
1750	.75	1.5	3.5	above 3.5
3450	.50	0.9	2.0	above 2.0

Interpolate for fan speeds other than shown above. The fan should not be operated unless the maximum vibration reading is in at least the "fair" range.

GREASE LUBRICATION SCHEDULE,

Lubrication guidelines for horizontal shaft, grease lubricated fan, blower or other high speed rotating equipment.
Lubrication guidelines for vertical shaft – see note below.

Link-Belt Ball Bearing Units – Series 200

Relubrication interval				6 mos	4 mos	2 mos	1 mo
Shaft size range		Amt. of Grease		Operating speed (RPM)			
Inches	MM	Cu. In.	Cu. Cm.				
1/2-1	17-25	0.12	2.0	3200	4800	7200	9600
1 1/16 - 1 7/16	30-35	0.30	4.9	2200	3400	5100	6800
1 1/2 - 1 3/4	40-45	0.45	7.4	1700	2600	4000	5300
1 7/8 - 2 3/4	50-55	0.52	8.5	1400	2100	3200	4300
2 1/4 - 2 7/16	60	0.56	9.2	1300	2000	3000	4000
2 1/2 - 3 (1)	65-75	1.36	22.3	1000	1600	2400	3200
3 1/16 - 3 1/2 (1)	85	2.24	36.7	900	1400	2100	2800
3 9/16 - 4 (1)	100	5.00	81.9	800	1200	1800	2300

Link-Belt Roller Bearing Units – Series 22400H & B22500

Relubrication interval				6 mos	4 mos	2 mos	1 mo	2 wk
Shaft size range		Amt. of Grease		Operating speed (RPM)				
Inches	MM	Cu. In.	Cu. Cm.	Up to	Up to	Up to	Up to	Up to
3/4-1	25	0.4	6.4	1400	2200	3400	5000	6800
1 1/16-1 1/4	30	0.5	7.7	1150	1800	2800	4500	5600
1 5/16-1 1/2	35	0.6	9.2	1000	1550	2400	3800	4800
1 9/16-1 3/4	40	0.8	13.1	870	1350	2100	3300	4200
1 13/16 - 2	45-50	0.9	14.6	700	1100	1700	2700	3400
2 1/16-2 1/4	55	1.1	17.9	630	1000	1500	2400	3000
2 5/16-2 1/2	60	1.3	21	580	910	1400	2250	2800
2 9/16 - 3	65-75	2.4	40	460	730	1100	1800	2200
3 1/16-3 1/2	80-85	3.9	64	410	640	1000	1550	2000
3 9/16 - 4	90-100	5.7	94	350	550	850	1350	1700
4 1/16-4 1/2	110-115	6.5	106	300	470	740	1150	1500
4 9/16 - 5	125	10.0	164	280	440	680	1050	1400

Link-Belt Spherical Roller Bearing Units Series P-LB6800

Relubrication interval				6 mos	4 mos	2 mos	1 mo
Shaft size range		Amt. of Grease		Operating speed (RPM)			
Inches	MM	Cu. In.	Cu. Cm.	Up to	Up to	Up to	Up to
1 7/16 - 1 1/2	40	0.3	4.9	2400	3600	5000	5500
1 11/16 - 1 3/4	45	0.3	5.0	2200	3300	4500	5000
1 15/16 - 2	50	0.4	6.6	2000	3000	4000	4500
2 3/16 - 2 1/4	60	0.8	12.7	1700	2500	3400	3800
2 7/16 - 2 1/2	65	0.8	12.3	1450	2200	3000	3400
2 11/16 - 2 3/4	70	0.9	14.3	1350	2000	2800	3200
2 15/16 - 3	75	1.2	19.7	1300	1900	2600	3000
3 3/16 - 3 1/4	80	1.7	27.4	1200	1800	2400	2700
3 7/16 - 3 1/2	90	2.3	37.7	1100	1650	2200	2300
3 11/16 - 4	100	3.1	50	1000	1500	1950	2100
4 3/16 - 4 1/4	110	4.3	70	900	1350	1850	1900
4 7/16 - 4 1/2	115	5.5	90.1	840	1250	1700	1800
4 15/16 - 5	125	6.4	105	780	1150	1600	1700
Clean and repack intervals				5 yrs	3 yrs	2 yrs	1 yr

Link-Belt Ball Bearing Units – Series 300

Relubrication interval				6 mos	4 mos	2 mos	1 mo
Shaft size range		Amt. of Grease		Operating speed (RPM)			
Inches	MM	Cu. In.	Cu. Cm.	Up to	Up to	Up to	Up to
3/4 - 1	20-25	0.3	4.1	2800	4400	6400	8400
1 1/16-1 7/16	30-35	0.7	10.7	2000	3100	4500	6000
1 1/2 - 1 3/4	40-45	1.0	16.4	1500	2400	3500	4600
1 13/16-2 3/16	50-55	1.7	28	1200	2000	2900	3800
2 1/4 - 2 7/16	60	2.1	34	1100	1800	2600	3500
2 11/16-2 15/16	70-75	3.3	54	900	1400	2100	2800
3 - 3 3/16	80	4.5	74	800	1300	2000	2600
3 7/16 - 3 1/2	85	6.6	108	800	1200	1800	2400
3 15/16	100	10	170	700	1100	1600	2100

Lubricate with a premium quality NLGI 2 or 3 multi-purpose ball bearing grease having corrosion inhibitors, anit-oxidant additives and mechanical stability for high speed operation. The grease should also have a minimum base oil viscosity of 500 SUS at 100° F and be suitable to operate continuously at 225° F. Do not use a heavy, long fibered grease.

The bearings on this fan shaft have been greased at the factory for the following application:
- General Purpose (Shell Oil-Shell Alvania EP#2)
- High Temperature (Shell Oil-Aeroshell #22 (-85° F. to 400° F.)
- Low Temperature (Shell Oil-Aeroshell #22 (-85° F. to 400° F.)
- Extreme Moisture (Shell Oil-Shell Alvania EP#2)
- Other:_____

These units are prelubricated(except Series P-LB6800) with a multi-purpose Lithium soap grease. When relubricating with greases of different types (especially synthetic soaps or oils), complete flushing is recommended.

The presence of dirt, moisture or chemical fumes around the bearings requires more frequent lubrication. Ambient temperatures below 20° F or above 200° F require special lubricants. Consult the machinery manufacturer for recommendations.

When vibration exceeds 0.15 in./sec., frequent lubrication is required and a need for equipment trim balancing may be indicated.

Fill bearings with lubricant prior to extended shutdown or storage. Rotate the shaft several revolutions each month during idle periods.

(1) It is possible to dislodge the seals on these units if grease is added too fast. It is preferable to use a hand gun or a regulated automatic system

VERTICAL SHAFTS: For vertical applications more frequent relubrication is required. The above relubrication interval (and clean and repack interval for the Series P-LB 6800 bearings,) should be reduced by half.

TABLE 2 FAN TROUBLE-SHOOTING CHART	
PROBLEM	POSSIBLE CAUSES
EXCESSIVE VIBRATION	-Accumulation of material on impeller -Worn or corroded impeller -Bent shaft -Impeller or sheaves loose on shaft -Motor out of balance -Impeller out of balance -Sheaves eccentric or out of balance -Bearing or drive misalignment -Mismatched belts -Belts too loose or too tight -Loose or worn bearings -Loose bearing bolts -Loose fan mounting bolts -Weak or resonant foundation -Foundation not flat and level -Structures not crossbraced -System pulsation (DANGER) - Fan operation in a stall -Blades at different angles on adjustable pitch fans
HORSEPOWER TOO HIGH	-Fan speed higher than design -Air density higher than design -Impeller rotating in the wrong direction -Angle set too high in adjustable pitch fans
AIRFLOW TOO LOW	-Impeller rotating in the wrong direction -Fan speed lower than design -Actual system is more restrictive (more resistance to flow) than expected. -Dampers or registers closed -Leaks or obstructions in duct work -Filters or coils are dirty or clogged -Inlet or outlet screens clogged -Restricted fan inlet or outlet -No straight duct at fan outlet -Sharp elbows near fan inlet or outlet -Improperly designed turning vanes
AIRFLOW TOO MUCH	-Actual system is less restrictive (less resistance to flow) than expected -Fan speed higher than design -Filter not in place -Registers, grilles and/or dampers not installed
FAN DOES NOT OPERATE	-Blown fuses -Broken belts -Loose pulleys -Electricity turned off or not wired properly -Wrong voltage -Motor too small and overload protector has broken circuit

(CONT.)

TABLE 2 (CONTINUED) FAN TROUBLE-SHOOTING CHART	
PROBLEM	**POSSIBLE CAUSES**
EXCESSIVE NOISE	-Accumulation of material on impeller -Worn or corroded impeller -Bent shaft -Impeller or sheaves loose on shaft -Impeller hitting housing -Impeller out of balance -Bearing or drive misalignment -Mismatched belts -Belts too loose or too tight -Belts hitting guard -Loose or worn bearings -Belts worn -Belts oily or dirty -Defective or bad bearings -Bearings need lubrication -Loose bearing bolts -Loose fan mounting bolts -System pulsation or surge -Electrical noises -Noise from high velocity air system -Duct work too small -Registers or grilles too small for application -Fan in stall condition (DANGER) -Rattle of components in high velocity airstream -Leaks in duct work -Vibrating duct work -Vibrating parts not isolated from building

It is recommended that the users and installers of this equipment familiarize themselves with AMCA publication #201, "Fans and Systems, and publication #202 "Trouble-Shooting", which are published by Air Movement and Control Association, 20 West
University Drive, Arlington Heights, Illinois 60004.

SPARE PARTS:

Spare parts are not needed for the first year of operation. If spare parts are desired, then it is suggested that a spare motor and impeller be ordered for direct drive fans. For belt drive fans, in addition to the motor and impeller, it is suggested that a spare set of bearings, shaft, sheaves, and belts be ordered.

When ordering spare parts, specify the parts desired, the fan model number, and the fan serial number. Contact your local sales representative for price and delivery.

HARTZELL FAN, INC., PIQUA, OHIO 45356
Phone# 937-773-8494
Fax# 937-773-8994